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(12) **Woytowitz**

(54) **SYSTEMS AND METHODS FOR PROVIDING POWER AND DATA TO DEVICES**

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(72) Inventor: **Peter John Woytowitz**, San Diego, CA (US)

(21) Appl. No.: **19/202,936**

(22) Filed: **May 8, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/587,209, filed on Feb. 26, 2024, now Pat. No. 12,302,474, which is a continuation of application No. 18/054,475, filed on Nov. 10, 2022, now Pat. No. 11,917,740, which is a continuation-in-part of application No. 17/098,752, filed on Nov. 16, 2020, now Pat. No. 11,503,694, which is a continuation of application No. 16/450,566, filed on Jun. 24, 2019, now Pat. No. 10,874,003, which is a continuation of application No. 16/172,438, filed on Oct. 26, 2018, now Pat. No. 10,375,793, which is a continuation of application No. 15/404,825, filed on Jan. 12, 2017, now Pat. No. 10,159,132, which is a continuation of application No. 14/700,961, filed on Apr. 30, 2015, now abandoned, which is a continuation-in-part of application No. 13/750,815, filed on Jan. 25, 2013, now Pat. No. 9,521,725, which is a continuation-in-part of application No. 13/230,665, filed on Sep. 12, 2011, now Pat. No. 8,710,770.

(60) Provisional application No. 61/511,934, filed on Jul. 26, 2011.

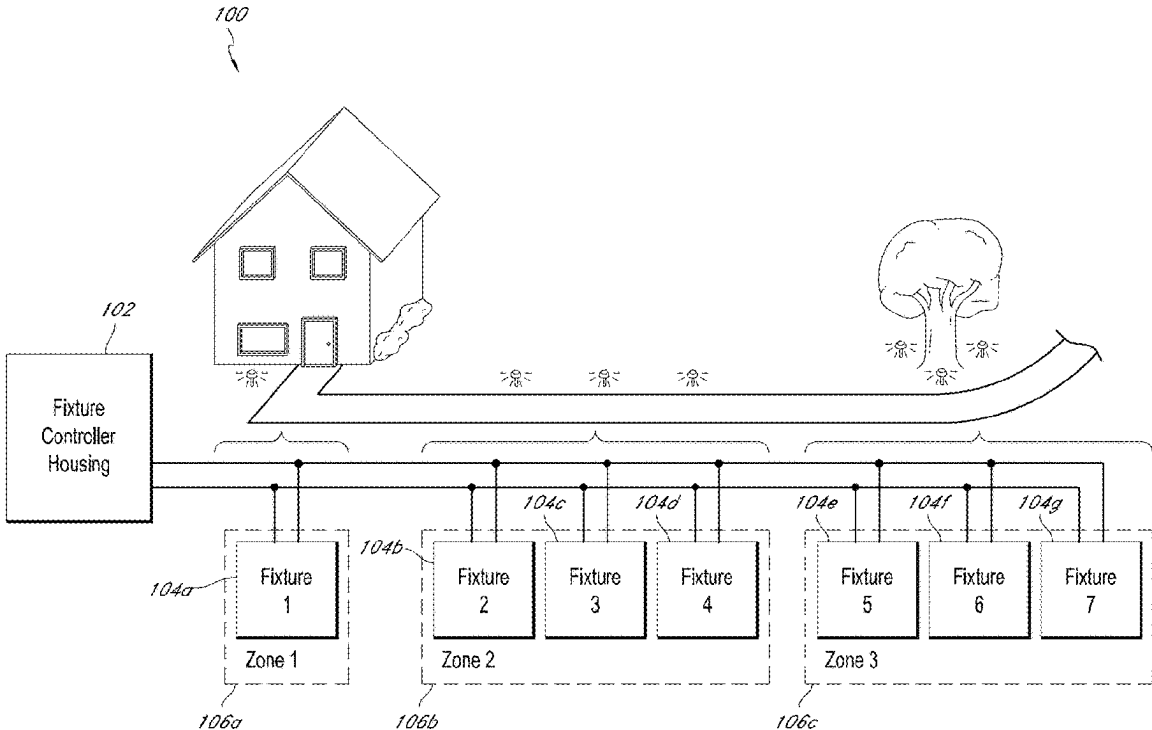
Publication Classification

(51) **Int. Cl.**
H05B 47/185 (2020.01)
H05B 45/20 (2020.01)
H05B 45/3725 (2020.01)
H05B 45/375 (2020.01)
H05B 45/382 (2020.01)
H05B 45/56 (2020.01)
H05B 47/19 (2020.01)

(52) **U.S. Cl.**
CPC **H05B 47/185** (2020.01); **H05B 45/20** (2020.01); **H05B 45/375** (2020.01); **H05B 45/382** (2020.01); **H05B 47/19** (2020.01); **H05B 45/3725** (2020.01); **H05B 45/56** (2020.01); **Y02B 20/30** (2013.01)

(57) **ABSTRACT**

Systems and methods are provided to transmit a data encoded power signal to addressable devices. A lighting controller sends the data encoded power waveform to a plurality of addressable lighting modules to power and control illumination intensity of lighting. The lighting modules are configured to receive the data encoded power signal via a two-wire communication network, decode illumination intensity data from the data encoded power signal, and generate a PWM signal responsive to the decoded illumination intensity data. The lighting modules use the data encoded power signal as a time base to synchronize the PWM signal to the data encoded power signal to reduce flickering when applying the PWM signal to control the illumination intensity of LEDs associated with the lighting modules.



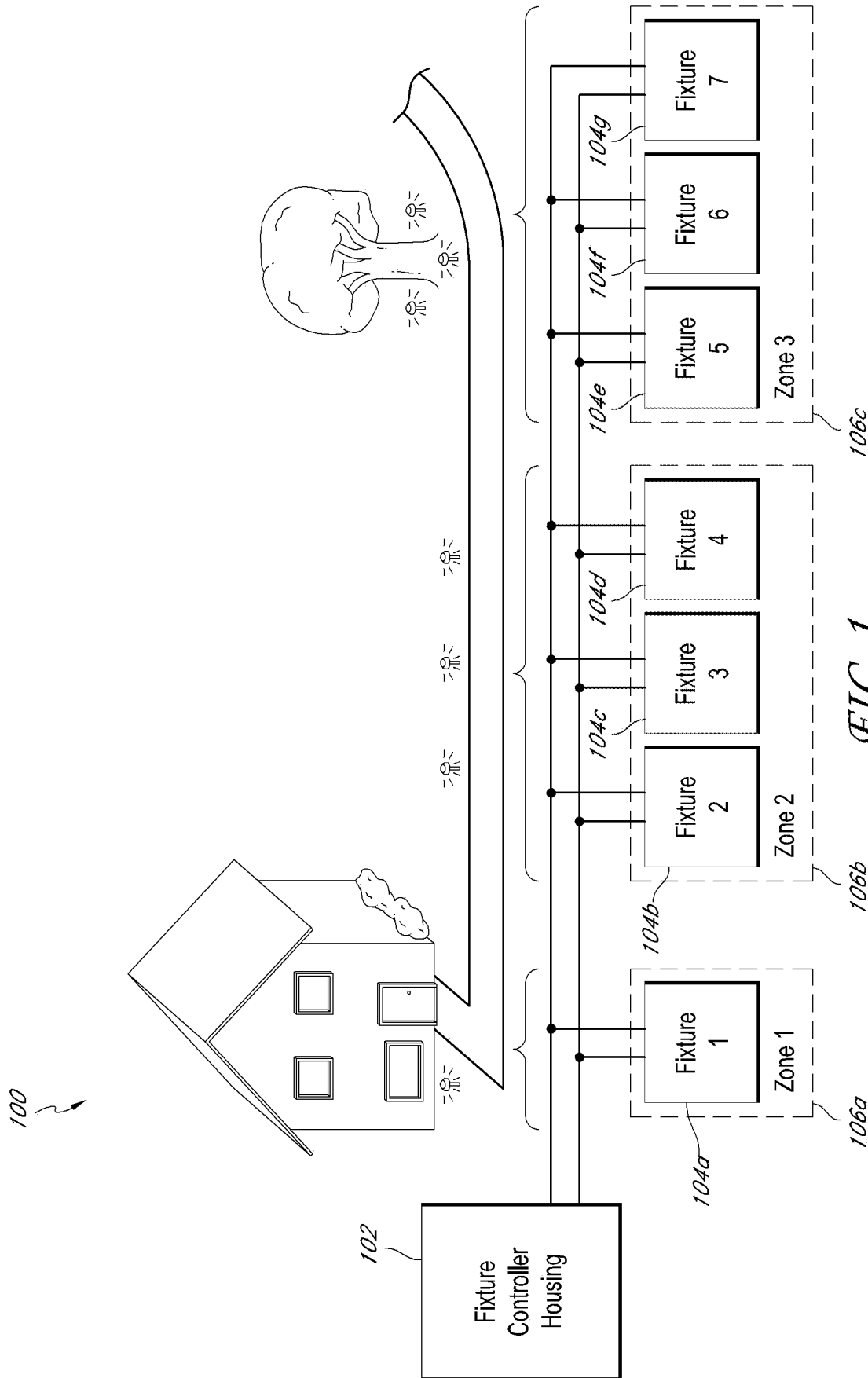


FIG. 1

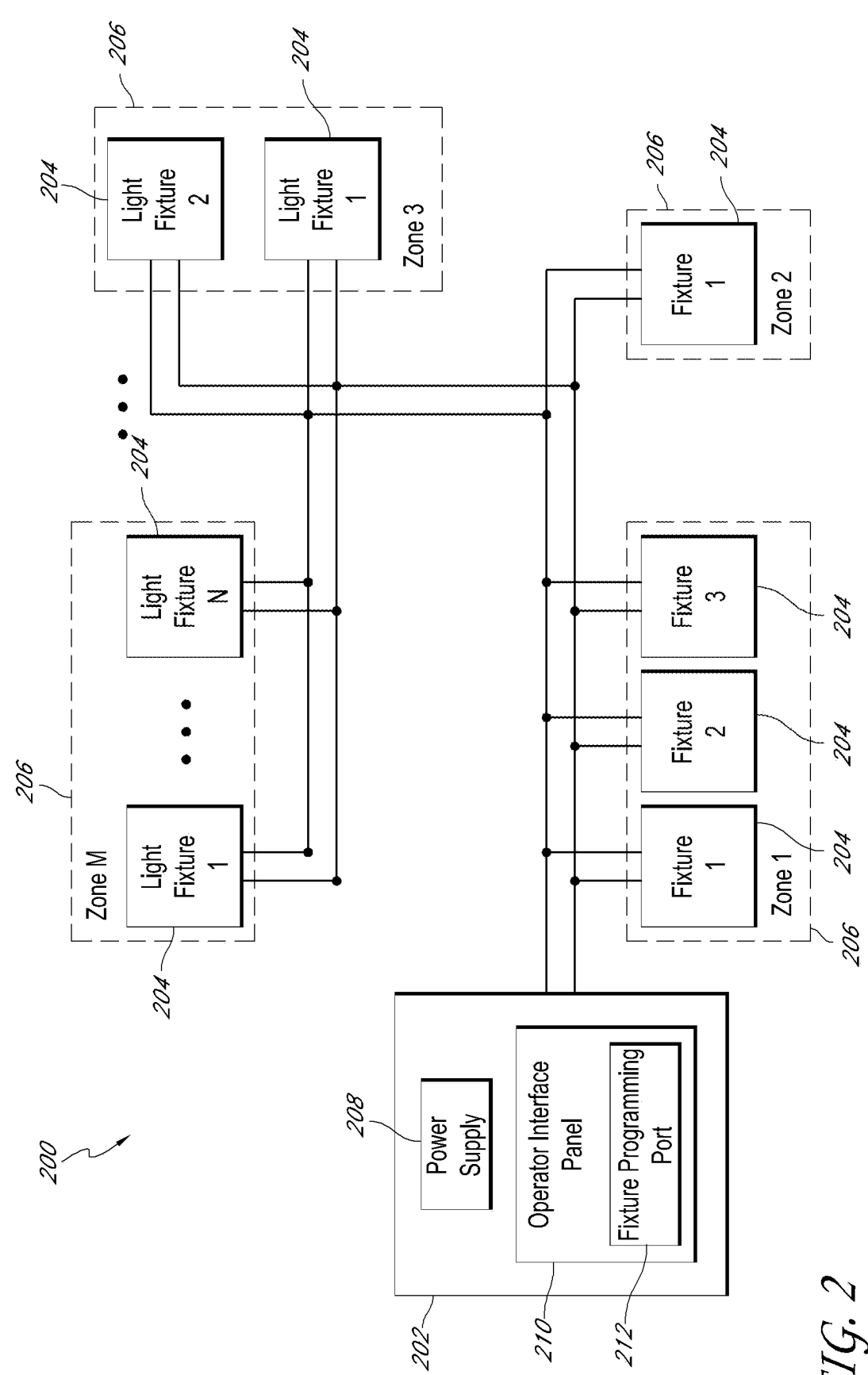


FIG. 2

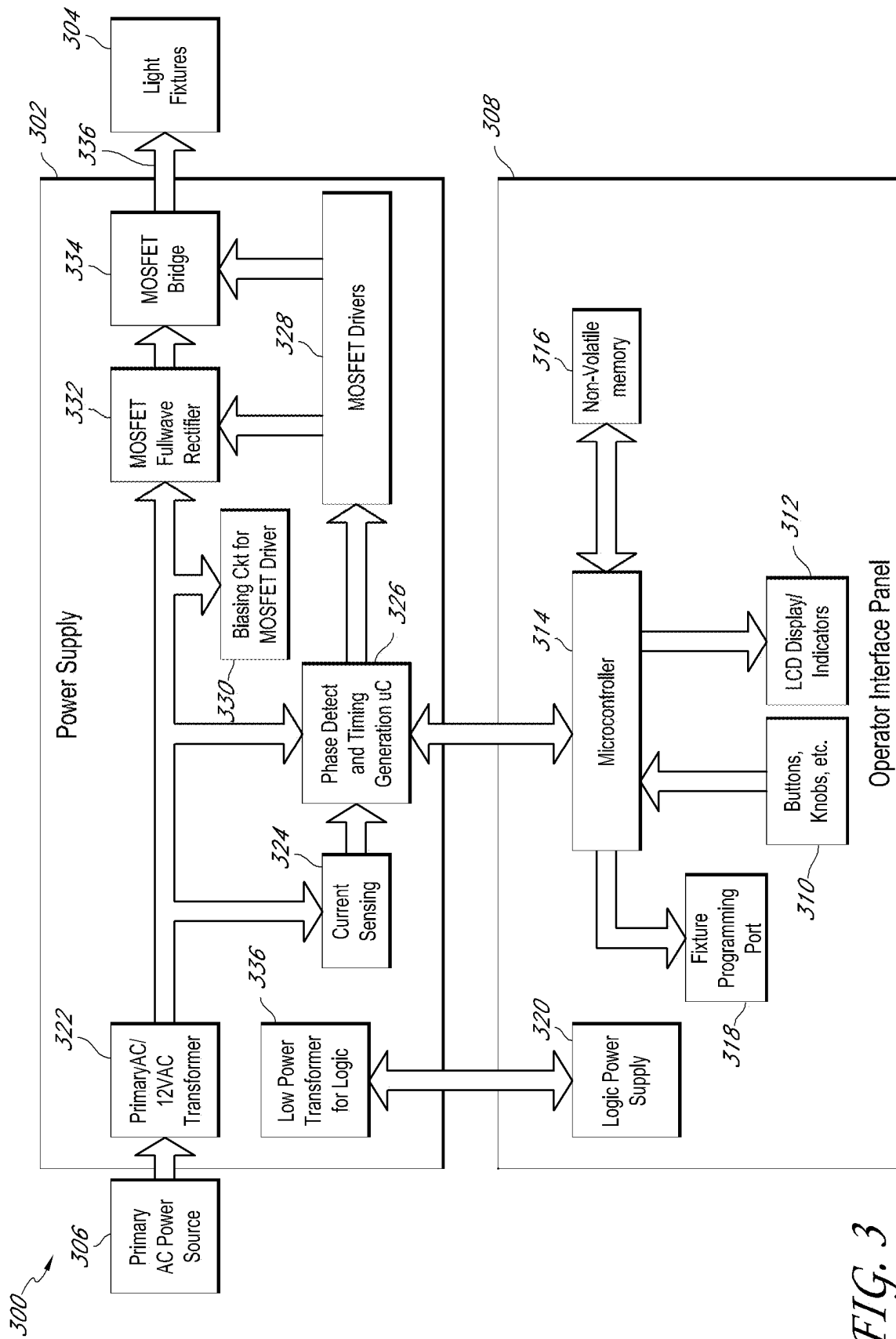


FIG. 3

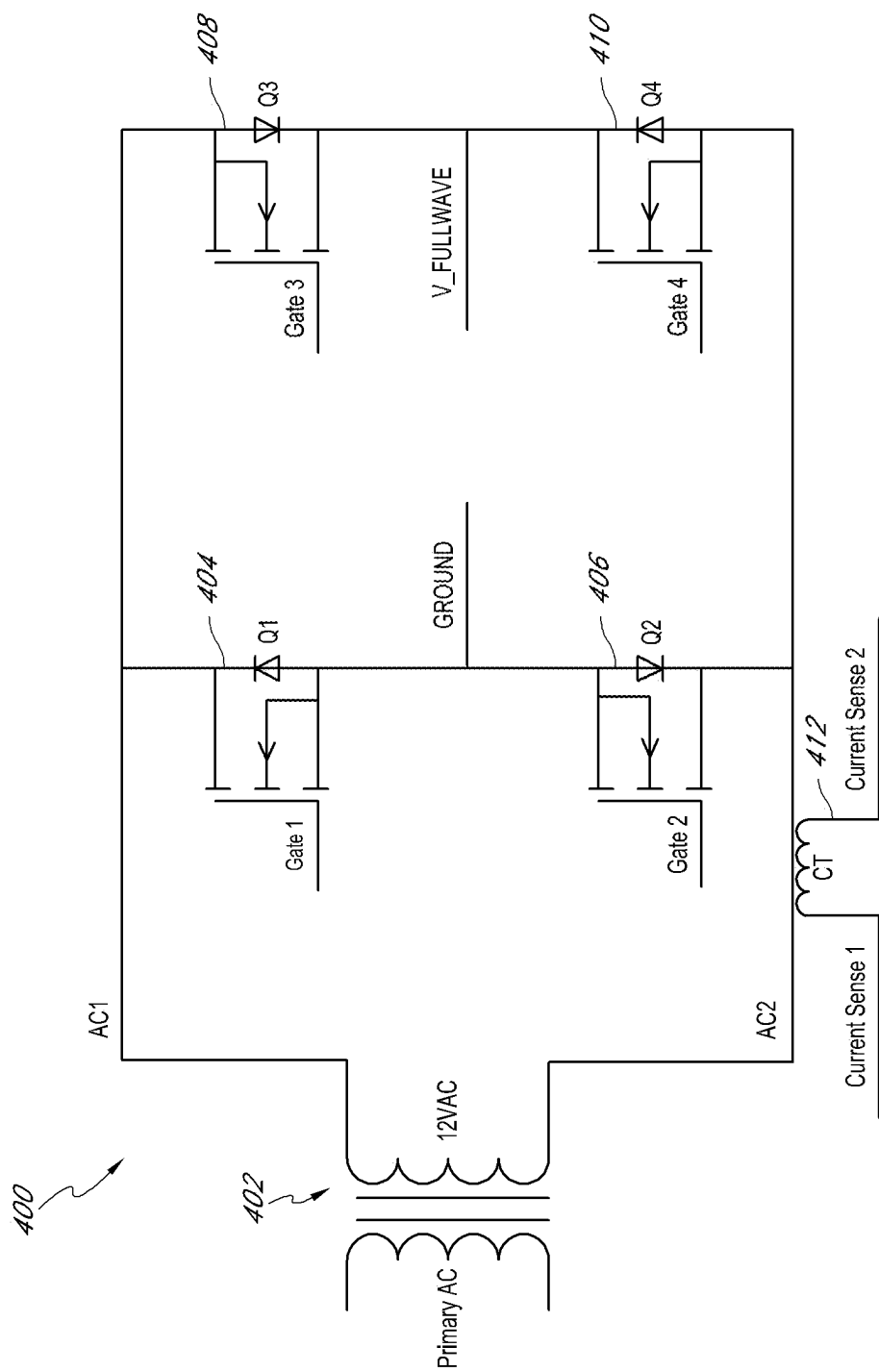
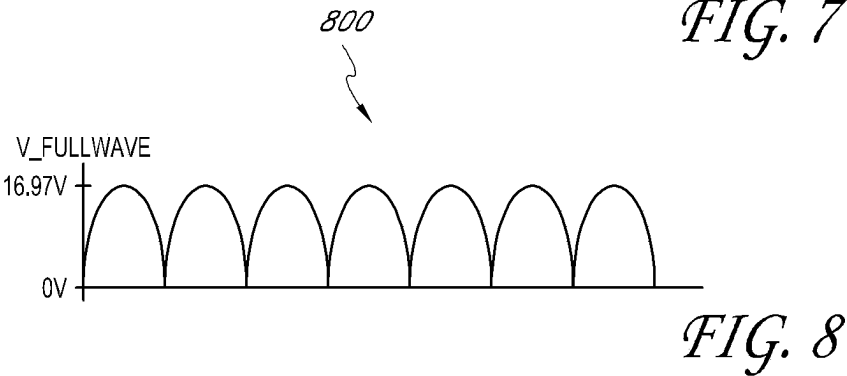
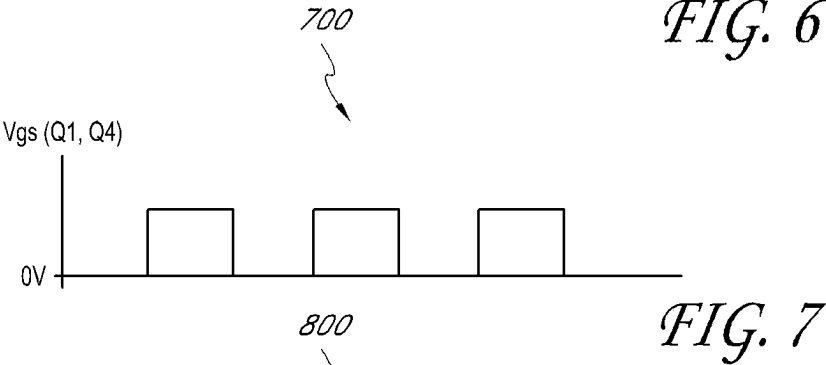
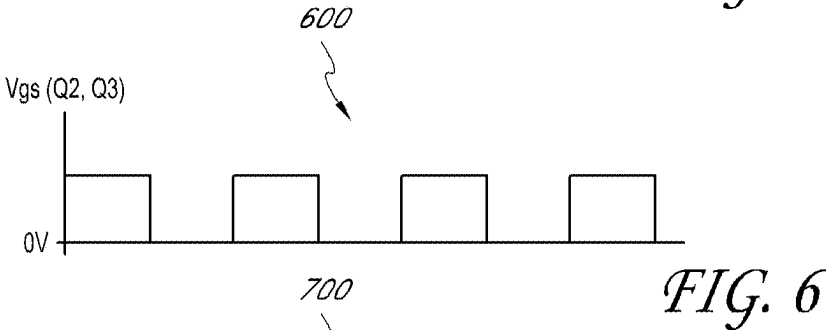
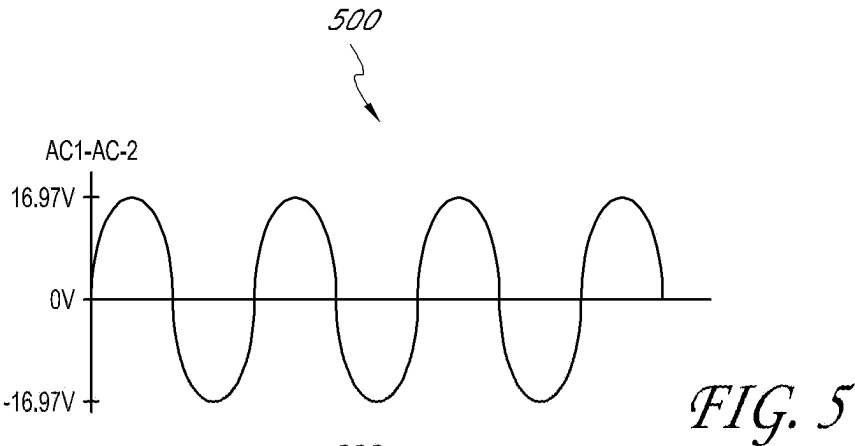


FIG. 4



MOSFET Full Wave Rectifier Waveforms

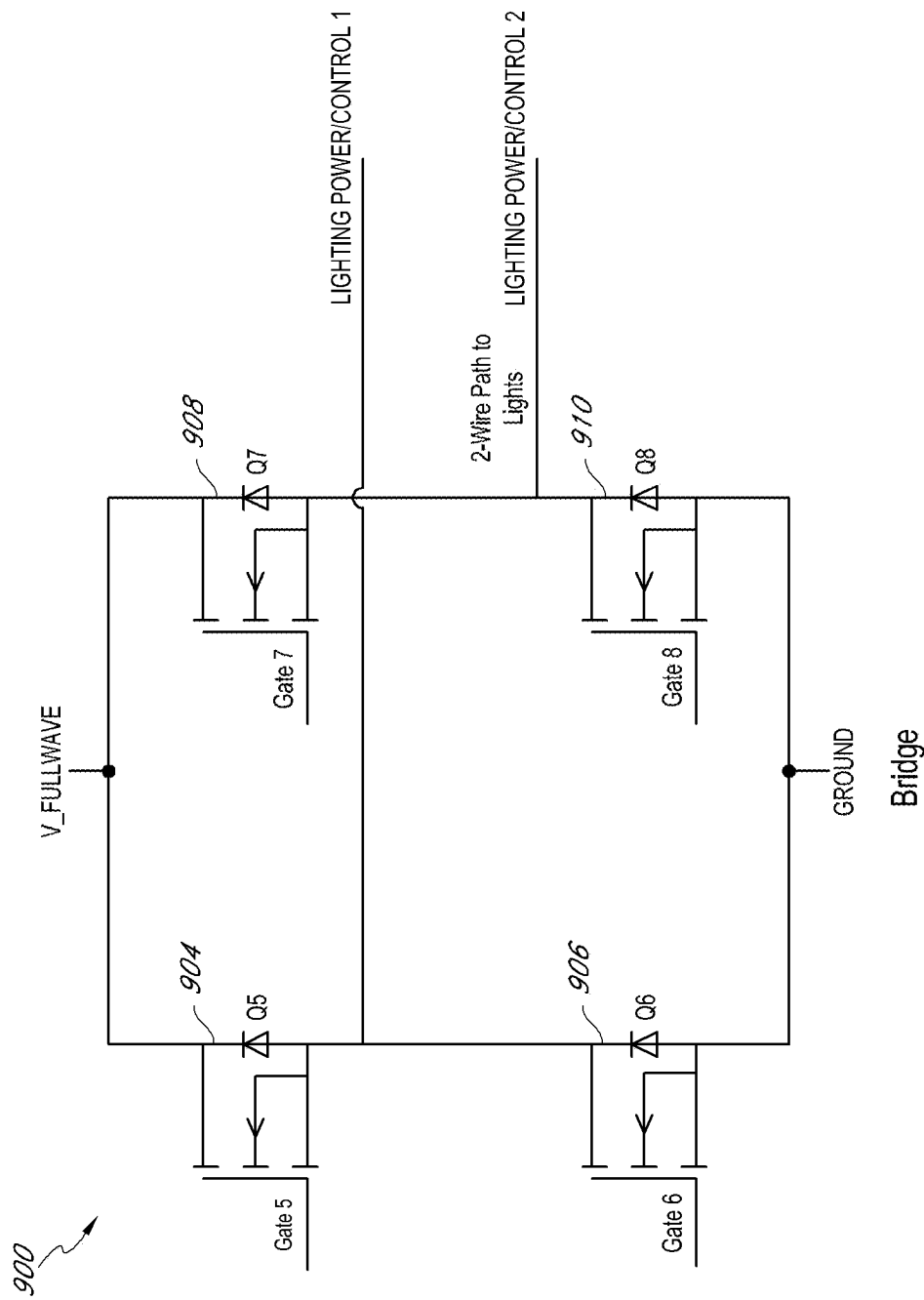
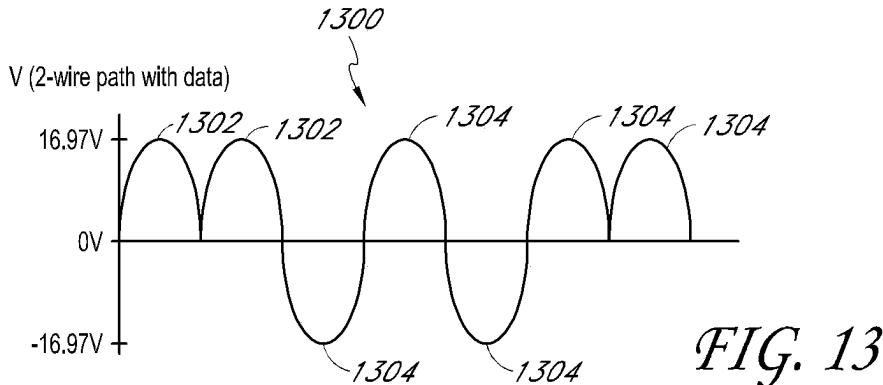
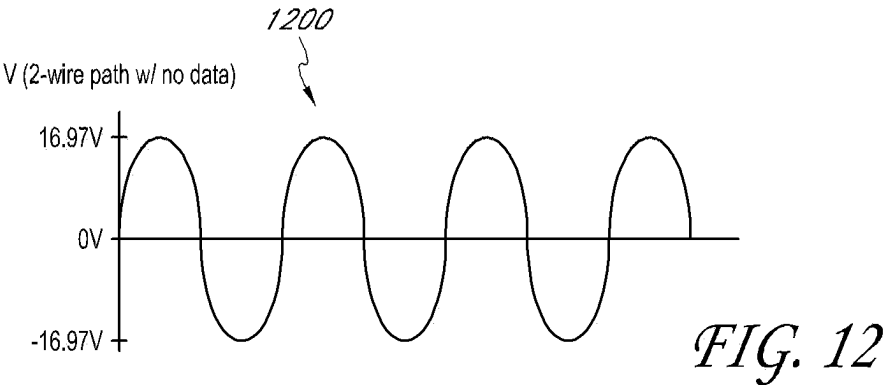
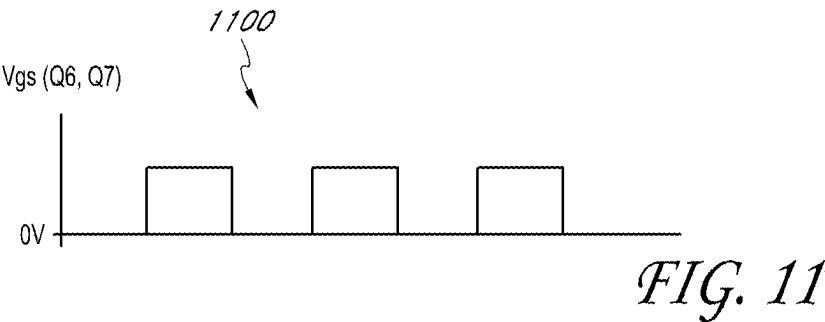
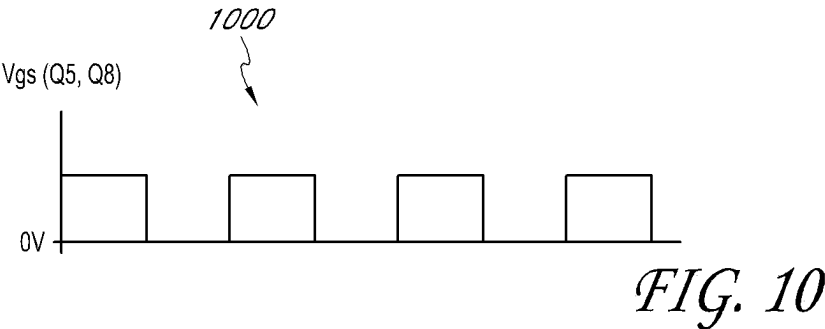
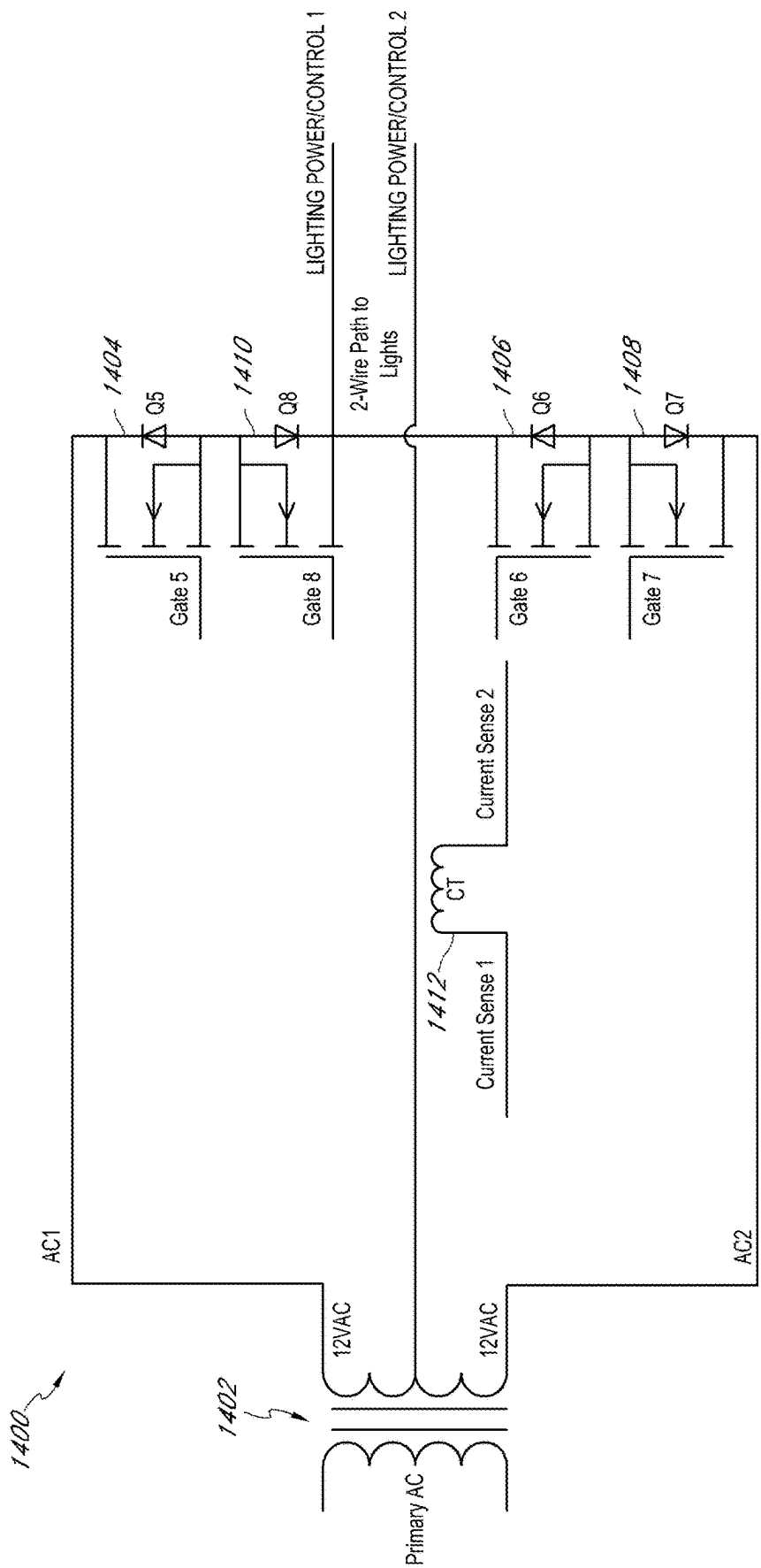


FIG. 9



Bridge Waveforms



Rectifier/Bridge

FIG. 14

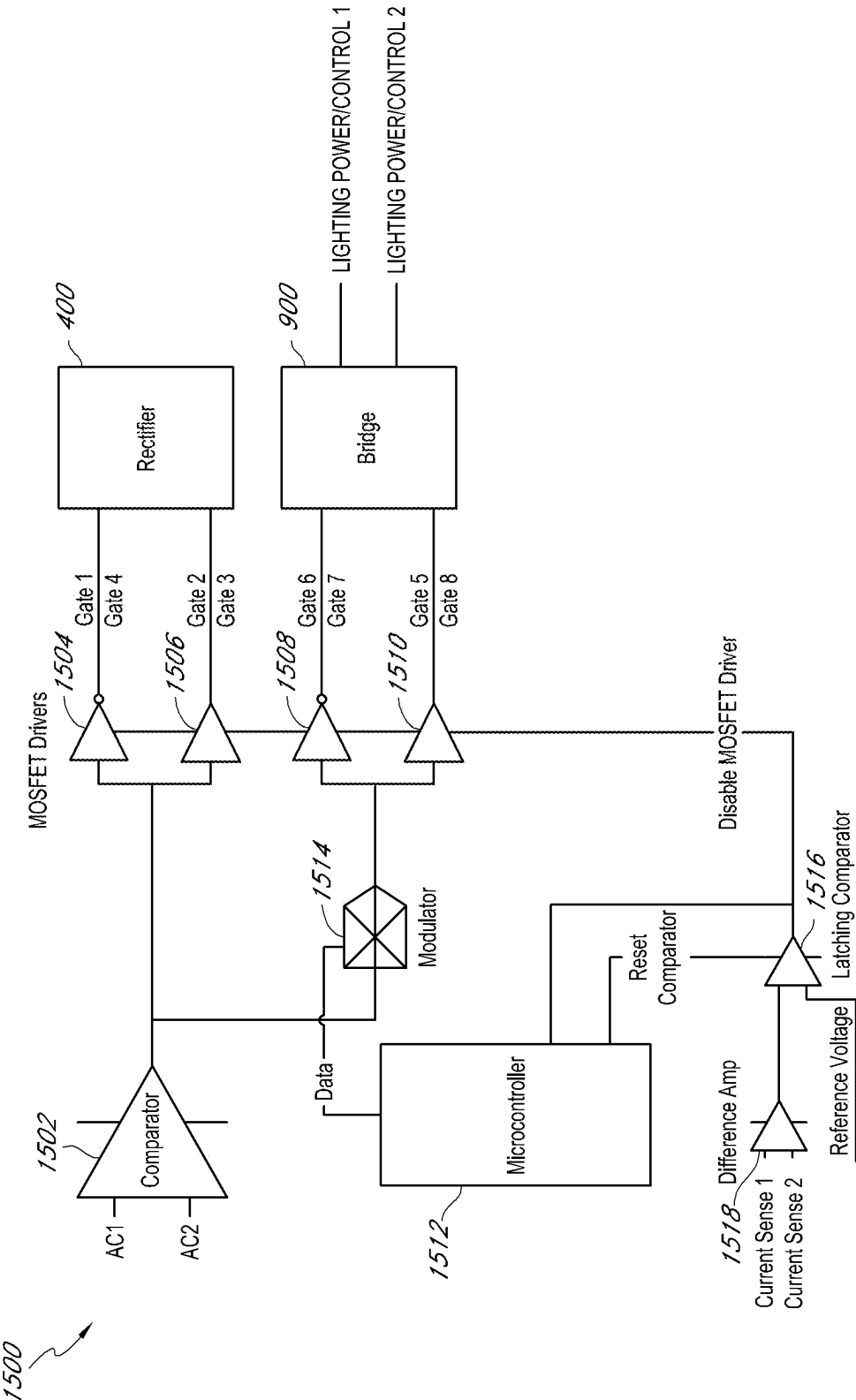


FIG. 15

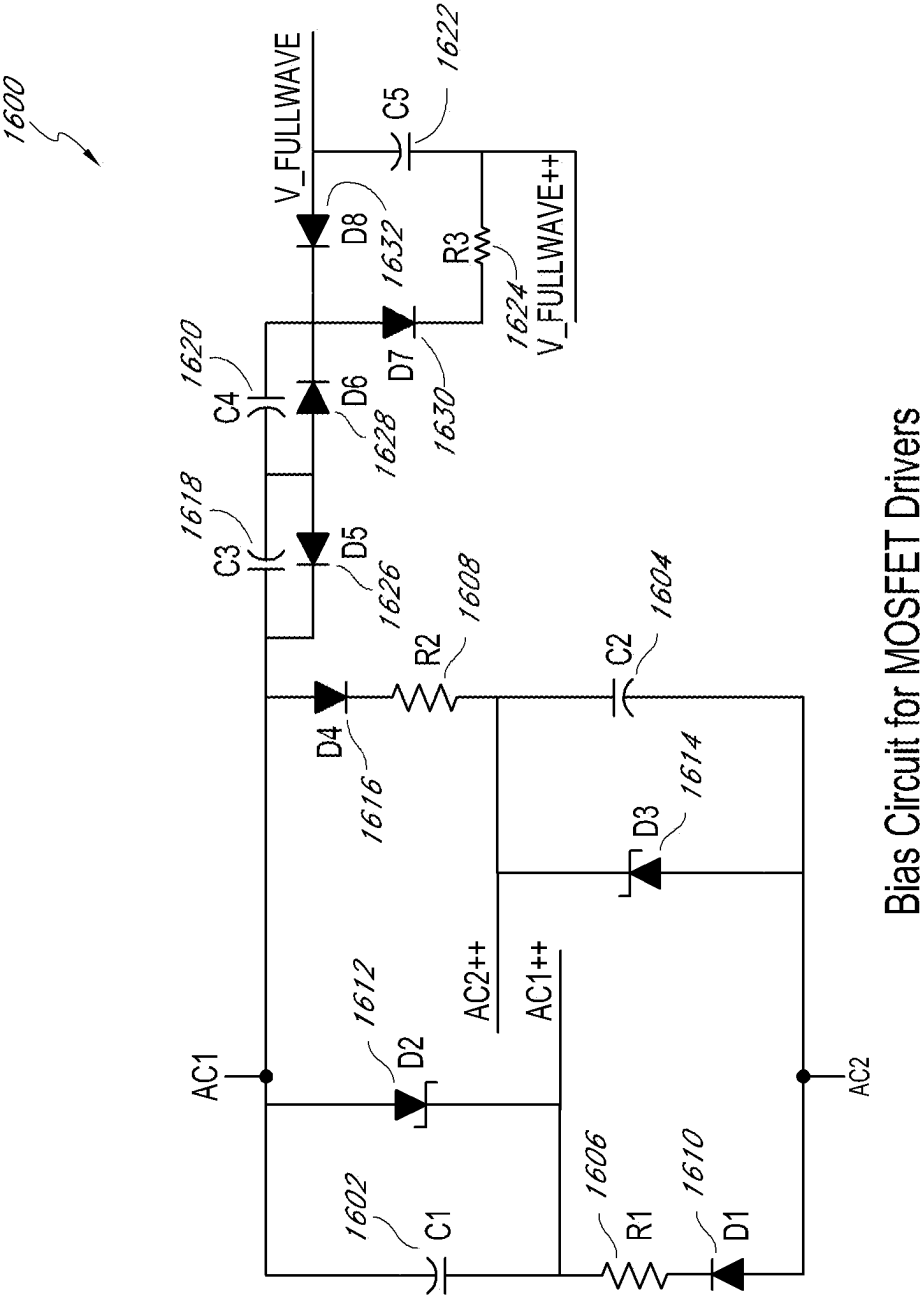


FIG. 16

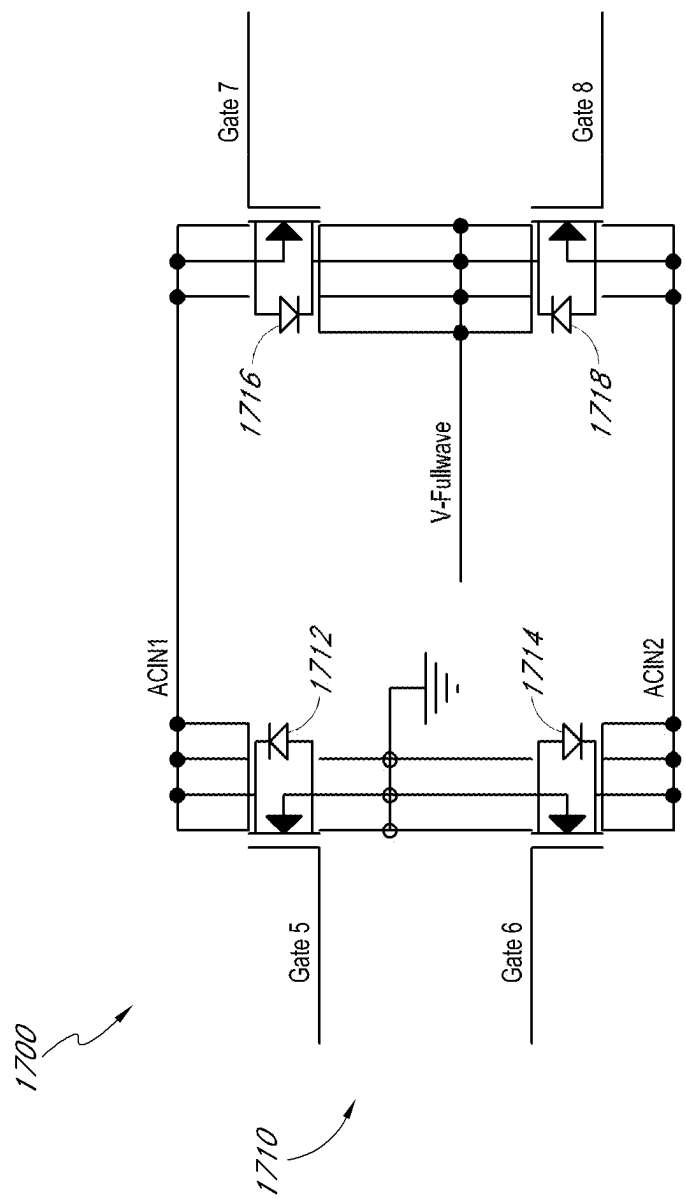


FIG. 17A-1

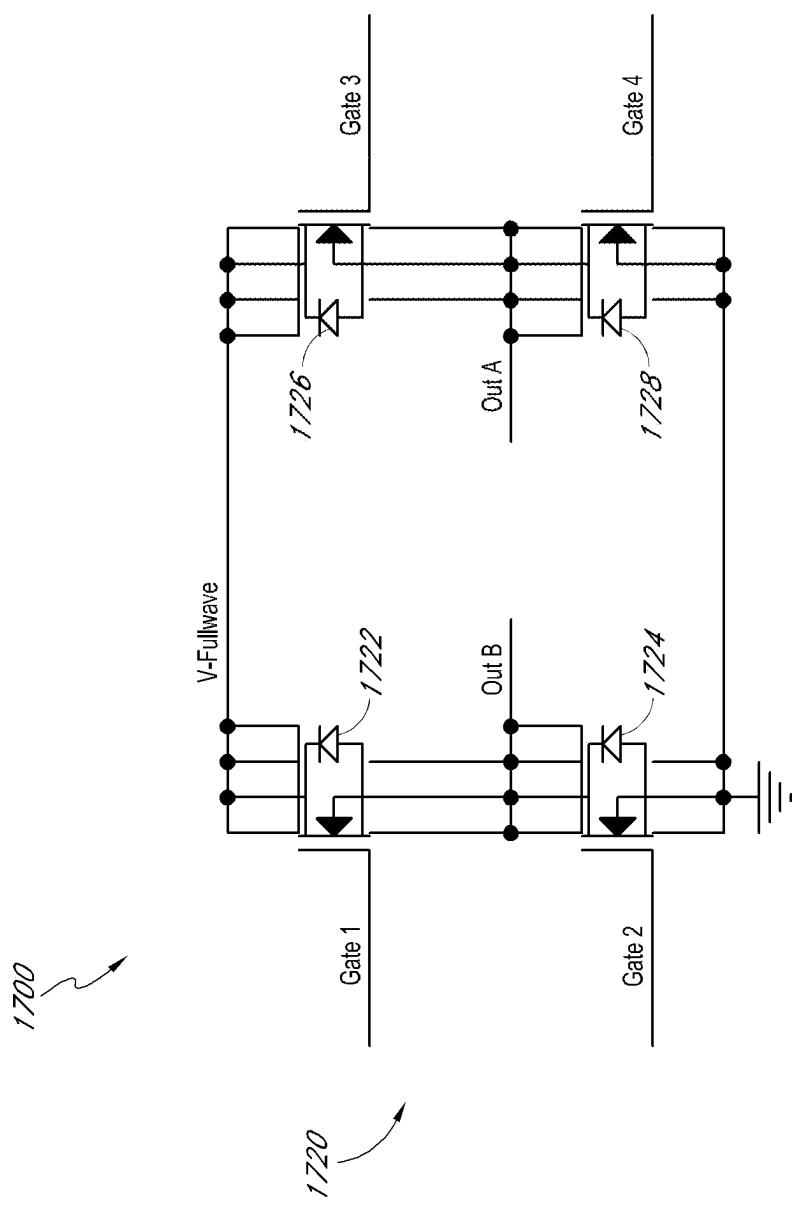


FIG. 17A-2

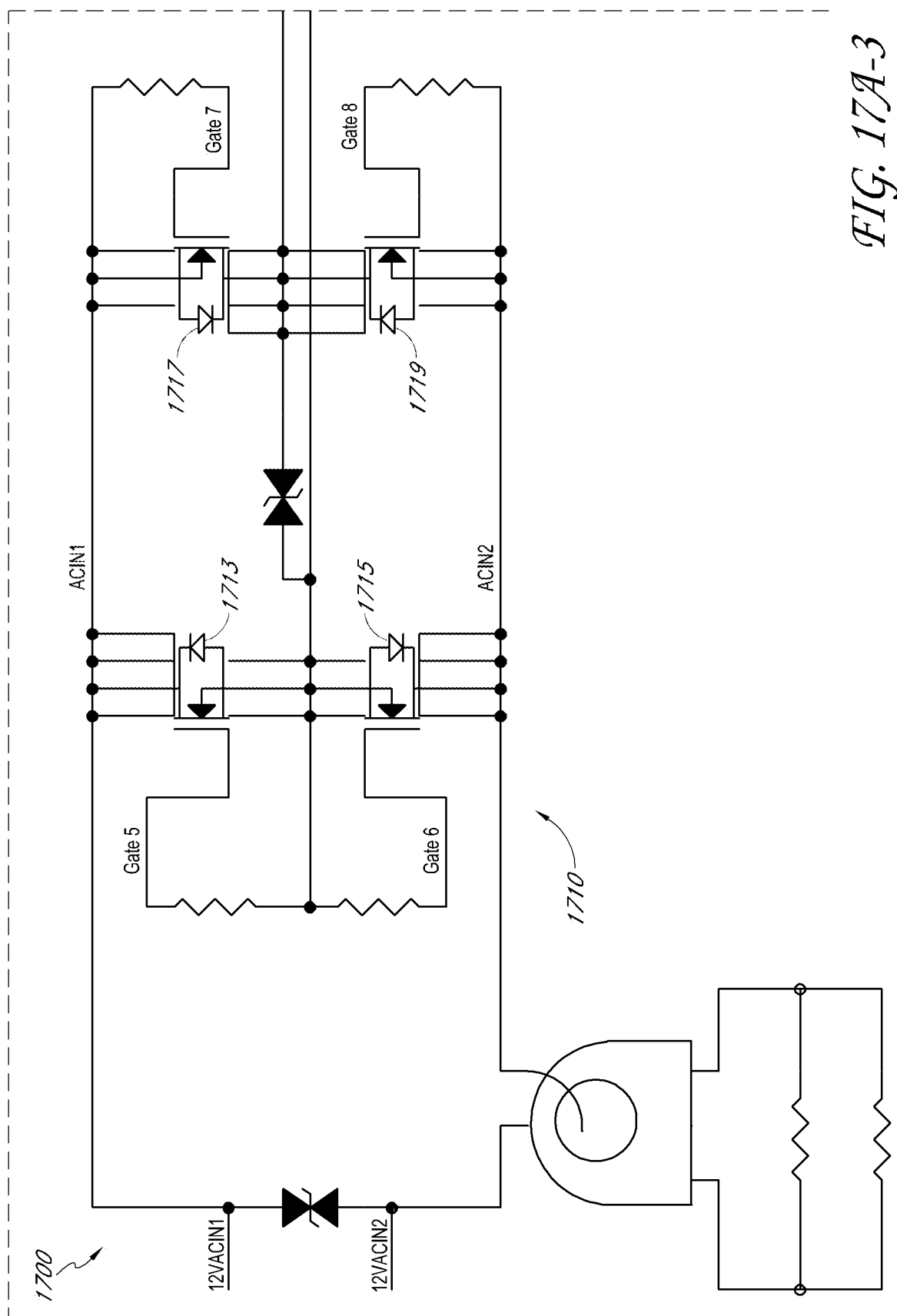


FIG. 17A-3

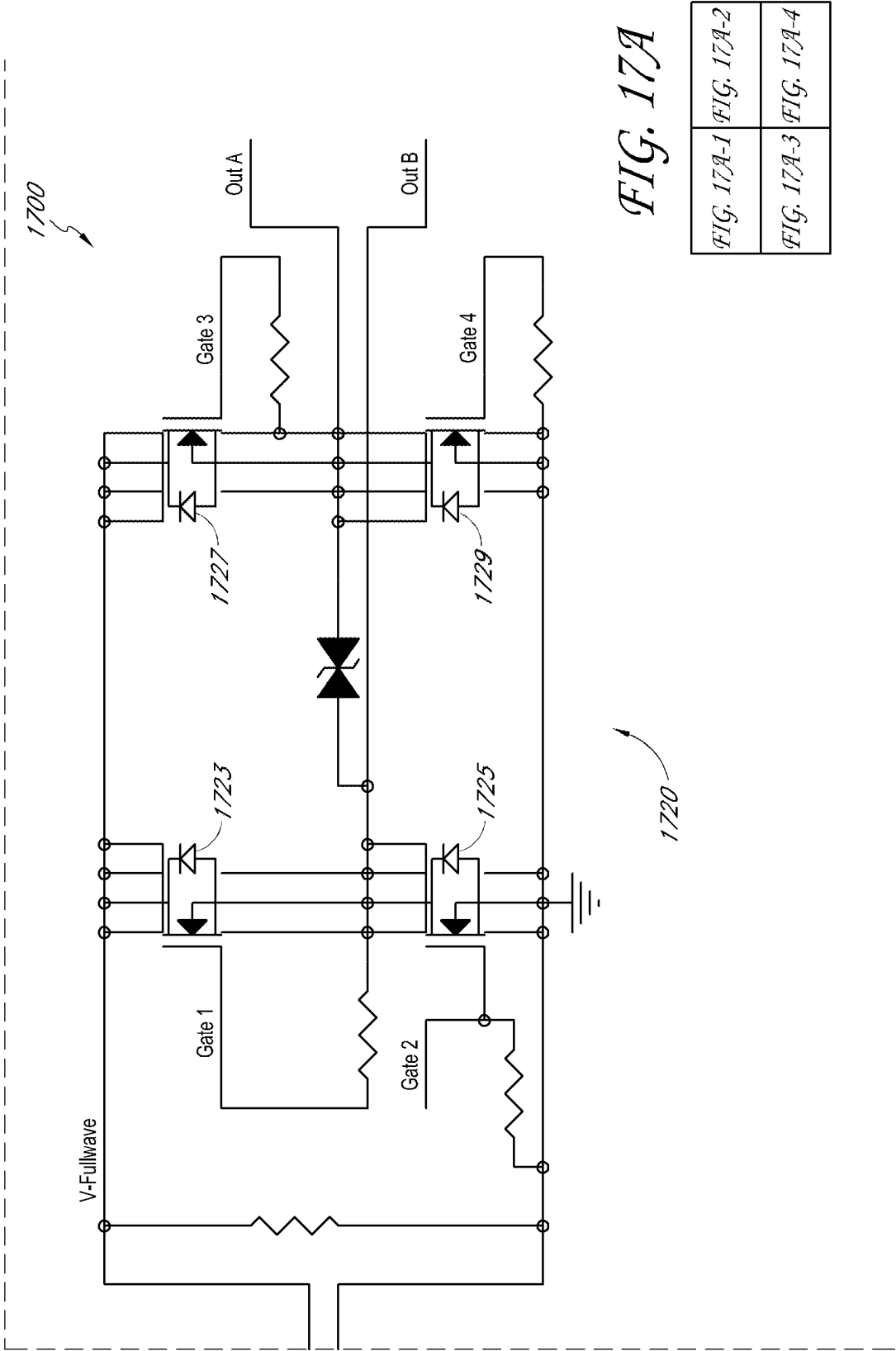


FIG. 17A

FIG. 17A-1	FIG. 17A-2
FIG. 17A-3	FIG. 17A-4

FIG. 17A-4

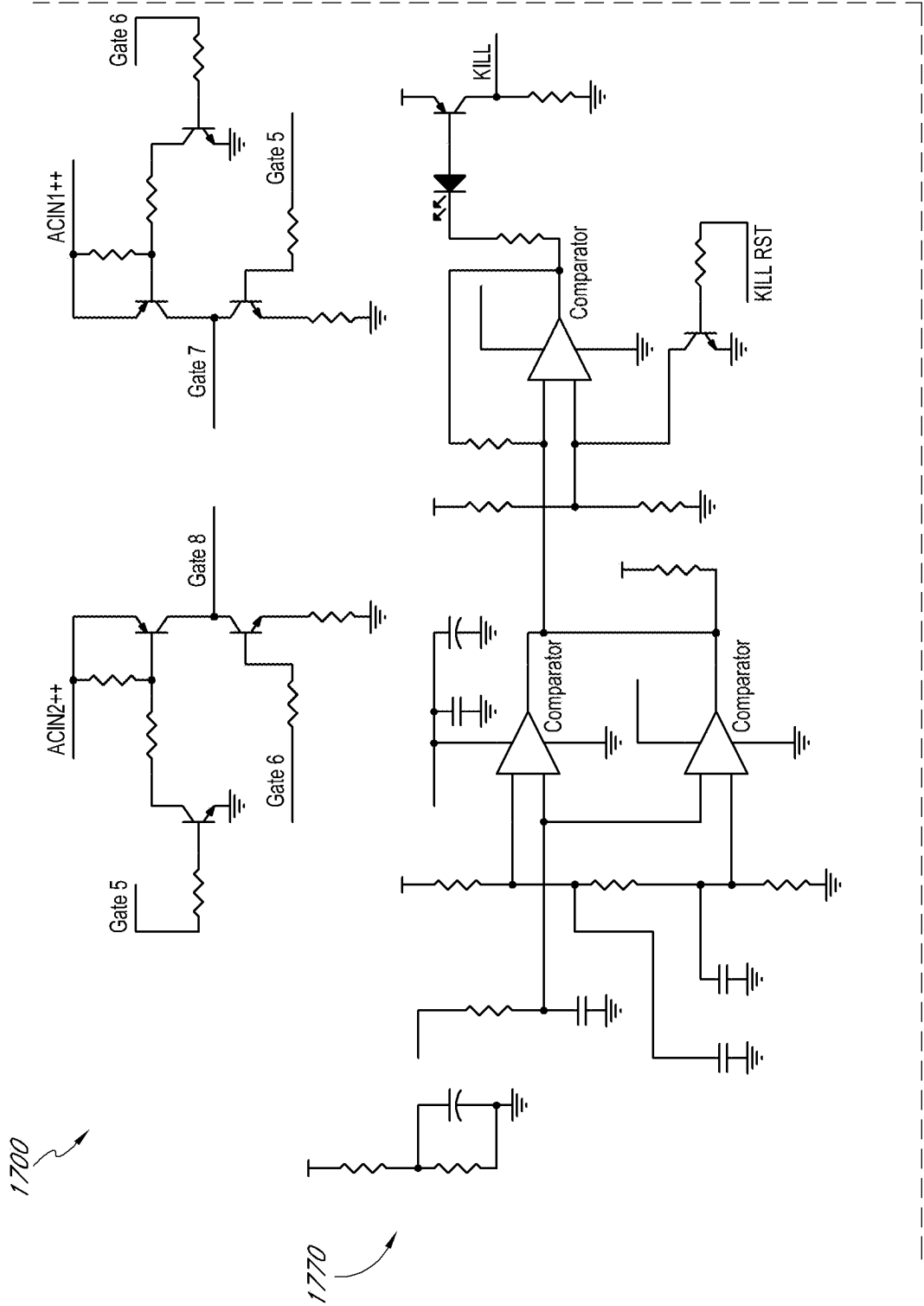
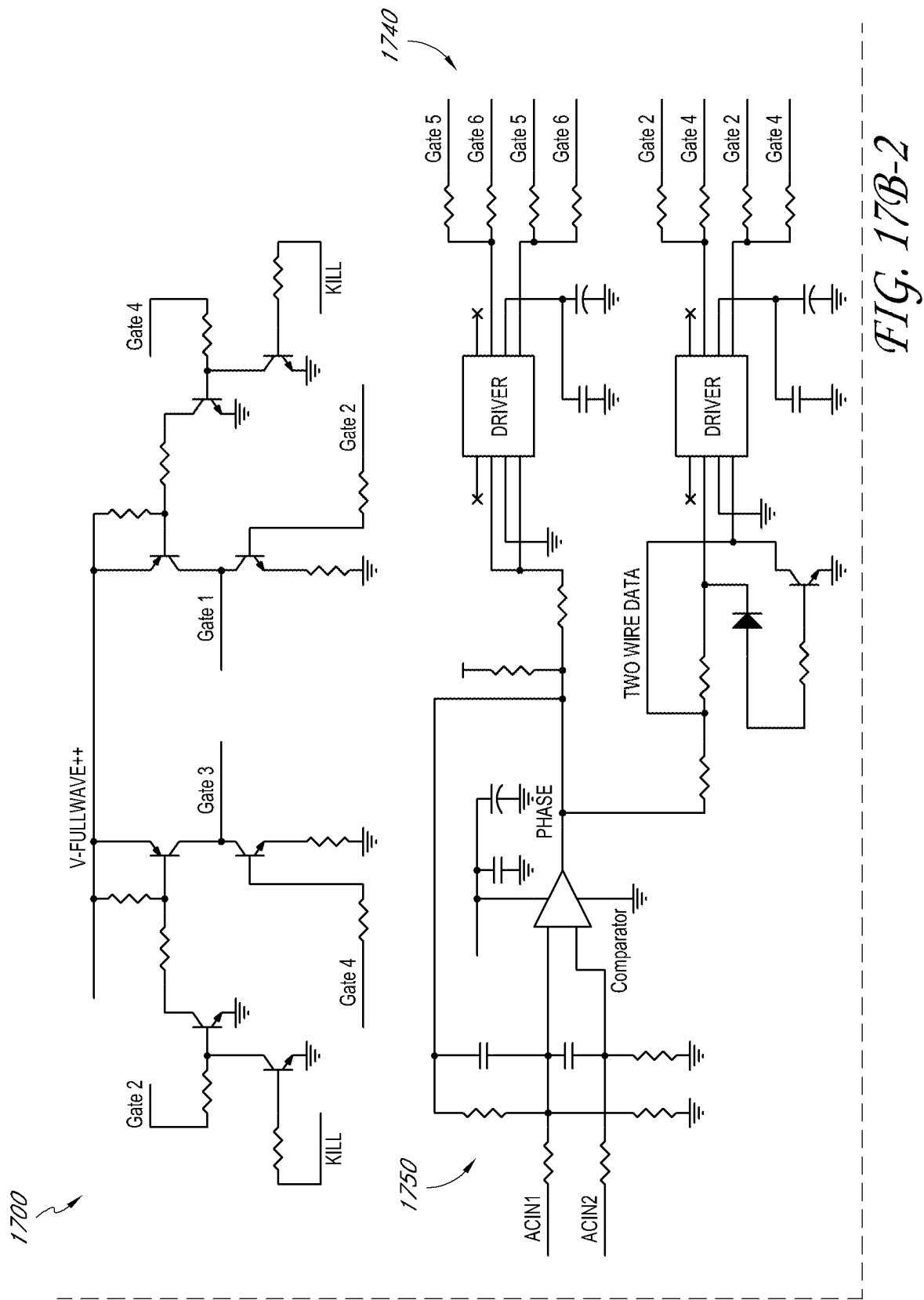


FIG. 17B-1



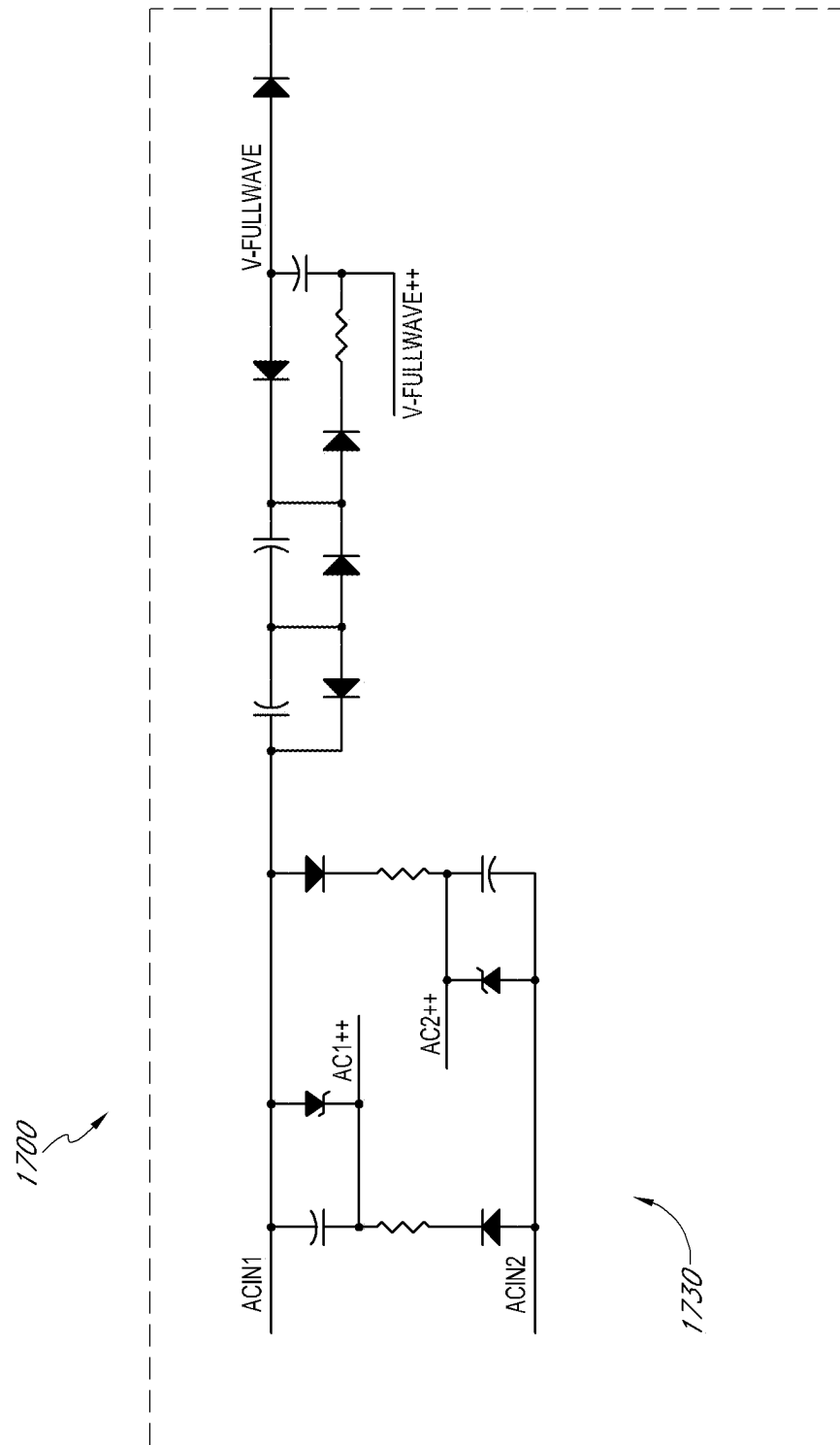
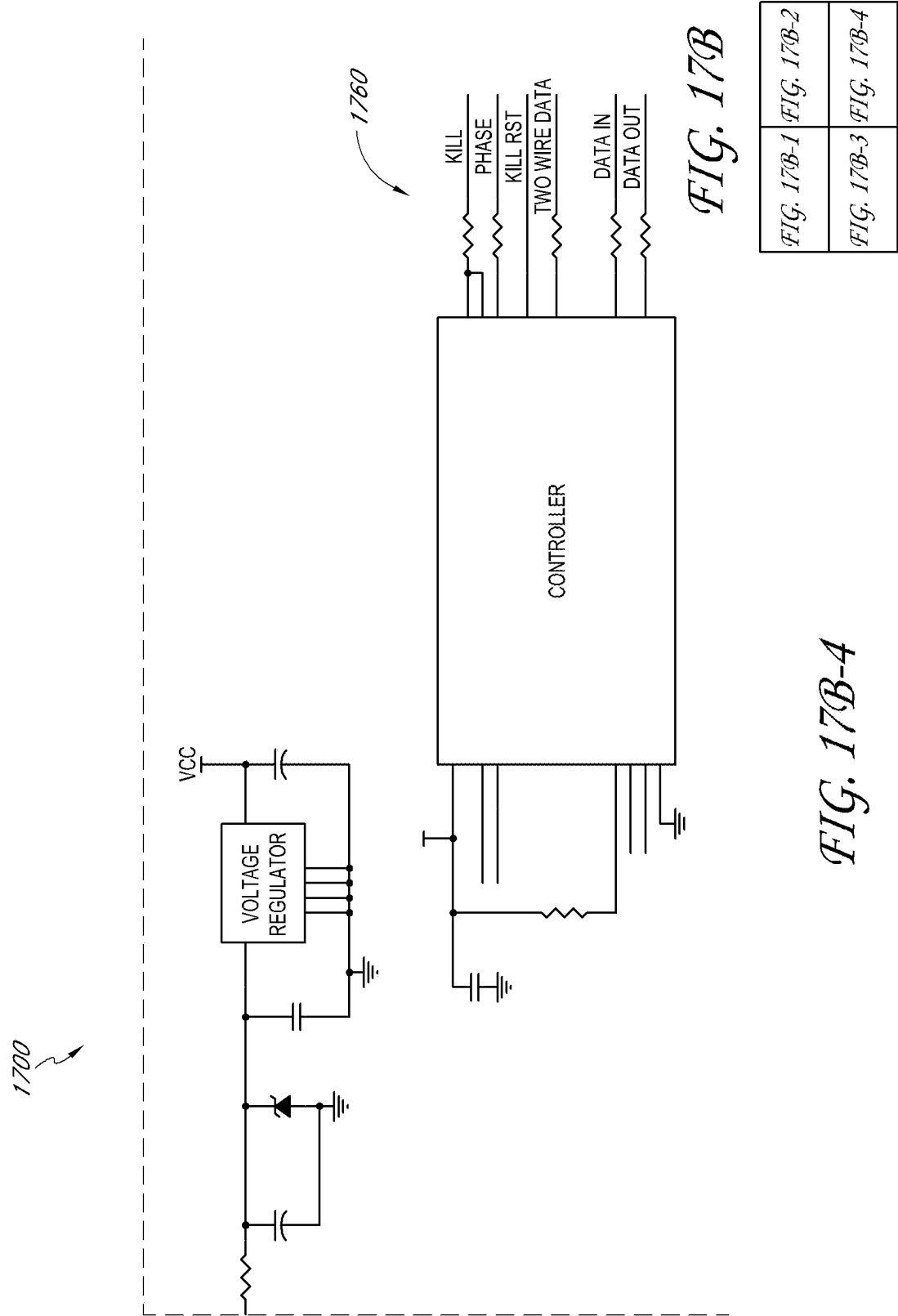
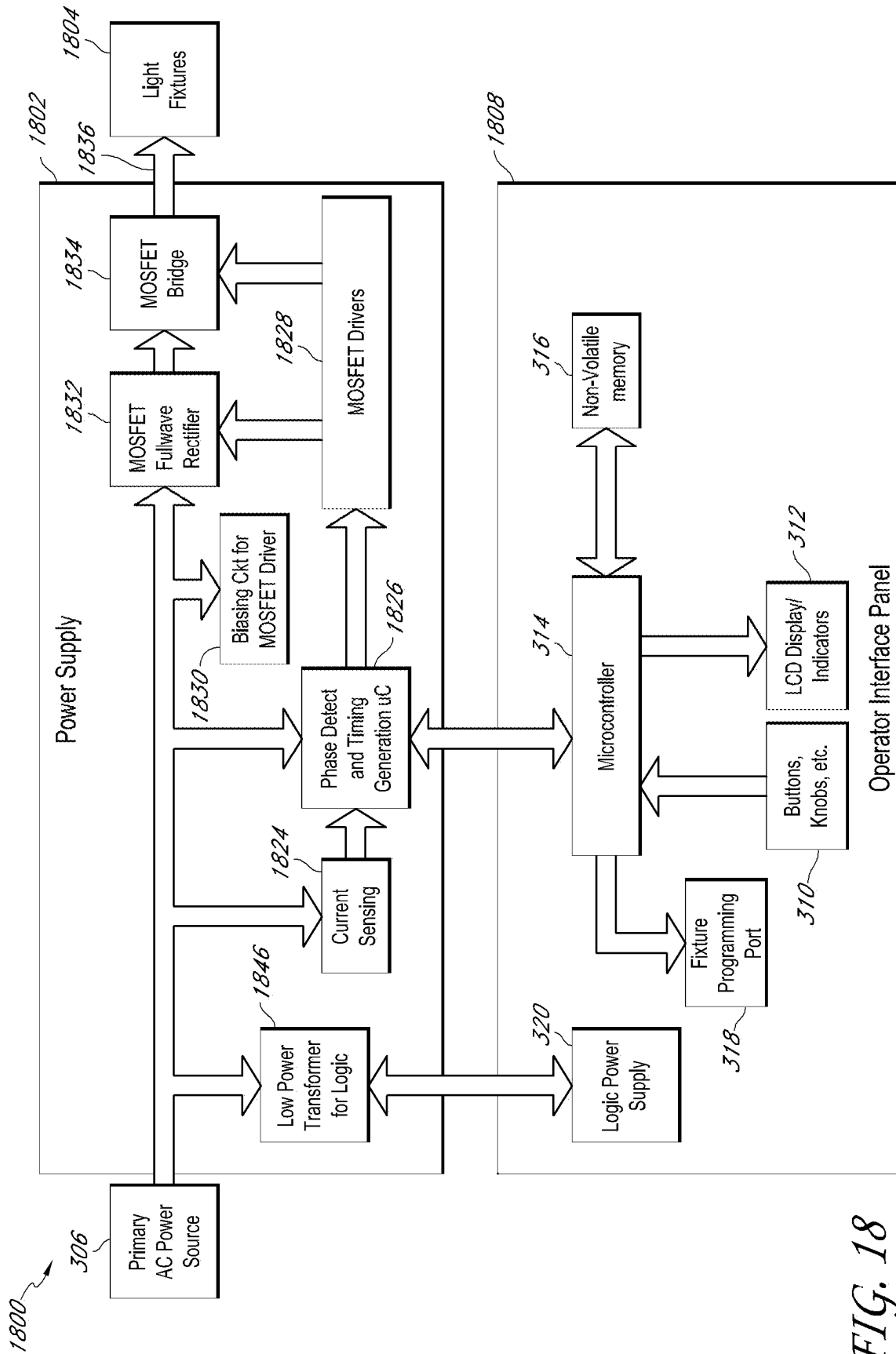


FIG. 17B-3





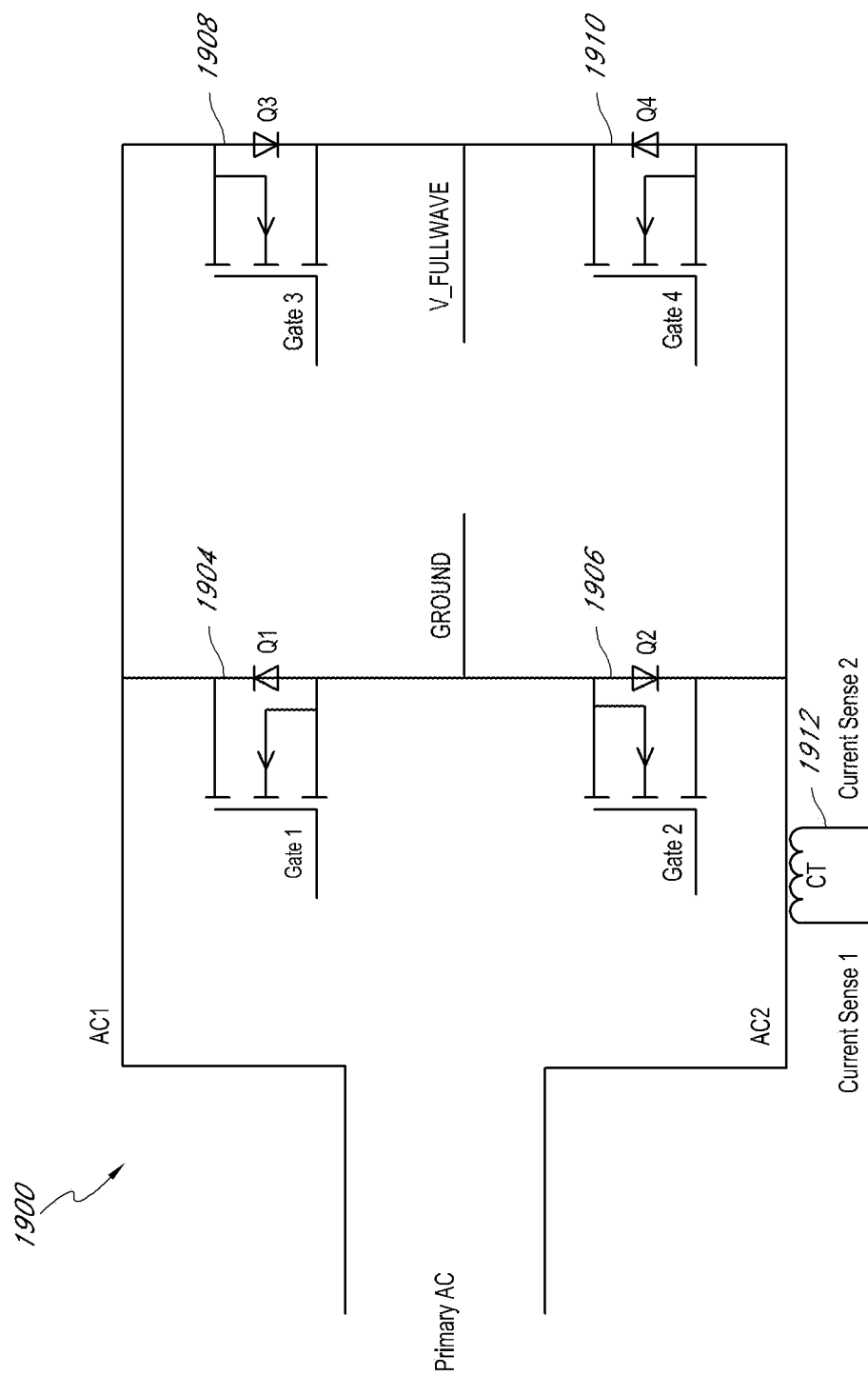


FIG. 19

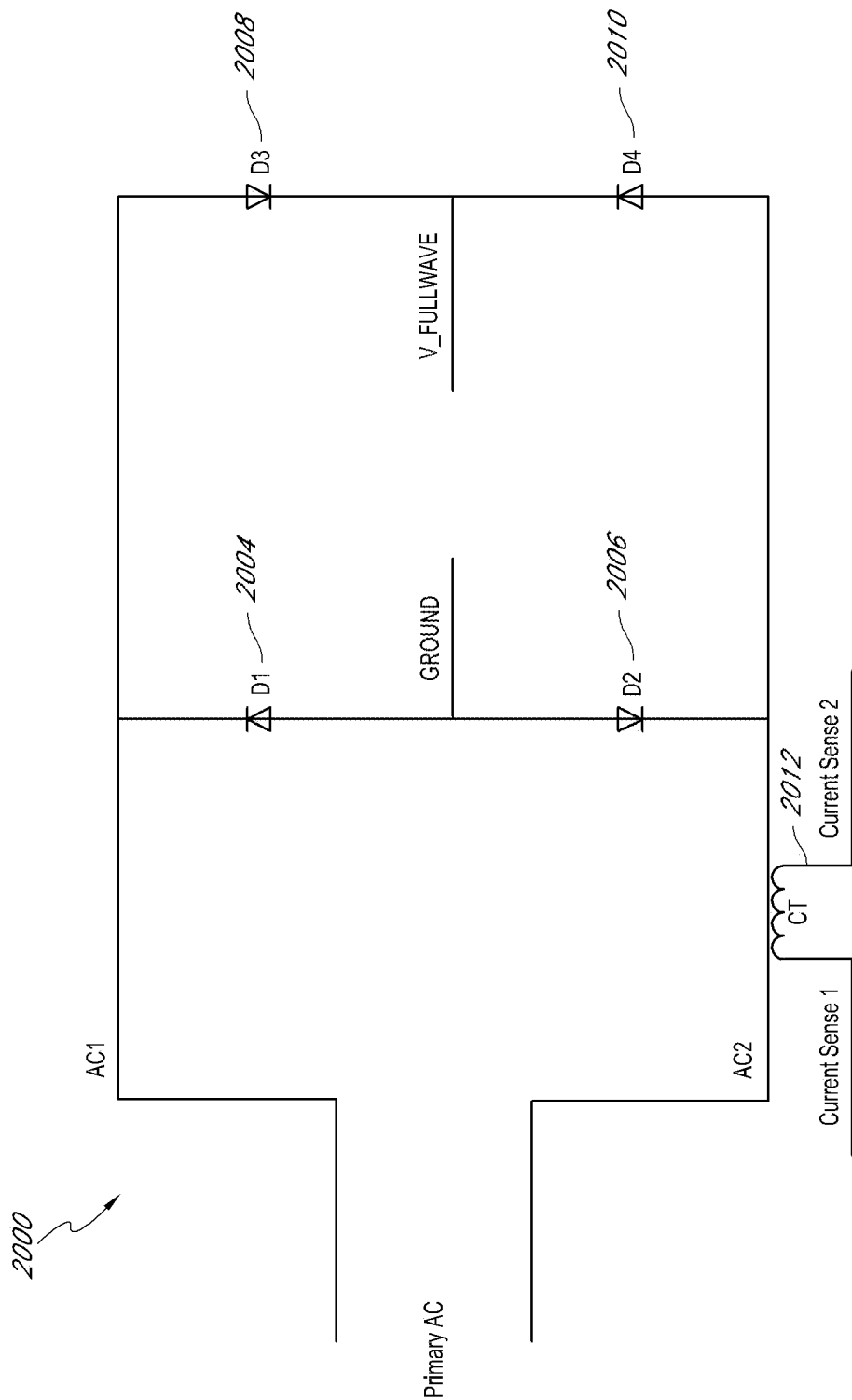
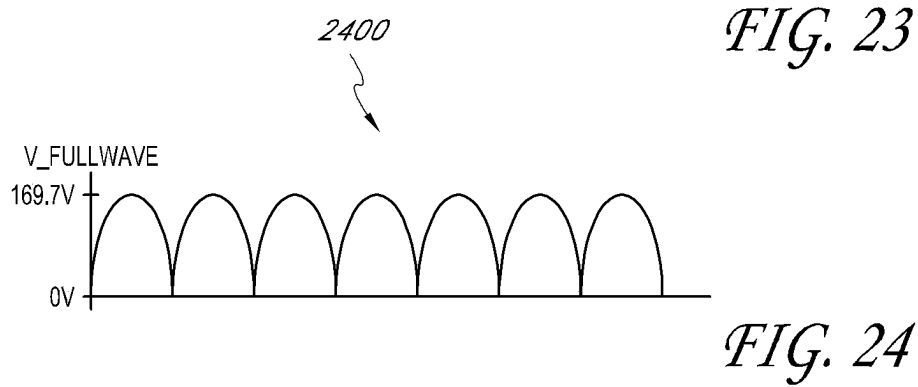
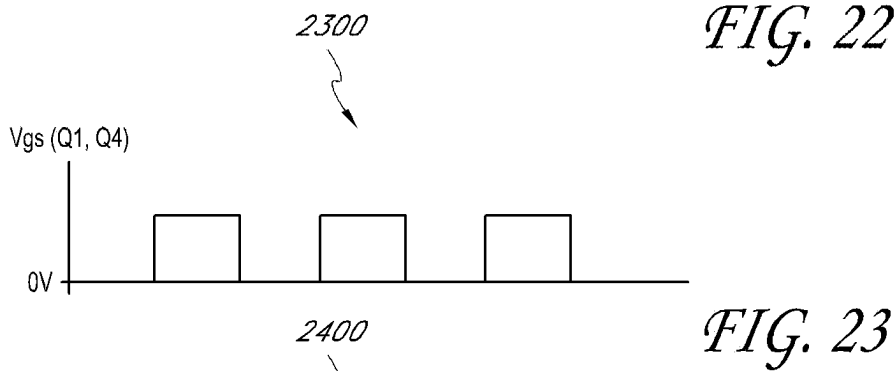
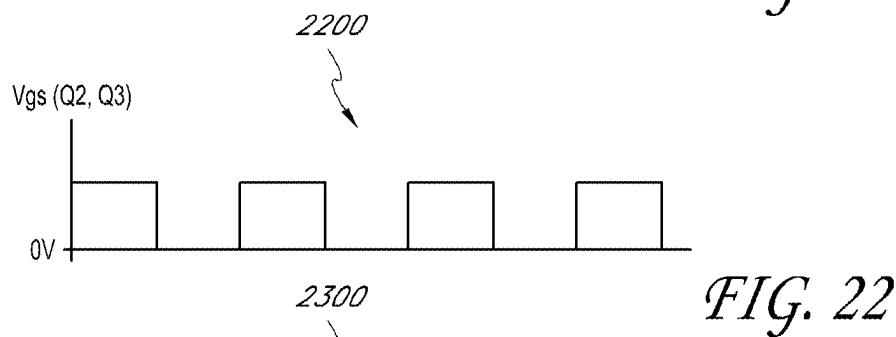
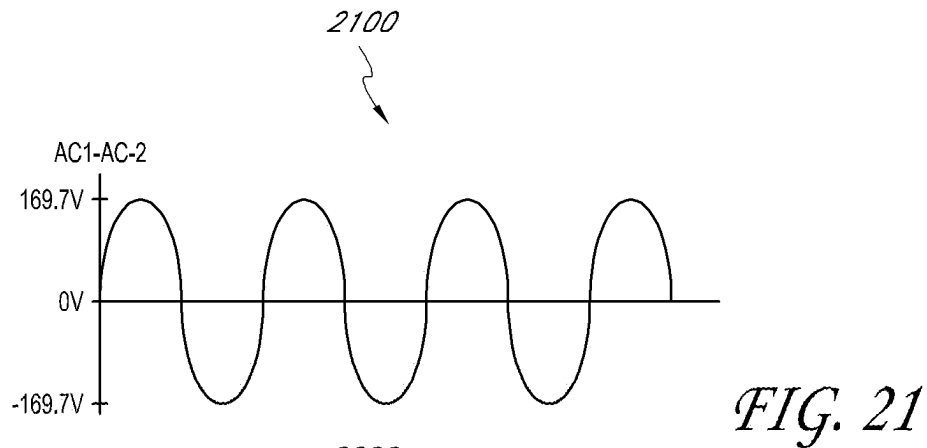
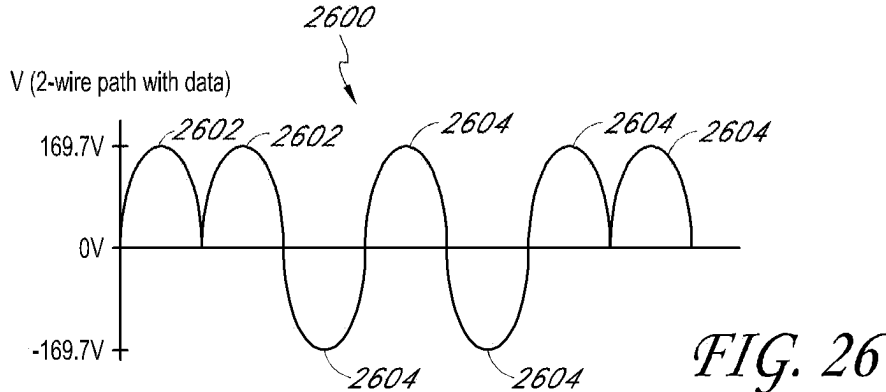
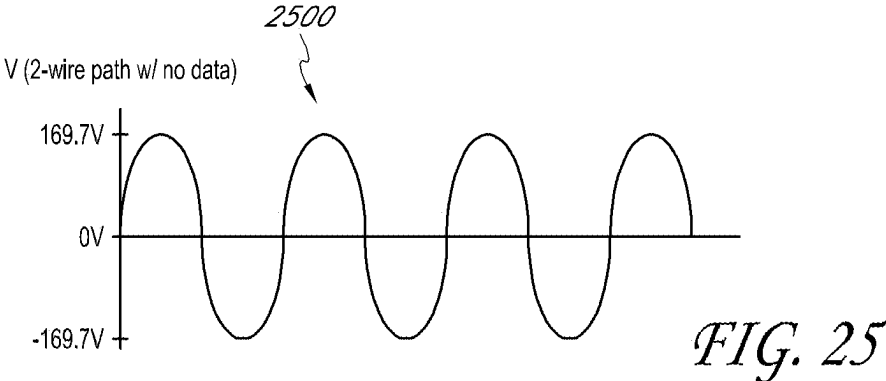


FIG. 20



MOSFET Full Wave Rectifier Waveforms



Bridge Waveforms

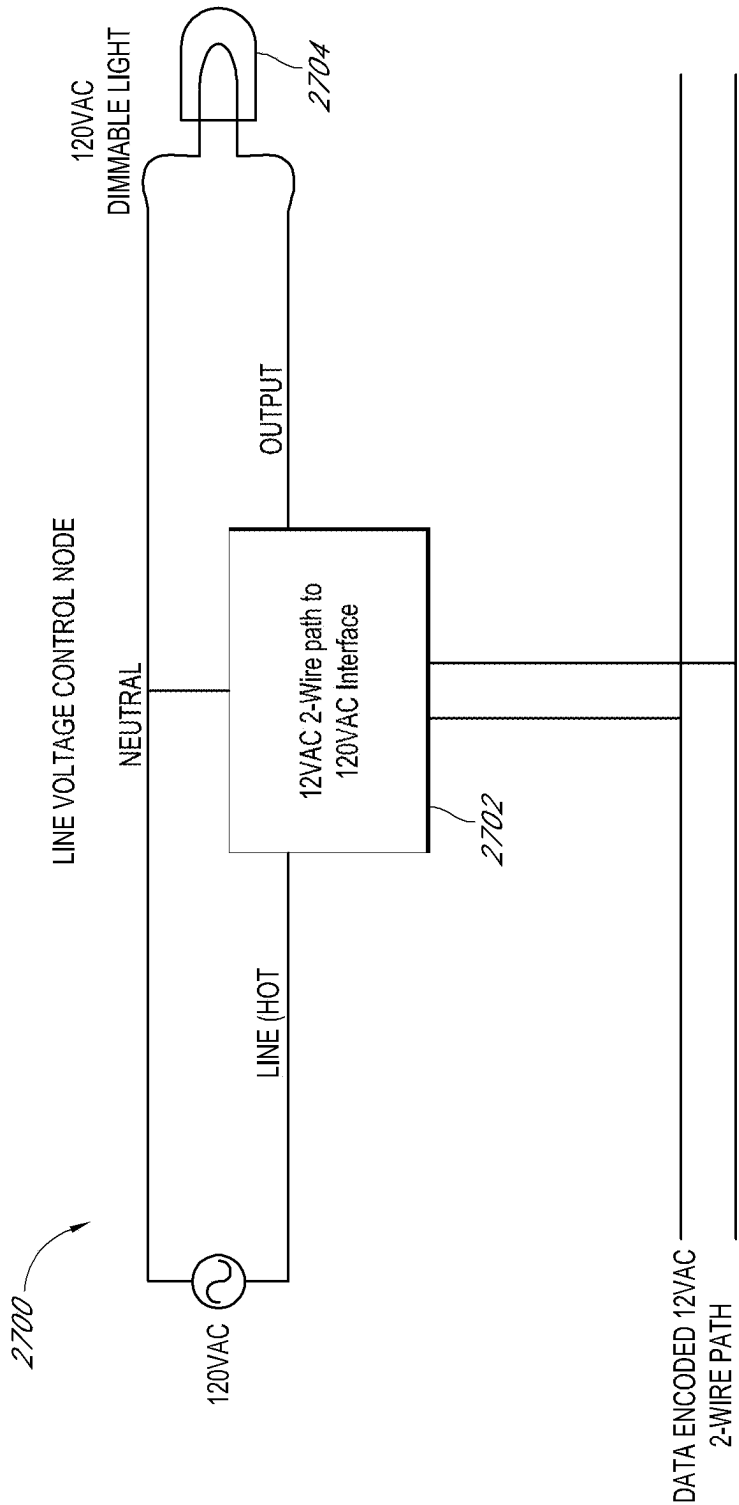


FIG. 27

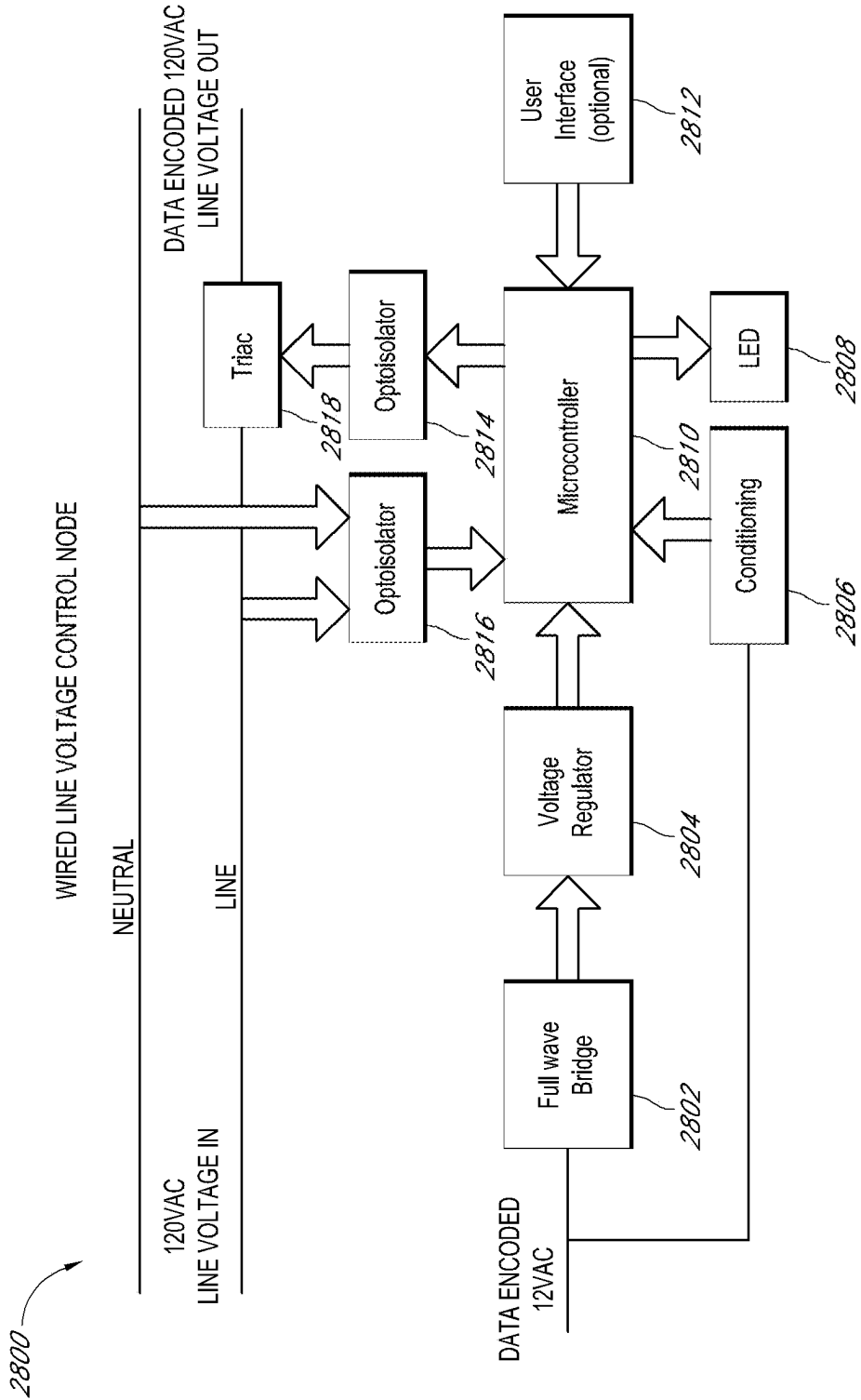
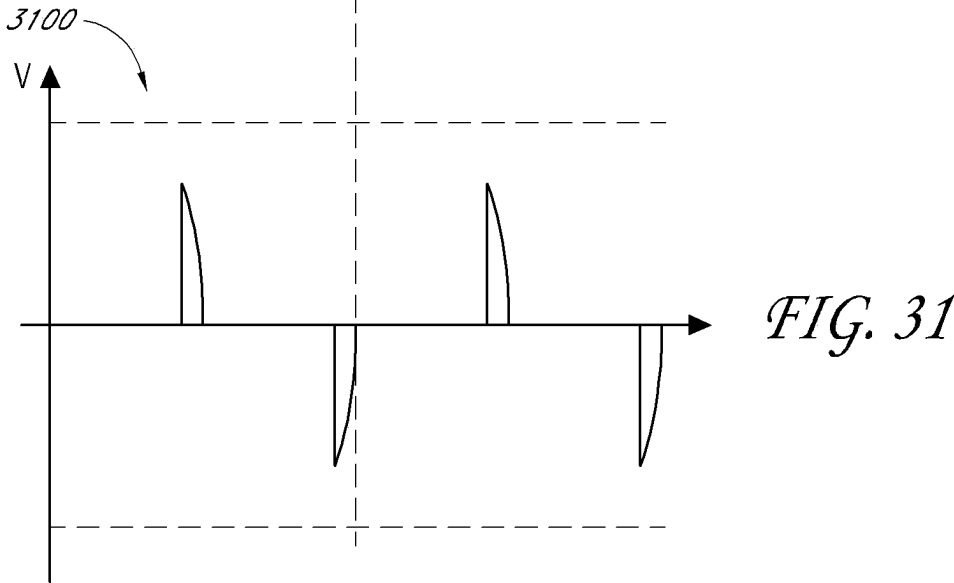
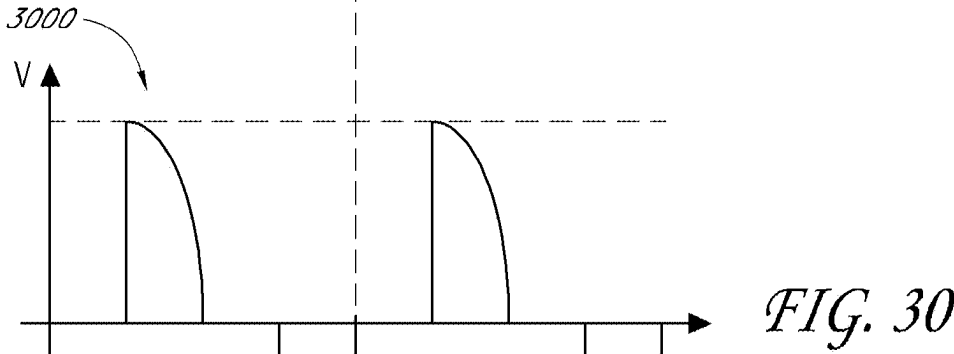
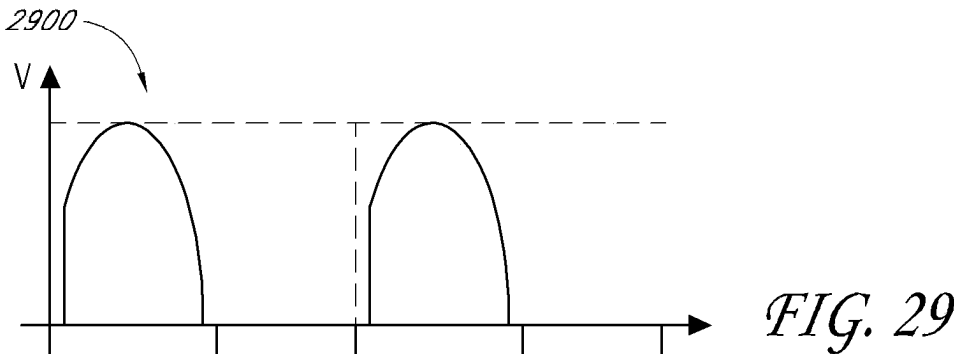


FIG. 28



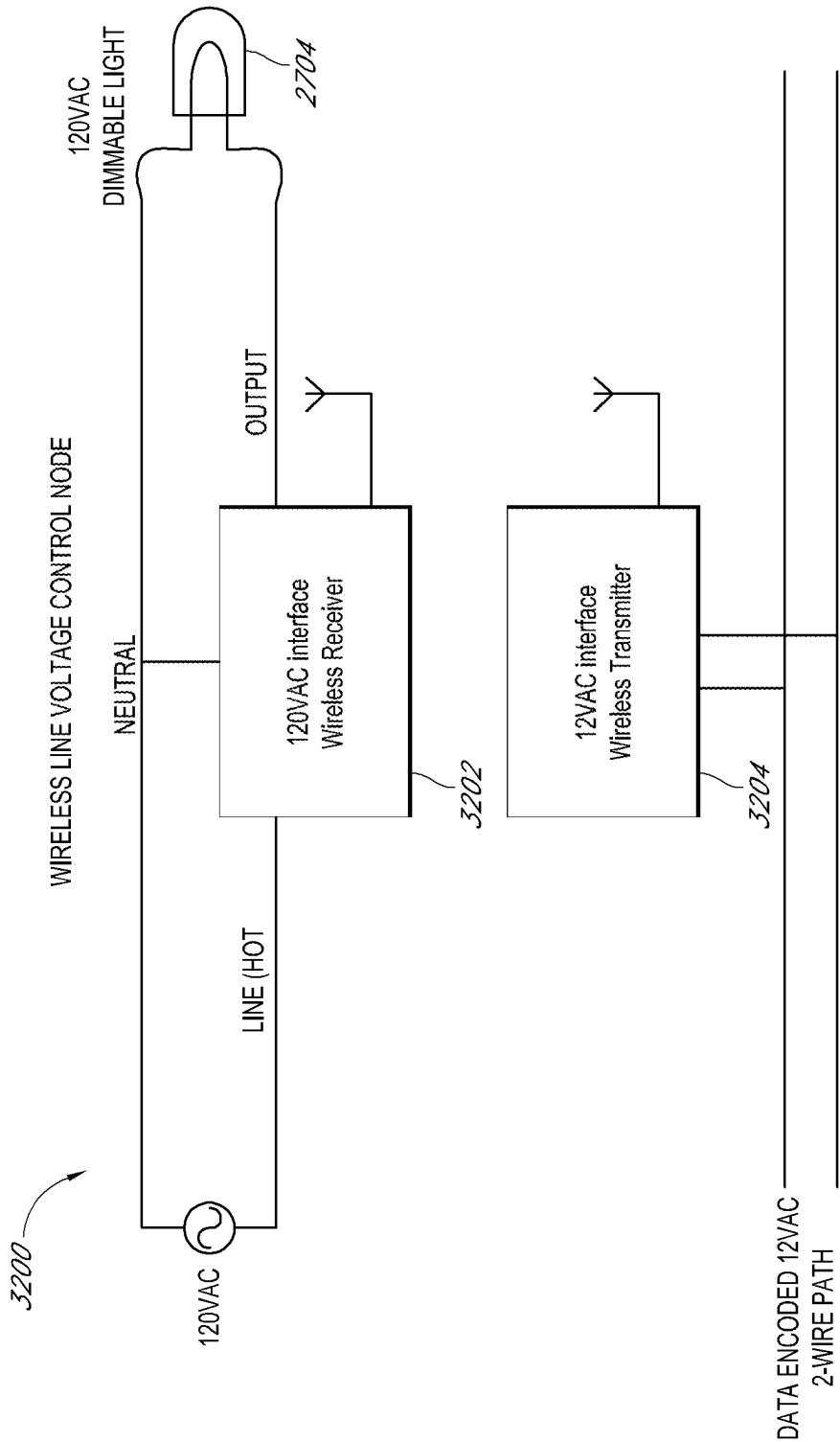


FIG. 32

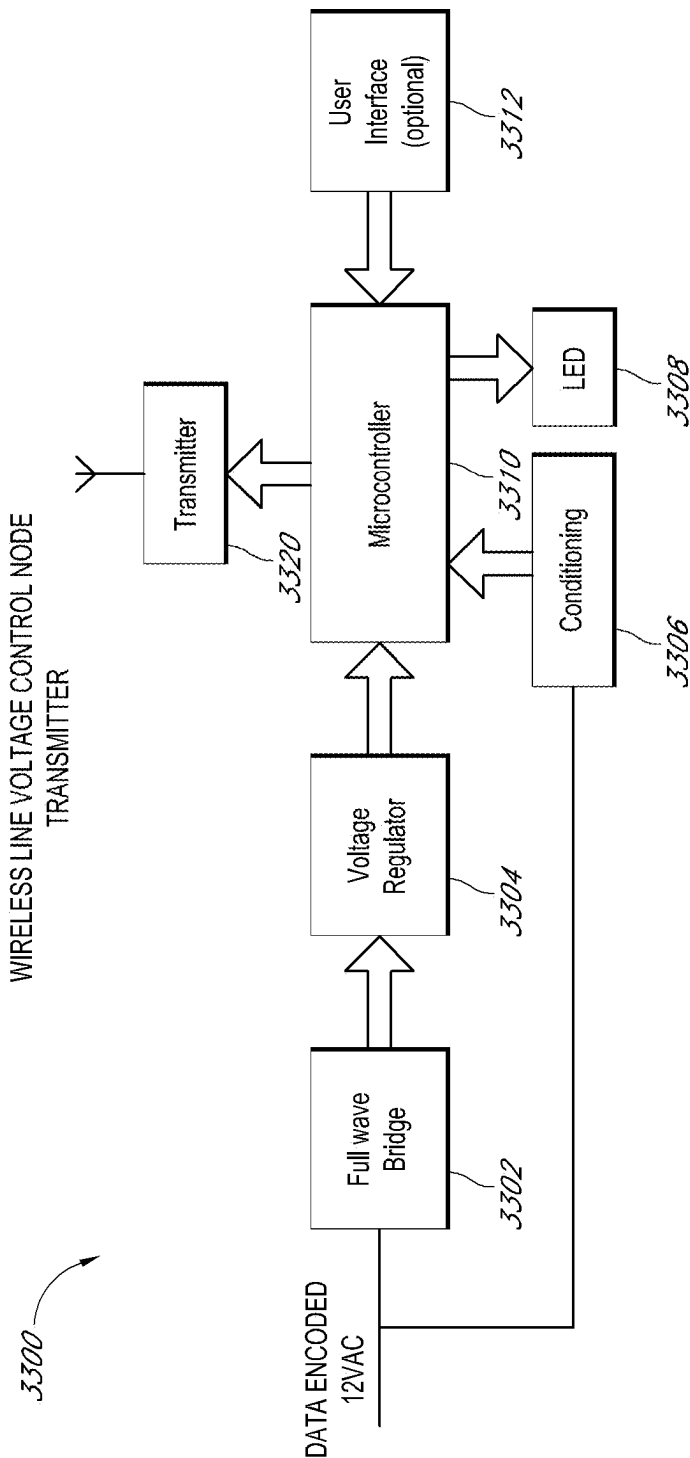


FIG. 33

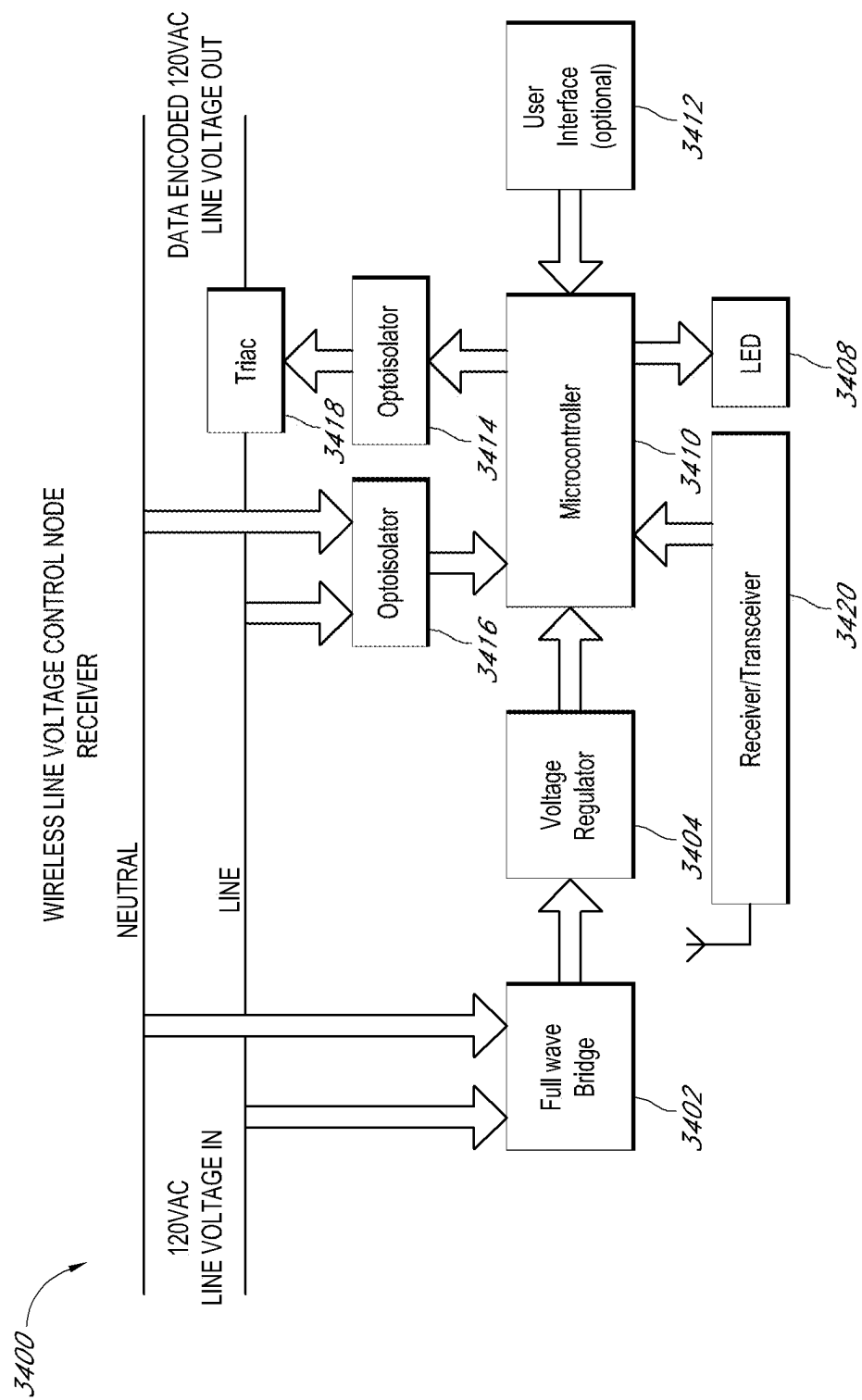


FIG. 34

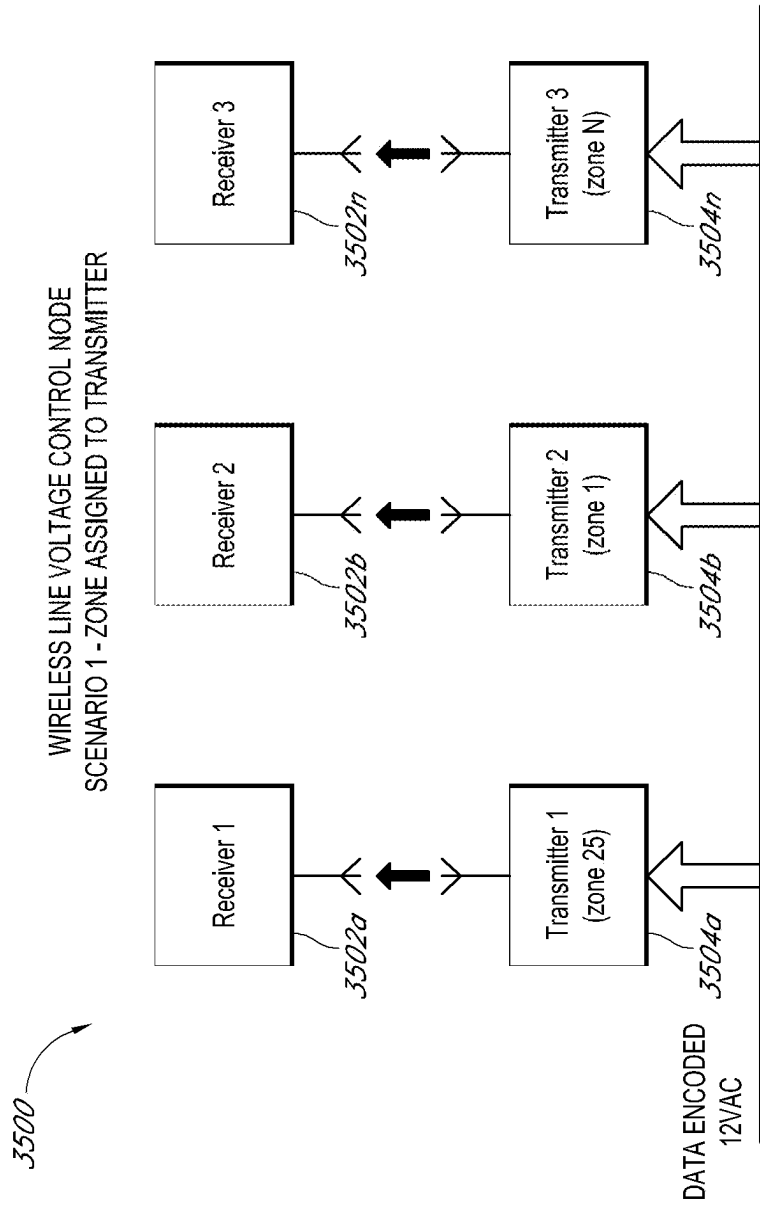


FIG. 35

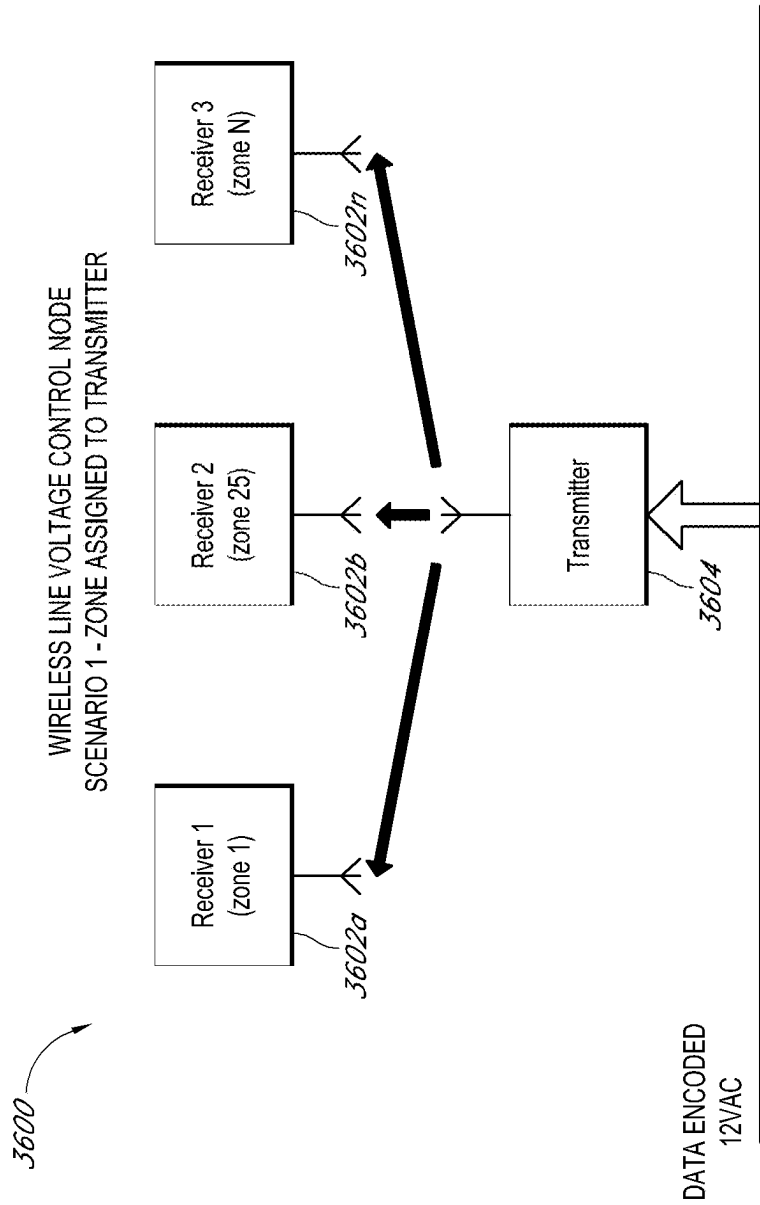


FIG. 36

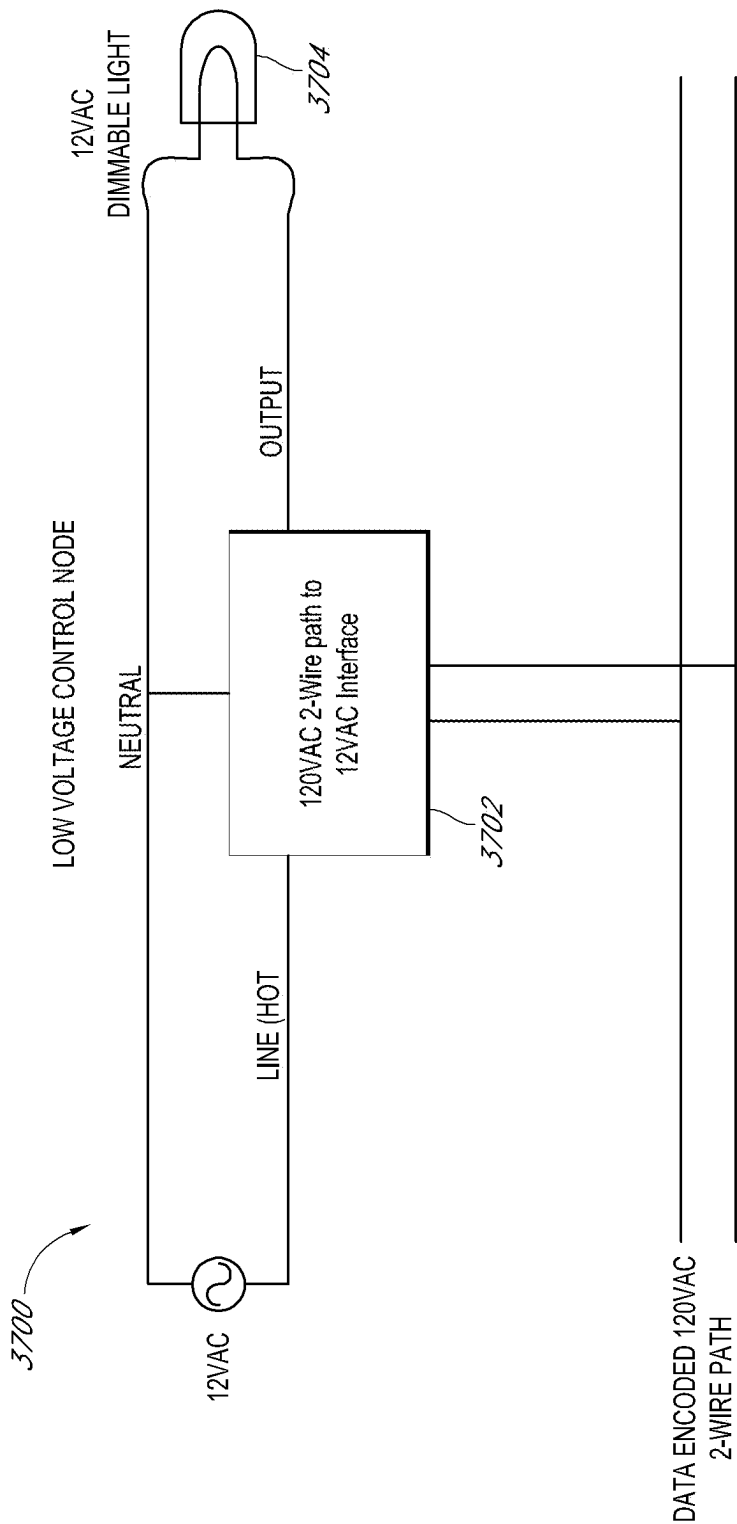


FIG. 37

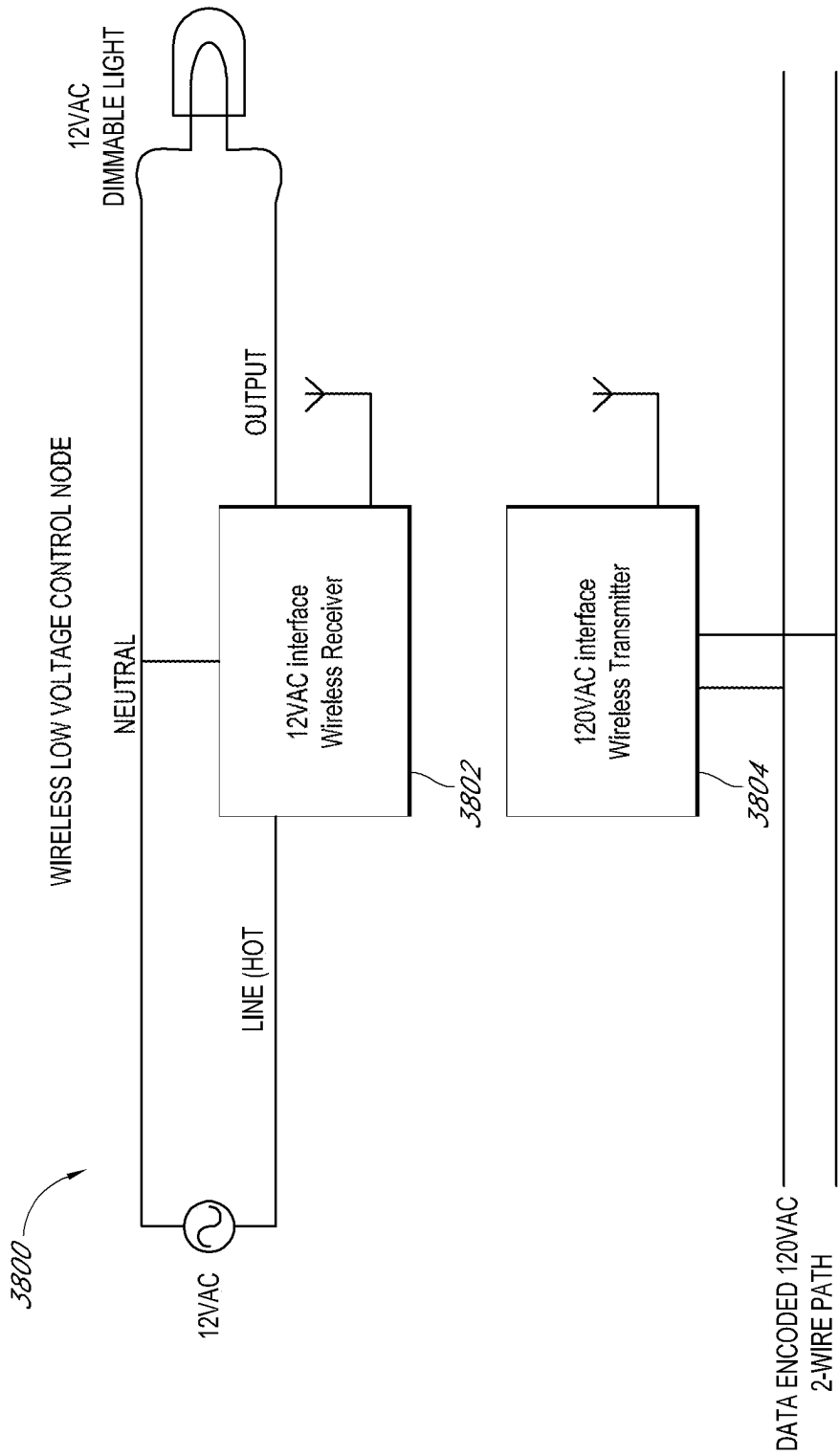


FIG. 38

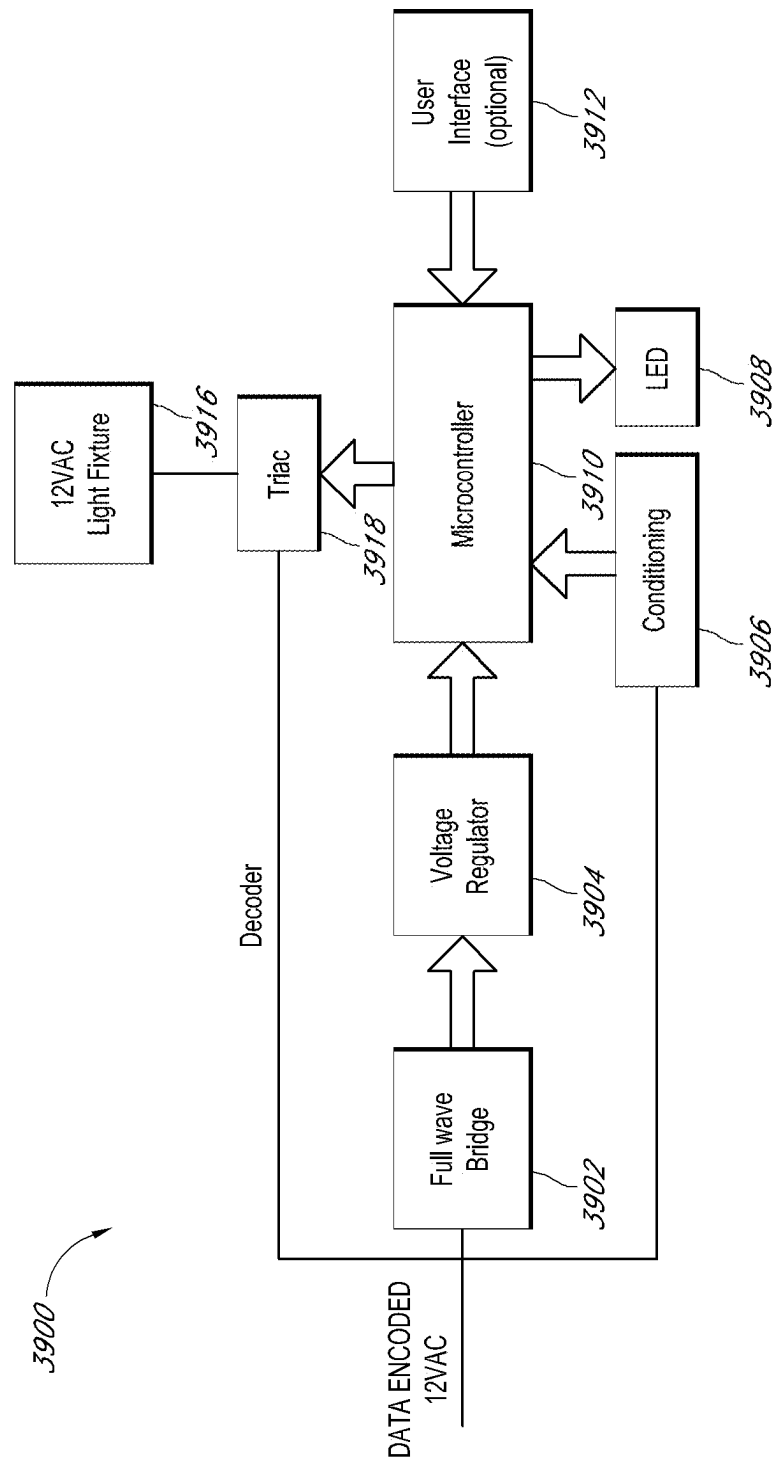
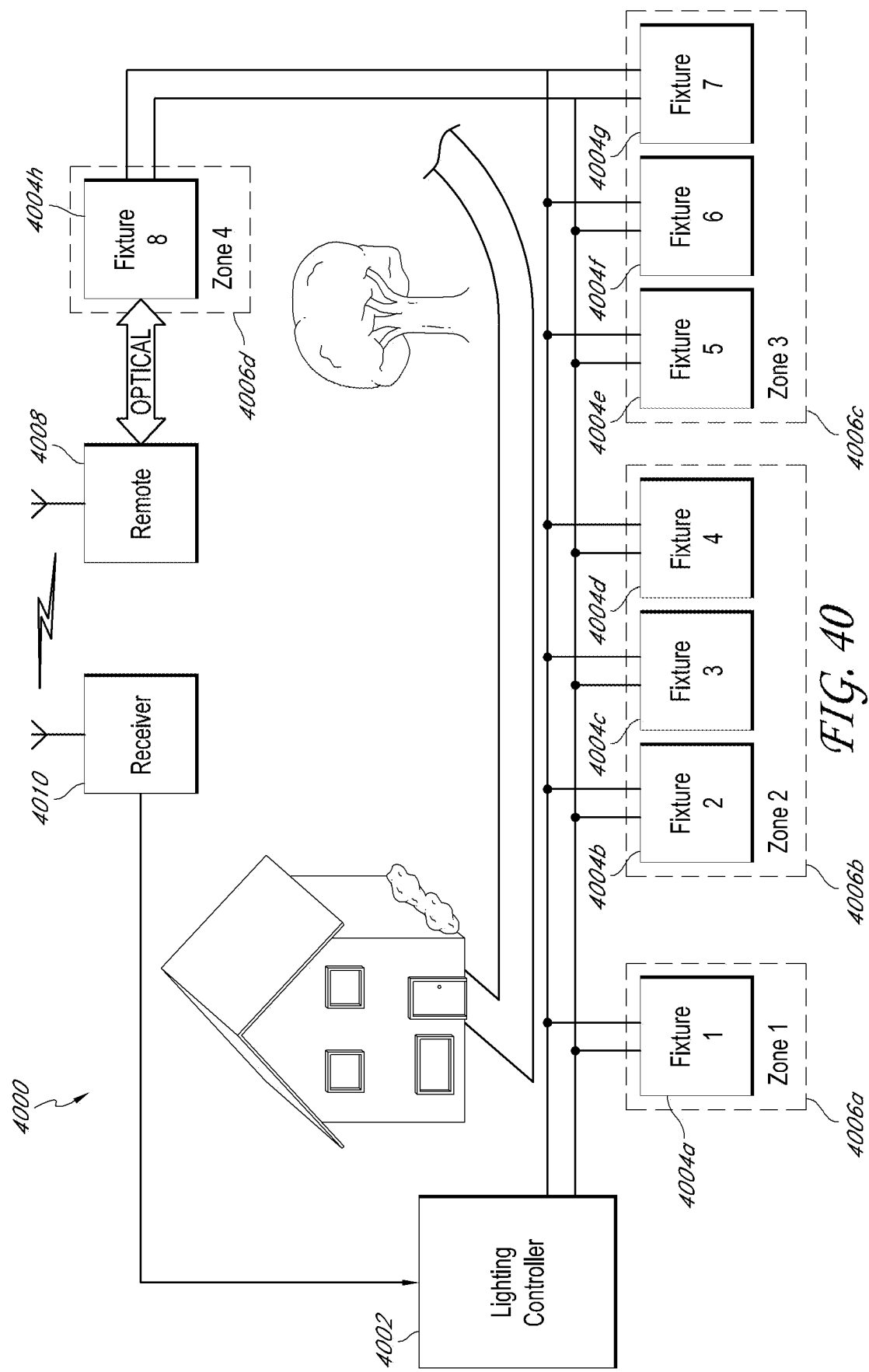


FIG. 39



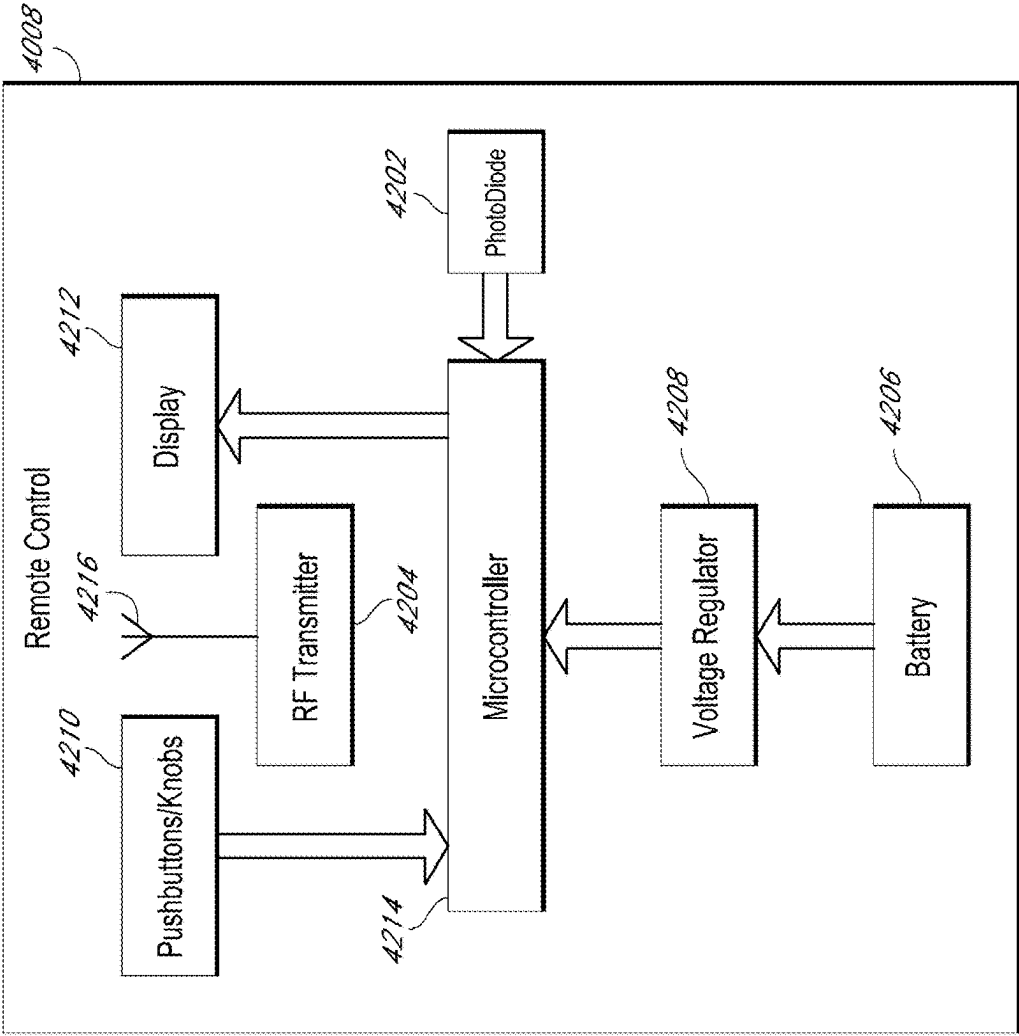


FIG. 42

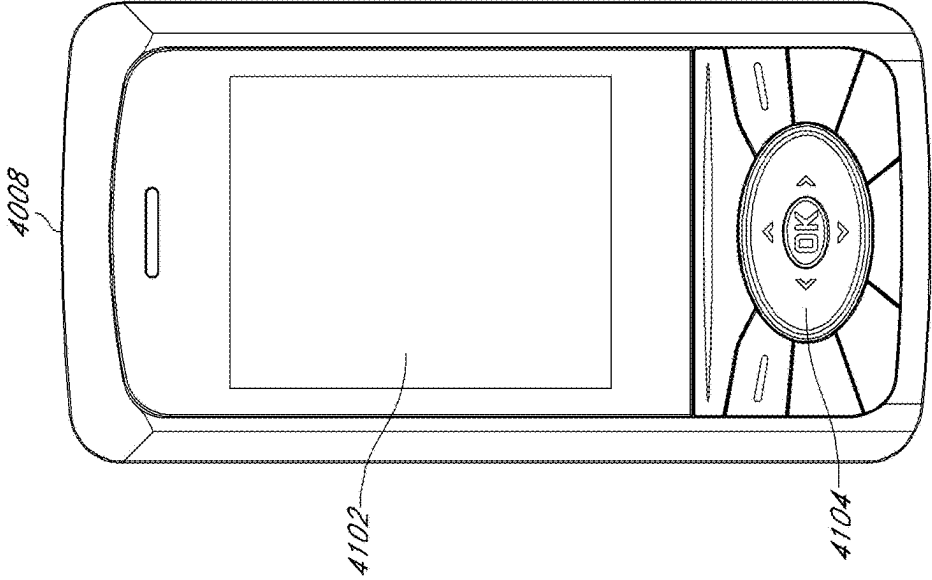


FIG. 41

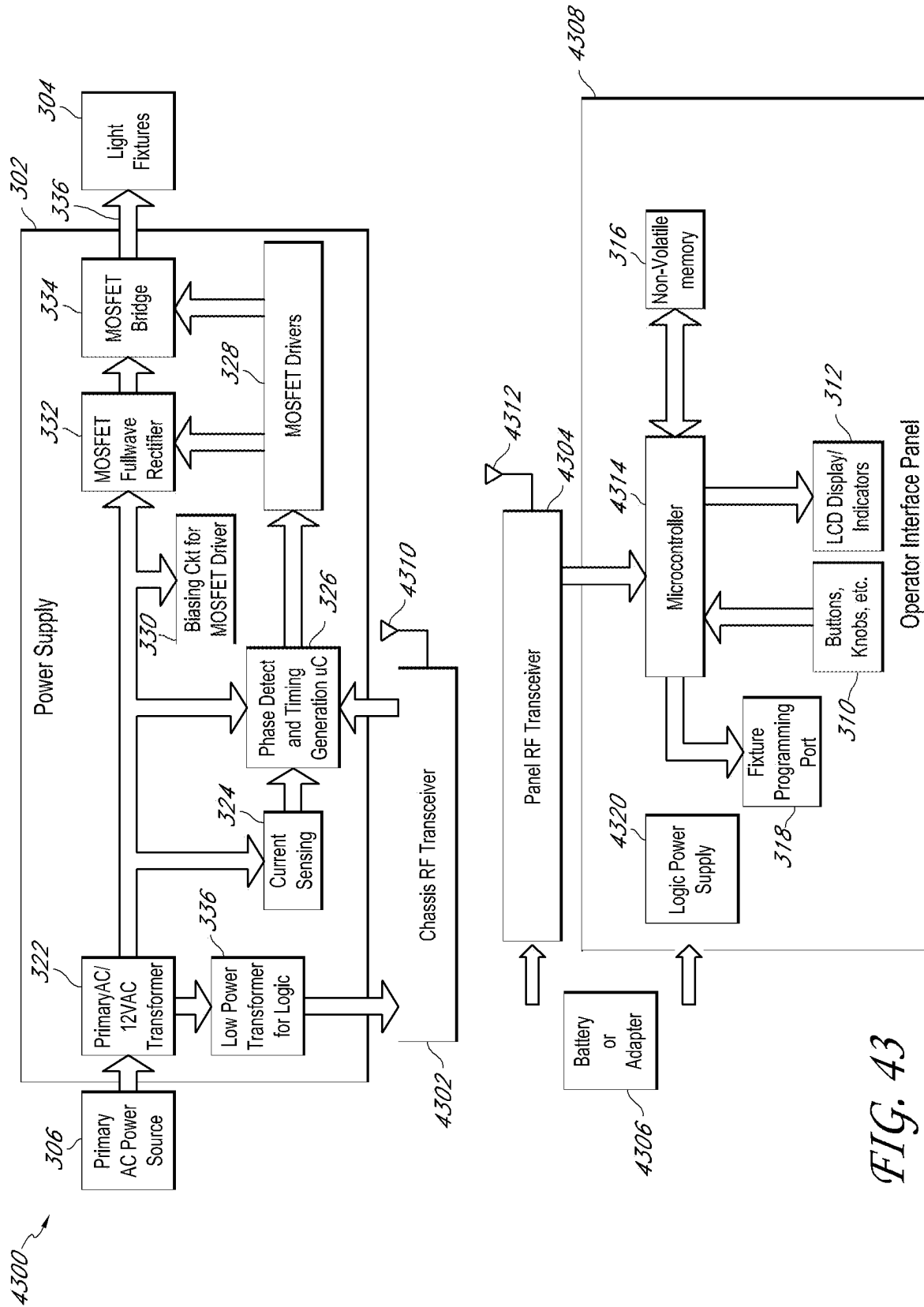


FIG. 43

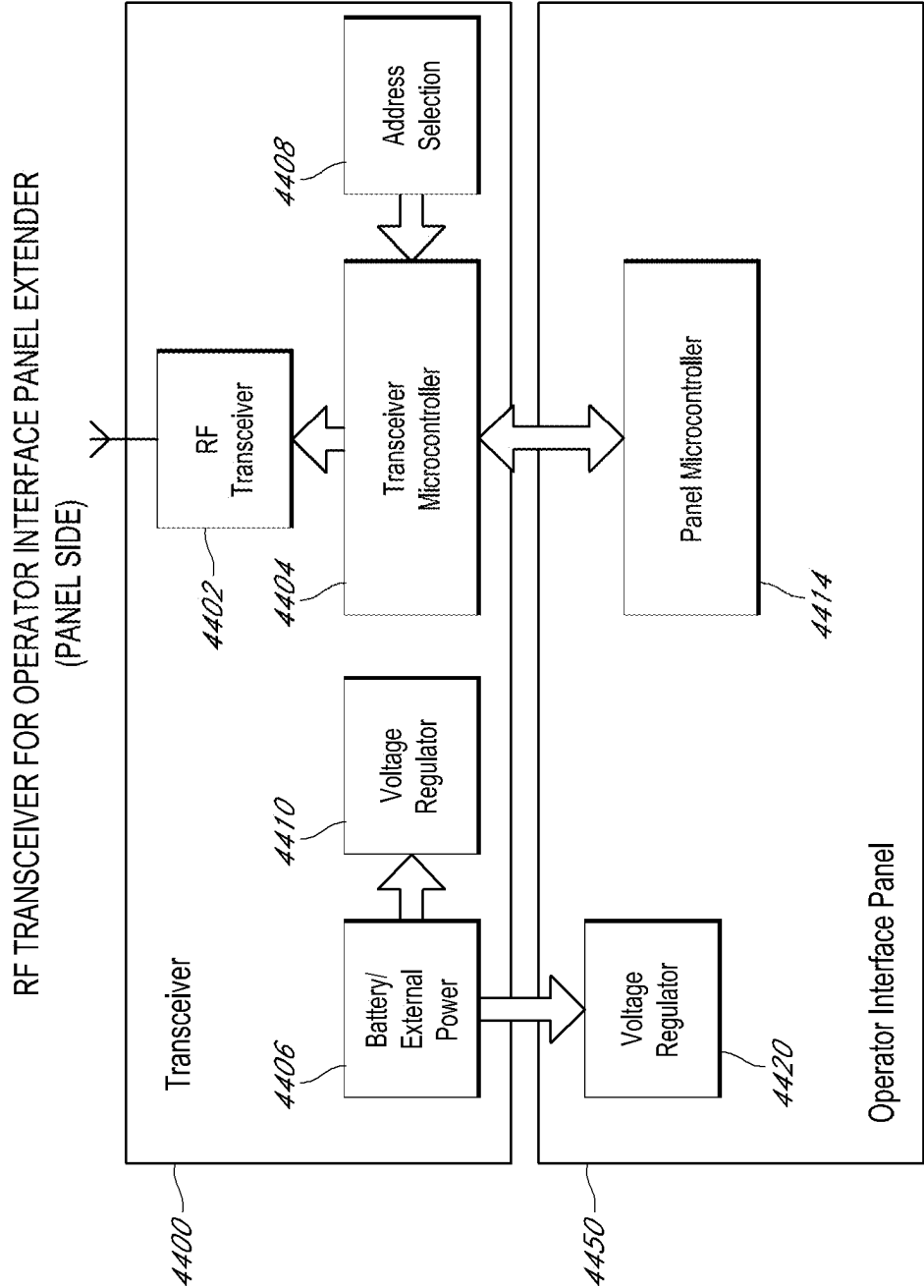


FIG. 44

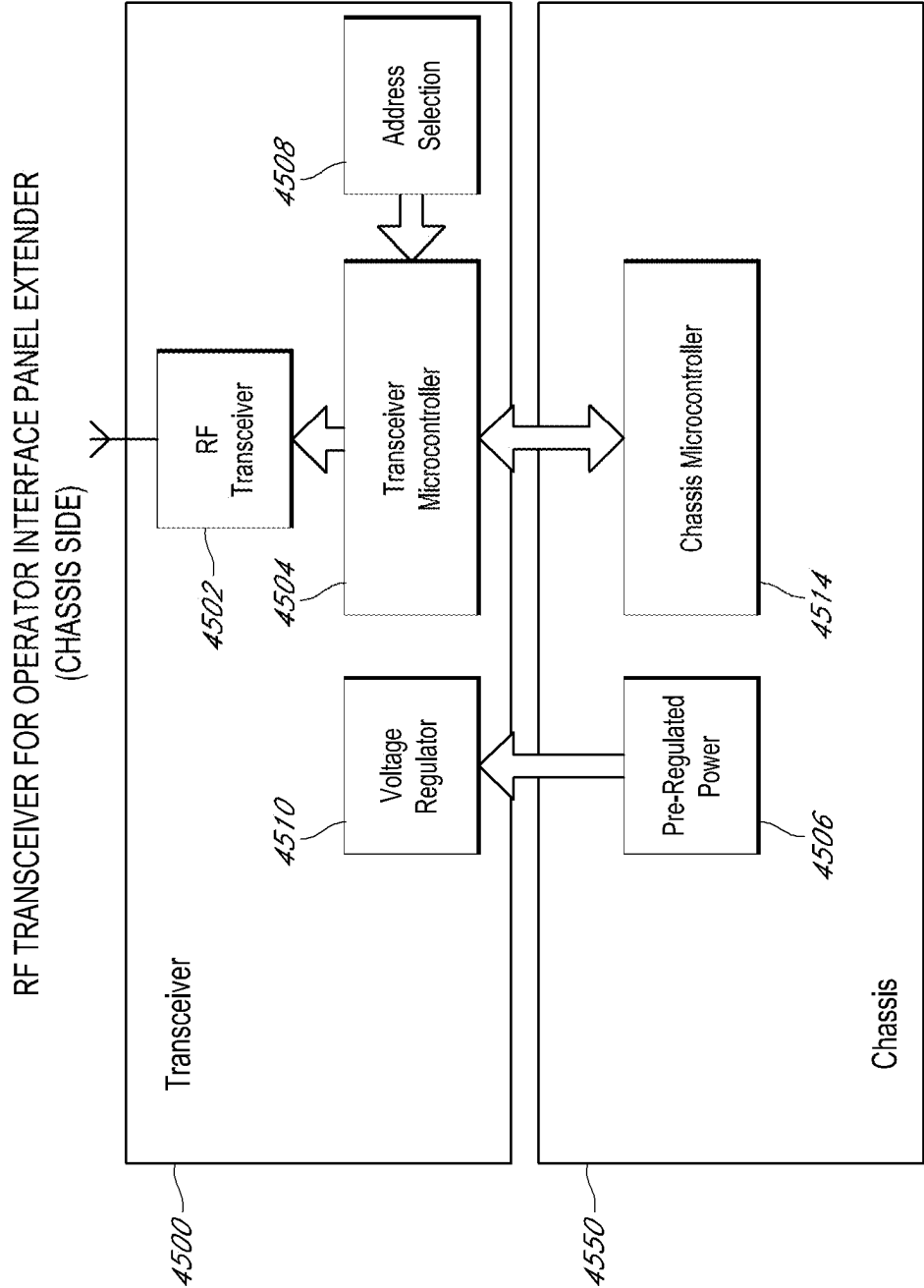


FIG. 45

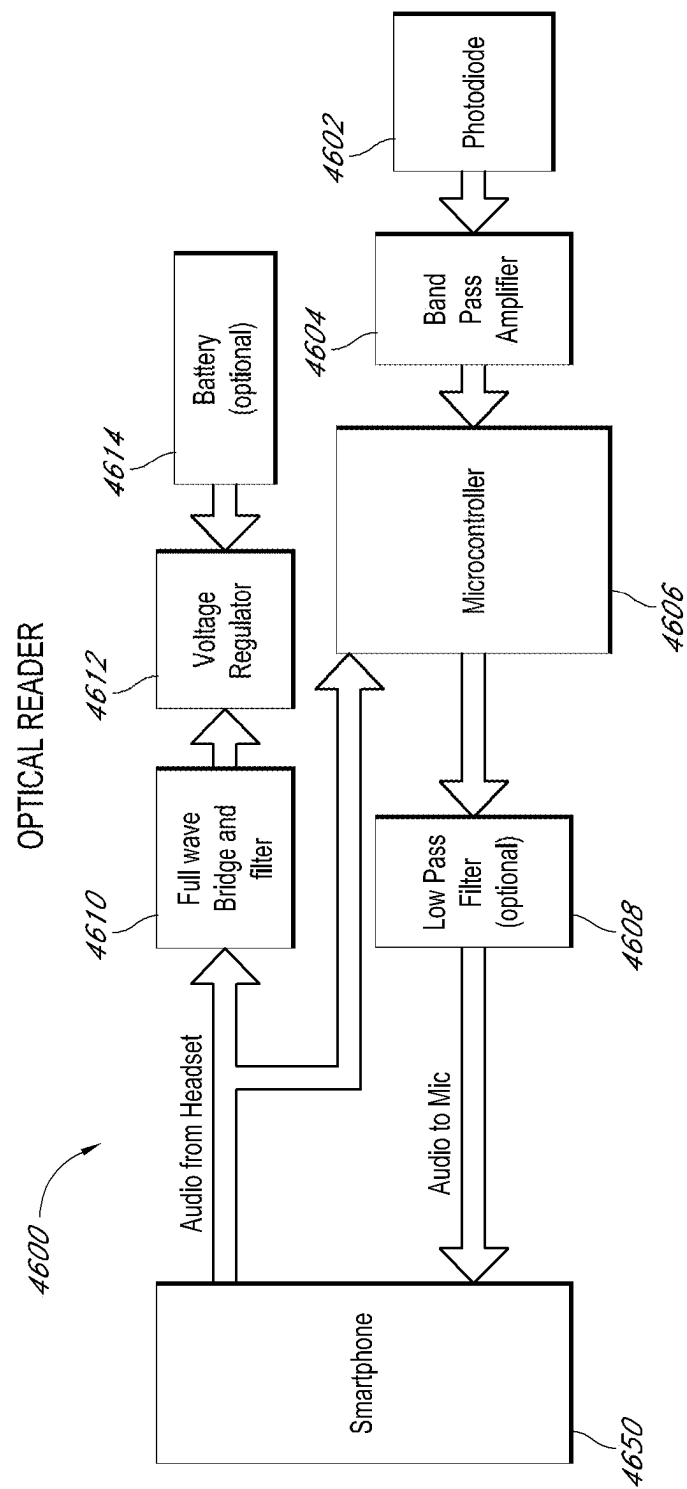
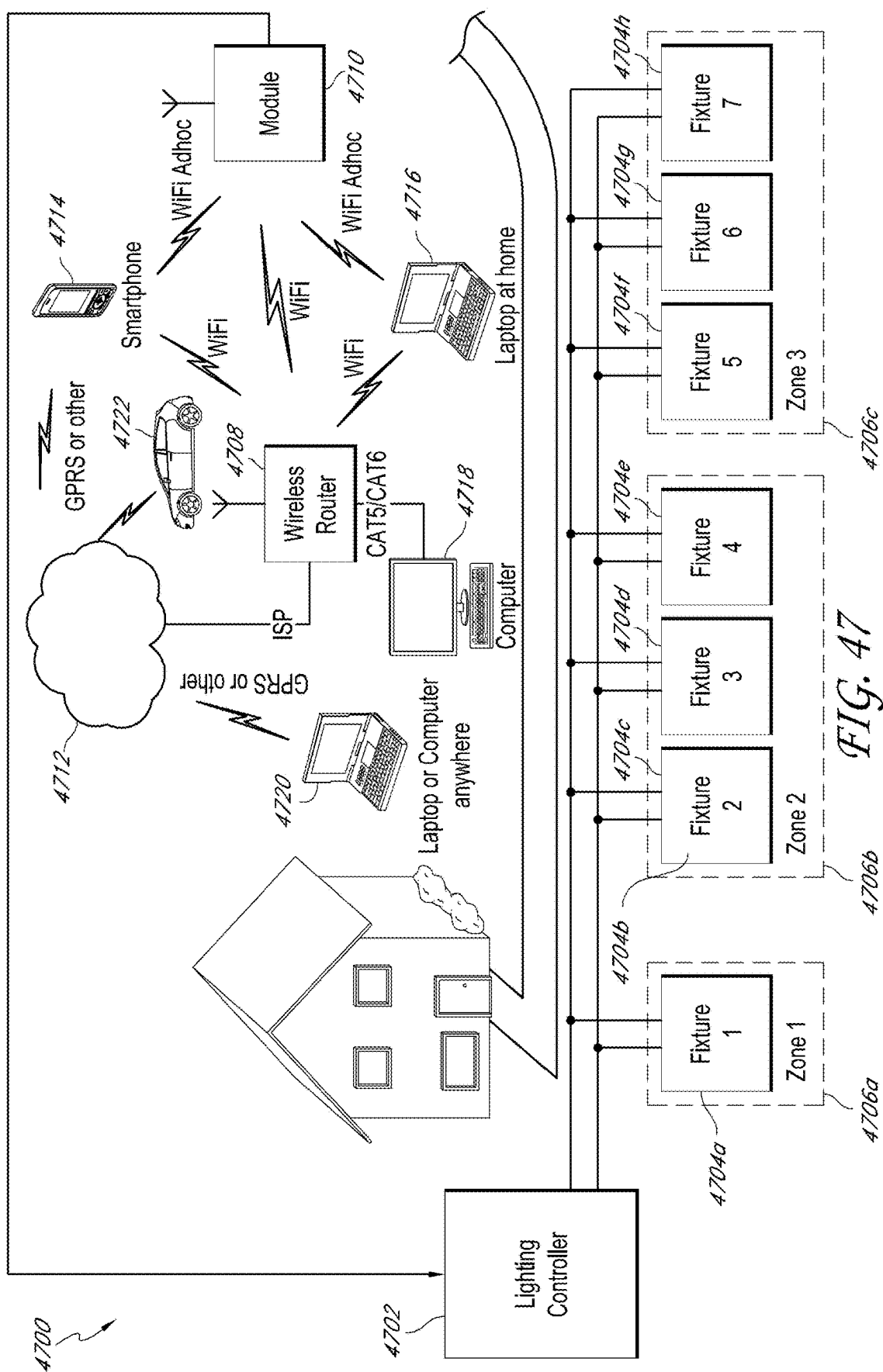
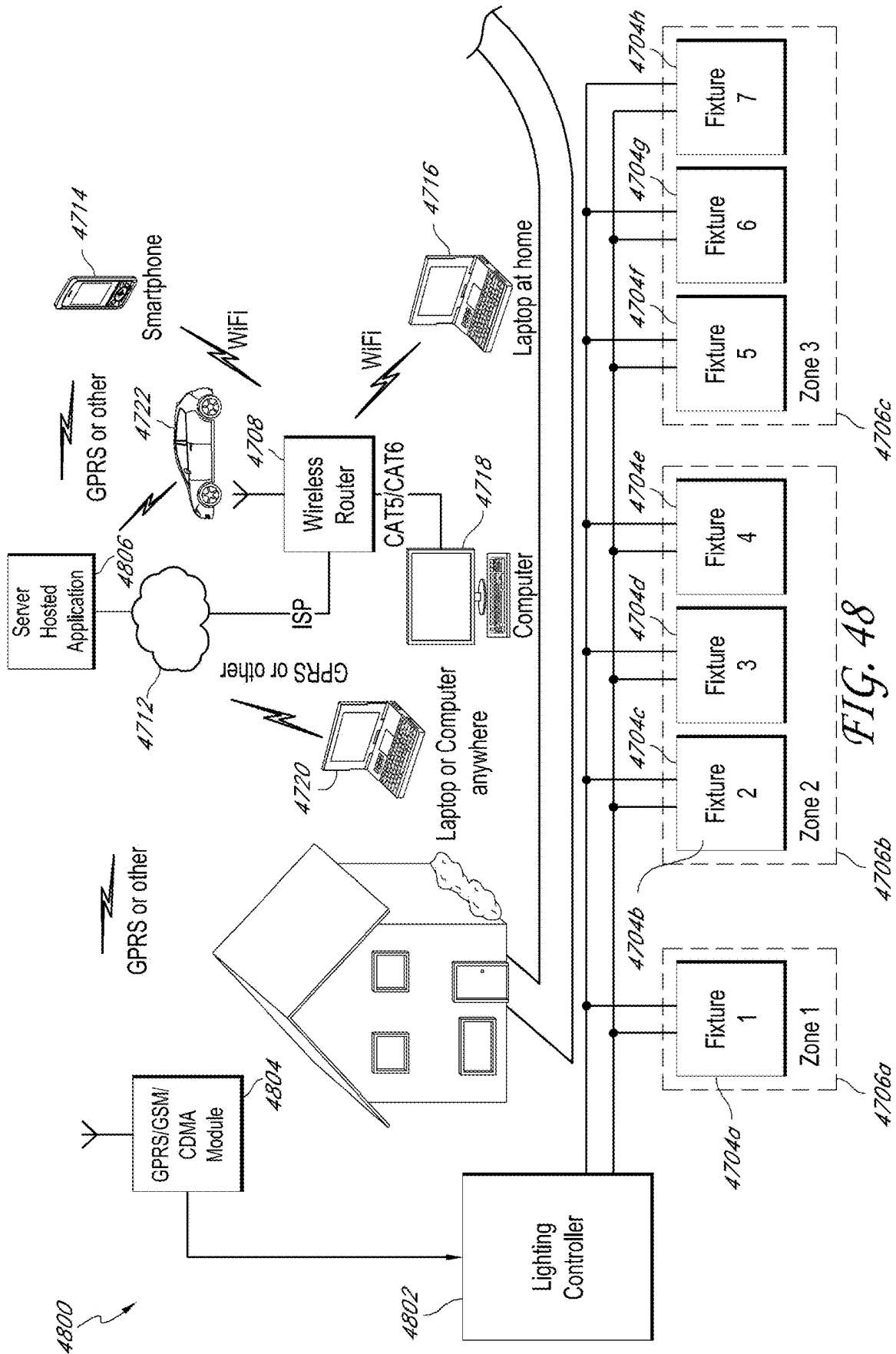


FIG. 46





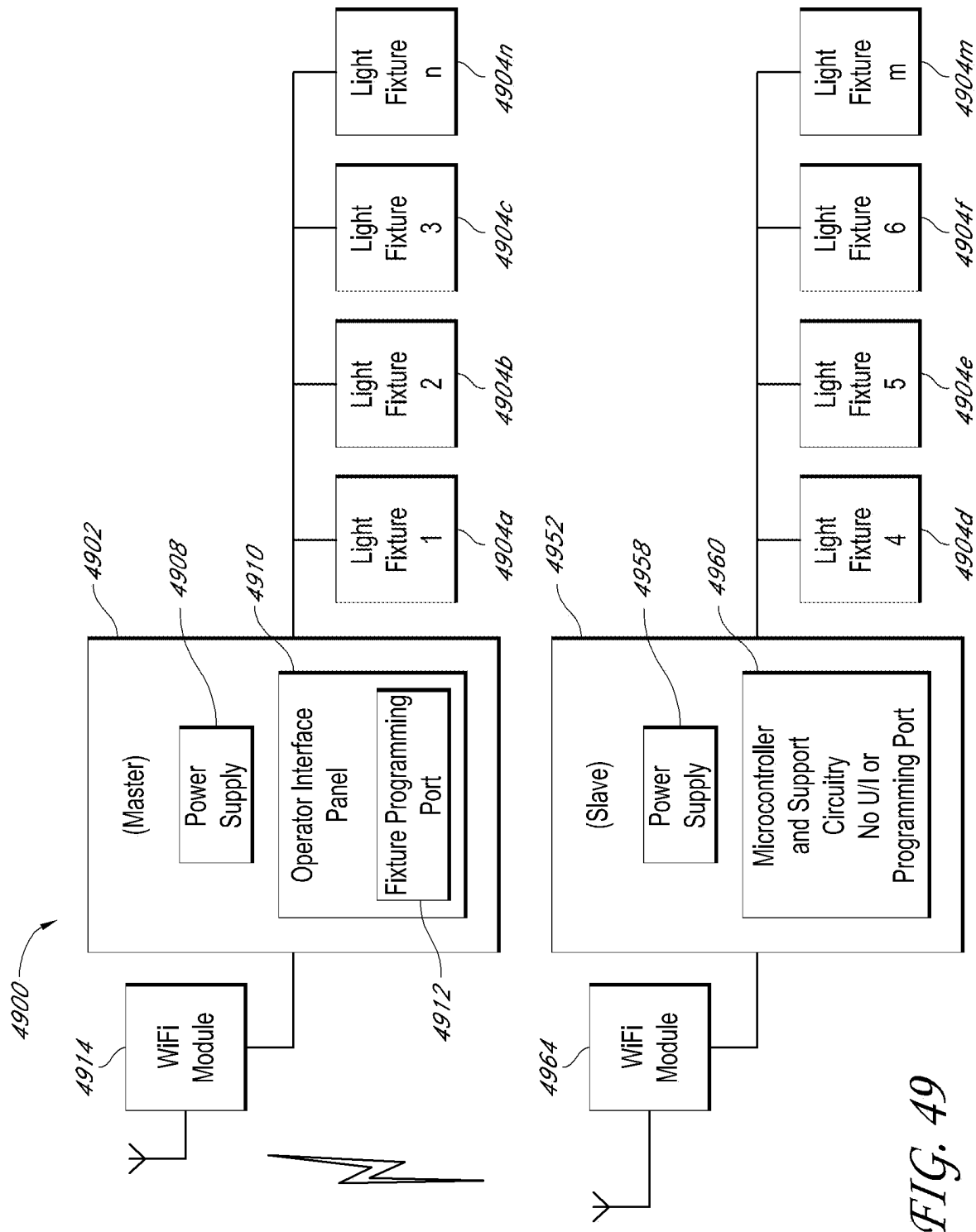


FIG. 49

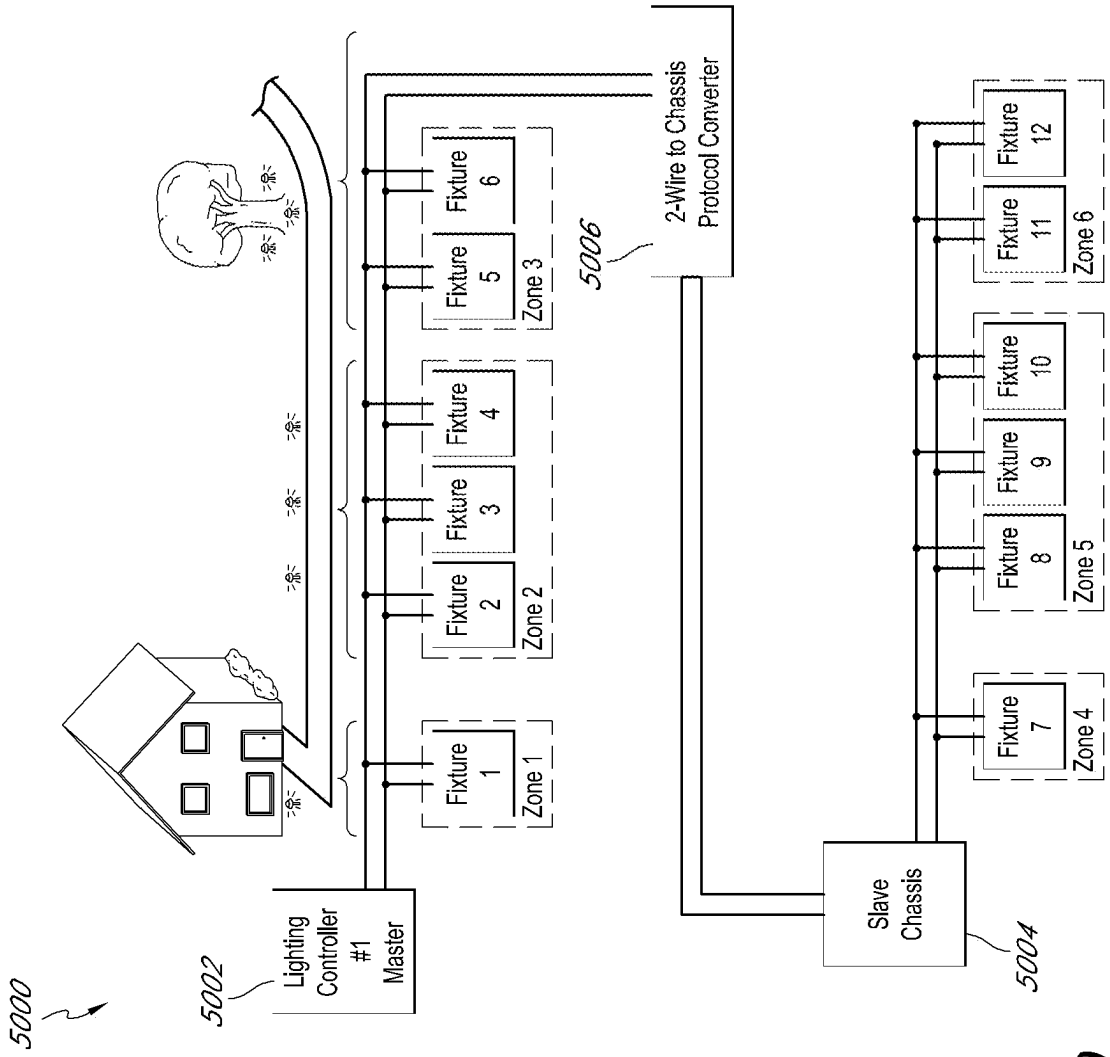


FIG. 50

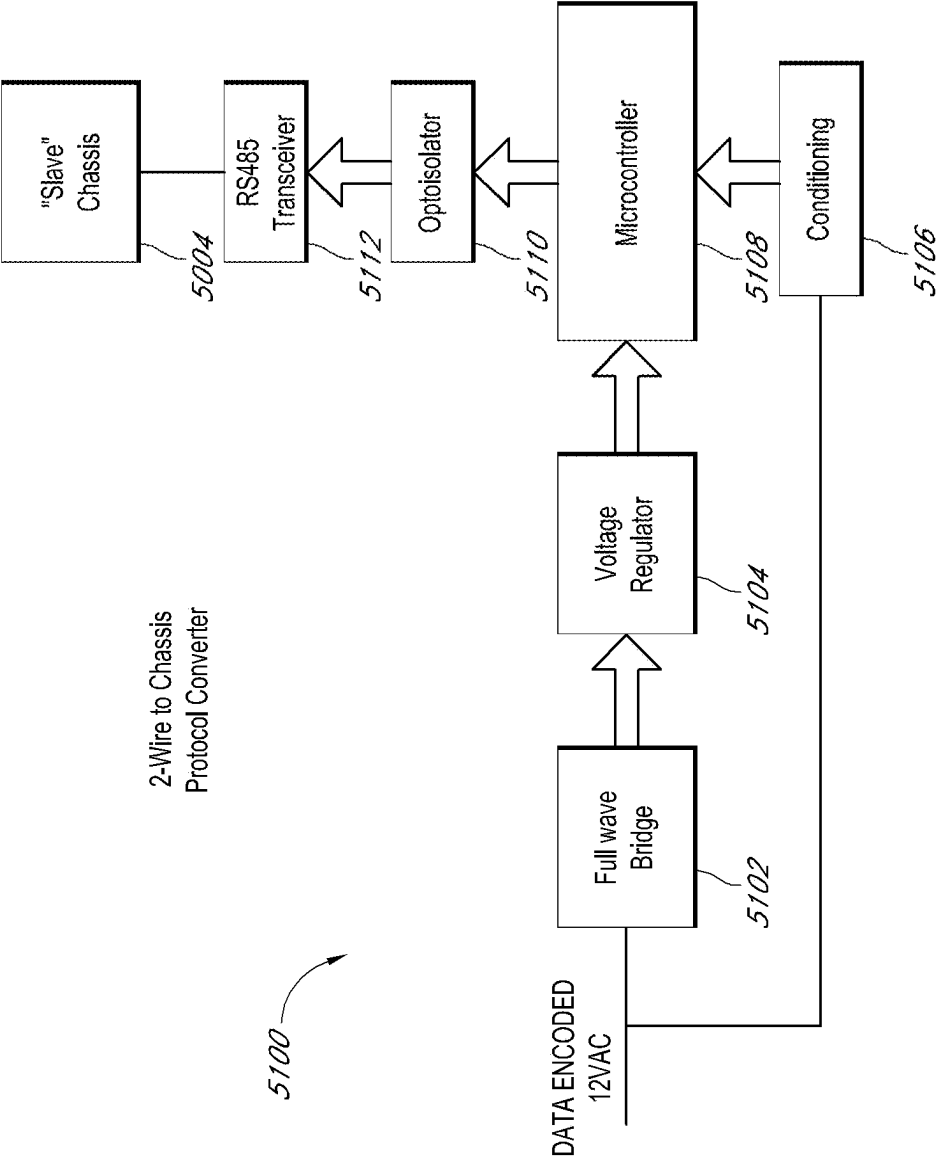


FIG. 51

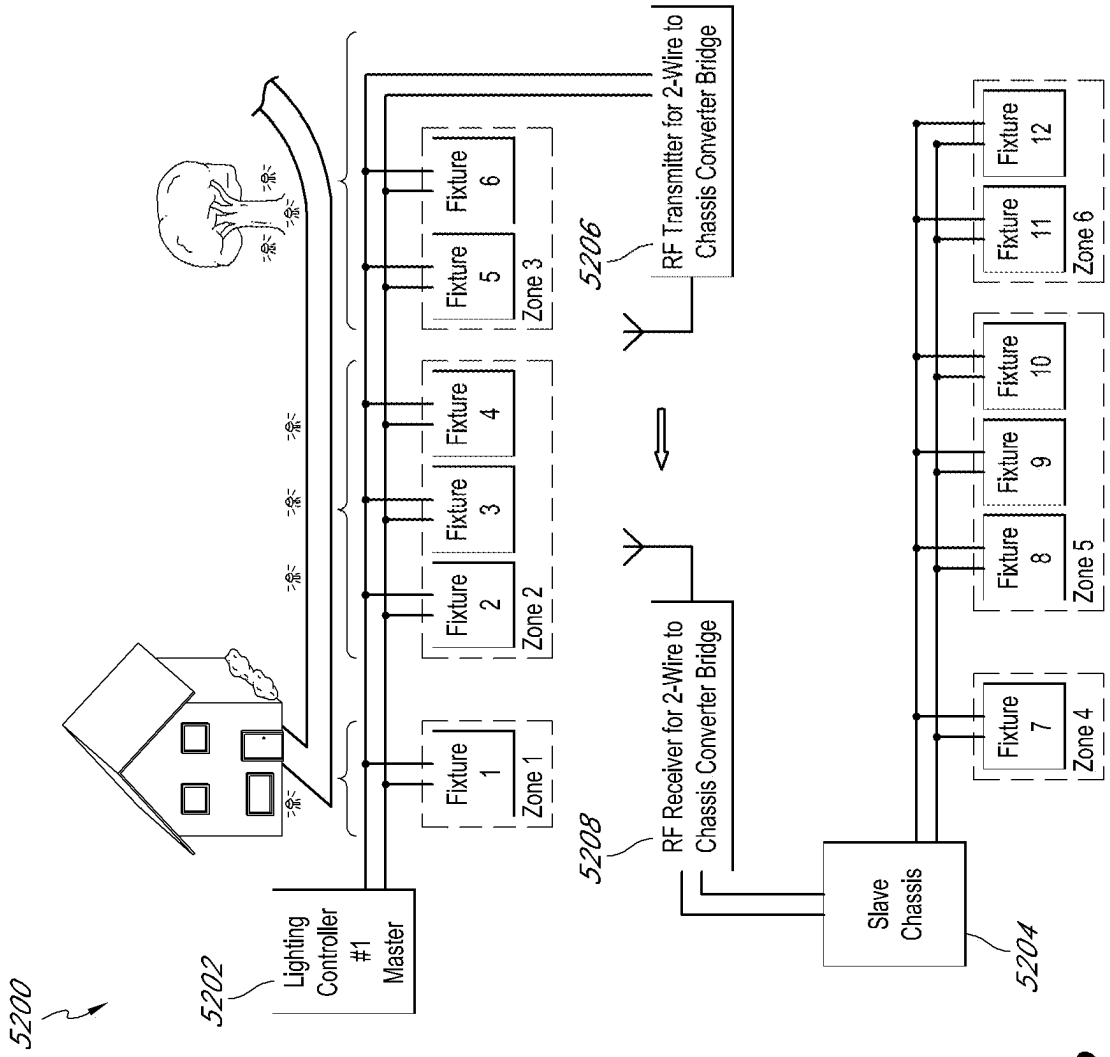


FIG. 52

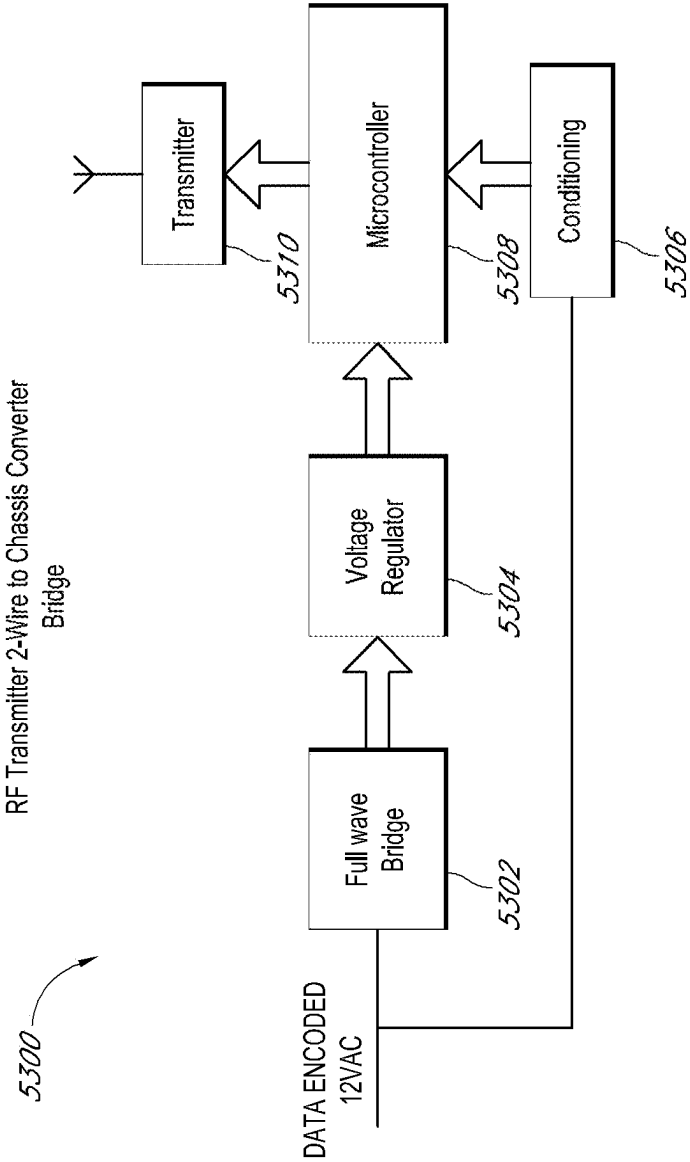


FIG. 53

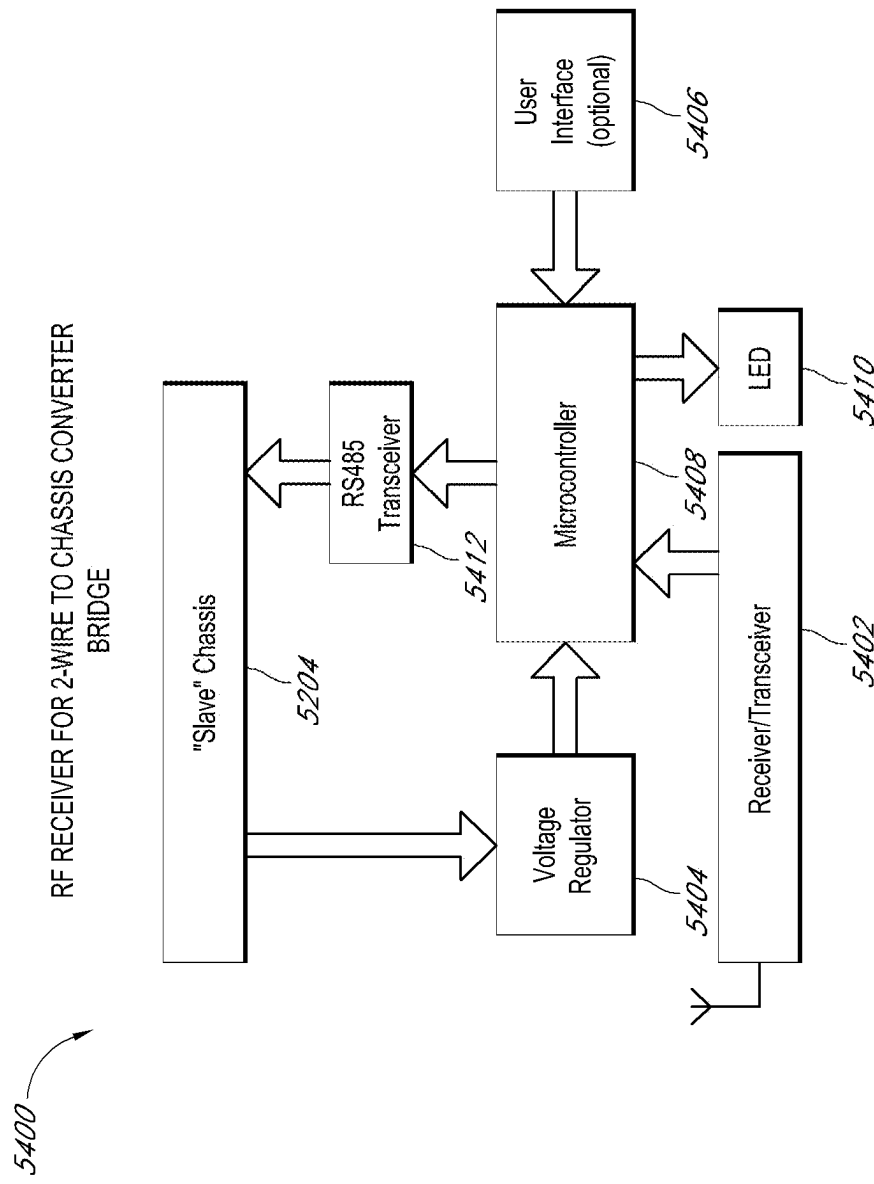
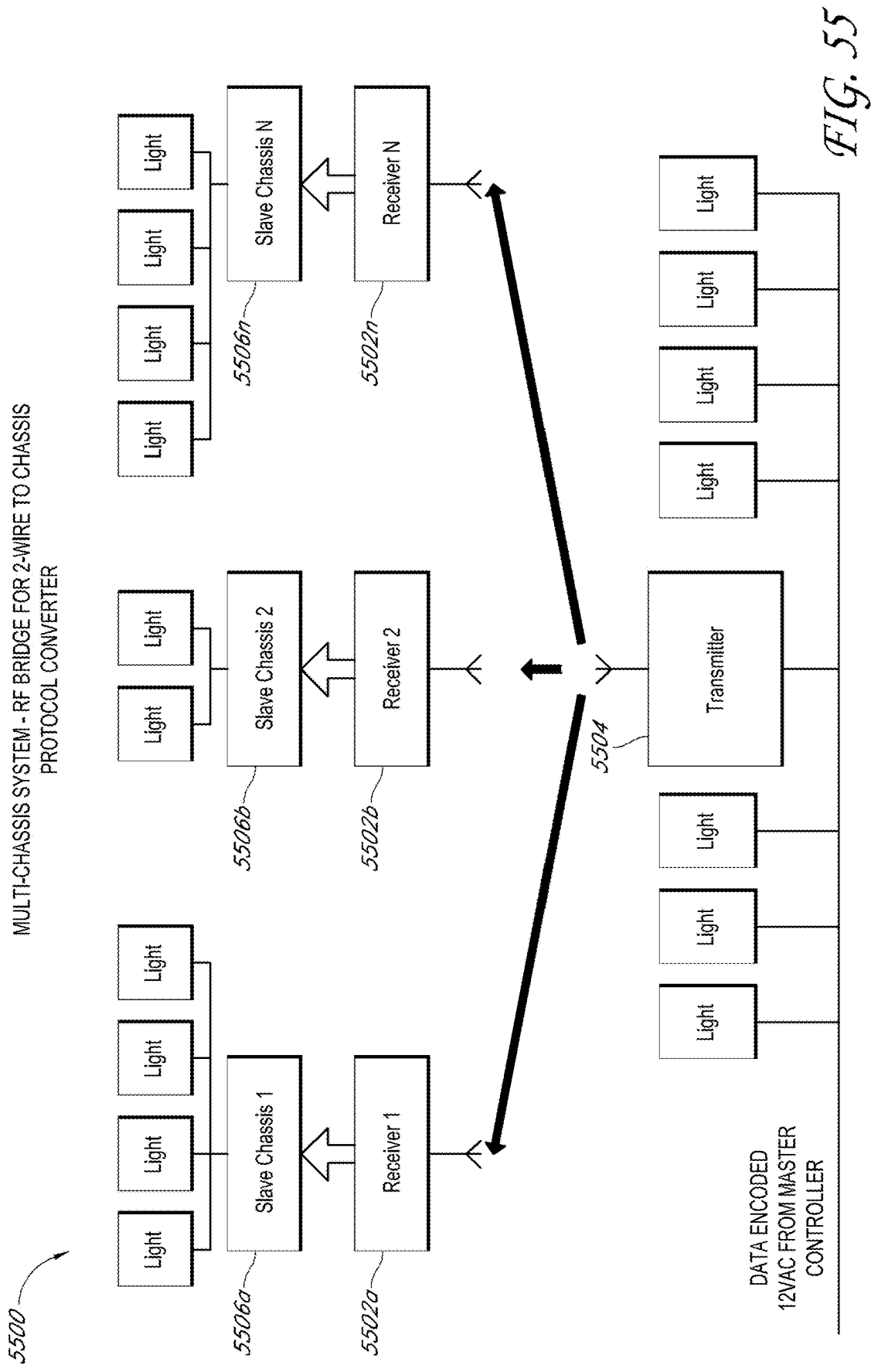


FIG. 54



5600

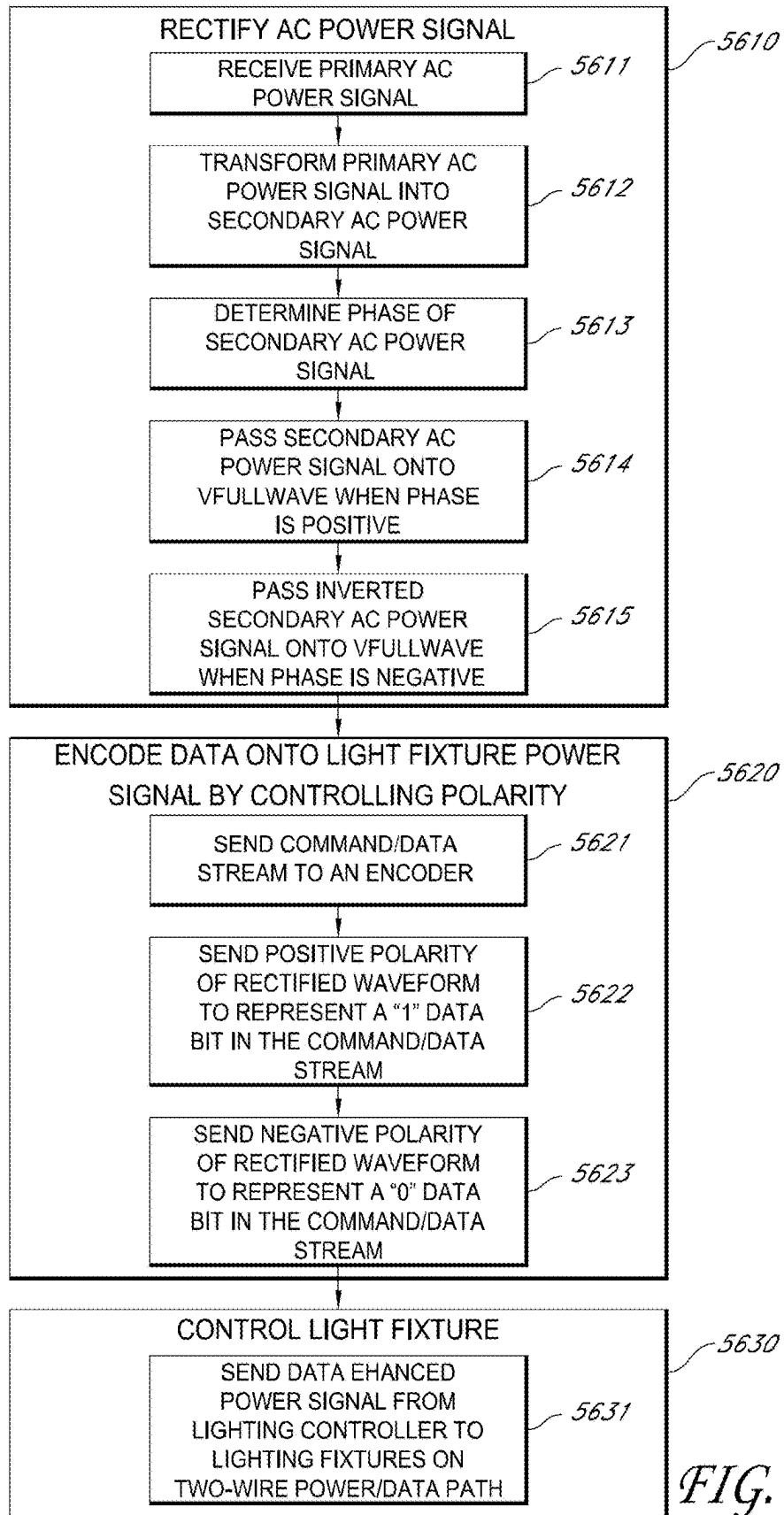


FIG. 56

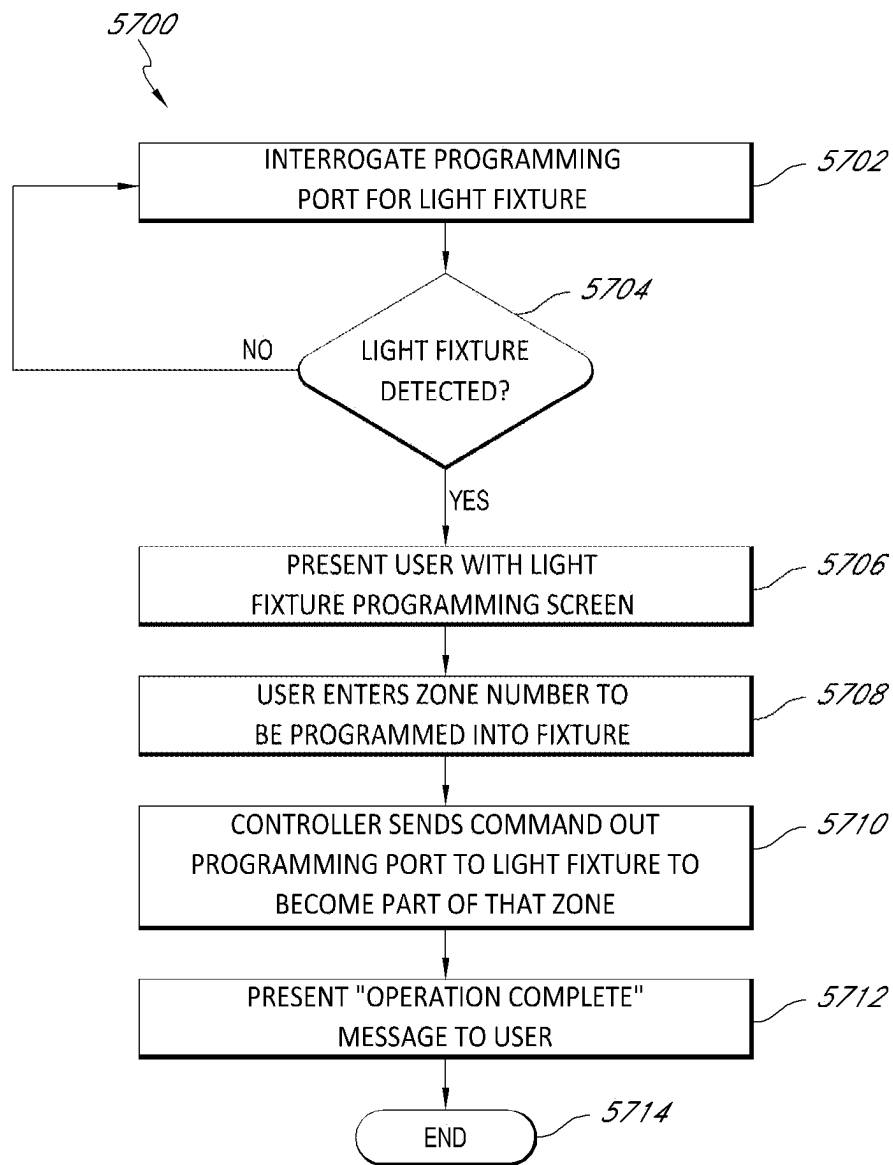
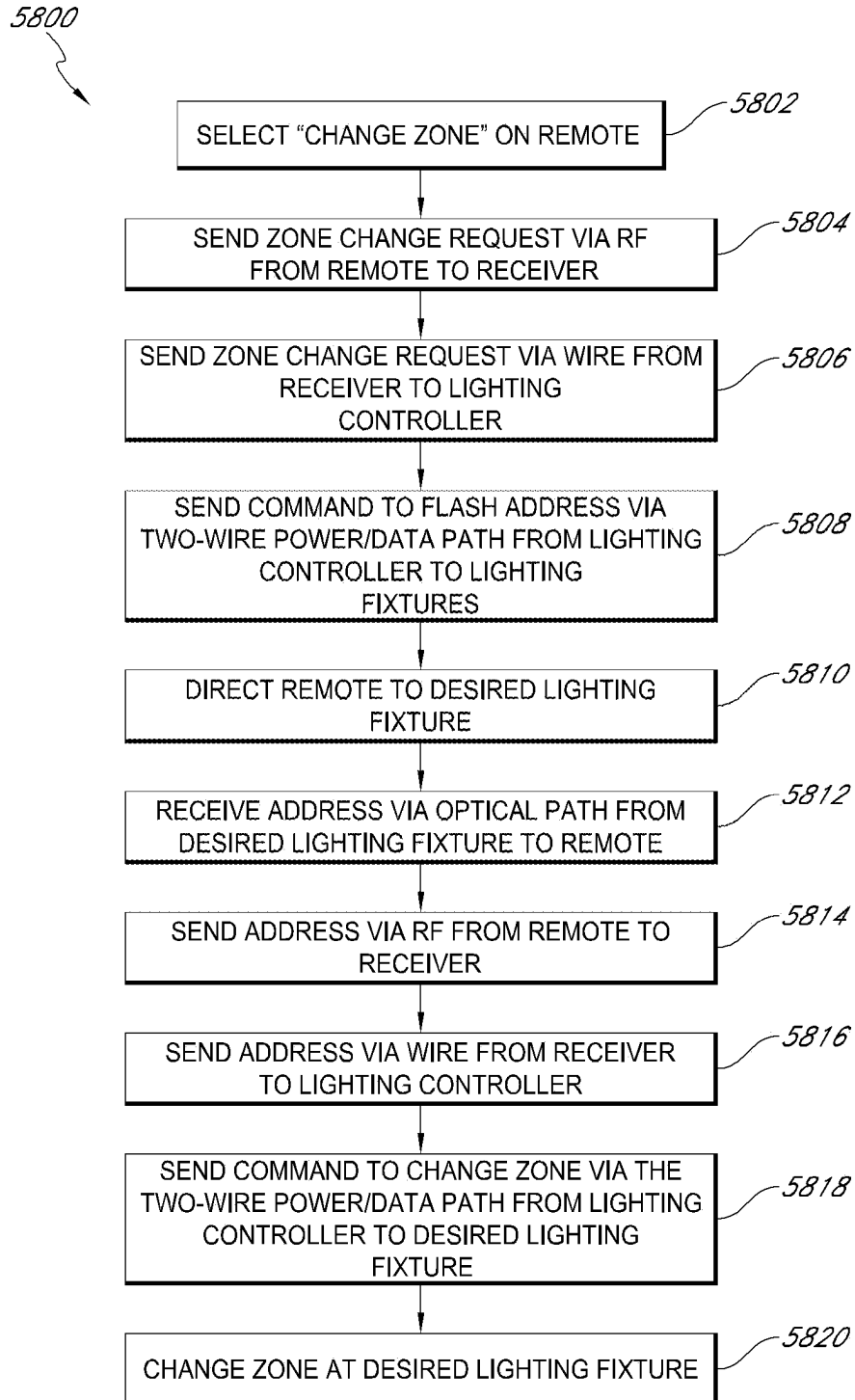


FIG. 57

*FIG. 58*

5900

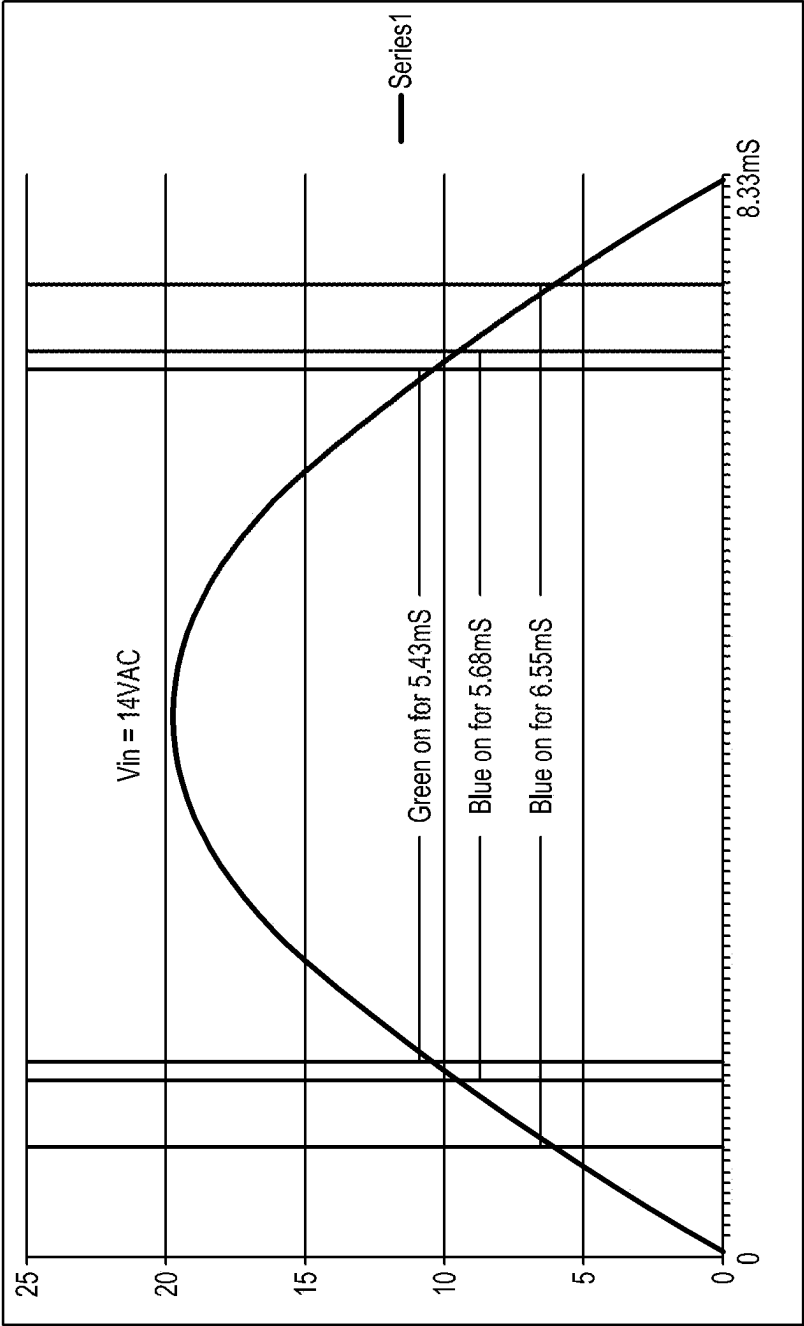


FIG. 59

6000

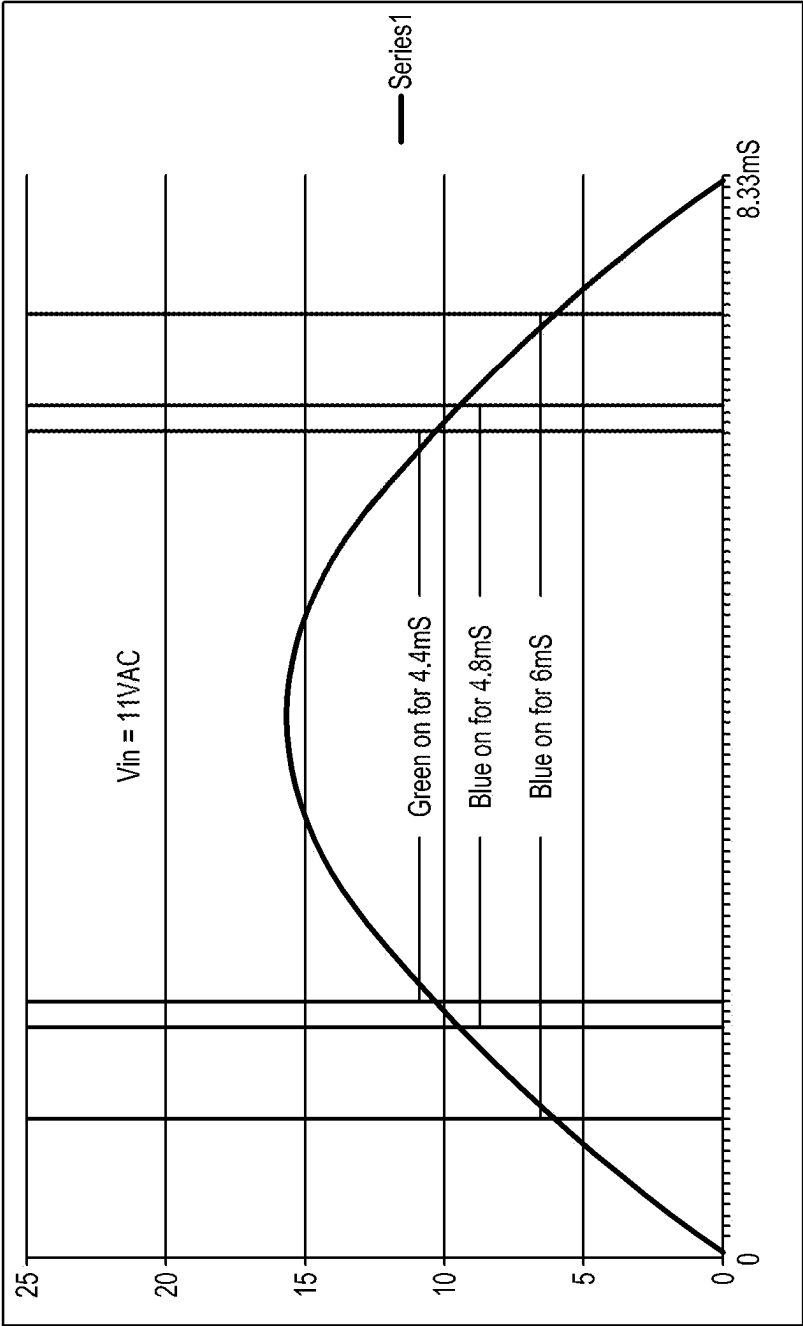


FIG. 60

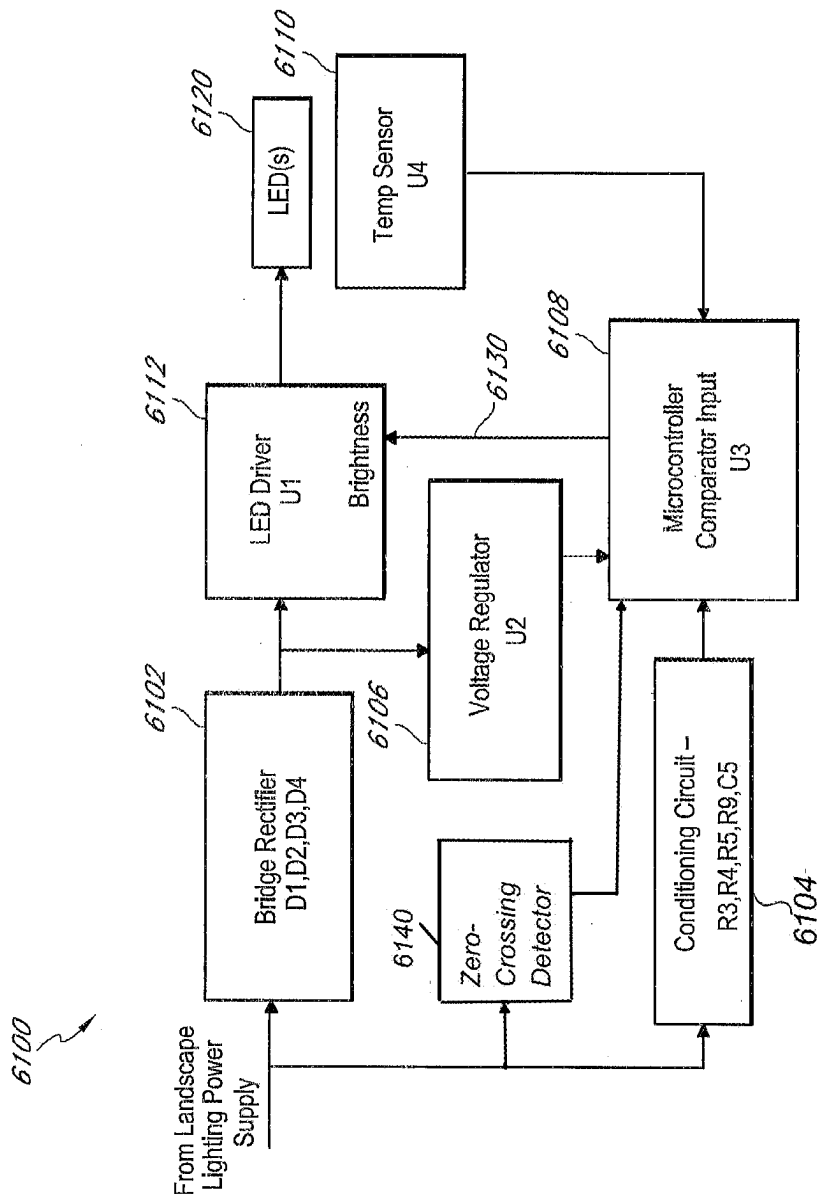
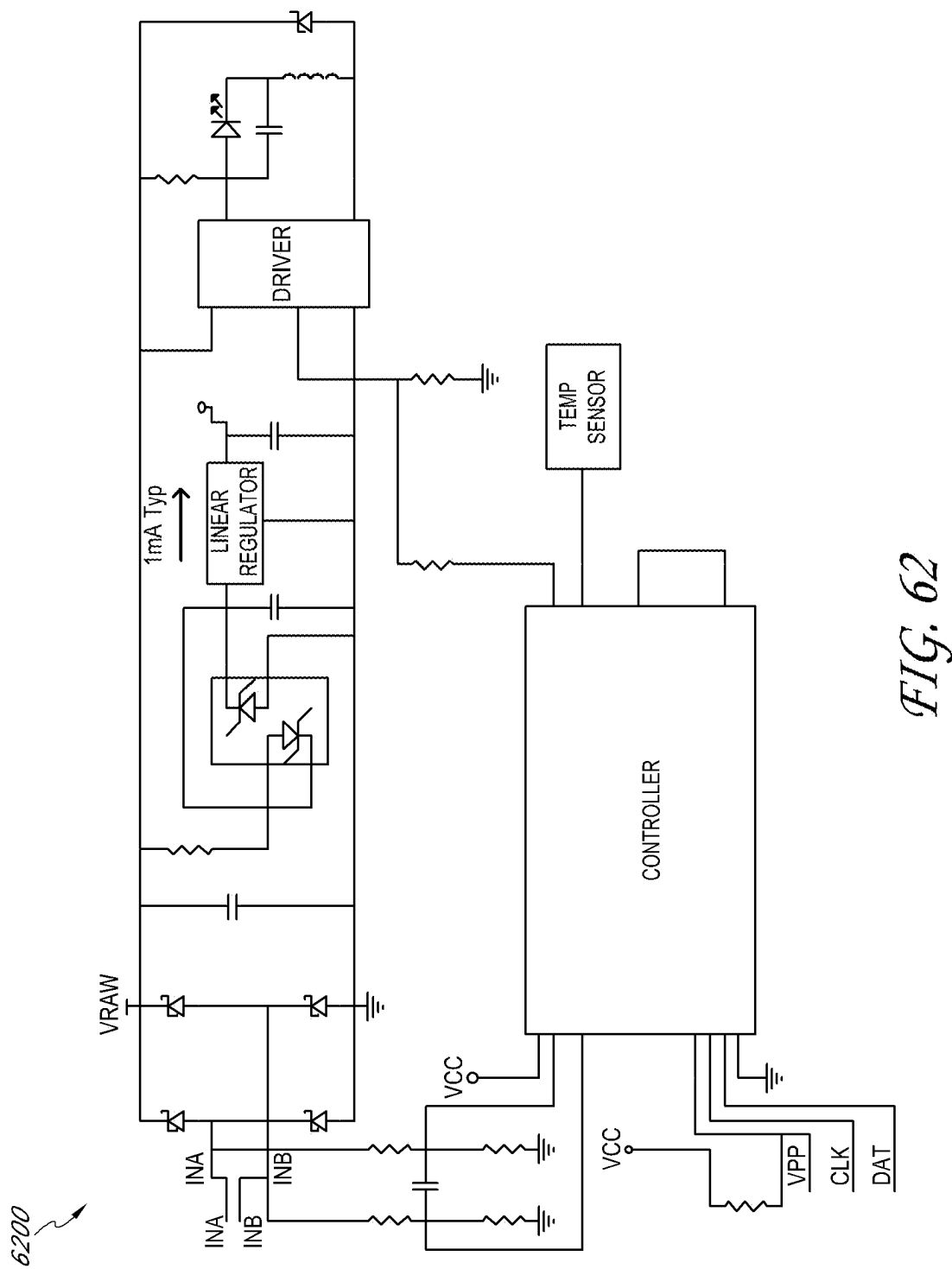


FIG. 61



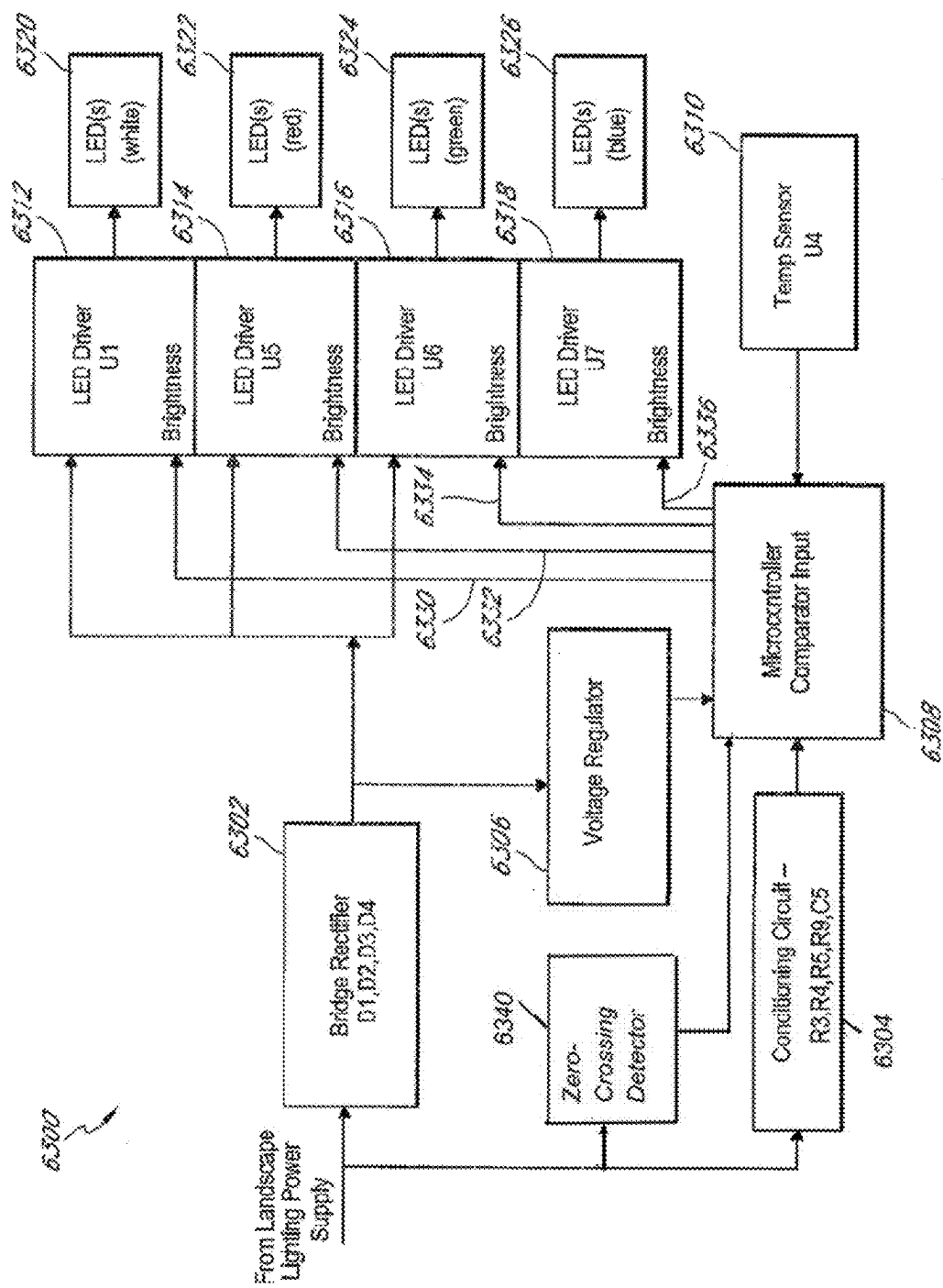


FIG. 63

6400

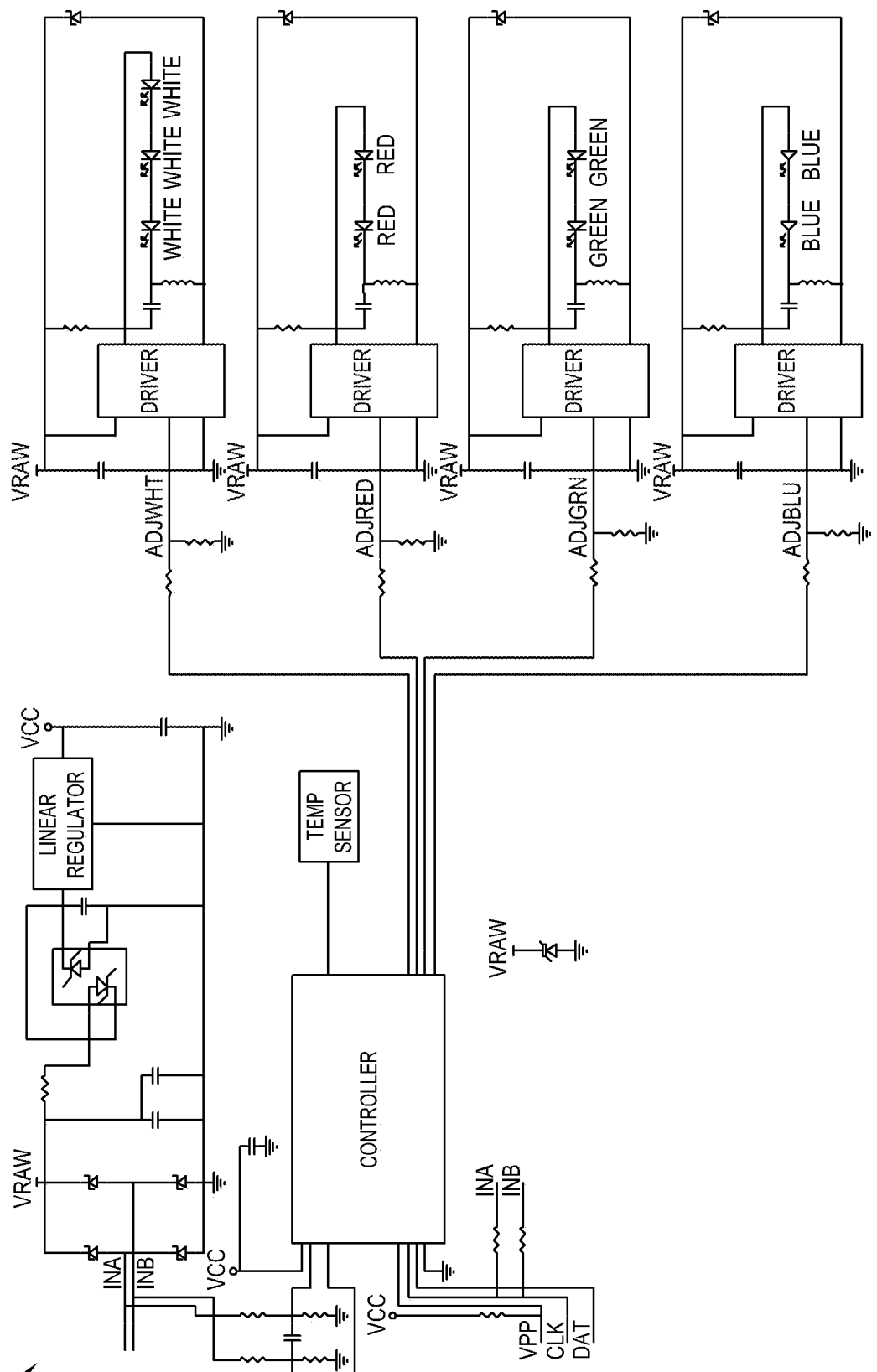
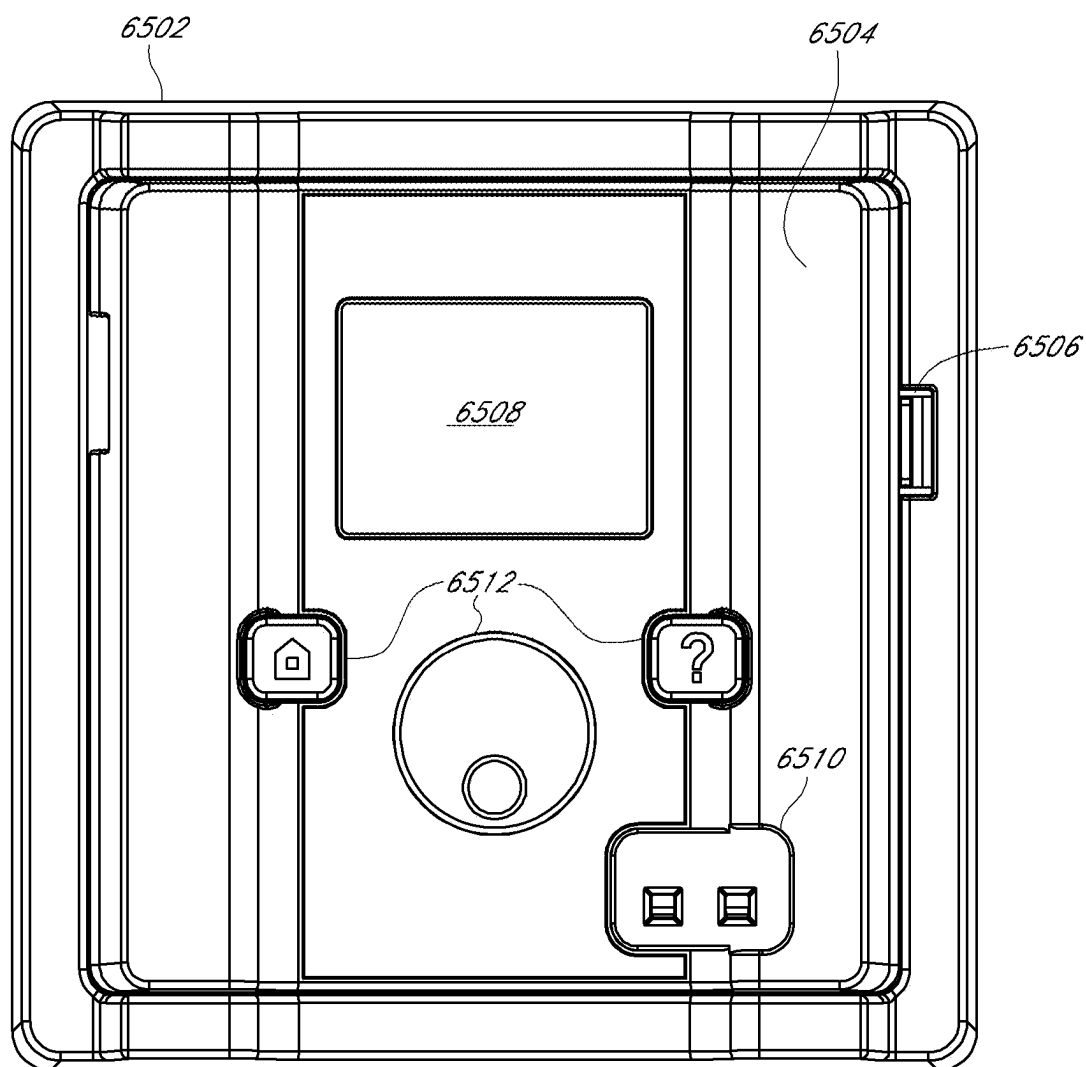
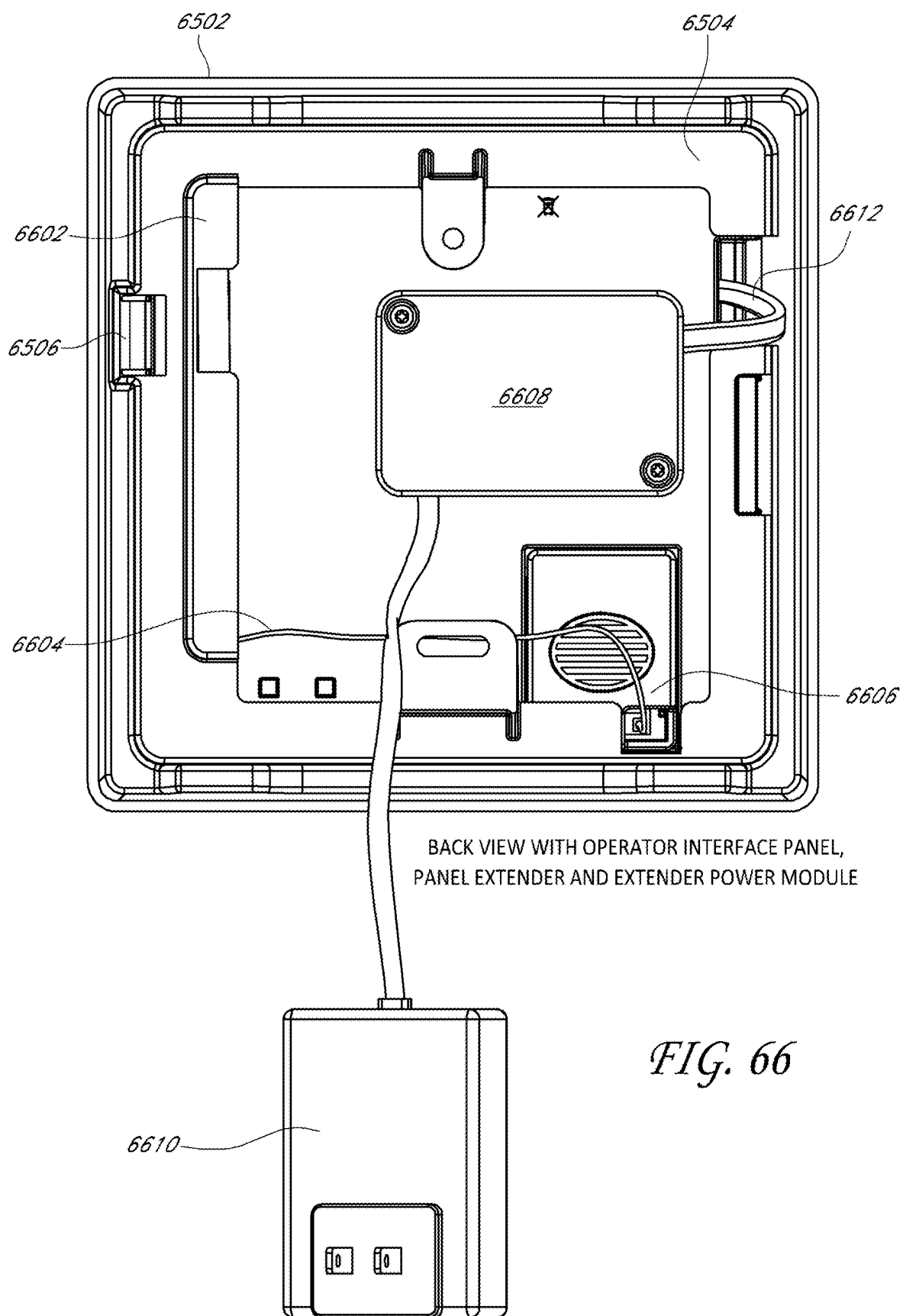


FIG. 64



FRONT VIEW WITH OPERATOR INTERFACE PANEL

FIG. 65



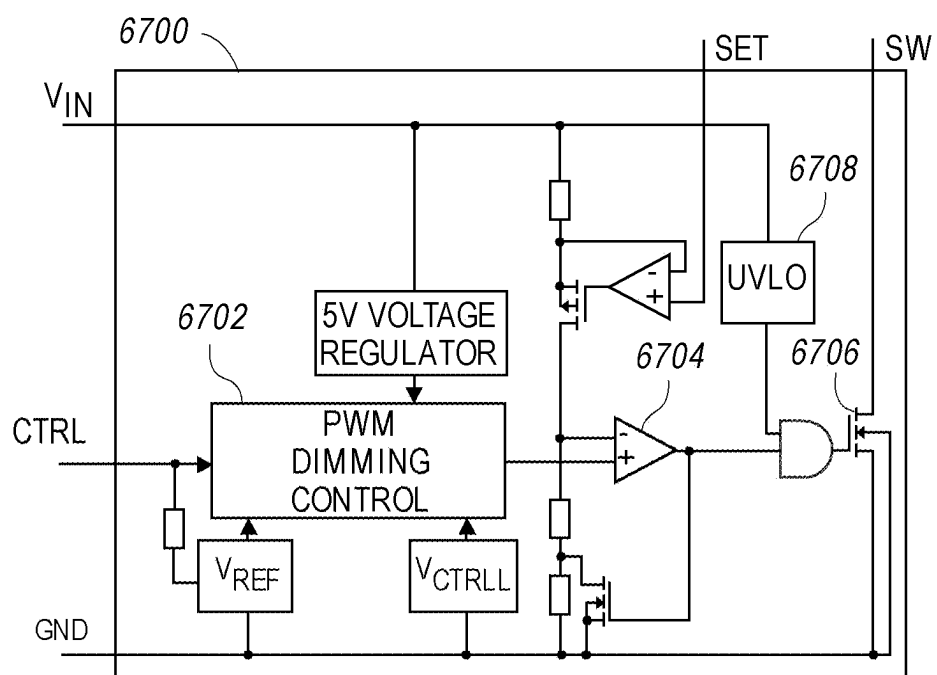


FIG. 67

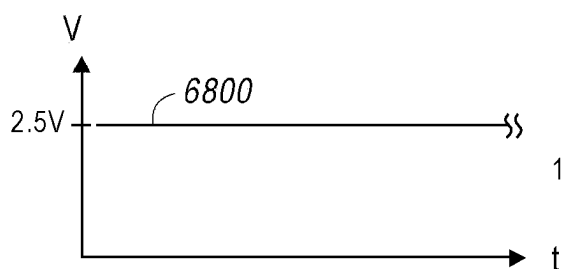


FIG. 68

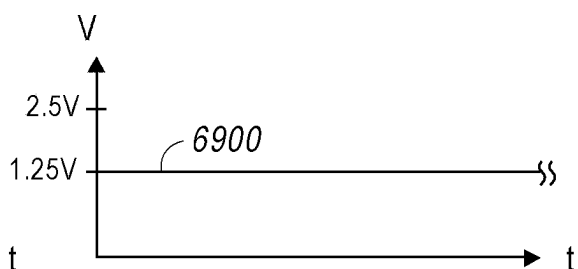


FIG. 69

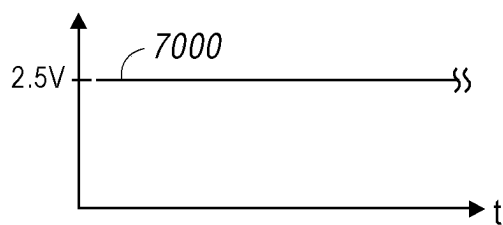


FIG. 70

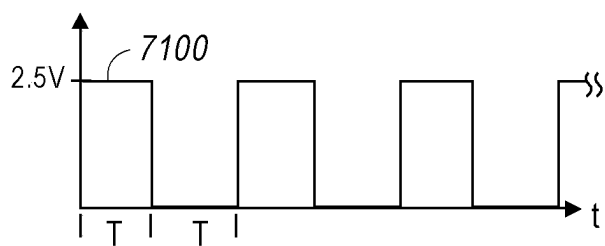


FIG. 71

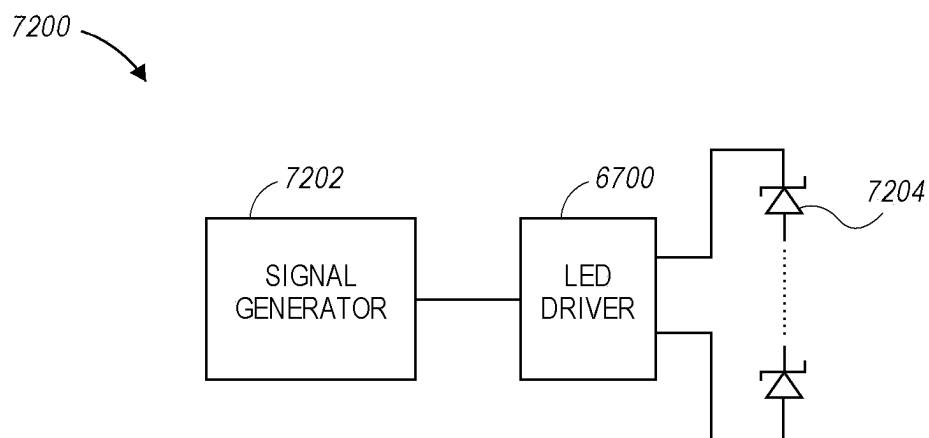


FIG. 72

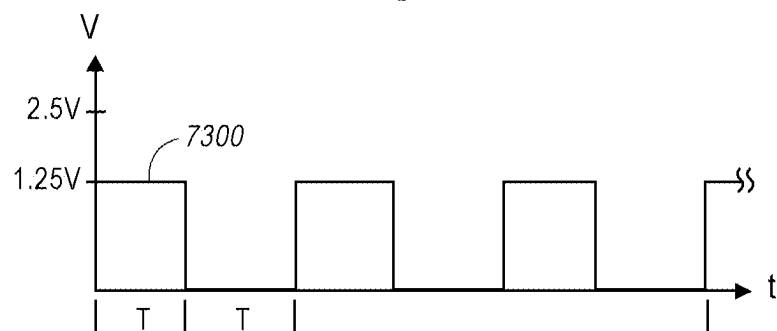


FIG. 73

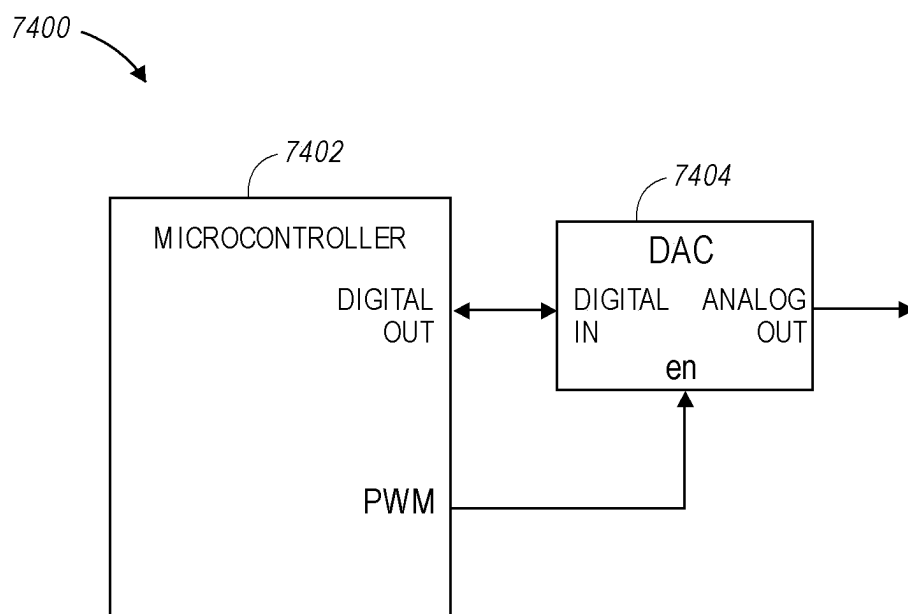


FIG. 74

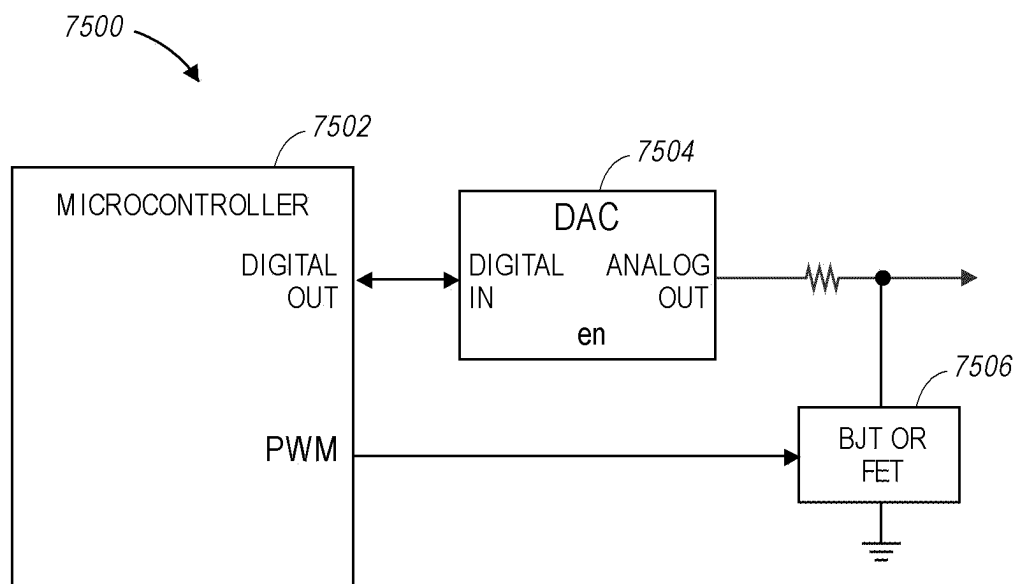


FIG. 75

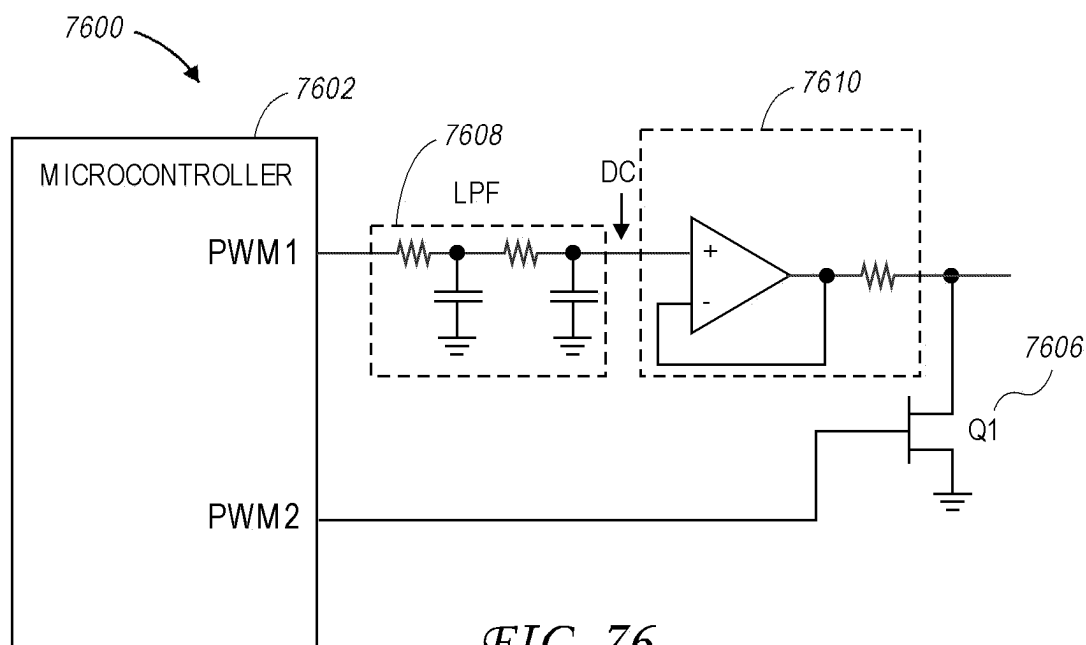


FIG. 76

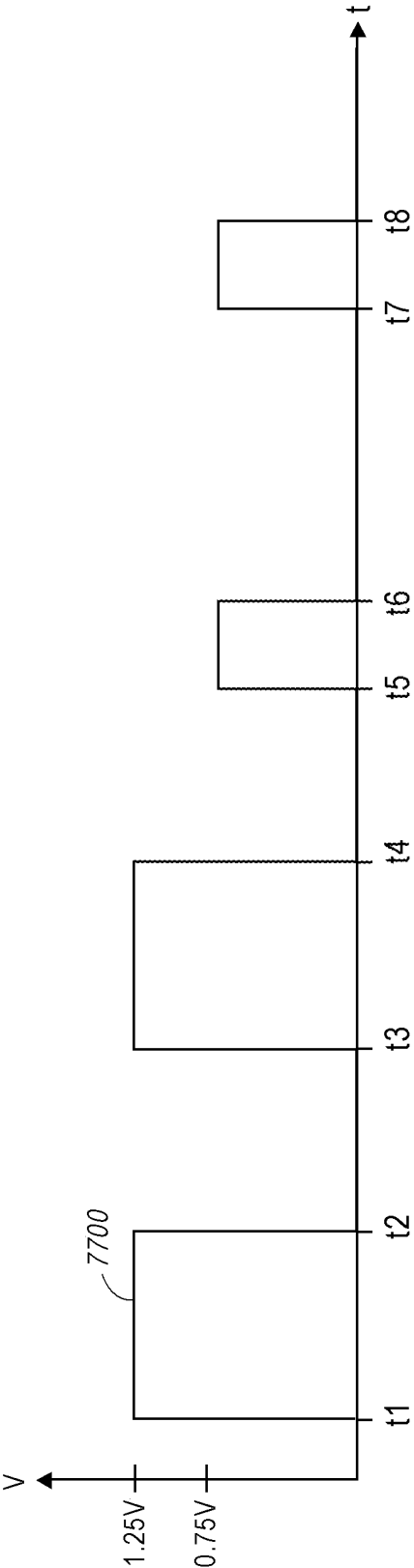


FIG. 77

7800

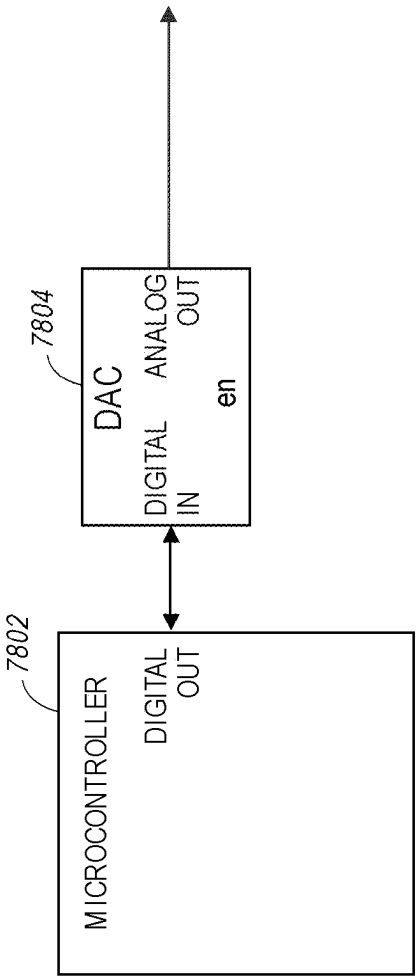


FIG. 78

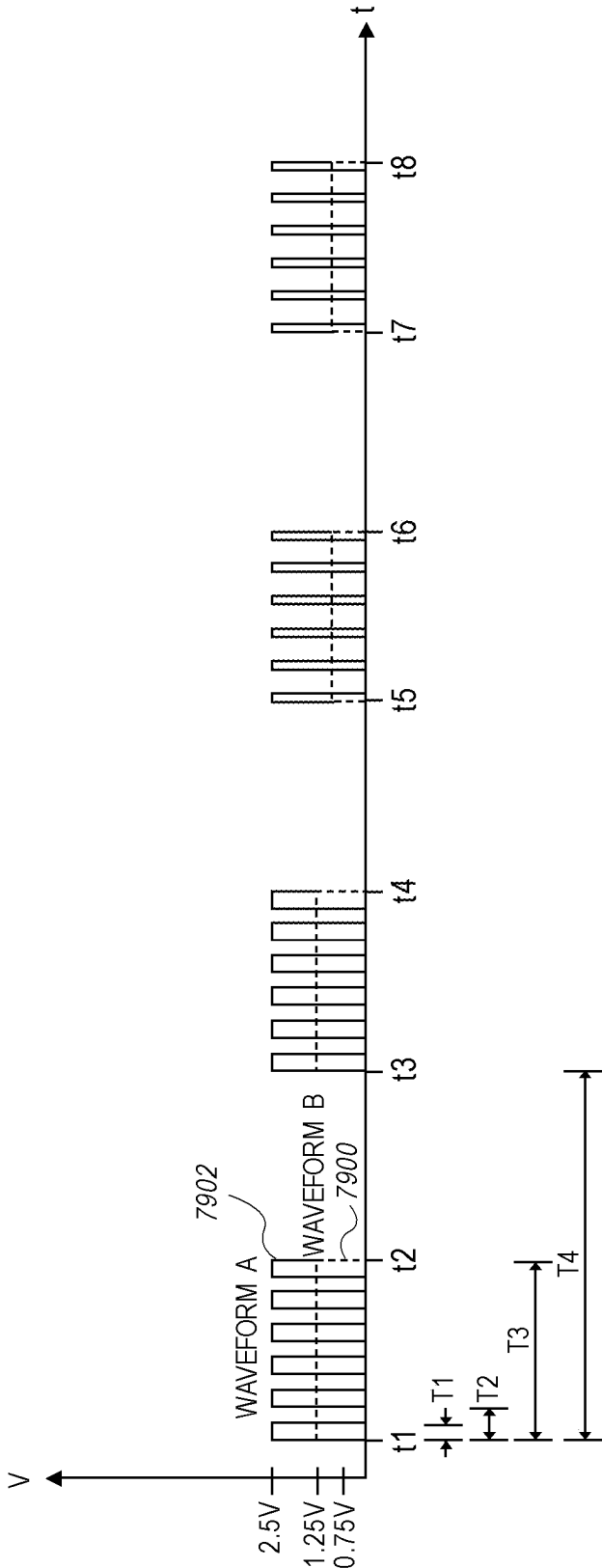


FIG. 79

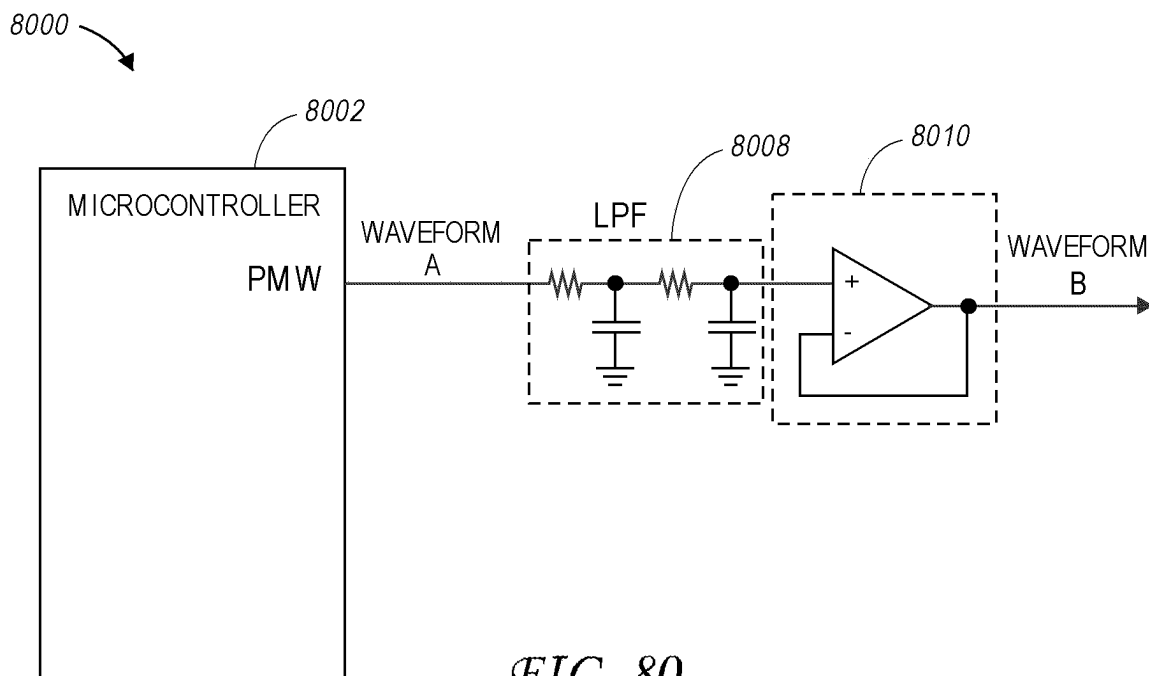


FIG. 80

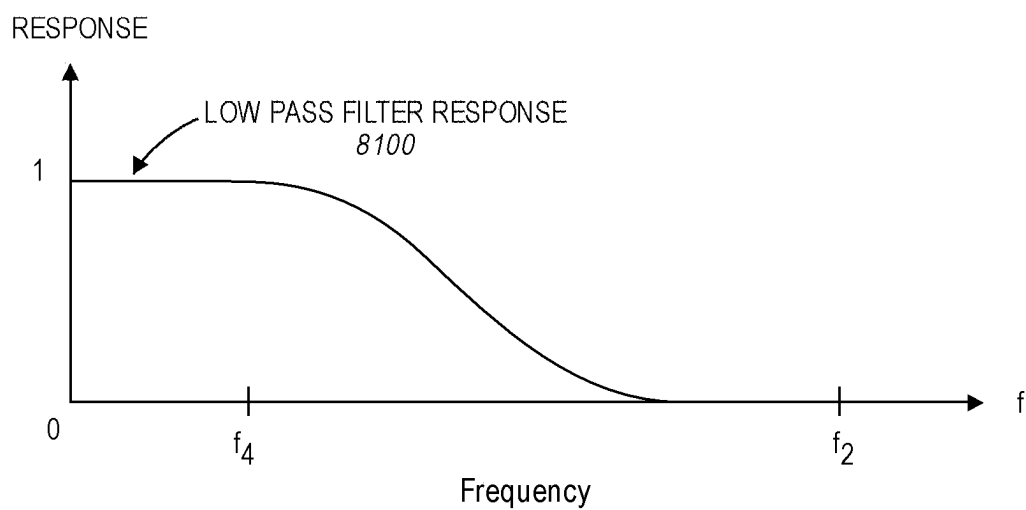


FIG. 81

SYSTEMS AND METHODS FOR PROVIDING POWER AND DATA TO DEVICES

RELATED APPLICATIONS

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

[0002] Traditionally, outdoor lighting systems include a plurality of lamps connected to a transformer. There may be one or more “legs” or sets of wires coming out of the transformer, each connected to at least one light. A timer box connects to the transformer. The user programs the on/off times and all of the lights energize in unison, such that all lights connected to a particular transformer turn ON or OFF together regardless of which leg they are on.

[0003] Some manufacturers provide lighting systems with addressable lighting modules. The timer box of the traditional lighting system is replaced with a lighting controller that supplies the lighting modules with a separate power and data signal. Each lighting module has an address and is independently addressable by the lighting controller via the data signal. These networked lighting systems provide the lighting modules with two sets of wires instead of the one or more legs. One set provides a power signal to illuminate the lamps or LEDs and a second set provides the lighting module with a data signal. The user programs the lighting controller to turn-on and turn-off lights at individual addresses such that a single light can turn-on or turn-off independently of the other lights in the network, when, for example, the data signal carries the address of a particular light.

[0004] In some instances, the power signal is the output of a low voltage power transformer, which is connected directly to the lighting modules to power the lamps or LEDs. For example, a primary AC to 12 VAC transformer accepts 120 VAC and outputs 12 VAC, where the 12 VAC power signal electrically couples directly to the lighting modules and powers the lamps/LEDs.

[0005] In other instances, the power signal is the output of a DC switching power supply. For example, a DC switching power supply accepts 120 VAC and outputs 12 VDC, where the 12 VDC power signal electrically couples directly to the lighting modules and powers the lamps/LEDs.

[0006] Other manufacturers of addressable lighting systems send power and data to the lighting modules on the primary power wires. The user programs the lighting controller to turn-on and turn-off lights at individual addresses such that a single light can turn-on or turn-off independently of the other lights in the network. In some instances, these lighting systems use a high frequency carrier, such as 125 kHz, and superimpose this signal on the power line. This approach requires large inductors, or complex Digital Signal Processors (DSPs) to decode the data contained in the carrier. One such commercially available system is the X10 control system originally developed by Pico Electronics of Glenrothes, Scotland.

[0007] In other instances, these lighting systems amplify the data signal to the level that can be used to power the lighting modules. For example, a PWM stepper motor driver chip can amplify a 0 volt to 5 volt transistor-transistor logic

(TTL) data signal to positive 24 volts to reflect a logical one and negative 24 volts to reflect a logical zero. The amplified data signal electrically couples to the lighting module, where the voltage is sufficient to supply power to the lamps/LEDs while maintaining the logical data values of the data stream.

SUMMARY

[0008] Based on the foregoing, each of the present manufacturing solutions suffers from a variety of drawbacks. In the context of individually addressable lighting networks with low voltage power transformers, they often employ special wiring or cabling. In particular, one wire and its return are needed for electrical power, while a second wire path comprising two or more wires is needed for data. For example, using a low voltage power transformer directly coupled to the lamps/LEDs to supply power prevents the data from being carried on the same power lines and, thus requires the two sets of wires. Accordingly, the owner of an existing set of lights must take significant effort to rewire in order to have a digitally controlled lighting environment.

[0009] In the context of lighting networks using a single wire for a power and a data signal, problems can occur when using a switching power supply to supply power to the lighting modules. Switching power supplies are inefficient when compared to a well-designed core and coil power transformer. The inefficient transformation of the primary AC power to a power waveform usable by the lighting modules creates heat. The heat, in turn, creates the need for a large enclosure to prevent the lighting controller circuitry from overheating. For example, a 300 watt switching power supply that has an efficiency of 85% wastes 45 watts in heat.

[0010] In contrast, in an embodiment of the present disclosure, a full-wave rectifier coupled to a bridge circuit provides a polarity controlled, sinusoidal power signal to power a plurality of lighting modules. The rectifier and bridge circuit include MOSFETs and each MOSFET has an integral body diode. When the full-wave rectifier MOSFETs are enabled at the appropriate point in time, such as when the body diodes would be conducting, they create a very low-loss switch. For example, for a MOSFET having a resistance of approximately 1 milliohm when it is enabled, conducting 25 amperes needed to power the plurality of lighting modules would lose approximately 25 millivolts of the signal. The corresponding power lost to heat is approximately 0.625 watts. In contrast, a standard rectifier would drop approximately 0.7 volts and dissipate approximately 17.5 watts.

[0011] In embodiments of the present disclosure using the output of a primary AC to 12 VAC 300 watt transformer to feed the circuitry, preferably the power lost to heat in the circuitry is less than approximately 2.0%. More preferably, the power lost to heat is between approximately 1% and approximately 2%. Even more preferably, the power lost to heat is between approximately 0.2% and approximately 1%, and most preferably, the power lost to heat is less than approximately 0.2%.

[0012] In other embodiments, the advantages of the rectifier and bridge of the present disclosure creating a very low-loss switch can be viewed from the drop in voltage across the rectifier. A transformer in the full-wave rectifier receives the primary AC signal and transforms the primary AC signal into a secondary AC power waveform. The full-wave rectifier coupled to a bridge circuit provides a polarity controlled, sinusoidal power signal to power a plurality of lighting modules. Preferably, the power wave-

form current is more than approximately 4 amperes and the power waveform voltage drop across the rectifier is less than approximately 0.2 volts and at full load the voltage drop across the rectifier, from the output of the transformer to the output of the rectifier, is approximately 25 millivolts. In another embodiment, the voltage drop across the rectifier is more preferably between approximately 0.1 volts and approximately 0.2 volts, yet more preferably between approximately 0 volts and approximately 0.1 volts, and most preferably between approximately 5 millivolts and approximately 30 millivolts. In yet other embodiments, the power waveform current is more preferably more than 10 amperes, yet more preferably more than 50 amperes, and most preferably more than 75 amperes. One basis for the above ratings is the wattage used for outdoor lighting systems. Typical systems are about 60 watts or higher. If such power requirements should be reduced due to technology advances, such as, for example, power requirements for lighting sources, or the like, one of ordinary skill will understand from the disclosure herein that the forgoing ranges may also change accordingly.

[0013] The low-loss full-wave rectified power waveform from the full-wave rectifier is communicated to the inputs of the bridge circuit. The bridge circuit outputs the full-wave rectified waveform with either a positive polarity or a negative polarity, thus having the ability to reconstruct the original sinusoidal output of the transformer, or alter its polarity to send data. The control signal from a processor in the lighting controller couples to the MOSFET drivers of the bridge circuit. The control signal enables certain of the gates in the bridge circuit at certain points in time to encode a data signal by varying the polarity of the power waveform.

[0014] In one embodiment, the control signal enables certain of the gates in the bridge circuit when the data is a logical 1-bit and others of the gates when the data is a logical 0-bit. This, in turn, causes the bridge circuit to output the positive polarity rectified waveform when the data stream is a 1-bit and causes the bridge circuit to output a negative polarity rectified waveform when the data stream is a 0-bit. In other embodiments, the bridge circuit outputs the negative polarity rectified power signal when the data is a 1-bit and outputs the positive polarity rectified power signal when the data is a 0-bit.

[0015] In one embodiment, the lighting system includes a controller having a data signal including data bits. The data bits have a first state and a second state for sending commands and addresses to at least one lighting module.

[0016] The lighting system further includes a MOSFET full-wave rectifier circuit for receiving a 12 VAC RMS power signal having first and second power waveforms and rectifying the 12 VAC RMS power signal. The MOSFET full-wave rectifier includes a first MOSFET coupled in series with a second MOSFET and a third MOSFET coupled in series with a fourth MOSFET where the series combination of the first and second MOSFETs electrically couple in parallel with the series combination of the third and fourth MOSFETs. Each MOSFET is associated with a gate signal and the gate signals electrically couple to an output of a comparator comparing the first and second power waveforms, via driver circuitry. The gates associated with the second and third MOSFETs are enabled when the first power waveform is greater than the second power waveform and

the gates associated with the first and fourth MOSFETs are enabled when the second power waveform is greater than the first power waveform.

[0017] The lighting system further includes a MOSFET bridge circuit for receiving the full-wave rectified waveform and providing a two-wire data/power signal to the at least one lighting module. The MOSFET bridge circuit includes a fifth MOSFET coupled in series with a sixth MOSFET and a seventh MOSFET coupled in series to an eighth MOSFET, where the series combination of the fifth and sixth MOSFETs couple in parallel with the series combination of the seventh and eighth MOSFETs. Each MOSFET is associated with a gate signal and the gate signals electrically coupled to the control signal. The gates associated with the sixth and seventh MOSFETs are enabled when the control signal is in the first state and the gates associated with the fifth and eighth MOSFETs are enabled when the control signal is in the second state, such that the MOSFET bridge circuit outputs the rectified waveform having a positive polarity when the control signal is in the first state and outputs the rectified waveform having a negative polarity when the control signal is in the second state. The two-wire data/power signal includes the positive and negative polarity rectified waveforms corresponding to the state of the control signal.

[0018] In another embodiment, a lighting system includes a controller having a data signal including data bits. The data bits have a first state and a second state for sending commands and addresses to at least one lighting module.

[0019] The lighting system further includes a MOSFET full-wave/bridge circuit for receiving a 12 VAC RMS power signal having first and second waveforms, rectifying the 12 VAC RMS power signal and providing a two-wire data/power signal to the at least one lighting module. The first and second power waveforms are provided by a transformer having a center tap. The MOSFET full-wave/bridge circuit includes a first MOSFET coupled in series with a second MOSFET and a third MOSFET electrically coupled in series with a fourth MOSFET where the series combination of the first and second MOSFETs electrically couple in parallel with the series combination of the third and fourth MOSFETs. Each MOSFET is associated with a gate signal and the gate signals electrically couple to the control signal. The gates associated with the third and fourth MOSFETs are enabled when the control signal is in the first state and the gates associated with the first and second MOSFETs are enabled when the control signal is in the second state, such that the MOSFET full-wave/bridge circuit outputs the rectified waveform having a positive polarity when the control signal is in the first state and outputs the rectified waveform having a negative polarity when the control signal is in the second state. The two-wire data/power signal includes the positive and negative polarity rectified waveforms corresponding to the state of the control signal.

[0020] In another aspect, systems and methods directed toward a user interface panel are disclosed. In an embodiment, a lighting controller includes an operator interface panel, which allows operator input to program the timing, dimming/brightness, color, and zones of the lighting system. In one embodiment, the user enters a chronologic schedule including a lighting group, a time, an intensity, a color, and the like. The program queues the user entered events and transmits the commands at the scheduled times.

[0021] With respect to color, in an embodiment, the colors are assigned a number and the user enters the number associated with the desired color. In another embodiment, the user designs a custom color by inputting the red, green and blue percentages. In some cases, a percentage of white can also be mixed with the red, green, and blue. Other user interfaces may include a color wheel with pointer sections, a scrollable list or color palette, or the like. The lighting controller then sends commands to the lighting modules with the user specified color percentages to create the custom color. In another embodiment, the lighting controller includes a thin film transistor liquid crystal display (TFT LCD) or the like, to display the color associated with the color number or the custom color. In another embodiment, the light controller may have a small red/green/blue LED, separate from the display that can be driven with the proper percentages to mimic the color emitted by the lighting fixtures.

[0022] In one embodiment, the user has the ability through the lighting controller to set on or off times around an event, such as create a lighting event around sunrise or sunset. For example, the user could use dusk as a reference time and have a zone of lights turn on at dusk minus two hours or dusk plus two hours. In one embodiment, the lighting controller includes a photocell and determines events such as dusk or dawn through the input from the photocell. In another embodiment, the user enters latitude and longitude information for his location. The lighting controller looks up or calculates the astronomical events based on the entered location values. In yet another embodiment, the lighting controller displays a map and the user indicates on the map his location. The lighting controller automatically displays the latitude and longitude and determines the astronomical events based on the displayed location values.

[0023] In another aspect, systems and methods relating to commanding the lighting modules through a remote device are disclosed. In another embodiment, the lighting system further includes a remote device and a wireless receiver. The remote device permits the user to adjust the lighting while in the illuminated area as an alternative to using the user interface panel in the lighting controller. The remote interacts with the lighting module via an optical or other link and interacts with the lighting controller via the receiver to allow the user to mix the color coefficients, assign lights to zones, control brightness, control on/off, or the like. The lighting controller receives the user requests through a wired or other connection to the receiver and sends commands to the lighting module through the two wire data/power path. For example, from the user's point of view, he points the remote at the desired lighting module and selects the change zone command. After a short time period, the selected lighting module is a member of a different lighting zone.

[0024] Certain embodiments relate to a lighting system including a lighting controller and at least one lighting module having an address and including a light emitting diode (LED). The LED is configured to transmit optically the address or other status information of the lighting module by turning on when transmitting a 1-bit and turning off when transmitting a 0-bit in the address. The lighting controller electrically couples to the lighting module through a two-wire path carrying a power/data signal.

[0025] The lighting system further includes a remote device including an optical sensor and an RF transmitter. The optical sensor is configured to receive the address from

the lighting module, and user request from the user interface of the remote device. The RF transmitter is configured to transmit an RF signal corresponding to the address and the request.

[0026] The lighting system further includes a wireless receiver electrically coupled to the lighting controller and configured to receive the RF transmission from the remote device. The wireless receiver down converts the RF transmission to a baseband signal corresponding to the address and request. The wireless receiver is further configured to electrically send the baseband signal corresponding to the address and the request to the lighting controller.

[0027] The lighting controller encodes a command corresponding to the user's request for the at least one lighting module associated with the address onto the power/data signal.

[0028] Other aspects relate to a method to increase a resolution of intensity control in a lighting module comprising an LED. The method comprises determining, at the lighting module, an intensity level for the LED, generating a first control signal that when coupled to an LED driver results in a first drive signal having a first resolution of intensity control, where the first control signal is based at least in part on the intensity level, generating a second control signal that when coupled to the LED driver results in a second drive signal having a second resolution of intensity control, where the second control signal is based at least in part on the intensity level, combining the first and second control signals to create a hybrid control signal, and coupling the hybrid control signal to the LED driver to produce a third drive signal having a third resolution of intensity control, where the third resolution of intensity control results in a greater resolution of intensity control than the first resolution of intensity control and the second resolution of intensity control.

[0029] In an embodiment, the hybrid control signal comprises a PWM signal having an amplitude and a duty cycle. In another embodiment, the amplitude is based at least in part on the first control signal and the duty cycle is based at least in part on the second control signal. In a further embodiment, creating the hybrid control signal comprises converting a digital value that is responsive to the amplitude to an analog signal, and modulating the analog signal according to the duty cycle. In a yet further embodiment, creating the hybrid control signal comprises filtering with a low-pass filter a first PWM waveform, where a duty cycle of the first PWM waveform is based on the amplitude and the filtered first PWM waveform approximates an analog signal having the amplitude, and modulating the analog signal with a second PWM waveform having the duty cycle of the hybrid control signal.

[0030] In an embodiment, creating the hybrid control signal comprises generating a digital value for predetermined intervals and converting the digital value to an analog value during the predetermined intervals, where the digital value is based on the amplitude and the predetermined intervals based on the duty cycle. In another embodiment, creating the hybrid control signal comprises averaging a PWM waveform, where a duty cycle of the PWM waveform is based on the amplitude and the averaged PWM waveform approximating an analog signal having the amplitude, and turning the PWM waveform ON and OFF at predetermined times, where the predetermined times is based on the duty cycle of the hybrid control signal. In a further embodiment,

averaging the PWM waveform comprises filtering the PWM waveform with a low-pass filter. In a yet further embodiment, components of the low-pass filter are selected so that a frequency of the PWM waveform is outside of the pass band of the low-pass filter and a frequency of the hybrid control signal is within the pass band.

[0031] Certain embodiments relate to a lighting module comprising an LED configured to illuminate when driven by an output of an LED driver circuit, a processor configured to determine an intensity level for the LED and to generate a first control signal that when coupled to the LED driver circuit results in a first drive signal having a first resolution of intensity control and to generate a second control signal that when coupled to the LED driver circuit results in a second drive signal having a second resolution of intensity control, where the first and second control signals are based at least in part on the intensity level, and a hybrid control signal generator configured to combine the first and second control signals to create a hybrid control signal. The LED driver circuit is configured to receive the hybrid control signal and to output a third drive signal having a third resolution of intensity control, where the third resolution of intensity control results in a greater resolution of intensity control than the first resolution of intensity control and the second resolution of intensity control.

[0032] In an embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value responsive to the amplitude and a PWM waveform responsive to the duty cycle, and a digital to analog converter (DAC) configured to enable a conversion of the digital value to an analog signal according to the PWM waveform. In another embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value responsive to the amplitude and a PWM waveform responsive to the duty cycle, a digital to analog converter (DAC) configured to convert the digital value to an analog signal, and a switch configured to modulate the analog signal according to PWM waveform. In a further embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a first PWM waveform, a low-pass filter configured to average the first PWM waveform to approximate an analog signal having the amplitude, where the microcontroller is further configured to generate a second PWM waveform having the duty cycle, and the hybrid control signal generator further comprises a switch configured to modulate the averaged first PWM waveform with the second PWM waveform. In a yet further embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value for predetermined intervals and a digital to analog converter (DAC) configured to convert the digital value to an analog signal during the predetermined intervals, where the digital value is based on the amplitude and the predetermined intervals based on the duty cycle.

[0033] Certain embodiments relate to a method to control color and brightness of a lighting module comprising one or more LEDs. The method comprises receiving at the lighting module a data encoded power signal over a two-wire communication network, where the data encoded power signal comprises data associated with a user-selected color and a user-selected brightness for the lighting module. The method further comprises decoding the data associated with the user-selected color and the user-selected brightness, where the decoded data comprises a color ratio and an intensity

level, determining, for each LED of the one or more LEDs, a corresponding amplitude and a corresponding duty cycle of a pulse width modulation (PWM) signal in response to the color ratio at the intensity level, and generating, for each LED of the one or more LEDs, a corresponding hybrid control signal having the corresponding amplitude and the corresponding duty cycle, where each LED of the one or more LEDs is further associated with a corresponding hysteretic LED driver. The method further comprises providing, for each LED of the one or more LEDs, the corresponding hybrid control signal to the corresponding hysteretic LED driver to generate a corresponding output signal based on the corresponding hybrid control signal, and driving each LED of the one or more LEDs with the corresponding output signal to cause the lighting module to illuminate in response to the user-selected color and the user-selected brightness, where the corresponding hybrid control signal comprises an analog component and a digital component, and permits the color ratio at varying intensities to be controlled with a level of resolution that is greater than the level of resolution provided by the analog component alone and the level of resolution provided by the digital component alone.

[0034] In an embodiment, the analog component comprises the corresponding amplitude and the digital component comprises the corresponding duty cycle. In another embodiment, the data encoded power signal forms a sinusoidal waveform between zero crossings.

[0035] In an embodiment, generating the corresponding hybrid control signal comprises converting a digital value that is responsive to the corresponding amplitude to an analog signal, and enabling the conversion of the digital value to the analog signal according to the corresponding duty cycle. In a further embodiment, generating the corresponding hybrid control signal comprises converting a digital value that is responsive to the corresponding amplitude to an analog signal, and modulating the analog signal according to the corresponding duty cycle. In a yet further embodiment, generating the corresponding hybrid control signal comprises filtering with a low-pass filter a first PWM waveform, where a duty cycle of the first PWM waveform is based on the corresponding amplitude and the filtered first PWM waveform approximates an analog signal having the corresponding amplitude, and where generating the corresponding hybrid control signal further comprises modulating the analog signal with a second PWM waveform having the corresponding duty cycle.

[0036] In an embodiment, generating the corresponding hybrid control signal comprises generating a digital value for predetermined intervals and converting the digital value to an analog value during the predetermined intervals, where the digital value is based on the corresponding amplitude and the predetermined intervals based on the corresponding duty cycle. In another embodiment, generating the corresponding hybrid control signal comprises averaging a PWM waveform, where a duty cycle of the PWM waveform is based on the corresponding amplitude and the averaged PWM waveform approximates an analog signal having the corresponding amplitude, where generating the corresponding hybrid control signal further comprises turning the PWM waveform ON and OFF at predetermined times, the predetermined times based on the corresponding duty cycle. In a further embodiment, averaging the PWM waveform comprises low-pass filtering the PWM waveform with a low-

pass filter. In a yet further embodiment, components of the low-pass filter are selected so that a frequency corresponding of the PWM waveform is outside of the pass band of the low-pass filter and a frequency of the corresponding hybrid control signal is within the pass band.

[0037] In another aspect, systems and methods relate to a lighting module comprising a processor configured to receive a data encoded power signal over a two-wire communication network, where the data encoded power signal comprises data associated with a user-selected color and a user-selected brightness. The processor is further configured to decode the data, where the decoded data comprises a color ratio and an intensity level. The processor further is configured to determine an amplitude and a duty cycle of a hybrid control signal in response to the color ratio at the intensity level. The lighting module further comprises a hybrid control signal generator configured to generate the hybrid control signal having the amplitude and the duty cycle, an LED driver configured to receive the hybrid control signal and generate an LED drive signal based at least in part on the hybrid control signal, and one or more LEDs configured to receive the LED drive signal and illuminate in response to the user-selected color and the user-selected brightness, where the hybrid control signal comprises an analog component and a digital component, and permits the color ratio at varying intensities to be controlled with a level of resolution that is greater than the level of resolution provided by an analog control signal alone and the level of resolution provided by a digital control signal alone.

[0038] In an embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value responsive to the amplitude and a PWM waveform responsive to the duty cycle, and a digital to analog converter (DAC) configured to enable a conversion of the digital value to an analog signal according to the PWM waveform. In another embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value responsive to the amplitude and a PWM waveform responsive to the duty cycle, a digital to analog converter (DAC) configured to convert the digital value to an analog signal, and a switch configured to modulate the analog signal according to PWM waveform. In a further embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a first PWM waveform, a low-pass filter configured to average the first PWM waveform to approximate an analog signal having the amplitude, where the microcontroller is further configured to generate a second PWM waveform having the duty cycle, and the hybrid control signal generator further comprises a switch configured to modulate the averaged first PWM waveform with the second PWM waveform. In a yet further embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a digital value for predetermined intervals and a digital to analog converter (DAC) configured to convert the digital value to an analog signal during the predetermined intervals, where the digital value is based on the amplitude and the predetermined intervals based on the duty cycle.

[0039] Other embodiments relate to a lighting system comprising a plurality of modules, where each module is associated with one or more LEDs and a processor, a two-wire communication network configured to address each of the plurality of modules and provide power to each of the plurality of modules, a user interface configured to

acquire user inputs responsive to a user-selected color and a user-selected brightness for the plurality of modules, a power supply including a rectifier circuit and a bridge circuit, where the power supply is configured to modify an input voltage of a power source to produce a data encoded power signal responsive to a polarity of the input voltage and responsive to the acquired user inputs. The power supply outputs the data encoded power signal over the two-wire communication network and the processor of an addressed module is configured to receive the data encoded power signal over the two-wire communication network and decode the data encoded power signal to provide decoded data comprising a color ratio and an intensity level. The processor is further configured to determine a first amplitude and a first duty cycle of a hybrid control signal in response to the color ratio at the intensity level. The lighting system further comprises a hybrid control signal generator configured to generate the hybrid control signal having the first amplitude and the first duty cycle, an LED driver configured to receive the hybrid control signal and generate an LED drive signal based at least in part on the hybrid control signal, where the one or more LEDs of the addressed module are configured to receive the LED drive signal and illuminate responsive to the user-selected color and the user-selected brightness, where the hybrid control signal comprises an analog component and a digital component, and permits the color ratio at varying intensities to be controlled with a level of resolution that is greater than the level of resolution provided by an analog control signal alone and the level of resolution provided by a digital control signal alone.

[0040] In an embodiment, the hybrid control signal generator comprises a microcontroller configured to generate a pulse width modulation (PWM) waveform having a second duty cycle based at least in part on the first amplitude, and a low-pass filter configured to average the PWM waveform, the microcontroller further configured to turn the PWM waveform ON and OFF at predetermined intervals that are based at least in part on the first duty cycle. In another embodiment, components of the low-pass filter are selected so that a frequency of the PWM waveform is outside of the pass band of the low-pass filter and a frequency of the hybrid control signal is within the pass band.

[0041] In an embodiment, a controller is configured to power and control a plurality of lighting modules. Each lighting module includes one or more lights and is serially addressable over a two-wire communication network. The controller comprises one or more processors configured to output command and address data to control a behavior of at least one light; a rectifier circuit communicating with a power signal and configured to form a rectified power waveform; a bridge circuit communicating with the rectifier circuit and the one or more processors and configured to receive the rectified power waveform and the command and address data, and to output to the two-wire communication network a data encoded power signal to power and control one or more lighting modules; current sensing circuitry operably coupled to the one or more processors and configured to monitor the power signal; and memory operably coupled to the processor and configured to store lighting module information and code executable by the one or more processors. The executable code causes the one or more processors to output first command and address data comprising an address of a first lighting module and a command

to turn ON the addressed lighting module, the bridge circuit receiving the first command and address data and outputting a first data encoded power signal responsive to the first command and address data; receive, after the first data encoded power signal is sent over the two-wire communication network, an indication of current associated with the monitored power signal from the current sensing circuitry; compare the received indication of the current with a threshold value; and determine, based on the comparison, whether the first lighting module responded to the command.

[0042] In an aspect, the controller further comprises a user input device communicating with the one or more processors and configured to accept user input and to output user information to the one or more processors. In an aspect, the user input device is further configured to receive the address of the first lighting module or an indication of the address of the first lighting module when the determination indicates that the first lighting module responded to the command.

[0043] In an aspect, the bridge circuit includes a plurality of transistors. The plurality of transistors communicate with the one or more processor to receive a control signal having first and second states. At least one of the plurality of transistors is enabled when the control signal is in the first state and at least one of the others of the plurality of transistors is enabled when the control signal is in the second state. The bridge circuit outputs the data encoded power signal responsive to the rectified power waveform having a higher polarity when the control signal is in the first state and responsive to the rectified power waveform having a lower polarity when the control signal is in the second state.

[0044] In an aspect, the current sensing circuitry includes a current transformer. In an aspect, executable code further causes the one or more processors to store in memory an indication of the comparison. The indication of the comparison includes an indication that the current is over the threshold, an indication that the current is under the threshold, or a value of the current. In an aspect, the executable code further causes the one or more processors to periodically send next command and address data for a next lighting module of the plurality of lighting modules to determine whether the next lighting module is responsive to the next command and address data. In an aspect, the address of the next lighting module is randomly or pseudo-randomly chosen from the addresses associated with the plurality of lighting modules. In an aspect, the first address and the address of the next lighting module are sequential.

[0045] In another embodiment, a method to power and control a plurality of lighting modules is disclosed. Each lighting module includes one or more lights and is serially addressable over a two-wire communication network. The method comprises generating, with one or more processors, command and address data comprising an address of a first lighting module and a command to turn ON the addressed lighting module; with a rectifier circuit, receiving a power signal and forming a rectified power waveform; with a bridge circuit, receiving the rectified power waveform and the command and address data, and providing, to the two-wire communication network, a data encoded power signal responsive to the command and address data; monitoring, with current sensing circuitry, the power signal; receiving, with the one or more processors and after the data encoded power signal is sent over the two-wire communication network, an indication of current associated with the monitored power signal from the current sensing circuitry; com-

paring, with the one or more processors, the received indication of the current with a threshold value; and determining, with the one or more processors and based on the comparison, whether the first lighting module responded to the command.

[0046] In an aspect, the method further comprises receiving user input from a user interface and outputting user information to the one or more processors, the user input received from a user to control the plurality of lighting modules. In an aspect, the method further comprises receiving, at the user interface, an indication of the location of the first lighting module when the determination indicates that the first lighting module responded to the command. In an aspect, the method further comprises receiving, at the user interface, an indication of the location of the first lighting module when the determination indicates that the first lighting module failed to respond to the command. In an aspect, the current sensing circuitry includes a current transformer. In an aspect, the method further comprises storing in memory an indication of the determination.

[0047] In an embodiment, a method to power and control a plurality of lighting modules is disclosed. Each lighting module includes one or more lights and is serially addressable over a two-wire communication network. The method comprises, with a power supply that includes a rectifier circuit and a bridge circuit, receiving a power signal; receiving command and address data that includes an address of a first lighting module and a command to be performed by the addressed lighting module; forming a rectified power waveform; and passing or inverting the rectified power waveform to provide, to the two-wire communication network, a data encoded power signal that is responsive to the command and address data; monitoring a current of the power signal; receiving, after the data encoded power signal is sent over the two-wire communication network, an indication of the monitored current; comparing the received indication of the monitored current with a threshold value; and based on the comparison, determining that the first lighting module responded to the command.

[0048] In an aspect, the method further comprises periodically generating second command and address data for a second lighting module of the plurality of lighting modules, providing, to the two-wire communication network, a second data encoded power signal that is responsive to the second command and address data, receiving, after the second data encoded power signal is sent over the two-wire communication network, a second indication of the monitored current, and determining that the second lighting module responded to the second command and address data based on a comparison of the second indication of the monitored current with the threshold. In an aspect, the address of the second lighting module is randomly or pseudo-randomly chosen from the addresses associated with the plurality of lighting modules. In an aspect, the address of the first lighting module and the address of the second lighting module are sequential.

[0049] For purposes of summarizing the disclosure, certain aspects, advantages and novel features of the embodiments have been described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment of the invention. Thus, the inventions may be embodied or carried out in a manner that achieves or optimizes one advantage or group

of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0050] Throughout the drawings, reference numbers are re-used to indicate correspondence between referenced elements. The drawings, associated descriptions, and specific implementation are provided to illustrate embodiments and not to limit the scope of the disclosure.

[0051] FIG. 1 illustrates an exemplary lighting system, according to certain embodiments.

[0052] FIG. 2 is a block diagram of an exemplary lighting system, according to certain embodiments.

[0053] FIG. 3 is a block diagram of an exemplary lighting controller, according to certain embodiments.

[0054] FIG. 4 is an exemplary schematic diagram of a rectifier circuit, according to certain embodiments.

[0055] FIG. 5 depicts an exemplary power waveform, according to certain embodiments.

[0056] FIG. 6 depicts an exemplary waveform of the transistor gate signal for a rectifier circuit, according to certain embodiments.

[0057] FIG. 7 depicts an exemplary waveform of another transistor gate signal for the rectifier circuit, according to certain embodiments.

[0058] FIG. 8 depicts an exemplary rectified power waveform, according to certain embodiments.

[0059] FIG. 9 is an exemplary schematic diagram of a bridge circuit, according to certain embodiments.

[0060] FIG. 10 depicts an exemplary waveform of the transistor gate signal for a bridge circuit, according to certain embodiments.

[0061] FIG. 11 depicts an exemplary waveform of another transistor gate signal for the bridge circuit, according to certain embodiments.

[0062] FIG. 12 depicts an exemplary power/data waveform without data, according to certain embodiments.

[0063] FIG. 13 depicts an exemplary power/data waveform with data, according to certain embodiments.

[0064] FIG. 14 is an exemplary schematic diagram of a rectifier/bridge circuit, according to certain embodiments.

[0065] FIG. 15 is an exemplary schematic diagram of circuitry for phase detect, timing generation and drivers, according to certain embodiments.

[0066] FIG. 16 is an exemplary schematic diagram of a bias circuit, according to certain embodiments.

[0067] FIGS. 17A comprising 17A1-17A4 and 17B comprising 17B1-17B4 are exemplary circuit diagrams for a lighting controller, according to one embodiment.

[0068] FIG. 18 is a block diagram of an exemplary lighting controller, according to other embodiments.

[0069] FIG. 19 is an exemplary schematic diagram of a rectifier circuit, according to other embodiments.

[0070] FIG. 20 is exemplary schematic diagram of a rectifier circuit, according to yet other embodiments.

[0071] FIG. 21 depicts an exemplary power waveform, according to other embodiments.

[0072] FIG. 22 depicts an exemplary waveform of the transistor gate signal for a rectifier circuit, according to other embodiments.

[0073] FIG. 23 depicts an exemplary waveform of transistor gate signal for the rectifier circuit, according to yet other embodiments.

[0074] FIG. 24 depicts an exemplary rectified power waveform, according to other embodiments.

[0075] FIG. 25 depicts an exemplary power/data waveform without data, according to other embodiments.

[0076] FIG. 26 depicts an exemplary power/data waveform with data, according to other embodiments.

[0077] FIG. 27 is an exemplary schematic diagram of a line voltage control node, according to certain embodiments.

[0078] FIG. 28 is a block diagram of an exemplary line voltage control node, according to certain embodiments.

[0079] FIG. 29 depicts a first exemplary output waveform to control dimming of a Line Voltage device, according to certain embodiments.

[0080] FIG. 30 depicts a second exemplary output waveform to control dimming of a Line Voltage device, according to certain embodiments.

[0081] FIG. 31 depicts a third exemplary output waveform to control dimming of a Line Voltage device, according to certain embodiments.

[0082] FIG. 32 is an exemplary schematic diagram of a wireless line voltage control node, according to certain embodiments.

[0083] FIG. 33 is a block diagram of an exemplary transmitter for a wireless line voltage control node, according to certain embodiments.

[0084] FIG. 34 is a block diagram of an exemplary receiver for a wireless line voltage control node, according to certain embodiments.

[0085] FIG. 35 is a block diagram of an exemplary lighting system for a wireless line voltage control node, according to certain embodiments.

[0086] FIG. 36 is a block diagram of an exemplary lighting system for a wireless line voltage control node, according to other embodiments.

[0087] FIG. 37 is a block diagram of an exemplary wired low voltage control node, according to certain embodiments.

[0088] FIG. 38 is a block diagram of an exemplary wireless low voltage control node, according to certain embodiments.

[0089] FIG. 39 is a block diagram of a low voltage decoder, according to certain embodiments.

[0090] FIG. 40 illustrates an exemplary lighting system for controlling and reassigning lighting zones using a remote device, according to certain embodiments.

[0091] FIG. 41 depicts a remote device, according to certain embodiments.

[0092] FIG. 42 is a block diagram of an exemplary remote device, according to certain embodiments.

[0093] FIG. 43 is a block diagram of an exemplary lighting controller comprising a remote operator interface panel, according to certain embodiments.

[0094] FIG. 44 is a block diagram of an exemplary transceiver for a lighting controller with a remote operator interface panel, according to certain embodiments.

[0095] FIG. 45 is a block diagram of an exemplary transceiver for a remote operator interface panel, according to certain embodiments.

[0096] FIG. 46 is a block diagram of an exemplary optical reader, according to certain embodiments.

[0097] FIG. 47 illustrates an exemplary lighting system controlled remotely, according to certain embodiments.

[0098] FIG. 48 illustrates another exemplary lighting system controlled remotely, according to certain embodiments.

[0099] FIG. 49 illustrates an exemplary lighting system with a master/slave configuration, according to certain embodiments.

[0100] FIG. 50 illustrates an exemplary lighting system with a master/slave configuration, according to other embodiments.

[0101] FIG. 51 is a block diagram of an exemplary two-wire to chassis protocol converter, according to certain embodiments.

[0102] FIG. 52 illustrates an exemplary lighting system with a wireless master/slave configuration, according to certain embodiments.

[0103] FIG. 53 is a block diagram of an exemplary transmitter for a wireless two-wire chassis protocol converter, according to certain embodiments.

[0104] FIG. 54 is a block diagram of an exemplary receiver for a wireless two-wire chassis protocol converter, according to certain embodiments.

[0105] FIG. 55 is a block diagram of an exemplary lighting system for a wireless two-wire chassis protocol converter with multiple slave chassis, according to certain embodiments.

[0106] FIG. 56 is a flowchart of an exemplary process for encoding data bits onto a power signal for lighting modules.

[0107] FIG. 57 is a flowchart of an exemplary process for assigning zones to addressable lighting modules in a networked lighting system, according to certain embodiments.

[0108] FIG. 58 is a flowchart of an exemplary process for modifying assigned zones in a lighting system using a remote controller, according to certain embodiments.

[0109] FIG. 59 depicts an exemplary power waveform illustrating ON/OFF voltages for colored LEDs, according to certain embodiments.

[0110] FIG. 60 depicts an exemplary power waveform illustrating ON/OFF voltages for colored LEDs, according to other embodiments.

[0111] FIG. 61 is a block diagram of an exemplary single channel lighting module, according to certain embodiments.

[0112] FIG. 62 is an exemplary schematic diagram of a single channel lighting module, according to certain embodiments.

[0113] FIG. 63 is a block diagram of an exemplary multichannel lighting module, according to certain embodiments.

[0114] FIG. 64 is an exemplary schematic diagram of a multichannel lighting module, according to certain embodiments.

[0115] FIG. 65 is a front view of an exemplary lighting controller chassis, according to certain embodiments.

[0116] FIG. 66 is a back view of an exemplary lighting controller chassis, according to certain embodiments.

[0117] FIG. 67 is a block diagram of an exemplary hysteretic driver, according to certain embodiments.

[0118] FIG. 68 is an exemplary analog input waveform for a hysteretic LED driver that results in approximately full (100%) brightness of the LED, according to certain embodiments.

[0119] FIG. 69 illustrates an exemplary analog input waveform for a hysteretic LED driver that results in approximately half (50%) brightness of the LED, according to certain embodiments.

[0120] FIG. 70 illustrates an exemplary PWM input waveform for a hysteretic LED driver that results in approximately full (100%) brightness of the LED, according to certain embodiments.

[0121] FIG. 71 illustrates an exemplary PWM input waveform for a hysteretic LED driver that results in approximately half (50%) brightness, according to certain embodiments.

[0122] FIG. 72 is a block diagram of an exemplary lighting control system to control color and brightness of LED lighting, according to certain embodiments.

[0123] FIG. 73 illustrates an exemplary hybrid control waveform for a hysteretic LED driver that results in approximately quarter (25%) brightness, according to certain embodiments.

[0124] FIG. 74 is a block diagram of an exemplary hybrid control waveform generator, according to certain embodiments.

[0125] FIG. 75 is a block diagram of an exemplary hybrid control waveform generator, according to certain embodiments.

[0126] FIG. 76 is a block diagram of an exemplary hybrid control waveform generator, according to certain embodiments.

[0127] FIG. 77 illustrates an exemplary hybrid control waveform for a hysteretic LED driver, according to certain embodiments.

[0128] FIG. 78 is a block diagram of an exemplary hybrid control waveform generator, according to certain embodiments.

[0129] FIG. 79 illustrates an exemplary hybrid control waveform for a hysteretic LED driver, according to certain embodiments.

[0130] FIG. 80 is a block diagram of an exemplary hybrid control waveform generator, according to certain embodiments.

[0131] FIG. 81 illustrates a filter response for an exemplary low pass filter for a hybrid control waveform generator, according to certain embodiments.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0132] The features of the inventive systems and methods will now be described with reference to the drawings summarized above.

[0133] FIG. 1 illustrates an exemplary lighting system 100. The lighting system 100 comprises a lighting controller housing 102 connected to a plurality of lighting fixtures or modules 104 through a two-wire interface. The lighting controller housing 102 houses a lighting controller including a power supply and user interface panel, as described in further detail below. The lighting fixtures 104 are grouped into zones 106.

[0134] In the example illustrated in FIG. 1, zone 1 106a comprises lighting fixture 1 104a, which provides illumination to a portion of the house exterior. Zone 2 106b comprises lighting fixtures 2, 3, 4 104b, 104c, 104d, respectively, which illuminate the path, while zone 3 106c comprises lighting fixtures 5, 6, 7, 104e, 104f, 104g, respectively, which provide accent lighting for the tree. In other embodiments, the lighting system 100 can be configured with more or less zones 106 and/or with more or less lighting fixtures 104 in each zone 106.

[0135] Typically, the lighting fixtures **104** in each zone **106** turn ON or OFF together, but unlike some traditional lighting systems, each zone **106** can be controlled independently of the other zones **106**. In one example for the lighting system **100** illustrated in FIG. 1, zone **1** **106a** turns ON at dusk and turns OFF at dawn to illuminate the front door of the house. Zone **2** **106b** turns ON at dusk and turns OFF at 9 PM to illuminate the path. Finally, zone **3** turns on at 7 PM and turns OFF at 10 PM to provide accent lighting in the yard.

[0136] In one embodiment, the lighting system **200** is a residential outdoor lighting system. In other embodiments, the lighting system **200** is used for outdoor commercial purposes to illuminate the outside of hotels, golf courses, amusement parks, and the like, and for indoor commercial purposes to illuminate hotel interiors, office-building interiors, airport terminals, and the like. In further embodiments, the lighting system **200** is used to illuminate housing developments. In yet further embodiments, the lighting system **200** is used to illuminate artwork in residences, in museums, or the like. Many possibilities exist for the lighting system **200** to one skilled in the art from the disclosure herein. The lighting functions ON/OFF include a plurality of lighting functions, such as, for example, timing control, dimming, brightness, color, hue, zone allocation, intensity, and the like.

[0137] FIG. 2 is a block diagram of an exemplary lighting system **200** comprising a lighting controller **202** and a plurality of lighting modules **204**. The lighting controller **202** comprises a power supply **208** and an operator interface **210**, which includes a fixture programming port **212**. A lighting controller housing houses the power supply **208** and the operator interface **210**. The size of the lighting controller housing depends on the size of the power supply **208** and the operator interface **210** contained within it. In an embodiment, the lighting controller housing has a height that ranges from approximately 11 inches to approximately 15 inches, a width that ranges from approximately 7 inches to approximately 9 inches, and a thickness that ranges from approximately 5 inches to approximately 7 inches. The lighting controller **202** electrically couples to the lighting modules **204** through a two-wire path carrying a power/data signal. The lighting modules **204** electrically connect in parallel to the two-wire path and are grouped into M zones **206**. In the illustrated embodiment, zone **1** comprises three lighting modules **204**, zone **2** comprises a single lighting module **204**, and zone **3** comprises two lighting modules **204**. Further, the lighting controller **202** controls up to M zones **206**, where in the illustrated embodiment, zone M includes N lighting modules **204**. Each zone **206** can be independently energized such that the lighting modules **204** in each zone **206** can turn ON or OFF independently of the lighting modules **204** in the other zones **206**.

[0138] Controller **202** is shown housing the power supply **208**, the operator interface **210**, and the fixture programming port **212**. In other embodiments, the power supply **208**, the operator interface **210**, and the fixture programming port **212** may be separate devices or any two of the power supply **208**, the operator interface **210**, and the fixture programming port **212** may be housed in the same housing.

[0139] FIG. 3 is a block diagram of an exemplary lighting controller **300** comprising a power supply **302** and an operator interface panel **308**. The power supply **302** receives AC power from a primary AC power source **306** and

addresses/data/commands from the operator interface panel **308** and provides a control signal to a plurality of lighting fixtures **304** through the two-wire path **336**.

[0140] The operator interface panel **308** comprises operator controls **310**, such as selection buttons, knobs, and the like, which the user uses to input the desired lighting effects to the lighting system **200**, and displays and indicators **312** to provide feedback to the user. The operator interface panel **308** further comprises a computer **314** and its associated memory **316**. The microprocessor **314** interfaces with the operator controls **310** to send the addresses/data/commands to the power supply **302** and interfaces with the displays and indicators **312** to display information received from the power supply **302**. The operator interface **308** can be buttons, virtual icons, or buttons on a touch screen, voice controlled, or any user interface recognizable to an artisan from the disclosure herein.

[0141] The computer **314** comprises, by way of example, processors, program logic, or other substrate configurations representing data and instructions, which operate as described herein. In other embodiments, the processors can comprise controller circuitry, processor circuitry, processors, general-purpose single-chip or multi-chip microprocessors, digital signal processors, embedded microprocessors, microcontrollers and the like. The memory **316** can comprise one or more logical and/or physical data storage systems for storing data and applications used by the computer **314**. The memory **316** comprises, for example, RAM, ROM, EPROM, EEPROM, and the like.

[0142] The operator interface panel **308** further comprises a fixture programming port **318** to provide unique addresses, a lighting group, and/or zone number to each of the plurality of lighting fixtures **304**, and a logic power supply **320** to provide a low voltage, such as +5 volts, for example, for the digital logic components of the operator interface panel **308**.

[0143] The power supply **302** comprises a primary AC transformer **322**, current sensing circuitry **324**, phase detect and timing circuitry **326**, driver circuitry **328**, a synchronous fullwave rectifier **332**, and a bridge **334**. The power supply **302** further comprises a low power transformer **346** to provide a low voltage, such as 9 VAC, for example, to a logic power supply which creates a regulated DC voltage for the digital logic components of the power supply **302** and biasing circuitry **330** to provide the proper voltage levels to operate transistors in the rectifier **332** and the bridge **334**.

[0144] The primary AC transformer **322** receives a primary AC power signal from the primary AC power source **306** and transforms the primary AC signal into lower voltage AC signal. In an embodiment, the primary AC signal is approximately a 120 volt 60 Hz power waveform. In other embodiments, the primary AC signal can be an approximately 110 volts 60 Hz, 220 volt 50 Hz, 220 volt 60 Hz, 230 volts 60 Hz, 240 volts 50 Hz, or the like, power waveform. In an embodiment, the primary AC transformer **322** is a primary AC to 12 VAC transformer **322**, and transforms the primary AC signal into an approximately 12 VAC RMS power signal. In other embodiments, the transformer **322** is a primary AC transformer with several taps. In an embodiment, the transformer has taps at approximately 11 VAC up to approximately 14 VAC. In other embodiments, the transformer **322** transforms the AC signal into an approximately 24 VAC.

[0145] In an embodiment, the transformer **322** is a high wattage transformer, such as a 300 watt transformer, or the

like, for example, in order to supply sufficient power to illuminate the plurality of lighting modules 304. The output of the transformer 322 electrically connects to the current sensing circuitry 324. The current sensing circuitry 324 senses the amount of current in the output of the transformer 322. The phase detect and timing circuitry 326 receives a signal proportional to the sensed current from the current sensing circuitry 324 and shuts off the power supply 302 when the sensed current exceeds a threshold. For example, if there is a short between the wires of the two-wire path 336, a 300 watt transformer can supply a large amount of power in the form of heat in a very short time. When the sensed current exceeds a threshold, the lighting controller 300 shuts off the power before the heat generated causes damage to the lighting system 200.

[0146] The phase detect and timing circuitry 326 further receives data and commands from the processor 314 and the power waveform from the transformer 322, and provides timing signals to the driver circuit 328. The timing signals control the driver circuitry 328 to encode a data signal onto the power signal by varying the polarity of the power waveform, as will be further discussed herein.

[0147] Further, the output of the transformer 322 electrically connects to the synchronous fullwave rectifier 332, which rectifies the power signal. The fullwave rectifier 332 electrically connects to the bridge 334 and the fullwave rectifier 332 and the bridge 334 electrically connect to the driver circuitry 328. Both the fullwave rectifier 332 and the bridge 334 receive drive signals from the driver circuitry 328. The bridge 334 receives the rectified power signal and outputs a control signal to the lighting fixtures 304. The control signal comprises a data encoded power waveform, which provides power to illuminate the lighting fixtures 304 and address/data/commands to individually control the lighting fixtures 304.

[0148] FIG. 4 is an exemplary schematic diagram of a synchronous rectifier circuit 400, according to an embodiment. The rectifier circuit 400 comprises a primary AC to 12 VAC transformer 402, a first transistor Q1 404, a second transistor Q2 406, a third transistor Q3 408, and a fourth transistor Q4 410. The primary AC to 12 VAC transformer 402 receives a primary AC power signal and outputs an approximately 12 VAC RMS power waveform having a first power waveform AC1 and a second power waveform AC2. FIG. 5 illustrates an exemplary 12 VAC RMS power waveform 500 having a peak-to-peak voltage of between approximately +16.97 volts to approximately -16.97 volts.

[0149] In an embodiment, the transistors Q1 404, Q2 406, Q3 408, Q4 410 are metal-oxide-semiconductor field-effect transistors (MOSFETs) with an integral body diode. The MOSFETs with the integral body diode advantageously function as a substantially loss-less switch when their gates are enabled at the appropriate point in time when their diodes would be conducting. For example, a MOSFET having a resistance of 1 milliohm conducting a current of 25 amps would attenuate a signal across it by approximately 25 millivolts. The synchronous rectifier 400 selectively turns on the MOSFETs when their body diodes would be conducting to create a highly efficient power supply 302.

[0150] In other embodiments, the transistors Q1 404, Q2 406, Q3 408, Q4 410 are P-channel or N-channel MOSFETs with or without an integral body diode. In yet other embodi-

ments, transistors, such as Bipolar Junction Transistors (BJTs), Isolated Gate Bipolar Transistors (IGBTs), or the like, can be used.

[0151] In another embodiment, each transistor Q1 404, Q2 406, Q3 408, Q4 410 comprises more than one transistor connected in parallel. In another embodiment, multiple MOSFETs may be packaged in a single module.

[0152] The first transistor Q1 404 is coupled in series with the second transistor Q2 406 across AC1 and AC2, such that a drain of the first transistor Q1 404 connects to the first power waveform AC1, and a drain of the second transistor Q2 406 connects to the second power waveform AC2. Further, a source of first transistor Q1 404 connects to a source of the second transistor Q2 406 and forms a third power waveform GROUND.

[0153] The third transistor Q3 408 is coupled in series with the fourth transistor Q4 410 across AC1 and AC2, such that a source of the third transistor Q3 408 connects to the first power waveform AC1, and a source of the fourth transistor Q2 410 connects to the second power waveform AC2. Further, a drain of the third transistor Q3 408 connects to a drain of the fourth transistor Q4 410 and forms a fourth power waveform V-FULLWAVE.

[0154] The series combination of the first transistor Q1 404 and the second transistor Q2 406 electrically couple in parallel with the series combination of the third transistor Q3 408 and fourth transistor Q4 410, such that the drain of the first transistor Q1 404 electrically couples to the source of the third transistor Q3 408, and the drain of the second transistor Q2 406 electrically couples to the source of the fourth transistor Q4 410.

[0155] Each transistor is associated with a gate signal and the gate signals electrically couple to an output of a comparator comparing the first and second power waveforms, AC1 and AC2, via driver circuitry. The gates of the second transistor Q2 406 and the third transistor Q3 408 enable when the first power waveform AC1 is greater than the second power waveform AC2. FIG. 6 depicts an exemplary waveform 600 of the transistor gate signal for the gates of the second transistor Q2 406 and the third transistor Q3 408, according to an embodiment. Referring to FIGS. 5 and 6, the gate signal Vgs (Q2, Q3) is enabled when AC1 is greater than AC2.

[0156] Further, the gates of the first transistor Q1 404 and the fourth transistor Q4 410 enable when the second power waveform AC2 is greater than the first power waveform AC1. FIG. 7 depicts an exemplary waveform 700 for the gates of the first transistor Q1 404 and the fourth transistor Q4 410, according to an embodiment. Referring to FIGS. 5 and 7, the gate signal Vgs (Q1, Q4) is enabled when AC2 is greater than AC1.

[0157] The rectifier 400 full wave rectifies a 12 VAC RSM signal creating the third power waveform GROUND and the fourth power waveform V-FULLWAVE. The rectified 12 VAC RMS signal, V-FULLWAVE, has a peak voltage of approximately 16.97 volts, which is approximately the same as the peak voltage of the power waveform at the output of the transformer 402. The small loss in signal is due to exemplary, but finite conduction of the transistors Q1 404, Q2 406, Q3 408, Q4 410 when their gates are enabled. FIG. 8 depicts an exemplary rectified 12 VAC RMS signal 800, according to an embodiment. As illustrated in FIG. 8, the rectifier 400 outputs a non-inverted 12 VAC RMS power

waveform **800** when **AC1** is greater than **AC2** and outputs an inverted 12 VAC RMS waveform **800** when **AC2** is greater than **AC1**.

[0158] Referring to FIG. 4, a current sensing element **412**, such as a current transformer, magnetically couples to the wire/trace carrying the 12 VAC RMS power waveform. In one embodiment, the current transformer **412** magnetically couples to the wire/trace carrying the power waveform **AC2**. In another embodiment, the current transformer **412** magnetically couples to the wire/trace carrying the power waveform **AC1**. Current flowing through wire/trace carrying **AC2**, in the illustrated embodiment, produces a magnetic field in the core of the current transformer **412**, which in turn induces a current in the winding wound around the core of the current transformer **412**. The induced current is proportional to the current of the power waveform **AC2**, in the illustrated embodiment, or to the current of the power waveform **AC1**, in another embodiment. The current transformer **412** outputs signals, Current Sense1 and Current Sense2, proportional to current flowing through the power waveforms **AC1** or **AC2**. The signals Current Sense1 and Current Sense2 are used to determine when the current flowing in the power waveforms **AC1** or **AC2** is greater than a threshold value, such that power supply **302** can be disabled before damage to the circuitry occurs. Accordingly, the rectifier **400** of FIG. 4 advantageously produces the V-FULLWAVE waveform **800** of FIG. 8 with minimal power loss and correspondingly, minimal heat generation.

[0159] FIG. 9 is an exemplary schematic diagram of a bridge circuit **900**, according to an embodiment. The bridge **900** comprises a fifth transistor **Q5 904**, a sixth transistor **Q6 906**, a seventh transistor **Q7 908**, and an eighth transistor **Q8 910**. The bridge **900** receives the rectified power waveforms V-FULLWAVE and GROUND from the rectifier **400**. In the illustrated embodiment, V-FULLWAVE is an exemplary rectified 12 VAC RMS signal as shown in FIG. 8. Advantageously, in a disclosed embodiment, the bridge **900** selectively outputs the rectified power waveforms V-FULLWAVE, GROUND with either a positive polarity or a negative polarity. By doing so, data or intelligence can be added to the presently described power signal. Thus, the rectifier **400** and the bridge **900** combine to produce a power signal with embedded data or logic.

[0160] The positive or negative polarity of V-FULLWAVE is, for example, the control signals, LIGHTING CONTROL1, LIGHTING CONTROL2 on the two-wire path to the lighting modules **304**. LIGHTING CONTROL1 and LIGHTING CONTROL2 comprise addresses/data/commands encoded within the power waveform V-FULLWAVE, to provide addresses/data/commands and power to the lighting modules **304**.

[0161] In an embodiment, the transistors **Q5 904**, **Q6 906**, **Q7 908**, **Q8 910** are metal-oxide-semiconductor field-effect transistors (MOSFETs) with an integral body diode. As described above, the MOSFETs with the integral body diode advantageously function as an almost or substantially lossless switch when their gates are enabled at the appropriate point in time when their diodes would be conducting.

[0162] In other embodiments, the transistors **Q5 904**, **Q6 906**, **Q7 908**, **Q8 910** are either P-channel or N-channel MOSFETs with or without an integral body diode. In yet other embodiments, transistors, such as Bipolar Junction Transistors (BJTs), Isolated Gate Bipolar Transistors (IGBTs), or the like, can be used.

[0163] In another embodiment, each transistor **Q5 904**, **Q6 906**, **Q7 908**, **Q8 910** comprises more than one transistor connected in parallel. In another embodiment, multiple MOSFETs may be packaged in a single module.

[0164] The fifth transistor **Q5 904** is coupled in series with the sixth transistor **Q6 906** across V-FULLWAVE and GROUND, such that a drain of the fifth transistor **Q5 904** connects to the power waveform V-FULLWAVE, and a source of the sixth transistor **Q6 906** connects to the power waveform GROUND. Further, a source of the fifth transistor **Q5 904** connects to a drain of the sixth transistor **Q6 906** and forms the first control signal, LIGHTING POWER/CONTROL1.

[0165] The seventh transistor **Q7 908** is coupled in series with the eighth transistor **Q8 910** across V-FULLWAVE and GROUND, such that a drain of the seventh transistor **Q7 908** connects to the power waveform V-FULLWAVE, and a source of the eighth transistor **Q8 910** connects to the power waveform GROUND. Further, a source of the seventh transistor **Q7 908** connects to a drain of the eighth transistor **Q8 910** and forms the second control signal, LIGHTING POWER/CONTROL2.

[0166] The series combination of the fifth transistor **Q5 904** and the sixth transistor **Q6 906** electrically couple in parallel with the series combination of the seventh transistor **Q7 908** and eighth transistor **Q8 910**, such that the drain of the fifth transistor **Q5 904** electrically couples to the drain of the seventh transistor **Q7 908**, and the source of the sixth transistor **Q6 906** electrically couples to the source of the eighth transistor **Q8 910**.

[0167] Each transistor **Q5 904**, **Q6 906**, **Q7 908**, **Q8 910** is associated with a gate signal. The gate signals electrically couple, via driver circuitry, to a control signal comprising data from the processor **314** associated with the operator interface panel **308** and the output of the comparator comparing the power waveforms **AC1**, **AC2**. The gates of the fifth transistor **Q5 904** and the eighth transistor **Q8 910** are enabled when the control signal is in a first state. When the gates of the fifth transistor **Q5 904** and the eighth transistor **Q8 910** are enabled, the bridge **900** outputs the power waveforms V-FULLWAVE and GROUND having a first polarity on the two-wire path as signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2. The gates of the sixth transistor **Q6 906** and the seventh transistor **Q7 908** are enabled when the control signal is in a second state. When the gates of the sixth transistor **Q6 906** and the seventh transistor **Q7 908** are enabled, the bridge **900** outputs the power waveforms V-FULLWAVE and GROUND having a second polarity on the two-wire path as signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2.

[0168] For example, in one embodiment, when the gates of the fifth transistor **Q5 904** and the eighth transistor **Q8 910** are enabled, the signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 comprise the power waveforms V-FULLWAVE and GROUND having a positive polarity. Further, when the gates of the sixth transistor **Q6 906** and the seventh transistor **Q7 908** are enabled, signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 comprise the power waveforms V-FULLWAVE and GROUND having a negative polarity.

[0169] In another embodiment, the polarities can be reversed, such that the signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 comprise

power waveforms V-FULLWAVE and GROUND having a negative polarity when gates of the fifth transistor Q5 904 and the eighth transistor Q8 910 are enabled and comprise power waveforms V-FULLWAVE and GROUND having a positive polarity when the gates of the sixth transistor Q6 906 and the seventh transistor Q7 908 are enabled.

[0170] As discussed above, the gate signals electrically couple, via driver circuitry, to a control signal comprising data from the processor 314 associated with the operator interface panel 308 and the output of the comparator comparing the power waveforms AC1, AC2. When there is no data present, the control signal follows the output of the comparator comparing the power waveforms AC1, AC2.

[0171] FIG. 10 depicts an exemplary waveform 1000 of the transistor gate signal for the gates of the fifth transistor Q5 904 and the eighth transistor Q8 910 with no data present. As shown in FIGS. 5 and 10, the gate signal Vgs (Q5, Q8) is enabled when AC1 is greater than AC2.

[0172] FIG. 11 depicts an exemplary waveform 1100 of the transistor gate signal for the gates of the sixth transistor Q6 906 and the seventh transistor Q7 908 with no data present. As shown in FIGS. 5 and 11, the gate signal Vgs (Q5, Q8) is enabled when AC2 is greater than AC1.

[0173] FIG. 12 depicts an exemplary bridge output waveform 1200 when there is no data present from the processor 314, in one embodiment. As illustrated in FIGS. 10, 11, and 12, the bridge 900 outputs V-FULLWAVE with a positive polarity when the gates of the fifth transistor Q5 904 and the eighth transistor Q8 910 are enabled and outputs V-FULLWAVE with a negative polarity when the gates of the sixth transistor Q6 906 and the seventh transistor Q7 908 are enabled, generating approximately a sine wave. As shown, without data on the power signal for the lights, the rectifier 400 and the bridge 900 take the 12 VAC RMS output of the transformer 402, which is illustrated as its 16.97 VAC peak-to-peak waveforms AC1 and AC2 in FIG. 5, fullwave rectify it, and change it back to its original form using substantially or almost loss-less circuitry. However, as described herein, the same rectifier 400 and bridge 900 accept control signals from the processor 314 according to user programming to selectively control one or more fixtures 104, 204, 304 in one or more zones 106, 206. The control signals activate the gates with the same or substantially similar almost loss-less process in a manner that embeds logic or data on the power signal 1200 of FIG. 12.

[0174] For example, when the control signal controlling the transistor gates comprises data from the processor 314 associated with the operator interface panel 308, the bridge 900 encodes the data onto the signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 such that the bridge 900 outputs V-FULLWAVE having one polarity when the control signal is in a first state and outputs V-FULLWAVE having the opposite polarity when the control signal is in the second state. FIG. 13 depicts an exemplary power/data waveform 1300 with data, according to an embodiment. FIG. 13 illustrates start bits 1302 comprising 1, 1, followed by data bits 1304 comprising 0, 1, 0, 1, 1. In other embodiments, other configurations of start bits can be used and opposite polarities can be used to represent the 0 and 1 data bits. For instance, the control signal may change state at the peaks or any point of V-FULLWAVE as opposed to at the point V-FULLWAVE is zero. In summary, the bridge 900 is used synchronously with the VAC power waveform from the transformer 302 to select either a posi-

tive or a negative peak or half-cycle of the power waveform and apply the selected half-cycle to the output signals, LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 to encode data within the power waveform for transmission to the lighting modules 304.

[0175] In an embodiment where the transformer 402 produces approximately a 12 VAC 60 hertz power waveform, the data rate is approximately 120 bits per second. In another embodiment, the lighting modules 304 comprise a comparator comparing the signals LIGHTING POWER/CONTROL1, LIGHTING POWER/CONTROL2 to detect the data and a full wave rectifier to rectify the signals LIGHTING POWER/CONTROL1, LIGHTING POWER/CONTROL2 to provide power to the lighting elements.

[0176] In an embodiment, the transistors Q5 904, Q6 906, Q7 908, Q8 910 are turned on at the zero crossing of the controls signal because advantageously, the lighting modules 304 draw less power. At that time, there is less voltage or current flowing and less EMI noise is generated. In other embodiments, the transistors Q5 904, Q6 906, Q7 908, Q8 910 are turned on and off at other than the zero crossing of the control signal and/or more frequently than every half cycle of the incoming power.

[0177] Another advantage of sending the data as either a positive polarity or a negative polarity rectified power waveform is that there is no DC bias on the two-wire data/power path. If a DC bias is present, moisture seeping through the wires can produce unwanted galvanic corrosion.

[0178] FIG. 14 is an exemplary schematic diagram of a rectifier/bridge circuit 1400, according to an embodiment, which is also capable of producing a power signal with embedded data the same or similar to those disclosed above. The rectifier/bridge circuit 1400 comprises a primary AC to 24 VAC center-tapped transformer 1402, a current transformer 1412, a fifth transistor Q5 1404, a sixth transistor Q6 1406, a seventh transistor Q7 1408, and an eighth transistor Q8 1410. The current transformer 1412 senses the current in the center tap of the transformer 1402 as described above with respect to FIG. 4.

[0179] The primary AC to 24 VAC transformer 1402 receives a primary AC power signal and outputs an approximately 12 VAC RMS between each end tap and the center tap. This waveform comprises a power waveform having the first power waveform AC1 and the second power waveform AC2. Referring to FIG. 5, the exemplary 12 VAC RMS power waveform 500 has a peak-to-peak voltage of between approximately +16.97 volts to approximately -16.97 volts. Further, the center tap of transformer 1402 electrically couples to one wire of the two-wire path and forms the signal LIGHTING POWER/CONTROL2.

[0180] In an embodiment, the transistors Q5 1404, Q6 1406, Q7 1408, Q8 1410 are metal-oxide-semiconductor field-effect transistors (MOSFETs) with an integral body diode. In other embodiments, the transistors Q5 1404, Q6 1406, Q7 1408, Q8 1410 are either P-channel or N-channel MOSFETs with or without an integral body diode. In another embodiment, each transistor Q5 1404, Q6 1406, Q7 1408, Q8 1410 comprises more than one transistor connected in parallel. In another embodiment, multiple MOSFETs may be packaged in a single module.

[0181] The transistors Q5 1404, Q6 1406, Q7 1408, Q8 1410 are coupled in series such that a source of the fifth transistor Q5 1404 connects to a source of the eighth transistor Q8 1410, a drain of the eighth transistor Q8 1410

connects to a drain of the sixth transistor Q6 1406 and couples to the other wire of the two-wire path and forms the signal LIGHTING POWER/CONTROL1, and a source of the sixth transistor Q6 1406 connects to a source of the seventh transistor Q7 1408. The series combination of the transistors Q5 1404, Q8 1410, Q6 1406, Q7 1408 connects to the power waveforms AC1, AC2 such that a drain of the fifth transistor Q5 1404 electrically connects to AC1 and a drain of the seventh transistor Q7 1408 electrically connects to AC2.

[0182] Each transistor Q5 1404, Q6 1406, Q7 1408, Q8 1410 is associated with a gate signal. The gate signals electrically couple, via driver circuitry, to the control signal comprising data from the processor 314 associated with the operator interface panel 308 and the output of the comparator comparing the power waveforms AC1, AC2, as described above with respect to FIG. 9.

[0183] As shown in FIG. 14, one of the wires in the two-wire path to the lighting modules is the center tap of the transformer 1402. Depending on whether the gates of transistors Q5 1404 and Q8 1410 or Q6 1406 and Q7 1408 are enabled, the positive half-cycle or the negative half-cycle of the power waveform AC1, AC2 is sent on the other wire of the two-wire path to the lighting modules 304. In this manner, the data from the controller 314 can be encoded within the power waveform sent to the lighting modules 304. The rectifier/bridge 1400 can transmit the same data and power to the lighting modules 304 as the combination of the rectifier 400 and the bridge 900, but advantageously with fewer MOSFETs.

[0184] FIG. 15 is an exemplary schematic diagram of circuitry 1500 comprising phase detect circuitry, timing generation circuitry, driver circuitry, and over current protection circuitry, according to certain embodiments. The circuitry 1500 comprises a comparator 1502, MOSFET drivers 1504, 1506, 1508, 1510, a computer 1512, a modulator 1514, a difference amplifier 1518, and a latching comparator 1516.

[0185] The comparator 1502 receives the power waveforms AC1, AC2 and electrically couples an output to the gates of the transistors Q1 404, Q2 406, Q3 408, Q4 410 in the rectifier 400 via the drivers 1504, 1506. The power waveforms AC1, AC2 received by the comparator 1502 have been preconditioned as is known to one of skill in the art to be within the acceptable input voltage range for the comparator 1502. The comparator 1502 compares AC1 and AC2 and, in one embodiment, outputs a positive pulse when AC1 is greater than AC2 and outputs a ground or negative pulse when AC2 is greater than AC1. While the input to the comparator is a sine wave, as shown in FIG. 5, the output is a square wave. The output of the comparator 1502 couples to the input of the inverting driver 1504, and the input of the non-inverting driver 1506.

[0186] The output of the non-inverting driver 1506 couples to the gates of transistors Q2 406 and Q3 408 on the rectifier 400. The waveform 600, in FIG. 6, illustrates an example of the transistor gate signal for the gates of the second transistor Q2 406 and the third transistor Q3 408. Referring to FIGS. 5 and 6, the output of the comparator, which is the input to the driver 1506, is positive and the gate signal Vgs (Q2, Q3) is enabled when AC1 is greater than AC2. Further, the output of driver 1504 is low and the transistors Q1 404 and Q4 410 are off when AC1 is greater than AC2.

[0187] The output of the inverting driver 1504 couples to the gates of transistors Q1 404 and Q4 410 on the rectifier 400. The waveform 700, in FIG. 7, illustrates an example of the transistor gate signal for the gates of the first transistor Q1 404 and the fourth transistor Q4 410. Referring to FIGS. 5 and 7, the output of the comparator 1502, which is the input to the inverting driver 1504, is negative or ground, and the gate signal Vgs (Q1, Q4) is enabled when AC2 is greater than AC1. Further, the output of driver 1506 is low and the transistors Q2 406 and Q3 408 are off when AC2 is greater than AC1.

[0188] The modulator 1514 receives the output of the comparator 1502 and receives a data signal from the computer 1512. The data signal comprises addresses/data/commands from the operator interface panel 308. In an embodiment, computer 1512 is computer 314. In another embodiment, computer 314 and computer 1512 are different computers. The computer 1512 comprises, by way of example, those devices or structures similar to computer 314.

[0189] An output of the modulator 1514 connects to the input of inverting driver 1508 and to the input of non-inverting driver 1510. The modulator 1514 passes the output of the comparator 1502 to the drivers 1508, 1510 when no data is present. The signal on the two-wire path to the lighting modules 304 is the sine wave 1200, shown in FIG. 12, when no data is present.

[0190] The output of the non-inverting driver 1510 couples to the gates of transistors Q5 904, 1404 and Q8 910, 1410 on the bridge 900 or the rectifier/bridge 1400. The waveform 1000, in FIG. 10, illustrates an example of the transistor gate signal for the gates of the fifth transistor Q5 904, 1404 and the eighth transistor Q8 910, 1410. Referring to FIGS. 5 and 10, the gate signal Vgs (Q5, Q8) is enabled when AC1 is greater than AC2 and data is absent.

[0191] The output of the inverting driver 1508 couples to the gates of transistors Q6 906, 1406 and Q7 908, 1408 on the bridge 900 or the rectifier/bridge 1400. The waveform 1100, in FIG. 11, illustrates an example of the transistor gate signal for the gates of the sixth transistor Q6 906, 1406 and the seventh transistor Q7 908, 1408. Referring to FIGS. 5 and 11, the gate signal Vgs (Q6, Q7) is enabled when AC2 is greater than AC1 and data is absent.

[0192] As shown, without data on the power signal for the lights, the rectifier/bridge 1400 takes the center tap of the transformer 1402, as one wire of the two-wire path to the lighting fixtures 104, 204. Depending on whether Q5 1404 and Q8 1410, or Q6 1406 and Q7 1408 are enabled, the rectifier/bridge 1400 sends the positive half-cycle or the negative half-cycle of the 12 VAC RMS output of the transformer 1402, which is illustrated as its 16.97 VAC peak-to-peak waveforms AC1 and AC2 in FIG. 5 on the other wire of the two-wire path, using substantially or almost loss-less circuitry. However, as described herein, the same rectifier/bridge 1400 accepts control signals from the processor 314 according to user programming to selectively control one or more fixtures 104, 204 in one or more zones 106, 206. The control signals activate the gates with the same or substantially similar almost loss-less process in a manner that embeds logic or data on the power signal 1200 of FIG. 12.

[0193] When data is present the modulator 1514 functions as a selective inverter, in an embodiment. The data signal inverts the signal between the comparator 1502 and the

drivers **1508**, **1510**. For example, when the data is high, the modulator **1514** acts as an inverter and inverts the signal from the comparator **1502** before the signal is received by the drivers **1508**, **1510**. When the data is low, the modulator **1514** passes the output of the comparator **1502** to the drivers **1508**, **1510**. This permits the phase of the signals LIGHTING POWER/CONTROL1, LIGHTING POWER/CONTROL2 output from the bridge **900** or rectifier/bridge **1400** on the two-wire path to the lighting modules **304** to be adjusted on a half-cycle basis to encode the data within the power waveform. Referring to FIG. 13, the waveform **1300** illustrates an example of a data encoded power waveform comprising start bits **1302** comprising 1, 1, followed by data bits **1304** comprising 0, 1, 0, 1, 1.

[0194] Referring to FIG. 15, the difference amplifier **1518** receives the signals, CURRENT SENSE1, CURRENT SENSE2, from the current transformer **412**, **1412**, which are proportional to the current flowing out of the transformer **402**. The difference amplifier **1518** subtracts CURRENT SENSE1, CURRENT SENSE2 to create a single ended current protection signal. The latching comparator **1516** receives the output of the difference amplifier **1518** and compares the current protection signal to a reference voltage or threshold. The output of the latching comparator **1516** couples to an enable signal common to the drivers **1504**, **1506**, **1508**, **1510**. When the peak voltage of the current protection signal exceeds the threshold, the output of the latching comparator **1516** disables the drivers **1504**, **1506**, **1508**, **1510** to prevent an overcurrent event from damaging the circuitry.

[0195] Further, processor **1512** receives the latched output of the latching comparator **1516** and the latching comparator **1516** receives a reset signal from the processor **1512**. In an embodiment, the processor **1512** can reset the latching comparator **1516**. In another embodiment, the processor **1512** can alert the user to the overcurrent event through communication with the processor **314**. The processor **314** could then display the information on the display **312**.

[0196] FIG. 16 is an exemplary schematic diagram of a bias circuit **1600**, according to an embodiment. In embodiments of the rectifier **400**, the bridge **900**, and the rectifier/bridge **1400**, the sources of some of the transistors Q1-Q8 are electrically connected to one of the two AC outputs, AC1, AC2, of the transformer **402**, **1402** or to the rectified power waveform V-FULLWAVE. When the transistor or MOSFET is turned on, nominally the gate voltage should be approximately 5 volts+/-about 4 volts to approximately 10 volts+/-about 5 volts more positive than the source voltage, for proper operation, as is known to one of skill in the art from the disclosure herein. However, this is a higher voltage than is present at the output of the transformer **402**, **1402**. The bias circuitry **1600** functions to provide the transistors Q1-Q8 in the rectifier **400**, the bridge **900** and the rectifier/bridge **1400** with the higher gate voltage.

[0197] The bias circuit **1600** receives the power waveforms AC1, AC2 from the transformer **402**, **1402** and generates the power waveforms AC1++, AC2++ that are at a higher DC level than AC1, AC2, but follow the AC1, AC2 waveforms, respectively. For example, AC1++ and AC2++ may have a DC offset of about 10 volts to about 20 volts above AC1, AC2, as they move up and down with AC1, AC2. AC1++, AC2++ power the MOSFET driver integrated circuits **1508**, **1510** that provide the gate signals for the

MOSFETs Q5 **904**, Q6 **406**, Q7 **908**, Q8 **910** in the bridge **900** and the MOSFETs Q5 **1404**, Q6 **1406**, Q7 **1408**, Q8 **1410** in rectifier/bridge **1400**.

[0198] The bias circuit **1600** comprises capacitors C1 **1602**, C2 **1604**, resistors R1 **1606**, R2 **1608**, and diodes D1 **1610**, D2 **1612**, D3 **1614**, D4 **1616**. AC2 electrically couples to an anode of diode D1 **1610** and the series combination of diode D1 **1610** and resistor R1 **1602** half-wave rectify AC2 with respect to AC1 and capacitor C1 **1602** stores the voltage. An anode of diode D2 **1612** couples to an end of capacitor C1 **1602**. Diode D2 **1612** is a zener or clamping diode and clamps the voltage at the clamping value. In an embodiment, diode D2 **1612** is a +18 volt zener diode. A cathode of diode D2 **1612** provides the power waveform AC1++.

[0199] Similarly, AC1 electrically couples to an anode of diode D4 **1616** and the series combination of diode D4 **1616** and resistor R2 **1608** half-wave rectify AC1 with respect to AC2 and capacitor C2 **1604** stores the voltage. An anode of diode D3 **1614** couples to an end of capacitor C2 **1604**. Diode D3 **1614** is a zener or clamping diode and clamps the voltage at the clamping value. In an embodiment, diode D3 **1614** is a +18 volt zener diode. A cathode of diode D3 **1614** provides the power waveform AC2++. In other embodiments, diodes D2 **1612**, D3 **1614** can have clamping values at other than +18 volts.

[0200] The bias circuit **1600** further receives the power waveform AC1 from the transformer **402** and V-FULLWAVE from the rectifier **400** and generates the power waveform V-FULLWAVE++. V-FULLWAVE++ is approximately AC half-wave rectified and at a DC level that is no lower than approximately one diode drop below V-FULLWAVE. V-FULLWAVE powers the MOSFET driver integrated circuits **1504**, **1506** that provide the gate signals for the MOSFETs Q1 **404**, Q2 **406**, Q3 **408** Q4 **410** in the synchronous rectifier **400**.

[0201] The bias circuit further comprises capacitors C3 **1618**, C4 **1620**, C5 **1622**, resistor R3 **1624**, and diodes D5 **1626**, D6 **1628**, D7 **1630**, D8 **1632**. AC1 electrically couples to a first end of capacitor C3 **1618** and a cathode of diode D5 **1626**. A second end of capacitor C3 **1618** connects to a first end of capacitor C4 **1620**, an anode of diode D5 **1626** and an anode of diode D6 **1628**. A second end of capacitor C4 **1620** and a cathode of diode D6 **1628** couple to an anode of diode D7 **1630** and a cathode of diode D8 **1632**. Capacitors C3 **1618**, C4 **1620**, diode D5 **1626**, and diode D6 **1628** form a charge pump circuit using the power waveform AC1. An anode of diode D8 **1632** electrically couples to V-FULLWAVE and clamps the AC signal passing through the capacitors C3 **1618**, C4 **1620** at approximately one diode drop below V-FULLWAVE at the cathode of diode D8 **1632**. The series combination of diode D7 **1630** and resistor R3 **1624** half-wave rectifies the clamped V-FULLWAVE signal with respect to V-FULLWAVE and capacitor C5 **1622** stores the voltage. An end of capacitor C5 **1622** couples to an end of resistor R3 **1624** and provides the power waveform V-FULLWAVE++.

[0202] FIGS. 17A1-17A4 and 17B1-17B4 are exemplary circuit diagrams for a lighting controller **1700**, according to one embodiment. FIGS. 17A1 and 17A3 are an example of a rectifier circuit **1710** where the MOSFETs **1712**, **1714**, **1716**, **1718** of FIG. 17A1 electrically couple in parallel with the MOSFETs **1713**, **1715**, **1717**, **1719** having the corresponding gate signals Gate5, Gate6, Gate7, Gate8 of FIG.

17A3 for increased current drive. FIGS. 17A2 and 17A4 are an example of a bridge circuit 1720 where the MOSFETs 1722, 1724, 1726, 1728 of FIG. 17A2 electrically couple in parallel with the MOSFETs 1723, 1725, 1727, 1729 having the corresponding gate signals Gate1, Gate2, Gate3, Gate4 of FIG. 17A4 for increased current drive. FIGS. 17B1-17B4 are examples of a bias circuit 1730, driver circuit 1740, phase detection circuit 1750, timing generation circuit 1760, and a current protection circuit 1770.

[0203] FIG. 18 is a block diagram of another exemplary lighting controller 1800 controlling lighting devices 1804 and comprising a power supply 1802 and the operator interface panel 308. The operator interface panel 308 comprises the operator controls 310, the displays and indicators 312, the computer 314, the memory 316, the fixture programming port 318, and the logic power supply 320 and operates as described above with respect to FIG. 3 to provide addresses/data/commands to the power supply 1802. The architecture of the power supply 1802 is similar to the architecture of the power supply 302 except that the power supply 1802 does not have the transformer 322 to transform the primary line voltage to a secondary AC voltage. Instead, the power supply 1802 drives the bridge and encoding circuitry from the primary line voltage. Further, the lighting devices 1804, described further herein, are powered by the primary line voltage.

[0204] The power supply 1802 comprises current sensing circuitry 1824, phase detect and timing circuitry 1826, driver circuitry 1828, a synchronous fullwave rectifier 1832, and a bridge 1834. The power supply 1802 further comprises a low power transformer 1846 to provide a low voltage, such as 9 VAC, for example, to a logic power supply which creates a regulated DC voltage for the digital logic components of the power supply 1802, and biasing circuitry 1830 to provide the proper voltage levels to operate transistors in the rectifier 1832 and the bridge 1834.

[0205] Similar to the power supply 302 illustrated in FIG. 3, the power supply 1802 receives AC power from the primary power source 306 and addresses/data/commands from the operator interface panel 308 and provides a control signal to a plurality of lighting fixtures 1804 through a two-wire path 1836. However, unlike the power supply 302, the primary power source 306 is electrically coupled to the power supply circuitry without being transformed to a lower voltage by the primary AC to 12 VAC transformer 322. In addition, while the lighting modules 304 operate at a low voltage signal, such as 12 VAC, for example, the lighting modules 1804 operate at approximately the primary AC power level, such as 120 VAC, for example.

[0206] In an embodiment, the primary AC signal is approximately a 120 volt 60 Hz power waveform. In other embodiments, the primary AC signal can be approximately 110 volts 60 Hz, 220 volt 50 Hz, 220 volt 60 Hz, 230 volts 60 or 50 Hz, 240 volts 60 or 50 Hz, or the like, power waveform.

[0207] The primary AC power source 306 electrically connects to the current sensing circuitry 1824. The current sensing circuitry 1824 senses the amount of current in the output of the primary AC power source 306. The phase detect and timing circuitry 1826 receives a signal proportional to the sensed current from the current sensing circuitry 1824 and shuts off the power supply 1802 when the sensed current exceeds a threshold, as described above.

[0208] The phase detect and timing circuitry 1826 further receives data and commands from the processor 314 and the power waveform from the primary AC power source 306, and provides timing signals to the driver circuit 1828. The timing signals control the driver circuitry 1828 to encode a data signal onto the power signal by varying the polarity of the power waveform, as will be further discussed herein.

[0209] Further, the primary AC power source 306 electrically connects to the synchronous fullwave rectifier 1832, which rectifies the power signal. The fullwave rectifier 1832 electrically connects to the bridge 1834 and the fullwave rectifier 1832 and the bridge 1834 electrically connect to the driver circuitry 1828. Both the fullwave rectifier 1832 and the bridge 1834 receive drive signals from the driver circuitry 1828. The bridge 1834 receives the rectified power signal and outputs a control signal to the lighting fixtures 1804. The control signal comprises a data encoded power waveform, which provides power to illuminate the lighting fixtures 1804 and address/data/commands to individually control the lighting fixtures 1804.

[0210] FIG. 19 is a schematic diagram of an exemplary synchronous rectifier circuit 1900 configured to operate in the power supply 1802. The rectifier circuit 1900 comprises a first switching device Q1 1904, a second switching device Q2 1906, a third switching device Q3 1908, and a fourth switching device Q4 1910. The rectifier circuit 1900 is similar to the rectifier circuit 400 in FIG. 4, but without the primary AC to 12 VAC transformer 402.

[0211] The rectifier circuit 1900 receives the primary AC power signal having a first power waveform AC1 and a second power waveform AC2. In one embodiment, the primary AC power signal is approximately 120 VAC and FIG. 21 illustrates an exemplary 120 VAC RMS power waveform 2100 having a peak-to-peak voltage of between approximately +169.7 volts to approximately -169.7 volts.

[0212] In an embodiment, the switching devices Q1 1904, Q2 1906, Q3 1908, Q4 1910 are metal-oxide-semiconductor field-effect transistors (MOSFETs) with an integral body diode, as described above with respect to FIG. 4. The synchronous rectifier 1900 selectively turns on the MOSFETs when their body diodes would be conducting to create a highly efficient power supply 1802.

[0213] In other embodiments, the switching devices Q1 1904, Q2 1906, Q3 1908, Q4 1910 are P-channel or N-channel MOSFETs with or without an integral body diode. In yet other embodiments, transistors, such as Bipolar Junction Transistors (BJTs), Isolated Gate Bipolar Transistors (IGBTs), or the like, can be used. In further embodiments, switching devices such as triode for alternating current devices (triacs), silicon-controlled rectifiers (SCRs), and the like can be used. In another embodiment, each switching device Q1 1904, Q2 1906, Q3 1908, Q4 1910 comprises more than one transistor connected in parallel. In another embodiment, multiple MOSFETs may be packaged in a single module.

[0214] The first switching device Q1 1904 is coupled in series with the second switching device Q2 1906 across AC1 and AC2 to form a third power waveform GROUND and the third switching device Q3 1908 is coupled in series with the fourth switching device Q4 1910 across AC1 and AC2, to form a fourth power waveform V-FULLWAVE, as described above with respect to FIG. 4. Also, as described above with respect to FIG. 4, the series combination of the first switching device Q1 1904 and the second switching device Q2

1906 electrically couple in parallel with the series combination of the third switching device **Q3 1908** and fourth switching device **Q4 1910**.

[0215] Each switching device **Q1 1904**, **Q2 1906**, **Q3 1908**, **Q4 1910** is associated with a gate signal and the gate signals electrically couple to an output of a comparator comparing the first and second power waveforms, **AC1** and **AC2**, via driver circuitry as described herein. The gates of the second switching device **Q2 1906** and the third switching device **Q3 1908** enable when the first power waveform **AC1** is greater than the second power waveform **AC2**. FIG. 22 depicts an exemplary waveform **2200** of the gate signal for the gates of the second switching device **Q2 1906** and the third switching device **Q3 1908**, according to an embodiment. Referring to FIGS. 21 and 22, the gate signal **Vgs (Q2, Q3)** is enabled when **AC1** is greater than **AC2**.

[0216] Further, the gates of the first switching device **Q1 1904** and the fourth switching device **Q4 1910** enable when the second power waveform **AC2** is greater than the first power waveform **AC1**. FIG. 23 depicts an exemplary waveform **2300** for the gates of the first switching device **Q1 1904** and the fourth switching device **Q4 1910**, according to an embodiment. Referring to FIGS. 21 and 23, the gate signal **Vgs (Q1, Q4)** is enabled when **AC2** is greater than **AC1**.

[0217] The rectifier **1900** full wave rectifies the primary AC signal, such as, for example, a 120 VAC RMS signal, creating a third power waveform **GROUND** and the fourth power waveform **V-FULLWAVE**. The rectified 120 VAC RMS signal, **V-FULLWAVE**, has a peak voltage of approximately 169.7 volts, which is approximately the same as the peak voltage of the power waveforms **AC1** and **AC2** at the input of the rectifier **1900**. The small loss in signal is due to exemplary, but finite conduction of the switching devices **Q1 1904**, **Q2 1906**, **Q3 1908**, **Q4 1910** when their gates are enabled. FIG. 24 depicts an exemplary rectified 120 VAC RMS signal **2400**, according to an embodiment. As illustrated in FIG. 24, the rectifier **1900** outputs a non-inverted 120 VAC RMS power waveform **2400** when **AC1** is greater than **AC2** and outputs an inverted 120 VAC RMS waveform **2400** when **AC2** is greater than **AC1**.

[0218] The waveforms illustrated in FIGS. 21-24 illustrate embodiments where the primary AC power source **306** is approximately 120 VAC. In other embodiments, the primary AC power signal can be an approximately 110 volts 60 Hz, 220 volt 50 Hz, 220 volt 60 Hz, 230 volts 50 or 60 Hz, 240 volts 50 or 60 Hz, or the like, power waveform, and the RMS and peak-to-peak voltages will vary correspondingly.

[0219] Referring to FIG. 19, a current sensing element **1912**, such as a current transformer, magnetically couples to the wire/trace carrying the primary AC power waveform. In one embodiment, the current transformer **1912** magnetically couples to the wire/trace carrying the power waveform **AC2**. In another embodiment, the current transformer **1912** magnetically couples to the wire/trace carrying the power waveform **AC1**. As described above with respect to FIG. 4, the current induced in the current transformer **1912** is proportional to the current of the power waveform **AC2**, in the illustrated embodiment, or to the current of the power waveform **AC1**, in another embodiment. The current transformer **1912** outputs signals, **Current Sense1** and **Current Sense2**, which are proportional to current flowing through the power waveforms **AC1** or **AC2**. The signals **Current Sense1** and **Current Sense2** are used to determine when the current flowing in the power waveforms **AC1** or **AC2** is

greater than a threshold value, such that power supply **1802** can be disabled before damage to the circuitry occurs. Accordingly, the rectifier **1900** of FIG. 19 advantageously produces the **V-FULLWAVE** waveform **2400** of FIG. 24 with minimal power loss and correspondingly, minimal heat generation. In other embodiments, the current flowing in **AC1** or **AC2** can be sensed with Hall-Effect based current sensors, which sense the magnet field produced by the current. In yet another embodiment, current can be sensed by a current sense resistor.

[0220] FIG. 20 is a schematic diagram of another exemplary rectifier circuit **2000** configured to operate in the power supply **1802**. In the embodiment illustrated in FIG. 20, the rectifier **2000** is a diode-based rectifier comprising diodes **D1 2004**, **D2 2006**, **D3 2008**, and **D4 2010**. Because the current drawn at 120 VAC is approximately one tenth of the current drawn at 12 VAC for the same amount of power delivered, the diode-based rectifier **2000** provides acceptable efficiency. The rectifier circuit **2000** rectifies the primary AC power signal to produce similar gate and **V-FULLWAVE** waveforms as illustrated in FIGS. 22-24. Because of the drop in voltage (diode drop) associated with the diodes **D1 2004**, **D2 2006**, **D3 2008**, **D4 2010**, the peak voltages may be approximately two diode drops or approximately 1.4 V less than the peak voltages illustrated in FIGS. 22-24.

[0221] Referring to FIG. 18, the bridge circuit **1834** receives the rectified power waveforms **V-FULLWAVE** and **GROUND** from the rectifier **1900**. In an embodiment, the bridge circuit **1834** is similar to the bridge circuit **900** described herein with respect to FIG. 9. The switching devices in the bridge circuit **1834** can be P-channel or N-channel MOSFETs with or without integral body diodes, BJT's, IGBTs, as described with respect to FIG. 9 and bridge circuit **900**. Because of the lower current, the switching devices in the bridge circuit **1834** can also be triacs, SCRs, and the like.

[0222] In an embodiment where the primary AC power signal is a 120 VAC signal, **V-FULLWAVE** is an exemplary rectified 120 VAC RMS signal depicted in FIG. 24. Advantageously, the bridge **1834** selectively outputs a control signal comprising the rectified power waveforms **V-FULLWAVE**, **GROUND** with either a positive polarity or a negative polarity. By doing so, data or intelligence can be added to the presently described power signal. Thus, the rectifier **1900** and the bridge **1834** combine to produce a 120 VAC power signal with embedded data, which provides power to illuminate the lighting fixtures **1804** and address/data/commands to individually control the lighting fixtures **1804**.

[0223] FIG. 25 depicts an exemplary bridge output waveform **2500** when there is no data present from the processor **314**. As shown, without data on the power signal for the lights **1804**, the rectifier **1900** and the bridge **1834** take the 120 VAC RMS input signal, which is illustrated as its 169.7 VAC peak-to-peak waveforms **AC1** and **AC2** in FIG. 21, fullwave rectify it, and change it back to its original form. However, as described above with respect to rectifier **400** and bridge **900**, the rectifier **1900** and bridge **1834** accept control signals from the processor **314** according to user programming to selectively control one or more fixtures **1804** in one or more zones. The control signals activate the gates of the rectifier **1900** and bridge **1834** with the same or substantially similar process as described above to embed logic or data on the power signal **2500** of FIG. 25.

[0224] For example, when the control signal controlling the gates of the switching devices comprises data from the processor 314 associated with the operator interface panel 308, the bridge 1834 encodes the data onto the signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 by outputting V-FULLWAVE having one polarity when the control signal is in a first state and outputting V-FULLWAVE having the opposite polarity when the control signal is in the second state. FIG. 26 depicts an exemplary power/data waveform 2600 with data, according to an embodiment. FIG. 26 illustrates a start bit sequence 2602 comprising 1, 1, followed by data bits 2604 comprising 0, 1, 0, 1, 1. This is similar to the exemplary power/data waveform depicted in FIG. 13 except that the positive peak waveform is approximately +169.7 V and the negative peak waveform is approximately -169.7 V. In summary, the bridge 1834 is used synchronously with the 120 VAC power waveform to select either a positive or a negative peak or half-cycle of the 120 VAC power waveform and apply the selected half-cycle to the output signals, LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 to encode data within the power waveform for transmission to the lighting modules 1804 over the two-wire path 1836. It should be noted that in FIG. 26, the control signals are switched at the zero crossing of the incoming line voltage waveform. It is also possible to encode data by switching at a point other than the zero crossing, or more frequently than once per half cycle of the incoming line voltage.

[0225] FIG. 27 is an exemplary schematic diagram of a line voltage control node system 2700. The line voltage control node 2702 receives the data encoded low voltage power signals LIGHTING POWER/CONTROL1 and LIGHTING POWER/CONTROL2 from the two-wire path 336. In an embodiment, the low voltage power signals are approximately 12 VAC. In other embodiments, the low voltage power signals are approximately 11 VAC, approximately 14 VAC, and the like.

[0226] The line voltage control node 2702 is assigned a lighting zone via the operator interface panel 308 as described above. Based on the information contained on the two-wire path 336 addressed to the line voltage control node 2702, the line voltage control node 2702 controls line/high voltage lighting devices 2704. In an embodiment, the devices 2704 are 120 VAC dimmable lighting devices.

[0227] FIG. 28 is a block diagram of an exemplary line voltage control node 2800 configured to dim the lighting device 2704. The line voltage control node 2800 comprises a full wave rectifier 2802, a voltage regulator 2804, a conditioning circuit 2806, a microcontroller 2810, a triggering optoisolator 2814, a detection optoisolator 2816, and a triac 2818. The full wave rectifier 2802, the voltage regulator 2804, the conditioning circuit 2806, the microcontroller 2810 are the same or substantially similar to the circuitry described herein with respect to FIGS. 61-64 for single and multi-channel lighting modules 6100, 6200, 6300, 6400. In the lighting modules 6100, 6200, 6300, 6400, the microcontroller controls the LED driver to vary the brightness of the LED, whereas in the line voltage control node 2800, the line voltage, and thus the dimming function, is controlled via the conduction angle of the triac 2818.

[0228] For safety, the 120 VAC line voltage signals are galvanically isolated from the 12 VAC low voltage signals by the triggering optoisolator 2814. The detection optoisolator

2816 senses the phase of the incoming line voltage signal so that the triac 2818 can be triggered at the appropriate point in time. In most embodiments the zero crossings of the 120 VAC line voltage signal occurs at substantially the same time as the zero crossing of the 12 VAC low voltage signal. However, in other embodiments, the zero crossings of the high and low voltage signals may not substantially coincide. For instance if a generator generates either of the 12 VAC low voltage signal or the 120 VAC line voltage signal, then the zero crossings would not substantially correspond to those of the utility grid. Because of this, the detection optoisolator 2816 is used to sense the actual zero crossing of the line voltage signal and to send this information to the microcontroller 2810. The triggering optoisolator 2814 can be, for example, an MOC3021 available from Fairchild Optoelectronics Group and the like. The detection optoisolator 2816 can be, for example, a TLP620 from Toshiba and the like. The triac 2818 can be, for example, a BTA208-800 from NXP Semiconductor and the like.

[0229] Once the zero crossing of the line voltage is known, the microcontroller 2810 triggers the triac 2818 with a delay time based at least in part on the desired output intensity of the 120 VAC lighting device 2704. FIGS. 29-31 are exemplary waveforms depicting various output voltages usable to dim the lighting device 2704. The microcontroller 2810 creates output voltages approximately equal to or less than the input line voltage by delaying the triac trigger.

[0230] FIG. 29 depicts an output waveform 2900 having a very short delay between the zero crossing and the trigger. Therefore, the output waveform 2900 is almost equal to the input waveform. FIG. 30 depicts an output waveform 3000 having a trigger delay such that the output waveform 3000 is approximately half of the input voltage. FIG. 31 depicts an output waveform 3100 that is a small percentage of the input voltage due to a larger delay of the trigger signal.

[0231] The above embodiment has been described with respect to dimming functionality for the lighting devices 2704. In other embodiments of the line voltage control node 2800, a Z-wave device and/or a relay device could be used for non-dimming applications. For example, the line voltage control node 2800 could control ON/OFF for a zone comprising a fountain or the like.

[0232] Referring to FIG. 28, the line voltage control node 2800 further comprises an optional user interface 2812 having buttons, slide switches, or other forms of user controls to permit the user to manually control the lighting device 2704, in addition to the commands originating from the user interface panel 308.

[0233] The line voltage control node 2800 further comprises an LED 2808. In an embodiment, the LED 2808 is a low power LED not intended for illumination. The LED 2808 has the ability to “flash out” its serial number in a manner the same as or substantially similar to that of lighting modules 4004 described in further detail herein with respect to FIGS. 40-42.

[0234] FIGS. 27 and 28 illustrate embodiments of the line voltage control node 2700, 2800 that is electrically connected (via wires) to the data encoded 12 VAC two-wire path 336. FIGS. 32-36 illustrate embodiments of a wireless line voltage control node. The wireless line voltage control node is substantially similar to the wired line voltage control node 2700, 2800 except that the connection from the 120 VAC node to the 12 VAC control signal 336 is wireless. This may

be particularly useful to control an existing lighting device to which there is no 12 VAC wiring run.

[0235] FIG. 32 is an exemplary schematic diagram of a wireless line voltage control node 3200 comprising a 120 VAC interface wireless receiver 3202 and a 12 VAC interface wireless transmitter 3204 communicating over a radio frequency (RF) wireless link. The 12 VAC interface wireless transmitter 3204 is electrically connected to the data encoded 12 VAC two-wire path 336 and receives control data from the signals LIGHTING POWER/CONTROL 1 and LIGHTING POWER/CONTROL2 via the two-wire path 336. The 12 VAC interface wireless transmitter 3204 sends over the wireless link the control commands to the 120 VAC interface receiver 3202. The 120 VAC interface receiver 3202 is electrically connected to the 120 VAC line voltage and the 120 VAC lighting device 2704, and receives the control data over the wireless link from the 12 VAC interface transmitter 3204. The 120 VAC interface receiver 3202 controls the dimming level of the lighting device 2704 base at least in part on the control data received from the 12 VAC interface transmitter 3204.

[0236] In other embodiments, the 12 VAC interface transmitter 3204 and the 120 VAC interface receiver 3202 can be replaced with transceivers. This may be useful to facilitate acknowledgment and retry algorithms to create a more robust link. For instance, the transceiver 3204 attached to the 12 VAC two-wire path would decode the signal on the 12 VAC line, and transmit its contents (if appropriate) to the transceiver 3202 attached to the 120 VAC line. The transceiver 3202, would in turn respond with an acknowledge transmission, indicating that the signal was received and the checksum was good. If the transceiver 3204 attached to the 12 VAC line does not receive the acknowledge signal, it would again transmit the command. This process could continue for a specified number of retries.

[0237] FIG. 33 is a block diagram of an exemplary wireless control node transmitter 3300 for the wireless line voltage control node 3200. The control node transmitter 3300 is assigned a lighting zone via the operator interface panel 308 as described herein. In the illustrated embodiment, the control node transmitter 3300 comprises a full wave rectifier 3302, a voltage regulator 3304, a conditioning circuit 3306, a microcontroller 3310, and a radio frequency (RF) transmitter or transceiver 3320. The full wave rectifier 3302, the voltage regulator 3304, the conditioning circuit 3306, and the microcontroller 3310 are the same or substantially similar to the circuitry described herein with respect to FIGS. 61-64 for single and multi-channel lighting modules 6100, 6200, 6300, 6400. In the lighting modules 6100, 6200, 6300, 6400, the microcontroller controls the LED driver to vary the brightness of the LED, whereas the control node transmitter 3300 controls the RF transmitter or transceiver 3320.

[0238] The control node transmitter 3300 further comprises an optional user interface 3312, which permits the user to manually adjust the lighting devices 2704 from the location of the transmitter 3300. Further, the user interface 3312 can also be used to determine whether the RF link is operational.

[0239] The control node transmitter 3300 further comprises a low power LED 3308, not intended for illumination. The LED 3308 has the ability to “flash out” its serial number

in a manner the same as or substantially similar to that of lighting modules 4004 described in further detail herein with respect to FIGS. 40-42.

[0240] FIG. 34 is a block diagram of an exemplary wireless control node receiver 3400 for the wireless line voltage control node 3200. In the illustrated embodiment, the control node receiver 3400 comprises a full wave rectifier 3402, a voltage regulator 3404, a microcontroller 3410, a triggering optoisolator 3414, a detection optoisolator 3416, a triac 3418, and a radio frequency (RF) receiver or transceiver 3420. The full wave rectifier 3402, the voltage regulator 3404, the microcontroller 3410, the triggering optoisolator 3414, the detection optoisolator 3416, and the triac 3418 are the same or substantially similar to the circuitry described herein for the wired line voltage control node 2700, 2800 with respect to FIGS. 27 and 28. The wired control nodes 2700, 2800 comprise circuitry to decode the data from the data encoded 12 VAC two-wire path 336, whereas the wireless control node receiver 3400 receives the control information over the RF link through the RF receiver or transceiver 3420. In addition, power for the microcontroller 3410 and RF receiver 3420 is derived from the 120 VAC line voltage. A simple linear regulation approach is shown. However, it should be noted that this is just one method to step down the rectified line voltage, which is well over 100 V, to a voltage acceptable to the microcontroller 3410. Other methods and devices such as switching converters, for example, can also be used as would be known to one of skill in the art from the disclosure herein.

[0241] Whereas the optoisolators 2814, 2816 in the wired line voltage control node 2800 were used for safety considerations, galvanic isolation is optional for the control node receiver 3400 because there is no low voltage input. Therefore, in an embodiment, the optoisolators 3414, 3416 are optional. Due to line voltage surges and other disturbances, the optoisolators 3414, 3416 may be useful to protect circuitry in the control node receiver 3400. The dimming method of controlling the output voltage via the conduction angle of the triac 2818 as described in the wired control node 2700, 2800 is the same as or substantially similar to the approach used with the control node receiver 3400. In the control node receiver 3400, the microcontroller 3410 delays the trigger signal to the triac 3418 to control the output voltage used to dim the lighting devices 2704.

[0242] The control node receiver 3400 further comprises an optional user interface 3412 which permits the user to manually adjust the lighting devices 2704 from the location of the control node receiver 3400. Further, the user interface 3412 can also be used to determine whether the RF link is operational.

[0243] The control node receiver 3400 further comprises a low power LED 3408, not intended for illumination. The LED 3408 has the ability to “flash out” its serial number in a manner the same as or substantially similar to that of lighting modules 4004 described in further detail herein with respect to FIGS. 40-42.

[0244] The above embodiment has been described with respect to dimming functionality for the lighting devices 2704. In other embodiments of the control node receiver 3400, a Z-wave device and/or a relay device could be used for non-dimming applications, such as zone control. For example, the wireless line voltage control node 3200 could control ON/OFF for a zone comprising a fountain or the like.

[0245] The wireless line voltage control node **3200** is assigned a zone number so that it can be controlled by the signals LIGHTING POWER/CONTROL **1** and LIGHTING POWER/CONTROL **2** via the two-wire path **336**. Because the wireless line voltage control node **3200** comprises the receiver **3400** and the transmitter **3300**, it is possible to have multiple control architectures.

[0246] FIG. **35** is a block diagram of a first exemplary lighting system **3500** for the wireless line voltage control node **3200**. The lighting system **3500** comprises a plurality of transmitters **3504a**, **3504b** . . . **3504n** and a corresponding plurality receivers **3502a**, **3502b** . . . **3502n**. Each receiver **3502** is paired to a corresponding transmitter **3504** to avoid crosstalk with other receivers. Each transmitter in the plurality of transmitters **3504** is assigned a zone number. Since the transmitters **3504** electrically couple to the 12 VAC data encoded two-wire path **336**, each transmitter **3504** can “filter” the commands from the incoming data by determining the zone associated with the command and send only those meant for its corresponding receiver **3502** over the RF link.

[0247] FIG. **36** is a block diagram of a second exemplary lighting system **3600** for the wireless line voltage control node **3200**. The lighting system **3600** comprises a transmitter **3604** and a plurality receivers **3602a**, **3602b** . . . **3602n**. Each receiver **3602a**, **3602b** . . . **3602n** is assigned a zone. The transmitter **3604** transmits all commands it receives, without filtering them, because it does not know which zone has been assigned to the receivers **3602**. If different zone numbers are assigned to the receivers **3602**, then a single transmitter **3604** could service multiple receivers **3602** by sending every command it receives over the RF link. The “filtering” would be done at each receiver **3602**. Advantageously, the star architecture embodiment illustrated in FIG. **36** avoids the user having to purchase transmitters **3604** for each receiver **3602** deployed.

[0248] To utilize the star architecture approach illustrated in FIG. **36**, the user would assign a zone number to each receiver **3602** by entering the zone number into the receiver **3602**. In a first embodiment, the user could enter the zone number using the optional user interface **3412** and LED **3408**. In a second embodiment, the receiver **3602** has a low voltage input provided to program the zone number using the same or substantially similar method as is used to program zone numbers in the single and multi-channel lighting modules **6100**, **6200**, **6300**, **6400** described herein with respect to FIGS. **61-64**. In a third embodiment, the receiver **3602** further comprises dip switches and the user selects the settings on the dip switches to program the receiver's zone number. In a fourth embodiment, the receiver **3602** learns the zone number contained in any command it receives as long as a “learn” button is selected on the receiver **3602**. In a fifth embodiment, the receiver **3602** receives a “teach zone” RF command that would be accepted for a short time after power up. Other approaches can be thought of by those skilled in the art from the disclosure herein.

[0249] In embodiment illustrated in FIG. **35**, the zone number is stored in each transmitter **3504** and the user would assign a zone number to each transmitter **3504** by entering the zone number into the transmitter **3504**. One approach is to provide the transmitter **3504** with the low voltage input to program the zone number using the same or substantially similar method as is used to program zone numbers in the

single and multi-channel lighting modules **6100**, **6200**, **6300**, **6400** described herein with respect to FIGS. **61-64**. In other embodiments, any of the methods described above with respect to programming the zone number in the receiver **3602** can be used for the transmitter **3504**.

[0250] Frequency of operation—In embodiments in the US, the most common frequencies for the RF link communications would be the ISM bands of 902-928 MHz, 2.4 GHz and 5.8 GHz. Transmission in the 260-470 MHz band is allowed but at a much lower power level. In most of Europe, the bands would be 433 MHz, 868 MHz, and 2.4 GHz. In other embodiments, other frequency bands may be allowed depending on the country.

[0251] Standards—In some embodiments, wireless standards such as the various parts of 802.11 could be used. These can include, for example, Zigbee®, Bluetooth®, and the like.

[0252] Pairing—As described herein, the embodiment illustrated in FIG. **35** comprises at least a first transmitter **3504** and a first receiver **3502** that are paired forming a first pair to reduce crosstalk between the other receivers **3502** and transmitters **3504** not in the first pair. In an embodiment, pairing can be achieved by embedding an address in the RF data that must match the address of a particular receiver **3502** in order for the receiver **3502** to accept the data. In another embodiment, each receiver/transmitter pair **3504/3502** may operate at a different frequency in the band. In an embodiment comprising a direct sequence spread spectrum device, each receiver/transmitter pair **3504/3502** may use a different correlation key. In an embodiment comprising a frequency hopping spread spectrum system, each receiver/transmitter pair **3504/3502** may use a different hopping sequence, speed, or set of frequencies. In the embodiment illustrated in FIG. **36**, pairing between the transmitter **3604** and the plurality of receivers **3602a**, **3602b** . . . **3602n** can be used to reduce crosstalk between the lighting system **3600** and another nearby lighting system.

[0253] The lighting controller **300** described with respect to FIGS. **3-17** transforms the line voltage, such as 120 VAC to a low voltage, such as 12 VAC, to control low voltage devices **304**. The lighting controller **1800** described with respect to FIGS. **18-26** uses the line voltage, such as 120 VAC, to control 120 VAC devices **1804**, **2704**. The control nodes **2700**, **2800**, **3200** described with respect to FIGS. **27-36** permit the low voltage controller, such as the lighting controller **300**, to control 120 VAC devices **1804**, **2704**. In other embodiments, a low voltage control node permits a high voltage system, such as the lighting controller **1800** to control low voltage devices **304**, such as 12 VAC lighting devices **304**. Both wired and wireless embodiments of a low voltage control node are described below with respect to FIGS. **37** and **38**.

[0254] FIG. **37** is a block diagram of an exemplary wired low voltage control node **3700**. The low voltage control node **3700** electrically couples to the two-wire path **1836** and receives the data encoded line voltage power signals from the two-wire path **1836**. In an embodiment, the line voltage power signals are approximately 120 VAC 60 Hz. In other embodiments, the data encoded line voltage power signals can be an approximately 110 VAC 60 Hz, 220 VAC 50 or 60 Hz, 230 VAC 50 or 60 Hz, 240 VAC 50 or 60 Hz, or the like.

[0255] The wired line voltage control node **3700** is assigned a lighting zone via the operator interface panel **308**

as described herein. Based on the information contained on the two-wire path **1836** addressed to the low voltage control node **3700**, the line voltage control node **3700** controls 12 VAC devices **304**. In an embodiment, the 12 VAC devices **304** are dimmable lighting devices.

[0256] The architecture of the wired low voltage control node **3700** is similar to the architecture of the wired line voltage control node **2800** described in FIG. 28. The wired line voltage control node **2800** receives commands from the data encoded 12 VAC two-wire path **336** and provides lighting control over the 120 VAC line voltage signals to 120 VAC devices **1804**, **2704**, whereas the wired low voltage control node **3700** receives commands from the data encoded 120 VAC two-wire path **1836** and provides lighting control over the 12 VAC power lines to 12 VAC lighting devices **304**. The triac **2818** in the line voltage control node **2800** receives the 120 VAC line voltage, whereas a corresponding triac in the low voltage control node **2700** receives the 12 VAC low voltage.

[0257] FIG. 38 is a block diagram of an exemplary wireless low voltage control node **3800** comprising a 12 VAC interface wireless receiver **3802** and a 120 VAC interface wireless transmitter **3804** communicating over an RF link. The 120 VAC interface wireless transmitter **3804** electrically connects to the data encoded 120 VAC two-wire path **1836** and receives control data via the two-wire path **1836**. The 120 VAC interface wireless transmitter **3804** sends over the wireless link the control commands to the 12 VAC interface wireless receiver **3802**. The 12 VAC interface wireless receiver **3802** electrically connects to the 12 VAC line voltage and the 12 VAC lighting device **304**, and receives the control data over the wireless link from the 120 VAC interface wireless transmitter **3804**. The 12 VAC interface wireless receiver **3802** controls the dimming level of the lighting device **304** based at least in part on the control data received from the 120 VAC interface wireless transmitter **3804**.

[0258] In an embodiment, the architecture of the 12 VAC interface wireless receiver **3802** is similar to the architecture of the wireless line voltage control node receiver **3400**, and the architecture of the 120 VAC interface wireless transmitter **3804** is similar to the architecture of the wireless line voltage control node transmitter **3300**. The wireless line voltage control node transmitter **3300** receives the control commands from the 12 VAC two-wire path **336** and transmits over the RF link an RF signal based at least in part on the control commands. The wireless line voltage control node receiver **3400** receives the RF signal over the RF link and provides power control over the 120 VAC lines based on the received commands to the 120 VAC lighting devices **2704**. In a complementary manner, the 120 VAC interface wireless transmitter **3804** receives the control commands from the 120 VAC two-wire path **1836** and transmits over the RF link an RF signal based at least in part on the control commands. The 12 VAC interface wireless receiver **3802** receives the RF signal over the RF link and provides power control over the 12 VAC lines to 12 VAC lighting devices **304**.

[0259] In other embodiments, the 120 VAC interface transmitter **3804** and the 12 VAC interface wireless receiver **3802** can be replaced with transceivers, as described herein with respect to FIGS. 32-34. In other embodiments, each 12 VAC interface wireless receiver **3802** can be paired with a corresponding 120 VAC interface wireless transmitter **3804**,

as described herein with respect to FIG. 35. In further embodiments, a plurality of 12 VAC interface wireless receivers **3802** can be paired with one 120 VAC interface wireless transmitter **3804** in a star architecture as described herein with respect to FIG. 36.

[0260] FIG. 39 is a block diagram of an exemplary low voltage decoder **3900** configured to permit the lighting controller **300** to dim a lighting device **3916**. In an embodiment, the lighting device **3916** comprises a wide variety of 12 VAC lighting devices, such as incandescent lights, fluorescent lights, LEDs, and the like. The low voltage controller **3900** comprises a full wave rectifier **3902**, a voltage regulator **3904**, a conditioning circuit **3906**, a microcontroller **3910**, and a triac **3918**.

[0261] The full wave rectifier **3902**, the voltage regulator **3904**, the conditioning circuit **3906**, the microcontroller **3910** are the same as or substantially similar to the circuitry described herein with respect to FIG. 28 for the wired line voltage control node **2800** and as described herein with respect to FIGS. 61-64 for single and multi-channel lighting modules **6100**, **6200**, **6300**, **6400**. In the lighting modules **6100**, **6200**, **6300**, **6400**, the microcontroller controls the LED driver to vary the brightness of the LED. Both the line voltage control node **2800** and the low voltage controller **3900** control the triac conduction angle to control dimming. In the line voltage control node **2800**, the triac **2818** receives the 120 VAC line voltage, whereas in the low voltage controller **3900** the triac **3918** receives the data encoded 12 VAC low voltage signals from the two-wire path **336**. Because the triac **3918** receives the low voltage signal as opposed to a high voltage signal, no optical isolation is used.

[0262] In a manner similar to that described in FIGS. 28-31, the microcontroller **3910** triggers the triac **3918** with a delay time based at least in part on the desired output intensity of the 12 VAC lighting device **3916**. The microcontroller **3910** creates output voltages approximately equal to or less than the input line voltage by delaying the triac trigger.

[0263] In the embodiment illustrated in FIG. 39, the triac **3918** is used as the dimming device. The triac **3918** can be, for example, a BTA208-800 from NXP Semiconductor and the like. In other embodiments, the dimming device can be, for example, MOSFETs, Bipolar Junction Transistors (BJTs), and Insulated Bipolar Transistors (IGBTs), and the like. In yet other embodiments, the triac **3918** could be replaced with a relay to provide simple ON/OFF control for a zone comprising a fountain, or the like.

[0264] The low voltage decoder **3900** further comprises an optional user interface **3912** having buttons, slide switches, or other forms of user controls to permit the user to manually control the lighting device **3916**, in addition to the commands originating from the user interface panel **308**.

[0265] The low voltage decoder **3900** further comprises an LED **3908**. In an embodiment, the LED **3908** is a low power LED not intended for illumination. The LED **3908** has the ability to "flash out" its serial number in a manner the same as or substantially similar to that of lighting modules **4004** described in further detail herein with respect to FIGS. 40-42.

[0266] FIG. 40 illustrates an exemplary lighting system **4000**. The lighting system **4000** comprises a lighting controller **4002** connected to a plurality of lighting modules **4004** through a two-wire interface. The lighting controller **4002** comprises the power supply **302** and the user interface

panel 308, which is the same or similar to that as described herein with respect to FIG. 3. The lighting fixtures 4004 are grouped into zones 4006.

[0267] In the example illustrated in FIG. 40, zone 1 4006a comprises lighting fixture 4004a, zone 2 4006b comprises lighting fixtures 4004b, 4004c, 4004d, zone 3 4006c comprises lighting fixtures 4004e, 4004f, 4004g, and zone 4 4006d comprises lighting fixture 4004h. In other embodiments, the lighting system 4000 can be configured with more or less zones 4006 and/or with more or less lighting fixtures 4004 in each zone 4006. Additional fixtures need not be wired to the end of the line. Instead, the user may elect to “branch” or “T” connect another leg of lights anywhere along the 2-wire path.

[0268] The lighting system 4000 further comprises a remote device 4008 and a wireless receiver 4010 to send addresses/data/commands to the lighting modules 4004. In an embodiment, the remote 4008 can be a digital device, a smart phone, an iPhone, an application for a smartphone, an application for an iPhone, or the like. The wireless receiver 4010 wirelessly connects to the remote 4008 through radio frequency (RF) transmissions and electrically connects through a wire to the lighting controller 4002.

[0269] In an embodiment, the remote 4008 sends addresses/data/commands to the receiver 4010 using a standard wireless protocol, such as, for example, Zigbee® or Bluetooth®. The receiver 4010, in an embodiment, operates in a license or a license-free band of frequencies. Examples of license-free bands in the United States are 270 MHz to 460 MHz; and the Industrial, Scientific, and Medical Band, 902 MHz to 928 MHz, and 2.4 GHz. The receiver 4010 can be a single or a dual-conversion receiver disclosed with reference to wireless technology as is known to one of skill in the art recognized from the disclosure herein. Other communication possibilities, like cell phone, applications for a cell phone or personal digital assistant (PDA) or other personal computing device, optical, wired, satellite or the like, can be used to communicate with the remote 4008.

[0270] The receiver 4010 receives the addresses/data/commands from the remote 4008 and transmits them to the lighting controller 300 via wire or other communication medium. The lighting controller 300 receives the addresses/data/commands from the receiver 4010, processes the commands and sends data and commands on the two-wire path to the addressed lighting modules 4004, where the commands are decoded and performed by the addressed lighting modules 4004.

[0271] For example, an operator can be standing in front of a lighting module 4004 or a zone 4006 can turn the lighting modules 4004 ON or OFF, adjust the brightness, determine what hue from the lights looks best, and the like. As the operator enters commands, the commands are translated to allow the program at the lighting controller 4002 to be responsive. The lighting controller 4002 then sends data embedded in the power signal to the fixtures 4004 or the zones 4006. Thus, the remote 4008 works interactively with the power supply 302, for example, via the receiver 4010, to mix the red, green, and blue coefficients of any particular lighting module 4004 or group of lighting modules 4006.

[0272] In another embodiment, the homeowner talks on the phone to a remote programmer who enters the information in a computing device, such as a browser or application, which through known Internet or other communication protocols, updates the lighting module behavior. Although

disclosed with reference to several embodiments, a skilled artisan would know from the disclosure herein many possible interactive methods of using remote computing devices to program module behavior.

[0273] FIG. 41 depicts an embodiment of the remote device 4008. In one embodiment, the remote 4008 is a key fob type device. In another embodiment, the remote 4008 is a larger hand-held device. The remote 4008 comprises a display 4102 to provide operator feedback and input buttons 4104 to receive operator input.

[0274] FIG. 42 is a block diagram of an exemplary remote device 4008, according to an embodiment. The remote 4008 comprises a photo diode 4202, an RF transmitter 4204, a battery 4206, a voltage regulator 4208, an operator interface 4210, a display 4212, and a computer 4214 with associated memory (not shown). In an embodiment, the operator interface 4210 comprises buttons, knobs, and the like, although touch screen, voice or other user interaction could be implemented. The photo diode 4202 optically couples to the lighting module 4004 and electrically communicates to the processor 4214. The processor 4214 also electrically communicates with the operator interface 4210, the display 4212 and the RF transmitter 4204.

[0275] In an embodiment, the photo diode is a PDB-C134 available from Advanced Photonix Inc., or the like. A phototransistor could also be used, but would have a slower response time. The RF transmitter 2004 can be a CC1050 available from Texas Instruments, or the like.

[0276] The computer 4214 comprises devices similar to those disclosed in the foregoing.

[0277] The battery 4206 provides a power signal to the voltage regulator 4208, which provides the proper power waveform to power the circuitry within the remote 4008, as is known to one of skill in the art from the disclosure herein.

[0278] Often, the lighting fixtures 4004 are assigned their address or their lighting zone 4006 before they are placed in a location. The fixture programming port 318 on the operator interface panel 308 can be used to program an address and/or zone 4006 into the lighting module 4004. Once the fixtures are located, such as in the ground, mounted to a wall, or the like, it can be cumbersome to disconnect or uninstall the fixture 4004 to bring it proximate to the fixture programming port 318 for zone reallocation. In an embodiment, the optical interface between the lighting modules 4004 and the remote 4008 can advantageously be used to change the lighting group 4006 of the fixtures 4004 without disconnecting or uninstalling it.

[0279] In an embodiment, the lighting modules 4004 comprise at least one light emitting diode (LED). The user sends a command to the lighting controller 300 to instruct every lighting module 4004 to flash or strobe its address using its at least one LED by selecting the appropriate button or knob on the remote's operator interface 4210.

[0280] Each lighting module 4004 comprises a unique address in addition to a group or zone number. In one embodiment, the lighting module address comprises a 16-bit address, having approximately 65,000 unique values. Other embodiments of the lighting module address can have more or less bits. Commands from the remote 4008 can target a specific lighting module 4004 using the unique address or a group of lighting modules 4004 using a zone address to turn the module 4004 ON/OFF, dim, brighten, adjust the color, adjust the hue, adjust the intensity, or the like.

[0281] As described above, the remote 4008 transmits the command to the wireless receiver 4010 using the wireless protocol. The wireless receiver 4010 receives the command and electrically converts the signal, which is then electrically sent to the power supply 302. In an embodiment, the receiver 4010 converts the RF signal to a baseband signal. The power supply 302 receives and interprets the command, and electrically sends a command to the lighting modules 4004 over the two-wire path to flash their addresses. For example, the LED could turn ON to represent a 1 address bit and turn OFF to represent a 0 address bit.

[0282] The user selects a lighting module 4004 to assign to a different zone 4006 by pointing the remote 4008 at the selected lighting module 4004 such that the photo diode 4202 receives the optical address from the flashing LED. The photo diode 4202 converts the optical address into an electrical signal and sends the address to the processor 4214.

[0283] In an embodiment where the remote 4008 is a smart phone comprising a camera, an iPhone comprising a camera, an application for a smartphone comprising a camera, an application for an iPhone comprising a camera, or the like, the camera receives the optical address from the flashing LED. The smartphone or iPhone and associated circuitry known to one of skill in the art from the disclosure herein converts the optical address into an electrical signal and sends the address to the processor 4214.

[0284] The processor 4214 sends the address to the RF transmitter 4204, where it is up converted and transmitted via an antenna 4216 on the remote 4008 to the wireless receiver 4010. The wireless receiver 4010 receives the RF transmission, down converts it, and transmits the address to the lighting controller 300. The power supply 302 in the lighting controller 300 receives the address and transmits a command to the selected lighting module 4004 to change its zone 4006. When the selected lighting module 4004 receives and executes the command, the lighting modules 4004 stop flashing their addresses.

[0285] Alternatively, in another embodiment, the module 4004 is numbered and the operator manually enters the number into the remote 4008. In yet another embodiment, where the remote 4008 is a smart phone comprising a camera, an iPhone comprising a camera, an application for a smartphone comprising a camera, an application for an iPhone comprising a camera, or the like, the address of the module 4004 is bar coded and the smartphone or iPhone camera reads the bar code from the module 4004.

[0286] In another embodiment, the lighting modules 4004 comprise a photo diode and the remote 4008 comprises an LED in addition to the RF transmitter 4204, the operator interface 4210, the display 4212, the processor 4214, the voltage regulator 4208 and the battery 4206. The remote 4008 optically sends commands and data by flashing or strobing its LED, which are received by the photo diode in the lighting module 4004, similar to the way a TV receives a signal from a handheld TV remote. The flashing would typically be so rapid, that it would not be perceived by the human eye. The remote 4008 also transmits data and commands to the RF receiver 4010 using the wireless protocol, which in turn sends the messages via wire to the lighting controller 300, as described above.

[0287] Referring to FIG. 3, the lighting controller 300 comprises the user interface panel 308 electrically coupled to the power supply 302 and mechanically attached to the chassis. In other embodiments, the user interface panel 308

can be remoted wirelessly. FIG. 43 is a block diagram of an exemplary lighting controller 4300 comprising a remote operator interface panel 4308, a power supply 302, and a pair of RF transceivers 4302, 4304 providing an RF communication link between the remote panel 4308 and the power supply 302. In an embodiment, the RF communication protocol is implemented so that the remote user interface panel 4308 functions the same as the integral user interface panel 308. The panel RF transceiver 4304 electrically connects to the panel 4308 and the chassis RF transceiver 4302 electrically connects to the power supply 302 in the chassis.

[0288] In an embodiment, the panel transceiver 4304 comprises an integrated battery 4306 to power the user interface panel 4308, permitting the user interface panel 4308 to be carried around for remote operation. In another embodiment, the user interface panel 4308 can be powered by a power supply at the remote location, via an adapter 4306, such as if it is permanently mounted indoors, remote from the power supply 302.

[0289] As illustrated in FIG. 43, the user interface panel 4308 is separated from the chassis/power supply 302, and the two are linked via an RF link between the panel RF transceiver 4304 and the chassis RF transceiver 4302. In an embodiment, the protocol used by the chassis RF transceiver 4302 emulates the user interface panel 4308 and the protocol used by the panel RF transceiver 4304 emulates the power supply 302. Thus, the communication between the remote user interface 4308 and the chassis/power supply 302 is the same or substantially similar to the communication between the integral user interface panel 308 and the chassis/power supply 302. This allows retrofitting into lighting controllers that were previous hardwired.

[0290] In addition to the RF circuitry, the transceivers 4302, 4304 also comprise processors. FIG. 44 is a block diagram of an exemplary panel transceiver 4400 for a remote operator interface panel 4450 and FIG. 45 is a block diagram of an exemplary chassis transceiver 4500 for a lighting system chassis 4550.

[0291] The panel RF transceiver 4400 comprises an RF transceiver 4402, a microcontroller 4404, address selection circuitry 4408, a battery/adaptor 4406, and a voltage regulator 4410. The microcontroller 4404 electrically couples to the address selection circuitry 4408 and the RF transceiver 4402, and the battery/adaptor 4406 electrically couples to the voltage regulator 4410. The operator interface panel 4450 comprises a microcontroller 4414 and a voltage regulator 4420. The microcontroller 4414 communicates with the microcontroller 4404 in the panel transceiver 4400 and the voltage regulator 4420 electrically couples to the battery/adaptor 4406.

[0292] Similarly, the chassis RF transceiver 4500 comprises an RF transceiver 4502, a microcontroller 4504, address selection circuitry 4508, and a voltage regulator 4510. The microcontroller 4504 electrically couples to the address selection circuitry 4508 and the RF transceiver 4502. The chassis 4550 comprises a microcontroller 4514 and pre-regulated power circuitry 4506. The chassis microcontroller 4514 communicates with the microcontroller 4504 in the chassis transceiver 4500 and the pre-regulated power circuitry 4506 electrically couples to the voltage regulator 4510.

[0293] In an embodiment, a common hardware platform for the microcontrollers 4404, 4414, 4504, 4514 is used. In

other embodiments, RF circuitry with no intelligence interfaces with the microcontrollers **4414**, **4514**.

[0294] The address selection circuitry **4408**, **4508** provides an address selection input to the respective transceivers **4404**, **4504**. The address selection circuitry **4408**, **4508** allows the transceivers **4404**, **4504** to communicate with each other while rejecting any RF information meant for others. In a first embodiment, the address selection circuitry **4408**, **4508** is a dial or one or more DIP switches, which are settable by the user. In a second embodiment, the address of the user interface panel transceiver **4404** is set using a display on the panel **4450** for user interface panel firmware that is aware of the RF transceiver **4404**. Once the address is set in the panel RF transceiver **4404**, it could be “taught” to the chassis RF transceiver **4504**. In a further embodiment, the chassis RF transceiver **4504** further comprises a “learn button”, and while the learn button is in a first state, the chassis RF transceiver **4504** learns the address of any valid message received. In yet other embodiments, one of the transceivers **4404**, **4504** enters the learn mode for a time period after power up. Other addressing scheme embodiments may be possible by those skilled in the art from the disclosure herein.

[0295] There are considerations of frequency of operation, wireless standards, and pairing to avoid crosstalk for the RF communication link between the transceivers **4400**, **4500** which are the same as or similar to those discussed herein with respect to FIGS. **32-36** for the RF communications for the wireless line voltage control node **3200**.

[0296] FIG. **46** is a block diagram of an exemplary optical reader **4600** configured to read the flashing address of the selected lighting device **4004** in order to change the zone number of an installed lighting device without wiring modifications to previously installed lighting. The method of changing the zone number of an installed lighting device is the same as or similar to the method described herein with respect to FIGS. **40-42** in the lighting system **4000** using the remote **4008**. The remote **4008**, in one embodiment as previously described, comprises the optical reception circuitry **4202** and RF transmitter circuitry **4204** and wirelessly transmits the optically received address to the receiver **4010**, which electrically couples to the lighting controller **4002**. The remote **4008**, in another embodiment as previously described, comprises a smartphone or other intelligent device with a camera where the camera is used to capture the flashing address, which is transmitted to the lighting controller **4002** via the receiver **4010**.

[0297] Whereas, in an embodiment, the optical reader **4600** electrically connects to an intelligent electronic device **4650**, such as a smartphone, tablet PC, or the like, through its headset jack, and sends the received address to an application running in the intelligent electronic device **4650**. The intelligent electronic device **4650** then sends the address to the lighting controller via a router and WiFi module, such as lighting controller **4702**, router **4708**, and WiFi® module **4710** as described herein with respect to FIG. **47**. The application resides in the intelligent electronic device/smartphone **4650** and is configured to change the zone number of a previously installed light **4004**, **4704** without wiring modifications. In an embodiment, the application is a proprietary application. In another embodiment, the intelligent electronic device/smartphone **4650** may communicate to the controller **4002**, **4702** via ad hoc or other peer to peer communication protocol.

[0298] Referring to FIG. **46**, the optical reader **4600** comprises a photodiode **4602**, a band pass amplifier **4604**, a microcontroller **4606**, an optional low pass filter **4608**, a full wave rectifier and filter **4610**, a voltage regulator **4612**, and an optional battery **4614**. The full wave rectifier and filter **4610**, the voltage regulator **4612**, and the battery **4614** comprise power supply options for the optical reader **4600**. Power can be derived from either rectifying, filtering, and regulating an audio signal from the intelligent electronic device/smartphone headset circuitry, or from the battery **4614**. For instance, prior to reading the data, the application could power the optical reader **4600** by sending a constant tone, such as 1 kHz or the like, for example, of high volume audio to the headset jack. The optical reader **4600** rectifies, filters, boosts if needed, and regulates the audio signal into a power source for the rest of the optical reader circuitry. In situations where the user has disabled the headset volume, the battery **4614** can supply power to the optical reader circuitry. Examples of the battery **4614** are a C **R2032** lithium coin cell and the like.

[0299] Once powered, the optical reader **4600** would look for a valid optical transmission signal, verify the checksum, and then send the information to the application via the microphone input of the headset jack. The flashing light from the lighting module **4004** is incident upon the photodiode **4602**, which outputs a current in response to the light. The band pass amplifier **4604** amplifies the current, converts the amplified current into a voltage signal, and filters the voltage signal to pass only the frequencies of interest.

[0300] In one embodiment, the amplifier **4604** has sufficient gain to “square up” the filtered voltage signal so that it can be input directly into a digital input of the microcontroller **4606**. In another embodiment, filtered voltage signal is input into a comparator input of the microcontroller **4606**, which squares the signal. In a further embodiment, the filtered voltage signal is input into an analog input of the microcontroller **4606**, read by an on-board A/D converter, and processed. The microcontroller **4606** receives the optical data as a digital signal and verifies its validity via a checksum. Once a valid message is received, the optical reader **4600** sends an audio frequency signal based on the valid message to the microphone input of the intelligent electronic device/smartphone **4650**.

[0301] The audio frequency signal can be modulated using AM, FM, FSK, OOK or any other modulation technique. In one embodiment, the optical reader **4600** frequency shift keys (FSK) an audio frequency signal into the microphone of the intelligent electronic device/smartphone **4650**. A logic zero comprises 1000 Hz signal and a logic one comprises 1500 Hz signal, for example. In another embodiment, ON-OFF key modulation (OOK) can be used. A logic one comprises a tone of 1 kHz and a logic zero comprises silence, for example. In another embodiment, the microcontroller **4606** sends each received bit to the intelligent electronic device/smartphone **4650** without validating the message and application validates the message.

[0302] FIG. **47** illustrates an exemplary lighting system **4700** controlled remotely, according to an embodiment. The lighting system **4700** comprises a lighting controller **4702**, and a plurality of lighting modules **4704** configured into a plurality of zones **4706**. In the illustrated embodiment, zone **1** **4706a** comprises one lighting fixture **4704a**; zone **2** **4706b** comprises three lighting fixtures **4704b**, **4704c**, **4704d**, and zone **3** **4706c** comprises three lighting fixtures **4704f**, **4704g**,

4704h. The lighting controller **4702** comprises the power supply **302** and the operator interface **308**. The lighting controller **4702** sends the data encoded power waveform to the plurality of lighting modules **4704** on the two-wire path, as described above.

[0303] The lighting system **4700** further comprises a wireless module **4710**, which electrically couples, via wire or other mediums, to the lighting controller **4702**. The wireless module **4710** communicates wirelessly to devices, such as a smartphone **4714**, a laptop computer **4716**, and other devices that have WiFi™ connection capability using a peer to peer communication mode such as ad hoc. In this communication mode, custom software, firmware, applications, programs, or the like, are written for both the wireless module **4710** and the communicating device **4714**, **4716**. In an embodiment, this proprietary communication approach is not constrained by conventional standards, such as the 802.11 standard and its versions, for example.

[0304] The user can send commands from the smart phone **4714**, the laptop computer **4716**, or other communicating devices within the range of the wireless module **4710** to remotely control the lighting system **4700**. For example, the user can send commands to turn ON/OFF, adjust the brightness, adjust the color, adjust the hue, and the like for the lighting system **4700**, a zone **4706**, or a specific lighting module **4704** from the remote device **4714**, **4716**. In an embodiment, the user views the web page being served by the wireless module **4710** by, for example, opening up the Internet Explorer® or other web browser on the smartphone **4714** or the laptop **4716**. The user then interacts with the web page to control the lighting system **4700**. In another embodiment, the web page is served from the computer in the lighting controller **4702**, and the wireless module **4710** provides the RF connectivity.

[0305] The wireless module **4710** wirelessly receives the commands using the ad hoc or other peer to peer protocol, electrically converts the signal and sends the lighting commands, via wire, to the lighting controller **4702**. In an embodiment, the module **4710** converts the signal to base band. The lighting controller **4702** receives the commands and sends the message to the addressed lighting modules **4704** or the lighting modules **4704** in the specified zones **4706** via the two-wire path.

[0306] In another embodiment, the lighting system further comprises a wireless router **4708** and the wireless module **4710** is a WiFi™ enabled device. WiFi™ enabled wireless devices, such as laptops or computers **4716**, **4720**, smartphones **4714**, WiFi™ enabled automobiles **4722**, or the like, communicate with the router **4708** using a standard communication protocol, such as 802.11. In other embodiments, a device, such as a computer **4718** is electrically connected, via wire or a cable, to the router **4708**. The user uses the devices **4714**, **4716**, **4718**, **4720**, **4722** to send commands to the lighting system **4700**. The devices **4714**, **4716**, **4718**, **4720**, **4722** send the commands through the router **4708** using a standard router protocol. The router **4708** connects to the World Wide Web **4712** using an Internet Service Provider (ISP) and an Internet connection. In another embodiment, the smartphone **4714** communicates through the Internet using a general packet radio service (GPRS) protocol.

[0307] In one embodiment, the wireless module **4710** comprises the router **4708**. In another embodiment, the lighting controller **4702** comprises the router **4708**.

[0308] The devices **4714**, **4716**, **4718**, **4720**, **4722** access the WiFi™ enabled wireless module **4710** through its Internet Protocol (IP) address. The module **4710** sends the commands to the lighting controller **4702**, where the lighting controller sends the command to the lighting modules **4704** through the two-wire path. In this manner, a user can access the lighting system **4700** from anywhere there is an Internet connection.

[0309] In a further embodiment, the module **4710** comprises an Ethernet module for communication using an Ethernet protocol via an Ethernet cable between the controller **4702** and the router **4708** and/or between two controllers **4702**. Examples of Ethernet modules are model MDL-2CE available from Texas Instruments or the like.

[0310] FIG. 48 illustrates another exemplary lighting system **4800** controlled remotely. The lighting system **4800** comprises the plurality of lighting fixtures **4704** configured in the one or more zones **4706** and the lighting controller **4802**. The lighting controller **4802** sends the data encoded power waveform to the plurality of lighting devices **4704** over the two-wire path, as described herein.

[0311] The lighting system **4800** further comprises a mobile carrier network module **4804**, which electrically couples, via wire or other mediums, to the lighting controller **4802**. The module **4804** communicates to the World Wide Web (WWW) **4712** via a mobile carrier's network. Depending on the location and carrier, various standards, such as GPRS, GSM, and CDMA, and the like may apply. A suitable GPRS and GSM module, for example, is model number MTSM C-G-F4 available from Multitech Systems Inc. and the like. A suitable CDMA module, for example, is model MTSM C-C1-IP-N3 available from Multitech Systems Inc.

[0312] The lighting controller **4802** can be accessed by devices, such as laptops or computers **4720**, smartphones **4714**, web-enabled automobiles **4722**, or the like, in communication with the WWW **4712** from any location. Further, the lighting controller **4802** can be accessed by a wireless router **4708** in communication with the WWW **4712** via an Internet service provider (ISP). Local devices, such as laptops or computers **4716**, typically in proximity to the wireless router **4708** and typically communicating with the router **2708** using a standard communication protocol, such as 802.11 can also access the lighting controller **4802**. In other embodiments, a device, such as the computer **4718** is electrically connected, via wire or a cable, to the router **4708**. In one embodiment, the lighting controller **4802** comprises the router **4708**.

[0313] The user uses the devices **4714**, **4716**, **4718**, **4720**, **4722** to send commands to the lighting system **4800**. In a first embodiment, firmware either inside the lighting controller **4802** or in the module **4804** serves up a webpage. As long as the module **4804** can be found on the World Wide Web **4712**, that webpage could be accessed by devices with a web browser, thus allowing control of the lighting controller **4802**. This is similar to the control provided by the WI-FI module **4710** discussed herein with respect to FIG. 47.

[0314] In a second embodiment, an application is provided for application-enabled devices, such as the control devices **4714**, **4716**, **4718**, **4720**, **4722**. The user interacts with the application, and the application communications with the module **4804** via the World Wide Web **4712**. In an embodiment, the application is written for various platforms, such as iPhone, Android, or the like.

[0315] In a third embodiment, a web based application 4806 is hosted on a server on the World Wide Web 4712. In an embodiment, this application 4806 is larger/more complex than could be stored in the lighting controller 4802 or the module 4804. The user interacts with this webpage 4806 using devices comprising a web browser and the application 4806 communicates with the lighting controller 4802.

[0316] There are some practical considerations when using mobile carrier networks. Most mobile carriers actually have far fewer IP addresses than they do subscribers. This is because at any given point in time, only a fraction of the subscribers is interacting with the web. Therefore, after sometime of inactivity, a mobile device will typically lose its IP address. If the mobile device goes online again, the network will issue a new (different) IP address. Furthermore, many times the IP addresses used by mobile carriers are private, not public, meaning they cannot be reached using the World Wide Web 4712. The significance of this is that if a user wants to connect with a device on a carrier's network, they must know the IP address of that device.

[0317] Understanding that people want to use their networks to communicate with, and control devices, most carriers have workarounds for this problem. For instance, they often allow companies to set up special servers that have access to the private IP address of the devices they sell. This sort of "proxy" server would itself have a fixed IP address and would be easily accessible from anyone on the WWW 4712. The server would use an authentication technique or password to allow a user in communication with it, to access only those remote (private IP) nodes associated with the users account. In a sense, the server is a "conduit" to reach the private IP device.

[0318] This approach may be combined with any of the three embodiments described above. For the third embodiment, the server may be the same device that hosts the application 4806.

[0319] FIG. 49 is a block diagram of an exemplary lighting system 4900 with a master/slave configuration, according to an embodiment. The lighting system 4900 comprises a first lighting controller 4902 and at least a second lighting controller 4952. Lighting Controller 4902 operates as a master controller and comprises a power supply 4908, an operator interface 4910, and a fixture programming port 4912. Lighting Controller 4952 operates as a slave to the master controller 4902 and comprises a power supply 4958 and a slave control panel 4960. The slave control panel 4960 comprises the processor 314 and support circuitry, such as the memory 316, the logic power supply 320, and the display and indicators 312. In an embodiment, the slave control panel 4960 may not have the fixture programming port 4912 and the operator interface devices, such as the buttons and knobs 310. In other embodiments, the slave controller 4952 is electrically the same as or similar to the master controller 4902.

[0320] Each lighting controller 4902, 4952 electrically connects to a plurality of lighting modules 4904 and to a WiFi™ enabled module 4914, 4964, respectively. In the illustrated embodiment, master controller 4902 electrically connects to lighting modules 4904a, 4904b, 4904c, and up to 4904n, and electrically connects to module 4914. Slave controller 4952 electrically connects to lighting modules 4904d, 4904e, 4904f, and up to 4904m, and electrically connects to module 4964.

[0321] In one embodiment, the WiFi™ enabled modules 4914, 4964 communicate with each other through an ad hoc or other peer to peer protocol, as described above with respect to FIG. 47. In another embodiment, the WiFi™ enabled modules 4914, 4964 can communicate with each other through a router 4708, also as described above with respect to FIG. 47.

[0322] For example, a user may have a lighting system 4900 that uses more than one lighting controller 4902 to control the lighting modules 4904. This may be caused by the transformer 322, 402 not being able to supply enough power to illuminate the plurality of lighting modules 4904. In this case, the user would connect some of the lighting modules to a first controller 4902 and others to a second controller 4952. In one embodiment, the first and second controllers 4902, 4952 each control the lighting modules 4904 associated with it, independent of the other controller 4902, 4952.

[0323] However, in another embodiment, the program to control all of the lighting modules 4904 executes in one lighting controller 4902, which acts as the master controller and communicates with the slave controller 4952. The master controller 4902 sends commands for the slave controller 4952 to the module 4914. Module 4914 communicates wirelessly with the module 4964 and module 4964 receives the commands from the module 4914 and sends the commands to the slave controller 4952. The slave controller 4952 receives the commands and sends the commands to the addressed lighting modules 4904 associated with it. Advantageously, the user can access all of the lighting modules 4904 by entering commands from the operator interface 4910 on the master controller 4902 or by communicating to the IP address of only the master controller 4902 instead of having to access two lighting controllers 4902, 4952. Another advantage is the reduced cost of the slave controller 4952, which does not include the buttons and knobs 310, the fixture programming port 4912, and other features not being used in the slave controller 4952.

[0324] In another embodiment, the lighting system 4000, 4700, 4900 further comprises a motion detector. The motion detector may be battery powered and communicate with the receiver/modules 4010, 4710, 4914. When the motion detector senses motion, it could send a message to the lighting controller 4002, 4702, 4902, which then turns ON the appropriate lighting modules 4004, 4704, 4904, as programmed by the user. In one embodiment, the motion detector receives power over the two-wire path connecting the plurality of lighting modules 4004, 4704, 4904.

[0325] In another embodiment, the data sent to the lighting controllers 300, 4002, 4702, 4902 is encrypted. In one embodiment, a proprietary encryption scheme is used. In another embodiment, a standard encryption protocol, such as TCP/IP, IPX/SPX, OSI, DLC, SNAP, exclusive or, and the like, is used to encode the data and commands.

[0326] FIG. 50 illustrates an exemplary lighting system 5000 with a master/slave configuration. The lighting system 5000 comprises a master lighting controller 5002, a slave chassis 5004, a two-wire to chassis protocol converter 5006, and a plurality of lighting modules. In an embodiment, the master lighting controller 5002 comprises an operator interface panel 308 while the slave chassis 5004 does not. In this embodiment, the master lighting controller 5002 controls lighting fixtures 1-6 in lighting zones 1-3 through data encoded power signals sent over a two-wire path from the

master lighting controller **5002**, and the slave chassis **5004** controls lighting fixtures **7-12** in zones **4-6** through data encoded power signals sent over a two-wire path from the slave chassis **5004**. The two-wire to chassis protocol converter **5006** receives the commands for all of the lighting fixtures **1-12** in the lighting system **5000** from the master lighting controller **5002** over the master lighting controller's two-wire path and relays the commands to the slave chassis **5004**. The protocol converter **5006** permits the user to add one or more slave chassis **5004** to the lighting system **5000** thereby increasing the number of lighting devices in the lighting system **5000** without firmware modification, network communication cables between master and slave controllers, and the like. In an embodiment, all of the lighting modules are programmed from the master lighting controller **5002** comprising the operator interface panel **308**.

[0327] In the embodiment illustrated in FIG. **50**, the lighting modules attached to the slave chassis **5004** comprise different zones than those attached to the master lighting controller **5002**. In other embodiments, the slave chassis **5004** can control lighting device belonging to the same zone(s) as those controlled by the master lighting controller **5002**. For example, the user may have more lights in a single zone than can be powered by the master lighting controller transformer. The protocol converter **5006** allows additional lights, belonging to the same zone, to be electrically connected to the two-wire path from the slave chassis **5004**. A II of the lights in the zone could turn on together. In another embodiment, more than one protocol converter **5006** and corresponding slave chassis **5004** can be controlled by one master lighting controller **5002**. In a further embodiment, a cascade or serial arrangement can be used such that a second protocol converter **5006** attaches to the slave chassis' output to control another slave chassis **5004**. In a further embodiment, multiple protocol converters **5006** and slave chassis **5004** can be chained to the output of the previous slave chassis **5004**, while maintaining one point of control at the master lighting controller **5002**.

[0328] FIG. **51** is a block diagram of an exemplary two-wire to chassis protocol converter **5100** comprising a full wave rectifier **5102**, a voltage regulator **5104**, a conditioning circuit **5106**, a microcontroller **5108**, an optoisolator **5110**, and a transceiver **5112**. The two-wire to chassis protocol converter **5100** receives the data encoded 12 VAC power signal from the master lighting controller **5002**. The full wave rectifier **5102**, the voltage regulator **5104**, the conditioning circuit **5106**, and the microcontroller **5110** are the same as or substantially similar to the circuitry described herein with respect to in FIGS. **61-64** for single and multi-channel lighting modules **6100**, **6200**, **6300**, **6400**. In the lighting modules **6100**, **6200**, **6300**, **6400**, the microcontroller controls the LED driver, whereas the microcontroller **5108** of the protocol converter **5100** takes each command and "translates" it into a command that the slave chassis **5004** understands, mimicking the operator interface panel **308** that would have been in that chassis. The commands are transmitted by the transceiver **5112** to the slave chassis **5004**. In an embodiment, the communication physical layer comprises an RS485 transceiver **5112**. In other embodiments other physical layers such as RS232, parallel data, controller area network (CAN) Bus, and the like can be used.

[0329] In an embodiment, optical isolation between the microcontroller **5108** and the RS485 transceiver **5112** is provided by the optoisolator **5110** to correct for differential

voltage between the master lighting controller **5002** and the slave chassis **5004**. If that voltage were to exceed the common mode voltage range of the transceiver **5112**, communication could fail. In an embodiment, the RS485 transceiver **5112** is powered from the slave chassis **5004** and the rest of the converter circuitry **5102**, **5104**, **5106**, **5108**, is powered from the data encoded two-wire path from the master controller **5002**.

[0330] In another embodiment, the two-wire protocol converter is wireless. FIG. **52** illustrates an exemplary lighting system **5200** with a wireless master/slave configuration. The lighting system **5200** comprises a master lighting controller **5202**, a slave chassis **5204**, a wireless two-wire to chassis protocol converter comprising a transmitter portion **5206** and a receiver portion **5208**, and a plurality of lighting modules. In this embodiment, the master lighting controller **5202** controls lighting fixtures **1-6** in lighting zones **1-3** through data/power signals sent over a two-wire path from the master lighting controller **5202**, and the slave chassis **5204** controls lighting fixtures **7-12** in zones **4-6** through data/power signals sent over a two-wire path from the slave chassis **5204**. The transmitter portion **5206** of the wireless two-wire to chassis protocol converter receives the commands for all of the lighting fixtures **1-12** in the lighting system **5200** from the master lighting controller **5202** over the master lighting controller's two-wire path. The transmitter portion **5206** broadcasts the commands over an RF communication link. The receiver portion **5208** of the wireless two-wire to chassis protocol converter receives the commands over the RF communication link, decodes the commands, and instructs the slave chassis to output the commands on its two-wire path. Advantageously, the wireless protocol converter **5206**, **5208** does not need proximity to the slave chassis **5204**, as does the wired protocol converter **5006**, **5100**. In addition, multiple receiver portions **5208** that are within the RF transmission range of the transmitter portion **5206** can receive commands and send the commands to their corresponding slave chassis **5204**.

[0331] FIG. **53** is a block diagram of an exemplary transmitter portion **5300** for the wireless two-wire chassis protocol converter. The transmitter portion **5300** comprises a full wave rectifier **5302**, a voltage regulator **5304**, a conditioning circuit **5306**, a microcontroller **5308**, and a transmitter **5310**. The circuitry is similar to wireless line voltage control node transmitter **3300** described herein with respect to FIG. **33**, except that the transmitter portion **5300** transmits every command it receives from the master lighting controller **5202**.

[0332] FIG. **54** is a block diagram of an exemplary receiver portion **5400** for the wireless two-wire chassis protocol converter. The receiver portion **5400** comprises a receiver/transceiver **5402**, a voltage regulator **5404**, a microcontroller **5408**, and a transceiver **5412**. The voltage regulator **5404** receives power from the power source in slave chassis **5204** that would have powered the operator interface panel **308**. The receiver/transceiver **5402** receives the commands via the RF link and sends the information to the microcontroller **5408**. The microcontroller **5408** instructs the slave chassis **5204** via the transceiver **5412** to output the commands over its two-wire path to the lighting modules. In an embodiment, the communication physical layer comprises an RS485 transceiver **5412**. In other embodiments other physical layers such as RS232, parallel data, controller area network (CAN) Bus, and the like can be used.

[0333] The receiver portion **5400** further comprises an optional user interface **5406** in case configured to provide manual control at the slave chassis **5204**, and an LED **5410** for feedback such as health and power, as described herein.

[0334] Similar to the wireless line voltage control node **3200** described herein with respect to FIGS. **34-36**, the wireless two-wire chassis protocol converter **5206**, **5208** supports a system where a single transmitter portion **5300** can be linked with one or more receiver portions **5400** as are in range. Each receiver portion **5400** is linked to a separate slave chassis **5204**, thus making it possible to expand the number of lights under control of the master lighting controller **5402** to very large numbers. Such an embodiment is illustrated in FIG. **55**.

[0335] FIG. **55** is a block diagram of an exemplary lighting system **5500** for a wireless two-wire chassis protocol converter comprising a transmitter portion **5504** and one or more receiver portions **5502a**, **5502b** . . . **5502n**. The transmitter portion **5504** electrically couples to the data encoded two-wire signal from the master lighting controller **5202**. Each receiver **5502a**, **5502b**, . . . **5502n** electrically couples to a corresponding slave chassis **5506a**, **5506b** . . . **5506n** and each slave chassis **5506a**, **5506b**, . . . **5506n** electrically couples to one or more lighting modules. The transmitter portion **5504** receives commands from the master lighting controller **5202** and broadcasts the commands over an RF link to the plurality of receiver portions **5502**. Each receiver portion **5502a**, **5502b** . . . **5502n** instructs its corresponding slave chassis **5506a**, **5506b** . . . **5506n** to output its commands on its two-wire path.

[0336] In another embodiment, there can be a one-to-one pairing between transmitter portions **5504** and receiver portions **5502** of the wireless two-wire protocol converter.

[0337] As discussed above with respect to the RF communications for the wireless line voltage control node in FIGS. **32-36**, there are considerations of frequency of operation, wireless standards, and pairing to avoid crosstalk for the RF communication link.

[0338] FIG. **56** is a flowchart of an exemplary process **5600** for encoding data onto a power signal for lighting modules **304**, **4004**, **4704**, **4904**. Beginning at block **5610**, the process **5600** rectifies an AC power signal to form a secondary VAC power waveform.

[0339] At block **5620**, the process **5600** encodes the data onto the rectified power signal by controlling the polarity of the rectified power signal, such that at least a portion of the rectified power waveform with a first polarity represents a 1-data bit **1304**, **2604** and at least portion of the rectified power waveform with a second polarity represents a 0-data bit **1304**, **2604**.

[0340] At block **5630**, the process **5600** sends the data encoded power waveform through the two-wire path to the lighting modules **304**, **4004**, **4704**, **4904**. The addressed lighting modules **304**, **4004**, **4704**, **4904** decode the commands and perform the lighting functions, such as turn ON/OFF, dim/brighten, change color/hue, and the like.

[0341] Looking at the process **5600** in more detail, at block **5611** the lighting controller **300**, **4002**, **4702**, **4902** receives the primary AC power signal. At block **5612**, the process **5600** transforms the primary AC power signal into a secondary VAC power signal. In an embodiment, the secondary VAC power signal is between approximately 11 VAC and 14 VAC. The process **5600** determines the phase of the secondary AC power signal at block **5613**. At blocks

5614 and **5615**, the process **5600** sends the secondary AC power waveform onto V-FULLWAVE when the phase is positive and sends the inverted secondary AC power waveform onto V-FULLWAVE when the phase is negative to generate the rectified secondary VAC power waveform.

[0342] At block **5621**, the process transmits the data stream as well as the phase information, to an encoder/modulator. The data stream comprises addresses, data, and commands. The bridge circuit **900** passes the rectified secondary power waveform onto the two-wire path to the lighting modules **304**, **4004**, **4704**, **4904** when the data bit **1304**, **2604** from the data stream has a first state. Further, the bridge circuit inverts the rectified secondary waveform when the data bit **1304**, **2604** has a second state. When no data is present, the bridge circuit reconstructs the sine wave of the secondary VAC power waveform from the rectified secondary waveform and sends the reconstructed secondary VAC power waveform.

[0343] At block **5631**, the process **5600** transmits the data enhanced power signal from the lighting controller **300**, **4002**, **4702**, **4902** to the plurality of lighting fixtures **304**, **4004**, **4704**, **4904** on the two-wire path. The addressed lighting modules **304**, **4004**, **4704**, **4904** receive the data encoded power waveform. An embodiment of a lighting module **304**, **4004**, **4704**, **4904**, its functionality, and its operation, is disclosed in FIGS. **13-55** and accompanying disclosure of U.S. application Ser. No. 12/564,840, filed Sep. 22, 2009, entitled "Low Voltage Outdoor Lighting Power Source and Control System", and are incorporated herein by reference. Other embodiments are described below in FIGS. **61-64**.

[0344] This waveform is first scaled and filtered, and is then passed through a comparator to determine the phase of the incoming signal which is used to decode the data bits **1304**, **2604** and perform the requested command. The data encoded power waveform is also rectified and used to power the lighting module. It should be noted that it is possible to store energy in the lighting module such that no power is being supplied at those instances in time when the actual bits of data are received.

[0345] FIG. **57** is a flowchart of an exemplary process **5700** for assigning zones **106**, **206**, **4006**, **4706**, **4906** to addressable lighting modules **104**, **204**, **4004**, **4704**, **4904** in the networked lighting system **100**, **200**, **4000**, **4700**, **4900**, according to an embodiment. In one embodiment, the user assigns the zone numbers into each lighting fixture **300**, **4002**, **4702**, **4902** through the fixture programming port **212**, **318**, **4912**. In one embodiment, the zone numbers comprise 8 bits and there can be up to 256 zones **106**, **206**, **4006**, **4706**, **4906**. In other embodiments, the zone numbers comprise more or less than 8 bits and there can be more or less than 256 zones **106**, **206**, **4006**, **4706**, **4906**.

[0346] At block **5702** and **5704**, the lighting controller periodically queries the programming port attempting to detect a lighting fixture that has been connected. At block **5706**, the lighting controller has detected a light fixture on the programming port and has presented the Lighting Fixture Programming screen to the user via the operator interface panel **210**, **308**, **4910** on the lighting controller **300**, **4002**, **4702**, **4902**. Next, at block **5708**, the user enters the zone number of the lighting fixture **104**, **204**, **4004**, **4704**, **4904** to be added to the entered zone **106**, **206**, **4006**, **4706**.

[0347] At block 5710, the process 5700 sends a command to assign the lighting fixture 104, 204, 4004, 4704, 4904 to the entered zone 106, 206, 4006, 4706, 4906.

[0348] At block 5712, the user is notified that the programming has completed and he removes the fixture from the programming port.

[0349] FIG. 58 is a flowchart of an exemplary process 5800 for modifying assigned zones 4006 in the lighting system 4000 using the remote controller 4008/optical reader 4600, according to an embodiment. At block 5802 and referring to FIGS. 40 and 46, the user selects the change zone selection on the remote device 4008, 4600 and enters the new zone number.

[0350] At block 5804, the remote device 4008, 4600 transmits the zone change request to the receiver 4010 via RF. The receiver 4010 sends the zone change request, via wire or other medium, to the lighting controller 4002 at block 5806. At block 5808, the lighting controller 4002 sends a command to the lighting modules 4004 via the two-wire path to begin flashing their addresses. The command is encoded onto the power waveform supplying power to the lighting modules 4004. After receiving the command, each lighting module 4004 flashes its address using an LED on the lighting fixture 4004.

[0351] At block 5810, the user directs the remote device 4008, 4600 to the selected lighting fixture 4004. The selected lighting fixture 4004 is the lighting fixture that the user wants to rezone. At block 5812, the remote device 4008, 4600 receives the address of the selected lighting fixture, via the optical path. The remote device 4008, 4600 sends the address of the selected lighting module 4004 to the receiver 4010 via RF at block 5814.

[0352] At block 5816, the receiver 4010 sends the selected address to the lighting controller 4002 via a wired path. The lighting controller 4002 receives the selected address and sends a command to the selected lighting fixture 4004 via the two-wire path. The command is encoded onto the power waveform sent via the two-wire path.

[0353] At block 5820, the lighting fixture 4004 decodes the command and changes zone 4006 to the new zone address.

[0354] In an embodiment, the lighting fixtures 104, 204, 4004, 4704, 4904 are advantageously constructed with a drive circuit, supervising functions, communication reception, and the like, within the fixture 104, 204, 4004, 4704, 4904 on a single printed circuit board to lessen the need for water tight splices, sealing, and other reliability concerns.

[0355] In another embodiment, the command protocol supports queued commands as well as immediate commands. The queued commands allow synchronized changes across multiple lighting groups or zones 106, 206, 4006, 4706, 4906. Several different queued commands could be sent to different lighting zones 106, 206, 4006, 4706, 4906. The lighting module 104, 204, 4004, 4704, 4904 remember the command but do not act on it until an “apply queued” command is received.

[0356] In a further embodiment, an accessory device having an optical sensor monitors the lighting fixtures when the fixtures are flashing or strobing their addresses. The accessory device reads the address and displays the address to the user. This is useful because while the fixtures would be marked with their address, the marking could be worn off or not visible after installation.

[0357] In a yet further embodiment, the lighting controller takes inventory of the lighting modules attached by sending, either one by one for each of the possible 65,000 unique addresses, or for a particular range of addresses, a command to turn ON lighting modules. Then the lighting controller monitors the current after the command is sent to determine whether a fixture responded to the command. Finally, the controller compiles a list of the fixture addresses detected to be presented to the user.

[0358] In a yet further embodiment, the lighting controller takes inventory of the lighting modules attached by sending, either one by one for each of the possible 65,000 unique addresses, or for a particular range of addresses, a command to turn ON lighting modules. Then the lighting controller monitors the current after the command is sent to determine whether a fixture responded to the command. Finally, the controller compiles a list of the fixture addresses detected to be presented to the user.

[0359] For example, when the address in the data encoded power waveform matches the address of a specific lighting module and the command included in the data encoded power waveform is a command to turn ON one or more lights of the addressed lighting module, the addressed lighting module responding to the command draws additional current by turning on the one or more lights.

[0360] The lighting controller can include current sensing circuitry to monitor the current. The current sensing circuitry can monitor the current flowing out of the data encoded power signal. In another aspect, the current sensing circuitry can monitor the current at the output of the 120VAC to 12VAC transformer. In another aspect, such as in embodiments of the lighting system where individual lights of the lighting module are powered with 120 VAC, the current sensing circuit can monitor the current at the output of the primary power source. The current sensing circuitry can be in communication with the processor of the lighting controller and send an indication of the monitored current to the processor.

[0361] The lighting controller can determine that the lighting module responded to the command by comparing, after the command is sent over the two-wire communication network, the monitored current with a threshold. In an aspect, the threshold is a reading of the monitored current just prior to sending the command. In another aspect, the threshold comprises an average of two or more readings of the monitored current, where the readings are taken at times prior to sending the command.

[0362] The lighting controller can store information about the lighting modules in memory. The memory can be one or more of ROM, RAM, EEPROM, FLASH or various other types. The lighting controller can monitor the current after the command is sent and modify the contents of the memory based on the current reading of the monitored current. The lighting controller can 1) set a bit in the memory when the monitored current is greater than the threshold value; 2) clear a bit in the memory when the monitored current is less than the threshold value; 3) set a bit in the memory when the monitored current is less than the threshold value; 4) clear a bit in the memory when the monitored current is greater than the threshold value; store the actual value of the monitored current; or 6) store a scaled value of the monitored current.

[0363] The lighting controller can periodically send a command targeting a specific address. The address can be selected randomly, pseudo-randomly, or sequentially. In an

aspect, the lighting controller can modify which addresses. The addresses of known, connected lighting modules may be checked more frequently to make sure they are still working and in communication with the controller.

[0364] In an aspect, the lighting controller can read the lighting controller information stored in the memory and present to the user the addresses of the lighting modules in communication with the lighting controller. In an aspect, the lighting controller can present a mnemonic associated with responding lighting controllers (e.g., Front walk #3). The list of lighting controller can be a list of the lighting controllers for which the monitored current reading after an ON command was sent is greater than the threshold. This indicates that the lighting module is in communication with the lighting controller. The list of lighting modules can be a list of the lighting controllers for which the monitored current reading after an ON command was sent is not greater than the threshold. This indicates that the lighting module is not in communication with the lighting controller.

[0365] In another embodiment, the lighting controller can command all of the lighting modules to turn ON, then sequentially, randomly, or pseudo-randomly select an individual lighting module address and send a command to turn OFF the addressed lighting controller. The lighting controller can determine whether the monitored current drops below a threshold. This indicates that the addressed lighting module is in communication with the lighting controller.

[0366] In another embodiment, the power supply has a detachable front panel with a slot designed to accept the accessory device. When the accessory device is installed, the user detaches the front panel, now powered and in communication with the accessory device, and walks around the yard. The user can perform more complex remote operations using the larger display and operator interface of the front panel. These operations relay back to the power supply via the RF transmitter of the accessory device. In this embodiment, the power supply comprises a second microcontroller to receive the RF commands and act on them.

[0367] In another embodiment, the lighting controller comprises two microcontrollers, where a first microcontroller is located in the power supply chassis and a second microcontroller is located in the operator panel. The two microcontrollers communicate via a wired link while the operator panel is installed in the power supply. When the operator panel is removed from the power supply chassis, the two microcontrollers communicate via a wireless link. In one embodiment, the operator panel is battery powered and portable. In another embodiment, a small plug-in power supply powers the operator panel. In this case, the panel could be mounted in a location that is more convenient for the user to access, such as a house's interior wall, for example, rather than the typical and less convenient exterior wall.

[0368] For years, landscape lighting systems have consisted of large, bulky and heavy transformers wired to 12 VAC incandescent bulbs. Typically, the transformer also has a timer either built into its enclosure, or next to it. The timer is used to switch power to the transformer ON and OFF to control all of the lights simultaneously. Recently, LEDs have begun to be used in landscape lights, but simply as long-life replacements for the incandescent bulbs that have historically been used.

[0369] In contrast, in an embodiment of the present disclosure, a lighting fixture receives the polarity controlled,

sinusoidal power signal from the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952**, decodes and performs the encoded commands, and uses the signal for power. In another embodiment, historical landscape lights could be fitted with special circuitry to receive this communications signal and use the information to control some aspect of the light.

LED Lighting

[0370] In a further embodiment, the light fixture comprises and controls LEDs of white, red, green, and blue color, or any subset. To control individual LED brightness levels, the controller receives a target brightness level. The brightness level is applied to the particular LED after several correction factors. First, the lighting controller applies the temperature correction factor. As the temperature of the printed wiring board of the lighting module increases, the light output of the LED changes. The relative color change depends on the color of the LED. If color mixing is done, an individual temperature correction factor is applied to each color LED or the overall hue will change as temperature changes. Second, the lighting controller applies an aging correction factor. The lighting module determines how many total hours of use of each LED and under the type of driving conditions. As LEDs age, their light output decreases. If color mixing is done, an individual age correction factor is applied to each LED or the overall hue will change as the LEDs age. The third correction factor is a temperature throttling factor that cuts back power to all LEDs when the printed circuit board temperature exceeds a predetermined threshold.

[0371] In yet a further embodiment, the lighting fixture uses a pulse width modulation (PWM) signal to dim the LEDs, where the PWM signal is synchronized to the incoming AC power signal. The synchronization is important to prevent the detrimental effect high PWM frequencies have on dimming linearity while maintaining a frequency high enough to avoid the visible flickering of the LEDs due to the PWM.

[0372] In addition to color correction for temperature and aging, other embodiments of the lighting module provide color correction based at least in part on input voltage. Different color LEDs have different forward voltage drops, and when driven with buck converters whose input is a full wave rectified waveform, the conduction angle is based at least in part on the forward drop of the LED and the input voltage. Therefore, a given ratio of intensities as determined by the PWM driver among red, green, and blue LEDs is only valid for a specific input voltage. As the input voltage changes, the conduction angle of all the colors change, but in varying amounts, thereby distorting the color ratio and changing the hue. This may occur in lighting module embodiments that drive the buck converter with a full wave rectified waveform, thereby allowing the LEDs to turn on and off at twice the line frequency. Advantageously, this approach does not use filter capacitors to maintain the voltage in between peaks of the incoming rectified signal. Filter capacitors are typically electrolytic capacitors due to high value required, and can exhibit life expectancies shorter than the other components in the system. Some are rated as low as approximately 1000 hours at elevated temperatures, which is usually the case for an LED fixture. At typical LED fixture can function approximately 50,000 hours or more. As a result, lighting fixtures without filter capacitors may be

preferable in certain embodiments, which then may subject to distorted color ratios and changing hues based at least in part on the input voltage.

[0373] The following example discusses a red, green, blue (RGB) LED fixture comprising three LEDs for each color, red, green, and blue. Other embodiments may comprise different numbers and colors of LEDs. Typical forward voltage drops for a red LED are approximately 2.2 V, for a green LED are approximately 3.5 V, and for a blue LED are approximately 3.2 V. Therefore, the string of three red LEDs will turn on at an input voltage of approximately 6.6 V, the string of three green LEDs will turn on at an input voltage of approximately 10.5 V, and the string of three blue LEDs will turn on at an input voltage of at approximately 9.6 V.

[0374] FIG. 59 depicts a first exemplary power waveform 5900 illustrating ON/OFF voltages for these colored LEDs receiving approximately a rectified 14 VAC, 60 Hz waveform. FIG. 60 depicts a second exemplary power waveform 6000 illustrating ON/OFF voltages for these colored LEDs receiving approximately a rectified 11 VAC, 60 Hz waveform. Table 1 shows the relative ratios of red, green, and blue ON time, normalized for the red LED for the waveforms 5900 (14 VAC), 6000 (11 VAC).

TABLE 1

V _{in}	RED (normalized)	GREEN	BLUE
11 VAC	1	0.73	0.80
14 VAC	1	0.83	0.88

[0375] As indicated in TABLE 1, the ratio of the colors has changed due to the change in the input voltage. In some embodiments, the result of uncorrected color based on input voltage could be one color for lights close to the transformer where the voltage is still relatively high, and a different color further down the run of wire where the voltage has dropped slightly due to the load along the line.

[0376] A first embodiment to correct color based at least in part on the input voltage prevents the LEDs from turning on until the instantaneous input voltage of the sine wave input exceeds the largest forward voltage drop of all the LED colors in the lighting fixture. In the above example, that would be the approximately 10.5 V drop of the three green LEDs. Most likely, in this embodiment, there is not a point on the input sine wave where one of the colors is on, but the others are not.

[0377] A second embodiment uses an A/D channel of the lighting fixture microcontroller to measure the input voltage, and then adjust the color ratio of all the LEDs to compensate for the color distortion that occurs as the input voltage varies. Input voltage could be scale factor, along with temperature and aging described herein.

[0378] FIG. 61 is a block diagram of an exemplary single channel lighting module 6100 that can be used with the lighting controller 202, 300, 4002, 4702, 4902, 4952 capable of encoding data on the low voltage power line. The lighting module 6100 comprises a bridge rectifier 6102, a conditioning circuit 6104, a voltage regulator 6106, a microcontroller 6108, a temperature sensor 6110, an LED driver 6112, and one or more lamps 6120. In the illustrated embodiment, the lamps 6120 comprise LEDs 6120. In other embodiments, the lamps 6120 can be other light emitting devices, such as, for example, incandescent bulbs, fluorescent bulbs, or the like.

[0379] The bridge rectifier 6102 receives the encoded power wave forms, LIGHTING CONTROL1 and LIGHTING CONTROL2 from the bridge 900 or the bridge/rectifier 1400. The bridge rectifier 6102 comprises a plurality of diodes, such as, for example, Schottky Rectifiers, part number SBR2A40P1 available from Diodes Inc., or the like. The bridge rectifier 6102 converts an input signal of any polarity into a DC signal to power the other circuits on the lighting board. This DC signal is fed into the LED Driver 6112, which can be a driver integrated circuit, part number AL8805 available from Diodes Inc., or an equivalent. The driver integrated circuit uses an efficient Buck Switching topology to generate a regulated output current, which is used to power the LED (s) 6120. In an embodiment, the LED 6120 can be a high-power LED, such as, for example, a CREE XP-E or an equivalent.

[0380] The DC voltage output from the bridge rectifier 6102 is also used to create a regulated logic supply voltage from the voltage regulator 6106. In an embodiment, the voltage regulator 6106 can be a 3-Volt regulator, such as, for example, part number TPS71530 available from Texas Instruments, or the like. The voltage regulator 6106 supplies power to the microcontroller 6108, such as, for example, part number PIC16F1824 available from Microchip Technology, or the like. The microcontroller 6108, and firmware that resides inside it, comprise a receiver for the data being sent from the lighting controller 202, 300, 4002, 4702, 4902, 4952. A conditioning network comprising a plurality of resistor and capacitors couples data from the power supply 302 to the microcontroller's comparator input while simultaneously limiting current into the microcontroller 6108. The output of the comparator (within the microcontroller 6108) is used to determine the nature of the data. The microcontroller 6108 then generates a signal 6130, which is coupled to the LED Driver 6112. This signal 6130 is used to vary the intensity of the light 6120 based on data received from the power supply 302.

[0381] In an embodiment, part of the data received is an address that is used to determine if the information being sent is intended for this light 6120, as each light will have a unique address. In other embodiments, it is also possible for certain commands to be intended for lighting "groups". A group may be defined as a certain type of light, for instance, a path light, or a group may be all lights in a certain location. In yet other embodiments, commands may be intended for all lights 6120. Therefore, using this addressing technique, commands may affect an individual light, a group of lights, or all lights. In another embodiment, the power supply 302 communicates an intensity pattern to the light 6120. This could be a pre-orchestrated pattern of varying intensities, for example. In an embodiment, the pattern may be "canned" or preset inside the lighting fixture, or for the details of it to be communicated from the lighting controller 202, 300, 4002, 4702, 4902, 4952. This feature may be useful, for example, for lighting "effects" which may be synchronized to music.

[0382] The output of the comparator (within the microcontroller 6108) also contains the phase information for the incoming power signal, LIGHTING CONTROL1, LIGHTING CONTROL2. In an embodiment, this is important because the brightness of the LED 6120 is determined by a pulse width modulation (PWM) waveform from the microcontroller 6108. Unless this PWM waveform is synchronized with the incoming power, visible "flickering" may be

seen as these two signals (power and PWM) are “mixed”. Therefore, it is important for the microcontroller **6108** to know the phase of the incoming power, and periodically reset a PWM counter in order to synchronize the PWM signal to the power signal.

[0383] In another embodiment, the microcontroller **6108** protects the light **6100** from overheating. In general, high-power LEDs **6120** generate heat. In an embodiment, the lighting fixture **6100** comprises the temperature sensor **6110** on the printed circuit board of the lighting fixture **6100**. The temperature sensor **6110** can be, for example, part number MCP9700 available from Microchip Technology, or the like. The temperature sensor’s output is an analog voltage, which is read by an A/D converter in the microcontroller **6108**. The microcontroller **6108** uses this information to “throttle back” the power to the LED **6120** when the temperature rises above threshold temperature. In an embodiment, the threshold temperature is chosen to keep the internal junction temperature of the LED **6120** within its rated specification. The throttling is achieved the same way the intensity variation is achieved, as described above.

[0384] Although this embodiment illustrates a single LED, other embodiments of the lighting fixture **6100** drive a plurality of LEDs **6120**.

[0385] FIG. 62 is an exemplary schematic diagram of a single channel lighting module **6200**, according to one embodiment.

[0386] Other embodiments of lighting devices, such as lighting device **1804**, can be used with the lighting controller **1800** which encodes data on the primary AC power line. The architecture of the lighting module **1804** can be similar to the low voltage version **6100**, **6200** disclosed above but using components that are rated at a voltage sufficiently higher than the line voltage. For example, diodes D1, D2, D3, D4 with sufficient ratings are 1N4007 silicon rectifier diodes from Diodes Inc., and the like. In another example, the zener diode clamping circuit (shown in FIG. 62) preceding the linear regulator U2 should be appropriately sized to handle the larger input voltage. In other embodiments, other methods to step down the rectified line voltage to a voltage usable by the microcontroller U3, such as switching converters and the like, can be used. Further, the conditioning circuit R3, R4, R5, R9, C5 should be modified to scale the input voltage for use by the microcontroller U3. An example of the LED Driver U1 that can accept a high input voltage is the AL9910 from Diodes Inc. Because of the high input voltage, this device uses an external MOSFET rather than an integral MOSFET as described in the AL8805 for the low voltage case.

[0387] In other embodiments of the line voltage LED fixture, an optional power factor correction IC can be placed between the full wave bridge D1, D2, D3, D4 and the rest of the circuit. A suitable device, for example, is an UCC28810 IC available from Texas Instruments, or the like. Advantageously, these devices drive the power factor of nonlinear loads, such as LED drivers, closer to unity.

[0388] In a further embodiment, the line voltage LED fixture **6100**, **6200** comprises an LED driver U1 that could be controlled from a microcontroller U3 having an input comprising a scaled and conditioned version of the input voltage. The microcontroller U3 deciphers the encoded data, and affects the LED driver U1 accordingly.

[0389] FIG. 63 is a block diagram of an exemplary multichannel lighting module **6300**, which receives the polarity

controlled, sinusoidal power signal from the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952**, decodes and performs the encoded commands, and uses the signal for power. The lighting module **6300** comprises a bridge rectifier **6302**, a conditioning circuit **6304**, a voltage regulator **6306**, a microcontroller **6308**, a temperature sensor **6310**, a plurality of LED drivers **6312**, **6314**, **6316**, **6318**, and one or more LEDs **6320**, **6322**, **6324**, **6326**. Each LED **6320**, **6322**, **6324**, **6326** may comprise one or more LEDs. The illustrated embodiment is a four channel lighting module **6300**, although other embodiments may have more or less than four channels.

[0390] The bridge rectifier **6302**, the conditioning circuit **6304**, and the voltage regulator **6306** are similar in construction and operation to the bridge rectifier **6102**, the conditioning circuit **6104**, and the voltage regulator **6106** of the single channel lighting fixture **6100**, respectively, as described above.

[0391] The four channel embodiment **6300** approximately quadruples the LEDs **6120** and LED driver **6112** on the single-channel embodiment **6100** with respect to the LEDs **6320**, **6322**, **6324**, **6326** and the LED drivers **6312**, **6314**, **6316**, **6318** for the four channel lighting fixture **6300**. Thus each LED **6320**, **6322**, **6324**, **6326** and each LED driver **6312**, **6314**, **6316**, **6318** is similar in construction and operation to the LED **6120** and LED driver **6112** of the single channel lighting fixture **6100**, respectively, as described above. Similarly, the microcontroller **6308** is similar in construction and operation to the microcontroller **6108** of the single channel lighting fixture **6100**, as described above, except the microcontroller **6308** controls multiple channels instead of a single channel. In conjunction with the microcontroller **6308**, the LED drivers **6312**, **6314**, **6316**, **6318** allow independent brightness control to four separate channels of LEDs. In a similar manner to microcontroller **6108**, which generates the signal **6130** to control the intensity of LED **6120**, microcontroller **6306** generates signals **6330**, **6332**, **6334**, and **6336** to control the intensities of LEDs **6320**, **6322**, **6324**, and **6326**, respectively. Each string of LEDs **6320**, **6322**, **6324**, **6326** may comprise one or more LEDs. In other embodiments, this approach could be used to add more channels, or to change the number of LEDs in each string. In yet other embodiments, each LED **6320**, **6322**, **6324**, **6326** may comprise several LED dies in a single package with a single lens, such as, for example, the CREE MC series of LEDs or the like.

[0392] Like the single-channel embodiment **6100**, the lighting fixture **6300** uses the microcontroller **6308** to receive information from the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952** and vary the LED intensity based on this information. Since each of the four channels can be independently controlled, the commands to a four-channel lighting fixture **6300** contain intensity level information for each of the four channels.

[0393] Advantageously, in the multi-channel embodiment **6300**, each channel may comprise a different color LED **6320**, **6322**, **6324**, **6326**. For instance, if the first channel comprises one or more white LEDs, the second comprises one or more red LEDs, the third comprises green LEDs and the fourth comprises blue LEDs, then a plurality of lighting colors could be generated by mixing the intensities in the correct ratios. For example, the white channel could create a brighter white light for general lighting needs, or slightly “wash out” the color created by the red, blue, and green

LEDs. This allows the user to formulate any color of light desired, and to vary that color, either abruptly, or by a gradual blending technique. Outdoor lights could also be modified to match a particular season or holiday. For instance, red, white, and blue colored lights could be used on the 4th of July; red and green lights could be used around Christmas; and orange lights could be used for Halloween and Thanksgiving.

[0394] In another embodiment, the multi-channel lighting fixture **6300** allows the user to adjust the shade of a white light. Perhaps, for example, the user is more of a “purest” and simply prefers white lights. The term “white” encompasses a wide range of shades from the more “blue” cool whites, to the more “yellow” warm whites. White LEDs by their nature are cool white. This is because a white LED is actually a blue LED with phosphor coating that glows white. For most people this is acceptable, but for some, a warmer white may be desired. If one of the three channels were populated with a red or yellow LED, then by varying the intensity of that channel, the user could vary the warmth, or color temperature as it is technically called, of the light. This is also important because different color temperatures are better at illuminating certain subject hues than others.

[0395] Control of individual lights or individual channels of LEDs within a single light is advantageous. Even more advantageous is to be able to achieve this control using the same set of wires that deliver power to the light. Lastly, integrating all of the decoder circuitry **6302**, **6304**, **6306**, **6308**, the driver circuitry **6312**, **6314**, **6316**, **6318**, and the temperature throttling **6310** on a single printed circuit board within the lighting fixture **6300**, results in a highly integrated, self-contained intelligent light fixture **6300**, which is no harder to install than a tradition landscape light.

[0396] FIG. **64** is an exemplary schematic diagram of a multichannel lighting module **6400**, according to one embodiment.

Color and Brightness Control of LED Lighting

[0397] LED-based lighting may comprise any of red, green, blue (RGB), red, green blue, white (RGBW), white, amber, cyan, red-orange, and the like, LEDs. A problem in the design of LED-based lighting is the resolution to maintain a specific or selected color while simultaneously dimming the light.

[0398] While the LEDs are ON, they produce light that acts as a continuous analog signal. The lighting modules provide the LED drivers with a signal, which is used by the LED drivers to create an LED drive signal to cause the LEDs, when optically mixed, to produce the specific or selected color of light at a specific or selected intensity. In an embodiment, the resolution of the control over the color and brightness of the LED is the smallest amount of change or discrete step in the signal provided to the LED driver that affects a change in the output of the LED. For example, if the intensity can be changed from 50% to 51%, then the resolution or resolution of intensity control is 1%. If the intensity can be changed from 50% to 50.1%, then the resolution or resolution of intensity control is 0.1%. A signal with 0.1% resolution has greater resolution than a signal with 1% resolution. The signal with 0.1% resolution can control the color and intensity of the LED with greater precision than the signal with 1% resolution. In an embodiment, the resolution is the number of output levels a signal can be quantized to. In another embodiment, a signal with

greater resolution has greater granularity than a signal with less resolution, where granularity is the relative size, scale, level of detail, or depth of penetration that characterizes an object or activity.

[0399] The following example uses RED, GREEN, and BLUE LEDs to illustrate the resolution issue of maintaining a lighting color and dimming the intensity of LED-based lighting. It is well understood that LED-based lighting may also comprise different combinations of colored LEDs.

[0400] In an embodiment, a lighting system controls the brightness of individual RED, GREEN, and BLUE LEDs with approximately 1% accuracy in order to mix the LED colors and create various hues. While accuracy of approximately 1% may provide a precise level of control to create specific hues, it is insufficient when both color mixing and overall dimming are simultaneously provided.

[0401] For example, color X is defined as RED LED driven at 84% brightness, GREEN LED driven at 27% brightness and BLUE LED driven at 3% brightness, which results in an RGB ratio of 28:9:1. The user is pleased with the hue of color X, but would like to dim the light to, for example, 10% of the overall brightness. Mathematically, to achieve this, the lighting controller would drive the RED LED at 8.4% brightness, drive the GREEN LED at 2.7% brightness, and drive the BLUE LED at 0.3% brightness. However, because the resolution is approximately 1%, rounding occurs. The lighting controller would drive the RED LED at 8% brightness, drive the GREEN LED at 3% brightness, and drive the BLUE LED at 0% brightness.

[0402] The desired RGB ratio of 28:9:1 for color X has changed to 8:3:0 due to dimming. The new RGB ratio creates an approximately 10% error in the balance of the RED and GREEN LEDs and eliminates the BLUE LED from the color mix. This example illustrates that the lighting system changes the hue of the LED-based lighting as the user dims the lighting because the RGB ratio changes due to the limited resolution of control.

[0403] Color mixing systems, in certain embodiments, use pulse width modulation (PWM) to control the brightness of the individual colors. While many microcontrollers can generate PWM waveforms with resolutions of approximately 0.1% or better, there may be other factors that prevent or inhibit this same level of control at the LED-based lighting output. Drivers or driver circuits that are used to drive the LEDs can prevent or inhibit driving the LEDs with greater resolution.

[0404] Linear regulators are a simple way to reduce the voltage of a DC supply but linear regulators waste energy as they operate by dissipating energy as heat. For efficiency reasons, high power LEDs, such as those used for lighting, are driven using switching regulators in certain embodiments.

[0405] A buck converter is a voltage step down and current step up switching regulator. When the input voltage is higher than the forward voltage of the string of LEDs, a switching regulator having a buck topology is often used to drive the LEDs. Furthermore, because of their simplicity, hysteretic buck regulators or converters, such as, for example, an AL8807 from Diodes Incorporated, or the like, are commonly used to drive LEDs with a relatively constant current.

[0406] These LED drivers comprising hysteretic buck regulators or converters use a high-frequency switched signal to drive a relatively constant current through the LED

at frequencies from approximately a few tens of kilohertz up to approximately a megahertz. In a hysteretic converter, the exact frequency of this high frequency signal depends on several factors, such as component values, input and output voltages, and output current, for example. Further, it takes a few cycles of this high-frequency signal to “ramp-up” and for the LED to turn ON. Again, the exact number of cycles to ramp-up and turn ON the LED depends on several factors, such as the component values, the input and output voltages, and the output current, for example.

[0407] This high frequency signal which drives the relatively constant current through the LEDs is switched ON and OFF based at least in part on the PWM signal that is input to the LED driver. The PWM signal that is input to the driver is generated by a microcontroller or microprocessor and is used to control the RGB ratios in color mixing for LED-based lighting, as described above.

[0408] Because the PWM input signal to the LED driver cannot be narrower than a few cycles of the high frequency signal used to drive the LED (due to the ramp-up and turn ON), the PWM input signal loses resolution through the LED driver.

[0409] In other words, the period of the high-frequency waveform that the LED driver is using to drive the LED is the “building block”. If 50% brightness is desired, the lighting system may drive the LED with 10 of these periods, then stop driving the LED for the next 10 periods, and so on until the desired brightness changes.

[0410] For example, consider a PWM signal with 0.1% resolution controlling a hysteric LED driver operating at approximately 250 kHz so that the waveform the LED driver uses to drive the LED has a period of approximately 4 microseconds. The narrowest pulse that could be sent into the PWM input of the LED driver to achieve accurate control would be about 20 microseconds. This approximately 20 microsecond pulse would correspond to approximately 0.1% of full brightness of the lighting system. A pulse width 1000 times greater (1/0.1%) would correspond to 100% brightness, or 20 milliseconds.

[0411] Unfortunately, a signal with an approximately 20 millisecond period corresponds to a 50 Hz signal, which when input into the LED driver may result in visual flicker.

[0412] Further, it is common for the hysteretic buck converter to be supplied power directly from a full wave bridge without the use of bulky, unreliable storage capacitors. For instance, in the example of a 12VAC input power signal and the AL8807 LED driver, the LED driver is turning ON for approximately 4-5 milliseconds then OFF for approximately 3-4 milliseconds for each half-wave of the input power. The LED driver turns ON when the full wave rectified signal goes above approximately 7V, which is the minimum operating voltage of the part. This corresponds to a frequency of approximately 120 Hz for a 60 Hz input signal.

[0413] The combination of the 120 Hz incoming power modulation and the 50 Hz PWM control signal modulation would result in cross-modulation components which may cause even more visual flicker.

[0414] In summary, the operating frequency of the LED driver limits how narrow the pulses of the PWM signal used at the PWM input of the LED driver can be. Stated another way, the operating frequency of the LED driver places a practical upper limit on the frequency of the PWM signal used at the PWM input of the LED driver.

[0415] Additionally, the human eye’s ability to perceive flicker along with the incoming power frequency places a practical lower limit on the PWM frequency that can be used. When taken together, these two limits usually result in a resolution much worse than 0.1%. Thus, the full accuracy of the PWM input signal to the LED driver is difficult to realize.

[0416] In some instances, increasing the operating frequency of the LED driver permits the upper limit of the frequency of the PWM input signal to be increased. However, this can create additional problems. Electromagnetic emissions are much more of a concern at these higher switching frequencies. Additionally, efficiency usually suffers since the MOSFETs that do the switching spend a greater percentage of time in their linear region.

[0417] A control method that increases resolution and operates with LED drivers operating in the hundreds of kilohertz range, such as 100 kHz or 250 kHz, and the like, would be advantageous.

[0418] FIG. 67 is a block diagram of an exemplary hysteretic LED driver 6700 that can be used to drive one or more LEDs electrically coupled between SET and SW. A low value resistor, for instance 0.1 ohms, would typically be connected between the SET pin and the positive power supply voltage. This allows the SET pin to monitor the current through the LED when the FET turns on. SET is therefore used to monitor the LED current and SW provides the high-frequency signal to drive the LEDs. UVLO 6708 controls the minimum operating voltage of the part.

[0419] A CTRL input comprises a dimming and ON/OFF control input for the LED driver 6700, and in an embodiment, receives the PWM input signal that operates the LED driver with increased resolution. In an embodiment, PWM control and analog control are used to create the PWM input signal.

[0420] In an embodiment, the LED driver 6112 associated with the lighting module 6100, described above with respect to FIG. 61, comprises the hysteretic LED driver 6700, and the signal 6130 from the microcontroller 6108 comprises the PWM input signal. In another embodiment, each of the LED drivers 6312, 6314, 6316, 6318 associated with the lighting module 6300, described above with respect to FIG. 63, comprise the hysteretic LED driver 6700, and the signals 6330, 6332, 6334, 6436 from the microcontroller U3 6308 comprises the PWM input signal.

PWM Control

[0421] Referring to FIG. 67, in an embodiment, the PWM input signal electrically couples to the CTRL input where it is fed through a PWM Dimming Control block 6702 into comparator 6704 which controls a MOSFET switch 6706. The duty cycle of the PWM input signal controls the ON/OFF states of the MOSFET switch and thereby the current that the LED is driven with. Moreover, controlling the LED drive current controls the brightness of the LED.

Analog Control

[0422] In an embodiment, the PWM Dimming Control block 6702 also receives a control voltage VCTRL and a reference voltage VREF. In an embodiment where the LED driver comprises an AL8807 from Diodes Incorporated, for example, the control voltage VCTRL is an approximately 0.45V signal, and the reference voltage VREF is an approxi-

mately 2.5V signal. In other embodiments of LED drivers, VREF may be more or less than 2.5V and VCTRL may be more or less than 0.45V.

[0423] For illustration purposes, in the embodiment of FIG. 67, when the CTRL input drops below approximately 0.45V, the comparator 6704 receives approximately 0V, such that the MOSFET switch 6706 does not turn ON, thereby disabling the driver 6700. Voltages on the CTRL input that are between approximately 0.45V and approximately 2.5V are passed through to the comparator 6704. Therefore, by driving a DC voltage into the CTRL input, the brightness of the LED can be controlled because the switching point (hysteresis) of the comparator 6704 can be controlled.

[0424] By controlling the voltage level of the PWM input signal, the reference voltage at the output of the PWM DIMMING CONTROL 6702 supplied to the comparator 6704 can be controlled, and in turn, the current supplied by the LED driver 6700 can be controlled, which controls the brightness of the LEDs.

[0425] While the topology of FIG. 67 is just one example of a hysteretic driver 6700, hysteretic drivers 6700 typically comprise the comparator 6704 (often called the hysteresis comparator) which controls the MOSFET switch 6706.

[0426] In summary, in an embodiment, hysteretic drivers 6700 can be controlled with either an analog signal or a digital (PWM) signal. In the analog approach, the current supplied by the driver 6700 is regulated to less than 100% by applying a lower voltage than the voltage that corresponds to 100% of the comparator reference. In the PWM or digital approach, the comparator reference switches between 2 states: 100% and 0%. The ratio of the time spent in these states determines the average current through the LED. As long as the frequency of the PWM signal is faster than the response of the human eye, the brightness is perceived as the average brightness.

Analog and PWM Control Signals

[0427] Embodiments of control signal generators generate a control signal that allows a hysteretic driver to simultaneously operate in both digital (PWM) and analog modes. Advantageously, when each mode alone allows a resolution of approximately 1%, then these two resolutions will compound resulting in a resolution approximately 0.01% ($1\% \times 1\%$). Furthermore, approximately 1% control is achievable using PWM control at driver frequencies as low as 50 kHz, while still avoiding the flicker described above.

[0428] FIGS. 68-72 comprise exemplary control waveforms to illustrate how analog and digital control methods can be used independently and then simultaneously in embodiments of the hysteretic LED driver 6700.

[0429] Consider an analog controlled system. FIG. 68 is an exemplary analog input waveform 6800 comprising an approximately 2.5V signal that when applied to the CTRL input results in approximately full (100%) brightness of the LED for the analog controlled system. FIG. 69 illustrates an exemplary analog input waveform 6900 that when applied to the CTRL input results in approximately half (50%) brightness of the LED for the analog controlled system. The waveform 6900 has an approximately constant level of approximately 1.25V, which is half of the 100% level of approximately 2.5V of the waveform 6800.

[0430] Consider a PWM controlled system. FIG. 70 illustrates an exemplary PWM input waveform 7000 that results in approximately full (100%) brightness of the LED for the

PWM controlled system. The waveform 7000 comprises an approximately 2.5V signal having a duty cycle of 100%. FIG. 71 illustrates an exemplary PWM input waveform 7100 that results in approximately half (50%) brightness for the PWM controlled system. The waveform 7100 comprises an approximately 2.5V periodic signal having a period of approximately 2T, where the amplitude of the waveform 7100 is approximately 2.5V for the first half of the period and approximately 0V for the second half of the period, or in other words, has a 50% duty cycle. The approximately half (50%) brightness can be achieved since half of the time the driver 6700 is working at approximately 100% brightness, and the other half of the time it is at approximately 0% brightness, which averages out to approximately 50% brightness.

[0431] Consider an analog controlled and a PWM controlled or hybrid controlled system. FIG. 72 illustrates a hybrid controlled lighting module 7200 comprising a hybrid signal generator 7202, the hysteretic LED driver 6700, and a string of one or more LEDs 7204. In an embodiment, the LEDs 7204 comprise the LEDs 6120, 6320, 6322, 6324, 6326. The hybrid signal generator 7202 generates a hybrid (analog and PWM) control waveform or signal which is received by the LED driver 6700 and used to control the brightness and color mixing of the LEDs 7204 with resolution provided by both analog and PWM control.

[0432] FIG. 73 illustrates an exemplary hybrid control waveform 7300 for the hysteretic LED driver 6700 that results in approximately quarter (25%) brightness. The waveform 7300 comprises an approximately 1.25V periodic signal having a period of 2T, where the amplitude of the waveform 7300 is approximately 1.25V for the first half of the period and approximately 0V for the second half of the period (50% duty cycle). The PWM aspect of the waveform 7300 enables the LED driver 6700 for approximately 50% of the time. However, while the LED driver 6700 is ON, analog dimming is employed, as evident from the voltage of the waveform 7300.

[0433] In an embodiment where the voltage to drive the LEDs 7204 to 100% brightness is approximately 2.5V, a voltage of approximately 1.25V drives the LEDs 7204 to approximately 50% brightness. When the hybrid signal generator 7202 generates the waveform 7300, the LED driver 6700 drives the LEDs 7204 at approximately 50% of maximum while the LED driver 6700 is ON. The effective brightness is therefore approximately 25% since the LEDs 7204 are being driven at approximately 50% brightness for approximately 50% of the time ($0.50 \times 0.50 = 0.25$). This illustrates the compounding effect of approximately simultaneous analog and PWM control of the LED driver 6700.

[0434] FIG. 73 illustrates just one example of a hybrid control waveform having simultaneous or approximately simultaneous analog and digital dimming with increased resolution. In other embodiments, the lighting module 7200 can achieve various dimming scenarios that do not alter or substantially alter the RGB ratio of the desired hue by supplying a PWM waveform of varying duty cycle and with varying amplitude.

Novel Control Signal Generators

Digital to Analog Converter Output Modulated by PWM Waveform

[0435] FIGS. 74-76 illustrate embodiments of the hybrid signal generator 7202 to generate hybrid control waveforms comprising one or more of varying duty cycles and varying amplitudes.

[0436] FIG. 74 is a block diagram of an embodiment of a hybrid control signal generator 7400 comprising a microcontroller 7402 and a digital to analog converter (DAC) 7404. The DAC 7404 receives digital data at a digital input and converts the digital data to an analog signal, which is output at an analog output. The DAC 7404 includes an enable input that enables or disables the analog output. The microcontroller 7402 is configured to output digital data at a digital output and a PWM waveform at a PWM output. The digital output of the microcontroller 7402 electrically couples to the digital input of the DAC 7404 and the PWM output of the microcontroller 7402 electrically couples to the enable input of the DAC 7404.

[0437] The microcontroller 7402 outputs a digital value which is converted to an analog signal and output by the DAC 7404. The microcontroller 7402 further generates the PWM signal, which, as the enable input to the DAC 7404, controls the analog output of the DAC 7404. When the enable signal is active, the output of the DAC 7404 is approximately the analog signal as set by the microcontroller 7402. When the enable signal is inactive, the output of the DAC 7404 is approximately 0V. The output of the DAC 7404 is modulated by the PWM waveform supplied by the microcontroller 7402. The output of the DAC 7404 is presented to the LED driver 6700.

[0438] Thus, the hybrid control signal generator 7400 generates a control waveform comprising a varying duty cycle and varying amplitude where the duty cycle is based at least in part on the PWM waveform from the microcontroller 7402 and the amplitude is based at least in part on the digital data from the microcontroller 7402.

[0439] FIG. 75 is a block diagram of an embodiment of a hybrid control signal generator 7500 comprising a microcontroller 7502, a digital to analog converter (DAC) 7504, and a switch 7506. In an embodiment, the switch 7506 comprises a transistor, such as a BJT or a FET. The microcontroller 7502 comprises a digital output and a PWM output and operates similar to the microcontroller 7402 as described above with respect to FIG. 74. DAC 7504 comprises a digital input, an enable input, and an analog output and operates similar to the DAC 7404 as described above with respect to FIG. 74.

[0440] The digital output of the microcontroller 7502 electrically couples to the digital input of the DAC 7504. The PWM output of the microcontroller 7502 electrically couples to a gate or a base of the switch 7506 and the analog output of the DAC 7504 electrically couples to a collector or a source of the switch 7506. The enable input is configured such that the DAC 7504 is always enabled, in an embodiment.

[0441] The microcontroller 7502 outputs digital data to the digital input of the DAC 7504. The DAC 7504 converts the digital data to an analog signal and outputs the analog signal. The microcontroller 7502 further generates a PWM waveform, which is received at the base or gate of the switch 7506. The analog output of the DAC 7504 is enabled through the enable input, but the analog output of the DAC 7504 is pulled to ground via the collector or source of the switch 7506 when the PWM waveform becomes active.

[0442] When the PWM waveform is inactive, the switch 7506 is open, such that the transistor is OFF, and the DAC analog output signal is presented to the LED driver 6700. In an embodiment, the PMW waveform is active when it is a logic high and inactive when it is a logic low. In another

embodiment the PMW waveform is active when it is a logic low and inactive when it is a logic high. The output of the DAC 7504 is modulated by the PWM waveform generated by the microcontroller 7502 and the amplitude of the output of the DAC 7504 is controlled by the digital data provided by the microcontroller 7502.

[0443] Thus, the hybrid control signal generator 7500 generates a control waveform comprising a varying duty cycle and varying amplitude where the duty cycle is based at least in part on the PWM waveform from the microcontroller 7502 and the amplitude is based at least in part on the digital data from the microcontroller 7502.

[0444] In the embodiments illustrated in FIGS. 74 and 75, the DAC 7404, 7504 generates the analog waveform and the PWM signal from the microcontroller 7402, 7502 is used to modulate this analog waveform. In other embodiments, the microcontroller 7402, 7502 comprises the DAC 7404, 7504, the switch 7506 comprises an analog switch, and the like. In an embodiment, the DAC 7404, 7504 comprises, for example, an MCP4728 available from Microchip Technology, or the like. In an embodiment, the microcontroller 7402, 7502 comprises, for example, a PIC16F1847 available from Microchip Technology, or the like. The switch 7506 comprises, for example, a transistor, such as a FET 2N7002 available from Diodes Inc., or the like.

PWM Waveform Modulated by PWM Waveform

[0445] If the microcontroller has multiple PWM channels, a first PWM waveform may be used to create an analog waveform by low-pass filtering (averaging) the first PWM waveform. When the cut-off frequency of the low-pass filter is much less than the PWM frequency of the first PWM waveform, the resulting signal will approximate an analog DC signal. The level of this analog DC signal can be controlled by the duty cycle of the first PWM waveform generated by the microcontroller. Once this analog DC signal is generated, it can be modulated by a second PWM waveform generated by the microcontroller using various methods.

[0446] FIG. 76 is a block diagram of an exemplary hybrid control signal generator 7600 comprising a microcontroller 7602, a switch 7606, a low-pass filter 7608, and a buffer circuit 7610. The microcontroller 7602 comprises at least two PMW channels, indicated by a PWM1 output and a PWM2 output and operates similar to the microcontroller 7402, 7502 as described above with respect to FIGS. 74 and 75. In an embodiment, the switch 7606 comprises a transistor, such as a BJT or a FET, an analog switch, and the like.

[0447] In an embodiment, the low-pass filter 7608 comprises a 2-stage RC network, as illustrated in FIG. 76. In other embodiments, other low-pass filters can be used, as is known to one of skill from the disclosure herein. In an embodiment, the buffer circuit 7610 comprises an operational amplifier, as illustrated in FIG. 75. In other embodiments, other circuits can be used to buffer or drive the output of the hybrid control signal generator 7600, as is known to one of skill from the disclosure herein.

[0448] The PWM1 output of the microcontroller 7602 electrically couples to an input of the low-pass filter 7608 and an output of the low-pass filter 7608 electrically couples to an input of the buffer circuit 7610. An output of the buffer circuit 7610 electrically couples to a source or emitter of the

switch **7606** and the PWM2 output of the microcontroller **7602** electrically couples to a gate or base of the switch **7610**.

[0449] The first PWM waveform from the PWM1 output of the microcontroller **7602** is filtered by the low-pass filter **7608** to approximate an analog signal where the amplitude of the analog signal is responsive to the average of the amplitude of the first PWM waveform. The buffer circuit **7610** buffers the output of the low-pass filter **7608**. The output of the buffer circuit **7610** is modulated in response to the second PWM waveform from the PWM2 output of the microcontroller **7602** using the switch **7606**. This is described above with respect to the microcontroller **7502** and the switch **7506** of FIG. 75. The output of the buffer circuit **7610** drives the LED driver **6700**.

[0450] In embodiments where the low-pass filter **7608** comprises a 2-stage RC network, the output of the low-pass filter **7608** is buffered by the buffer circuit **7610** because the 2-stage RC network effectively raises the source impedance of the signal, which lowers its drive capability. In other embodiments, when the input impedance of the LED driver **6700** is high, the buffer circuit **7610** may be eliminated.

[0451] In an embodiment, the microcontroller **7602** comprises, for example, a PIC16F1847 available from Microchip Technology, or the like. The switch **7606** comprises, for example, a transistor, such as a FET 2N7002 available from Diodes Inc., or the like.

Digital to Analog Converter Output Controlled by Digital Data

[0452] In an embodiment, a digital to analog converter is used to generate the hybrid control signal using digital data from a microcontroller. Advantageously, this solution uses fewer components than the embodiments illustrated in FIGS. 74, 75, and 76.

[0453] FIG. 77 illustrates an exemplary hybrid control waveform **7700** for the hysteretic LED driver **6700** where the analog and PWM components of the hybrid control waveform change at $t=t_5$. FIG. 78 is a block diagram of an exemplary hybrid control signal generator **7800** comprising a microcontroller **7802** and a digital to analog converter (DAC) **7804** that can be used to generate the hybrid control signal **7700**.

[0454] A digital output of the microcontroller **7802** electrically couples to a digital input of the DAC **7804** and the microcontroller **7802** provides digital data to the DAC **7804**. The microcontroller **7802** writes or sends new digital values to the DAC **7804** at determined intervals to generate the hybrid control waveform **7700**.

[0455] For example, to generate the hybrid control waveform **7700**, the microcontroller **7802** writes a digital value corresponding to a DAC output of approximately 1.25V at $t=t_1$, and a digital value corresponding to a DAC output of approximately 0V at $t=t_2$. At $t=t_3$, the microcontroller **7802** again writes a digital value corresponding to the DAC output of approximately 1.25V, and at $t=t_4$ the microcontroller **7702** again writes a digital value corresponding to the DAC output of approximately 0V. At this point, the user, through the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952**, decides to change the intensity of the light. So at $t=t_5$, the microcontroller **7802** writes a digital value to the DAC **7804** that causes the DAC **7804** to output a signal with a value of approximately 0.75V. Then at $t=t_6$, the microcontroller **7802** writes a digital value corresponding to a DAC output of

approximately 0V. This process continues. The output of the DAC **7804** is presented to the LED driver **6700**.

[0456] The microcontroller **7802** writes the digital data at specific time intervals in order for the desired outcome to occur. In an embodiment, the time intervals between “writes” can vary, which also affects the color and brightness of the LEDs **7204**.

[0457] In contrast, the microcontrollers **7402**, **7502**, write the digital data to the corresponding DAC **7404**, **7504** and to the PWM controller in the microcontroller **7402**, **7502** once. These values do not need to be re-written unless the user, through the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952**, changes the color and/or brightness of the LED lighting **7204**. And if the color and the brightness changes, in an embodiment, a single pair of writes to affect the changes is needed until there is another change.

Filtered PWM Waveform

[0458] In an embodiment, a PWM waveform from a microcontroller is filtered and buffered to provide the hybrid control waveform. The microcontroller provides a new PWM waveform with a different duty cycle to change the hybrid control waveform. Advantageously, this solution uses fewer components than the embodiments illustrated in FIGS. 74, 75, and 76.

[0459] FIG. 79 illustrates an exemplary PWM waveform **7902** (Waveform A) from a microcontroller that is filtered to provide a hybrid control waveform **7900** (Waveform B). FIG. 80 is a block diagram of an exemplary hybrid control waveform generator **8000** comprising a microcontroller **8002**, a low-pass filter **8008**, and a buffer circuit **8010** that can be used to generate the hybrid control signal **7900**.

[0460] A PWM output of the microcontroller **8002** electrically couples to an input of the low-pass filter **8008** and an output of the low-pass filter **8008** electrically couples to an input of the buffer circuit **8010**. An output of the buffer circuit **8010** provides the hybrid control waveform **7900** for the LED driver **6700**.

[0461] In an embodiment, the low-pass filter **8008** comprises a 2-stage RC low-pass filter. In other embodiments, other configuration of low-pass filters **8008** can be used, as is known to one of skill from the disclosure herein. In an embodiment, the buffer circuit **8010** comprises an operational amplifier. In other embodiments, other configurations of buffer circuits **8010** can be used, as is known to one of skill from the disclosure herein.

[0462] Waveform A **7902** is an example of the PWM waveform generated by the microcontroller **8002**. To generate Waveform A, the microcontroller **8002** writes digital data corresponding to, for example, amplitude, period T_2 , and duty cycle (T_1/T_2), to the PWM controller in the microcontroller **8002**. Waveform A **7902** is averaged by the low-pass filter **8008** and buffered by buffer circuit **8010** to provide a PWM waveform with varying amplitude as the hybrid control waveform **7900** (Waveform B). During the period in between the pulses, both waveforms **7900**, **7902** are at approximately 0V.

[0463] To generate the hybrid control signal **7900**, for example, at $t=t_1$, the microcontroller **8002** generates the PWM Waveform A **7902** having an amplitude of approximately 2.5V and a duty cycle of approximately 50% which corresponds to a low-pass filter output of approximately 1.25V. At $t=t_2$, the microcontroller **8002** writes new digital data to its PWM peripheral corresponding to a low-pass

filter output of approximately 0V. At $t=t_3$, the microcontroller **8002** again writes digital data to its PWM peripheral that causes the microprocessor **8002** to generate the PWM Waveform A **7902** having an amplitude of approximately 2.5V and a duty cycle of approximately 50% and at $t=t_4$, the microcontroller **8002** writes digital data corresponding to a low-pass filter output of approximately 0V.

[0464] This process continues until $t=t_5$, where the user, through the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952**, decides to change the intensity of the light. At $t=t_5$ the microcontroller **8002** writes digital data with a new duty cycle corresponding to a low-pass filter output of approximately 0.75V. At $t=t_6$, the microcontroller **8002** again writes digital data corresponding to a low-pass filter output of approximately 0V. The times ($t=t_2$, t_4 , t_6) when the PWM output of the microcontroller **7802** is OFF, corresponding to a low-pass filter output of approximately 0V, are chosen to provide the hybrid control signal **7900** with a period of T_4 and a duty cycle of T_3/T_4 . The process continues.

[0465] FIG. 79 illustrates an exemplary hybrid control waveform **7900** with varying amplitude and an approximately constant period. In other embodiments of the hybrid control signal generator **8000**, the microcontroller **8002** can write digital data at differing times to generate the PWM waveform **7902** that when filtered, provides the hybrid control waveform **7900** having varying amplitude and/or varying modulation periods.

[0466] FIG. 81 illustrates an exemplary filter response **8100** for the low-pass filter **8008**. Referring to FIGS. 79 and 80, the PWM output (Waveform A) **7902** has a period of T_2 , which corresponds to a frequency of f_2 and the hybrid control waveform (Waveform B) **7900** has a period of T_4 corresponding to a frequency of f_4 . The values of the components of the low-pass filter **8010** are chosen such that the low-pass filter **8010** readily passes waveforms having the frequency f_4 , and filters or averages waveforms having the frequency f_2 .

[0467] Embodiments of the lighting systems **100**, **200**, **1800**, **4000**, **4300**, **4700**, **4800**, **4900**, **5000**, **5200** further comprise the hybrid controlled lighting module **7200** to control color and dimming of LED lighting. Embodiments of the lighting controller **202**, **300**, **4002**, **4702**, **4902**, **4952** further comprise any of the hybrid control signal generators **7400**, **7500**, **7600**, **7800**, **8000**. Embodiments of the lighting modules **6100**, **6200**, **6300**, **6400** comprise any of the hybrid control signal generators **7400**, **7500**, **7600**, **7800**, **8000**. Embodiments of the microcontroller **6102**, **6302** comprise any of the microcontrollers **7402**, **7502**, **7602**, **7802**, **8002**.

[0468] The embodiments disclosed herein illustrate non-limiting examples of the concept of combining two control signals, each with a limited control resolution, such that the combined control signal has a resulting resolution with a much finer resolution than either of the two control signals alone could achieve.

[0469] The examples herein describe the generation and use of a hybrid control signal having an analog component and a digital component in conjunction with a hysteretic buck converter. In other embodiments, the two components of the hybrid control signal could also both be analog, or both be digital. Furthermore, the driver need not be a hysteretic buck topology driver, but may also be buck, boost, buck-boost, single-ended primary-inductor converter (sepic), linear, or various other driver types. Additionally, the two control signals need not be combined prior to

presenting them to the converter, but may also be combined by the converter itself. In such a design, the converter comprises more than one dimming input.

[0470] Lastly, the concept of dimming a particular hue was used as an example of one of the benefits associated with increased resolution of control. However, even if dimming were not a consideration, the techniques described could be used to give more resolution for hue control, since the ratios of the colors to be mixed can be controlled with greater precision.

Color Wheel Feature

[0471] Colored LED-based lighting, such as an RGB lighting fixture, not only has the ability to render nearly any hue of light and/or intensity using, for example, one or more of the hybrid control signals described herein, it also has the ability to change that hue autonomously if desired. In an embodiment, an intelligent RGB lighting fixture receives a command, which comprises a starting hue, an ending hue, and a transition time, where the transition time is the time it takes the lighting fixture to transition from the starting hue to the ending hue. When the lighting fixture receives this command, it outputs the starting hue, and begins transitioning from the starting hue to the ending hue, at a speed that is based at least in part on the transition time.

[0472] In another embodiment, the command further comprises cycle information. The cycle information may comprise additional information such as whether after the ending hue is reached, the lighting fixture should transition back to the starting hue, or remain at the ending hue. The lighting fixture may, in a further embodiment, be instructed to start at one hue and continually transition through all possible hues perpetually.

[0473] This feature is useful when lighting fountains or water features. For example, a user wants to illuminate a 3-tiered fountain with three separate groups of lights, such that a first group illuminates the lowest tier, a second group illuminates the middle tier, and a third group illuminates the highest tier. He could start each tier at a different hue, and instruct the lighting fixtures to step or cycle through all hues with approximately the same speed. This results in a visually appealing display since each of the three fountain tiers is illuminated with a different color, which is changing. This example shows the importance of specifying a starting hue as well as a transition speed.

[0474] In an embodiment, for such a feature to be useful, it is important for lighting fixtures of different groups, or lighting fixtures of the same group to use a time base that is synchronous among the lighting fixtures. When multiple fixtures using the same or approximately the same time base are placed in this "color wheel" mode, they would maintain synchrony with each other. In an embodiment, accurate crystals or oscillators can be used in each lighting fixture to minimize variations in each lighting fixture's time base. Unfortunately, accurate crystals or oscillators can be expensive. In another embodiment, without using expensive highly accurate crystals in each lighting fixture, the incoming AC power is used as a time base. Because the lighting fixtures receive power from a common source of power, using the timing information from the common power source is a simple, inexpensive method to maintain the lighting fixtures in sync with one another. In an embodiment, the timing information of the common power source comprises the frequency of the power source.

[0475] In an embodiment, the zero-crossing of the incoming power generates an interrupt to the microcontroller in the lighting fixture. For example, a zero-crossing detector could be used to generate the interrupt. Depending at least in part on the transition speed that of the lighting fixture received in the command, the microcontroller may take one “step” toward the next color or hue every interrupt, or every second interrupt, or every third interrupt, etc. For example, a counter could be used to count the interrupts generated by the zero-crossing detector. When the number of received interrupts correlates with to the transition speed, the microcontroller generates control signals to control the LEDs in the lighting fixture to produce the next color.

[0476] In an embodiment, a LED-based lighting fixture is configured to receive a command comprising a starting hue, an optional ending hue, and a transition speed. In another embodiment, the LED-based lighting fixture is configured to use timing information from the power signal as a time base for color transitions defined by the starting hue, the optional ending hue, and the transition speed.

Chassis

[0477] Embodiments of the lighting systems 100, 200, 1800, 4000, 4300, 4700, 4800, 4900, 5000, 5200 further comprise a frame that would mount inside the home or anywhere an aesthetic installation was desired. In one embodiment, the frame holds the operator interface panel 308 and could further comprise one or more of the following:

[0478] An extension of the RS485 connection from the chassis 302 to the frame, where the frame has wiring channels for the cable, thus providing a very clean installation;

[0479] An integrated “nest” for the WI-FI antenna so that it would not be seen;

[0480] Work with the wireless operator interface panel extender described herein by invisibly housing the RF transceiver and providing power to the operator interface panel 308; and

[0481] Power the operator interface panel 308 such that the frame could also recharge an integrated battery. This would allow users to have the wall mounted option, be able to remove the operator panel 308, which is now powered by the battery and in communication with the chassis via the radio, and walk around to adjust their lighting, seeing the affects in real time.

[0482] FIG. 65 is a front view of an exemplary lighting controller comprising a frame 6502 and an operator interface panel 6504 which comprises a display 6508, fixture programming ports 6510, and user interface devices 6512, such as button, switch, knobs, and the like. In the illustrated embodiment, latch 6506 is configured to releasably hold the operator interface panel 6504 in the frame 6502.

[0483] FIG. 66 is a back view of an exemplary lighting controller comprising the frame 6502, the operator interface panel 6504 and the latch 6506. The illustrated lighting controller further comprises a WIFI module 6606 electrically coupled to an antenna 6602 through antenna wire 6604, and an operator interface panel extender 6608 detachable affixed to the lighting controller. The operator interface panel extender 6608 supplies power to the power supply of the operator interface panel 6504 through a cable, such as a 6-conductor flat modular cable, when the operator interface panel extender 6608 is affixed to the operator interface panel

6504. Referring to FIG. 43, the operator interface panel extender 6608 comprises the panel RF transceiver 4304, the battery/adaptor 4306, and the antenna 4312. For remote operation, the operator interface panel extender 6608 wirelessly communicates with the chassis/power supply 302 via the chassis RF transceiver 4302 and antenna 4310.

[0484] An adapter 6610 configured as a wall plug-in module, provides power to the power operator interface panel extender 6608. The operator interface panel extender 6608 may optionally comprises a battery and a battery charger, which provide power to the extender 6608 and operator panel 6504, allowing the panel 6504 and extender 6608 to be removed from the frame 6502 and operated while moving about the property.

Terminology

[0485] Depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the algorithm). Moreover, in certain embodiments, acts or events can be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially.

[0486] The various illustrative logical blocks, modules, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

[0487] The various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0488] The steps of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM,

or any other form of computer-readable storage medium known in the art. An exemplary storage medium can be coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The processor and the storage medium can reside in an ASIC.

[0489] Conditional language used herein, such as, among others, “can,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or states. Thus, such conditional language is not generally intended to imply that features, elements, and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding whether these features, elements, and/or states are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

[0490] While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments of the inventions described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of certain inventions disclosed herein is indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A remote programming device for programming at least one lighting module that is configured to be assigned to a lighting zone, each lighting module comprising a unique module address and each lighting zone comprising a zone address, the at least one lighting module installed in a lighting system and receiving a data encoded power signal through a two-wire path, the remote programming device comprising:

- a portable housing;
- a user interface housed by the portable housing; and
- a processor housed within the portable housing and responsive to the user interface, the processor configured to wirelessly communicate with a selected lighting module of the at least one lighting module and configured to receive the unique module address of the selected lighting module from the selected lighting module and to provide instructions to encode the unique module address onto the data encoded power signal in order to reassign the selected lighting module to a selected lighting zone without wiring modifications to the lighting system.

2. The remote programming device of claim 1, wherein each lighting module stores its unique module address and its zone address.

3. The remote programming device of claim 1, wherein the remote programming device comprises a smartphone executing one or more appropriate applications.

4. The remote programming device of claim 3, wherein the selected lighting module includes at least one light emitting diode (LED) and the smartphone includes an optical receiver, the at least one LED flashing an address of the selected lighting module and the optical receiver detecting the address of the selected lighting module from the flashing at least one LED.

5. The remote programming device of claim 3, wherein the selected lighting module includes at least one light emitting diode (LED) and the smartphone includes a camera, the at least one LED flashing an address of the selected lighting module and the camera detecting the address of the selected lighting module from the flashing at least one LED.

6. The remote programming device of claim 5, wherein a rate of flashing of the at least one LED is less than approximately 30 Hz.

7. The remote programming device of claim 3, wherein the selected lighting module includes a bar code and the smartphone includes a camera, the bar code encoded with the unique address of the selected lighting module and the camera reading the bar code to determine the unique address of the selected lighting module.

8. A method to program at least one lighting module that is configured to be assigned to a lighting zone, each lighting module comprising a unique module address and each lighting zone comprising a zone address, the at least one lighting module installed in a lighting system and receiving a data encoded power signal through a two-wire path, the method comprising:

- with a remote programming device that includes a portable housing, a user interface, and a processor that is responsive to the user interface,
- wirelessly communicating with a selected lighting module of the at least one lighting module;
- receiving the unique module address of the selected lighting module from the selected lighting module; and
- providing instructions to encode the unique module address onto the data encoded power signal in order to reassign the selected lighting module to a selected lighting zone without wiring modifications to the lighting system.

9. The method of claim 8 further comprising storing, by the selected lighting module, the unique module address of the selected lighting module and the zone address of the selected lighting zone.

10. The method of claim 8, wherein the remote programming device comprises a smartphone executing one or more appropriate applications.

11. The method of claim 10 further comprising flashing, with at least one light emitting diode (LED), an address of the selected lighting module, and detecting, with an optical receiver, the address of the selected lighting module from the flashing at least one LED, wherein the selected lighting module includes the at least one LED and the smartphone includes the optical receiver.

12. The method of claim 11, wherein a rate of flashing of the at least one LED is less than approximately 30 Hz.

13. The method of claim **10** further comprising flashing, with at least one light emitting diode (LED), an address of the selected lighting module, and detecting, with a camera, the address of the selected lighting module from the flashing at least one LED, wherein the selected lighting module includes the at least one LED and the smartphone includes the camera.

14. The method of claim **10** further comprising reading, with a camera, a bar code encoded with the unique address of the selected lighting module, wherein the selected lighting module includes the bar code and the smartphone includes the camera.

15. A lighting system for reassigning lighting zones without wiring modifications, the lighting system comprising:

- a plurality of lighting modules, each lighting module of the plurality of lighting modules including a unique module address and one or more light emitting diodes (LEDs), each lighting zone including a unique zone address;
- a two-wire communication network configured to transmit a data encoded power signal to at least one lighting module of the plurality of lighting modules; and
- a remote programming device comprising a user interface and a processor responsive to the user interface, the processor configured to wirelessly communicate with a selected lighting module of the at least one lighting module and configured to receive the unique module address of the selected lighting module from the selected lighting module and to provide instructions to

encode the unique module address onto the data encoded power signal in order to reassign the selected lighting module to a selected lighting zone without the wiring modifications to the lighting system.

16. The lighting system of claim **15**, wherein the remote programming device further comprises a portable housing that houses the user interface and the processor.

17. The lighting system of claim **15**, wherein the remote programming device comprises a smartphone executing one or more appropriate applications.

18. The lighting system of claim **17**, wherein the selected lighting module includes at least one LED and the smartphone includes an optical receiver, the at least one LED flashing an address of the selected lighting module and the optical receiver detecting the address of the selected lighting module from the flashing at least one LED.

19. The lighting system of claim **17**, wherein the selected lighting module includes at least one LED and the smartphone includes a camera, the at least one LED flashing an address of the selected lighting module and the camera detecting the address of the selected lighting module from the flashing at least one LED.

20. The lighting system of claim **17**, wherein the selected lighting module includes a bar code and the smartphone includes a camera, the bar code encoded with the unique address of the selected lighting module and the camera reading the bar code to determine the unique address of the selected lighting module.

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